

Louis Vervoort

Problem Solving in Philosophy

How to Do Philosophy in the Age
of Ultra-Intelligent AI

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of Ultra-Intelligent AI

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The point of philosophy is to start with something so simple as not to seem worth stating, and to end with something so paradoxical that no one will believe it.

Bertrand Russell

Preface

This book starts with a provocative idea, namely, that the times when philosophers could claim that philosophical problems are too complex to be solved, are over. Artificial intelligence has reached, or will soon reach, such a level of performance that computers can excellently and in a matter of seconds solve philosophical problems—even problems that are vividly debated in the expert literature. This is not just a prophecy, but the outcome of experiments the author has done, notably in the context of causation. Identifying causes of events in somewhat complex situations is far from easy, as witnessed by the plethora of theories proposed in the professional literature; no overarching definition of the concept of cause exists. And yet advanced AI easily solves cases over which hundreds of the best experts debate¹.

Anyone who has followed, even from a distance, the rapid evolution of AI will surely agree that there is no reason why machines could not learn to answer philosophical questions in an entirely professional manner. Even if we just extrapolate from the way current artificial neural nets (ANNs) work—in essence, by looking for correlations in data—it is clear that they can, to the least, provide answers that are an excellent ‘common denominator’ of expert positions published in the literature—of which there are often so many that a human philosopher easily loses track. Thus, even if the functioning of Large Language Models—creating a planetary AI-hype recently—is not entirely transparent, we should assume that they are capable of proposing a (highly?, most?) probable solution to a vast array of philosophical problems. Especially, disruptive innovations in the functioning of ANNs and AI in general might lead to problem-solving capacities that are indistinguishable from, or outperform, those of the best human philosophers. There are solid reasons to believe that this is a highly likely evolution. To the point that one might wonder whether the profession of philosopher will keep any relevance, or whether it soon will be an occupation of a bygone age.

This book comes with an optimistic message. It argues that there is a way for philosophers to keep ahead of machine intelligence. I will argue that this virtuous

¹ Cf. Vervoort and Nikolaev (J. General Philosophy of Science 2025, published online, cf. <https://doi.org/10.1007/s10838-025-09735-7>), see also arXiv preprint <https://arxiv.org/abs/2506.14239>.

path takes us to a new method of doing philosophy, a method that can be aided by AI. To help this approach make its mark I will put it at the centre of a new philosophy, baptised ‘synthetic philosophy’, which I consider the natural prolongation of analytic philosophy. (I call the baby synthetic philosophy, because ‘synthetic analytic philosophy’ sounds decidedly bad.) In times of an AI explosion, *it seems clear that analysis alone does not suffice anymore*. In a nutshell, the main message of this book is that, if problem-solving is their goal, philosophers should strive to build synthetic theories, and derive solutions within such encompassing theories. Theory-building may well be the last skill in which we outperform machines². The really good news is that there is a semi-quantitative rule that allows us to know whether we are on track of the best theory and the best solution to a given problem. ‘Best solution’ should be understood here as the solution that solves the problems of preceding attempts. Needless to say, synthetic philosophy lies in the continuation of existing philosophical methods and schools; but it takes theory-building more seriously than is usually done.

As a side remark, initially I had subtitled this book “Introduction to synthetic philosophy”. But I decided to change the subtitle when in 2023 commercial Large Language Models made spectacular progress in a wide array of problem-solving tasks. However, I wish to be clear: this book is much more about using a synthetic methodology for problem-solving than about using AI. I claim that AI can, for the time being, function *as a great supporting tool, in a precise way to be specified*, not yet as the synthetic agent.

I believe there is a second important reason why problem-solving should receive wider attention from philosophers than is habitual. It is difficult to deny that human society presently faces problems of an unprecedented urgency: think only of the climate change and the inequality in income and capabilities among people worldwide, exacerbated by depleting natural resources. The analysis (say, of causes and remedies) of any societal problem beyond what science tells us brings us inevitably to philosophy, to ethics and practical philosophy in the first place. But in philosophy, as in scientific disciplines, all problems are related. This idea is nicely captured by Quine’s concept of the ‘web of belief’, the interconnected network of theories and disciplines in which revisions of one theory can have implications for all others, at least in principle (we will have much to say about this beautiful idea in this book). In sum, I believe that even the most practical problems have a strong link with more fundamental philosophical questions, for instance related to truth, knowledge, causation/causality, probability, free will, etc. —some of which will be studied in the next chapters. In other words, *many practical questions cannot be answered with good confidence if the theoretical ones are not satisfyingly answered first*. As an example, a theory of ethics that does not cohere with an adequate model of free will—a narrower and solvable problem, as I will argue in Chapters 6 and 7 —, cannot be the real deal. Therefore, not only practically oriented philosophers but also theorists can contribute

² Cf. Vervoort et al. (2021) “Artificial consciousness, superintelligence and ethics in robotics: how to get there?”, *Mind and Matter*, 19(1), 7–43.

to proposing solutions for the great challenges of our time: to basing solutions on solid grounds.

Sure, this is a decidedly optimistic vision of our profession, one that contemporary academics rarely proclaim, it seems, for a variety of reasons, one reason being, perhaps, the fear of sounding quixotic. But is sounding quixotic not an inevitable task of innovative philosophy? Another reason may be that philosophers have lost the synthetic drive, the conviction that it is possible to connect their highly specialised research to the greater web of knowledge. The overwhelming majority of publications in contemporary analytic philosophy deals with problems that are studied on the analytic micro-scale, detached from the broader view (surely for understandable and not always unjustified reasons). But exclusively focussing attention on micro-problems is leading philosophy astray from its vocation to be more immediately relevant for society at large. Again, this book comes with a message of hope: it claims that analytic philosophy is now mature enough for a natural next phase, i.e. for synthetic steps; that it is possible to integrate smaller-scale problems into a more synthetic whole. These synthetic, solid models and theories are then ready to be used in practical philosophy *and* in science. Of course, this final step is not the main focus of this book, which is dedicated to theoretical problems mainly. But in the Epilogue, Chapter 9, I will indicate, as an example, how a coherent account of free will can serve as a principle for practical ethics (the Epilogue is dedicated to more speculative problems). And throughout this book I will give examples of how fundamental philosophy could inspire and guide scientific research, in neuroscience (Chapter 7§7), computer science (Chapter 7§7), and physics (e.g. Chapter 9§1). One may even doubt whether some of the toughest problems of the foundations of physics can be solved without a good understanding of the philosophical concepts involved, notably those of cause, causation, and probability (more in Chapter 9§1).

Of course, philosophers will and should be eminently sceptical when hearing the intrepid claim that *philosophical problems can be solved*. But I will substantiate my claim by showing in detail how the method I recommend leads to answers to concrete and well-known open problems; I will argue that these solutions maximise an objective indicator of solidity. Thus, the readers will be able to judge for themselves whether the promise is met. (Of course, the ‘problems’ I consider in this book are tractable problems of analytic philosophy, not the questions that are sometimes called the ‘big’ or ‘ultimate’ ones, say about the existence of God, the meaning of life, the ultimate nature of the ultimate constituents of the universe, etc.)

This book is intended for a wide public; I hope to trigger the interest of expert philosophers but also of philosophical neophytes and scientists. Most chapters in the book will start with an introduction for beginners, where I have tried to phrase the problem under scrutiny in an as simple and clear way as possible. I do this for two reasons. First, to allow the newcomers to think for themselves. Few things are more pleasurable than creative thinking; and people who have independently thought about a problem are more apt to evaluate the soundness of a proposed solution. Second, because I believe that expressing problems in the simplest way possible is the best starting ground for solving them. This is an old wisdom, but it seems frequently neglected in the contemporary professional literature, which often uses jargon of an

extraordinary sophistication. I would like to emphasise: the philosophy I promote is one that uses transparent language, and that does not create new concepts when notions that are known in the wider intellectual community largely suffice. I believe that the task of philosophy is to make simple what seems complicated; certainly not the other way around. Furthermore, clarity should be conducive to remain relevant for society at large.

To put this book in context, an autobiographical note may be helpful. The method of synthetic philosophy is, in its core, the same as that of theoretical science, physics in particular. Before embarking on a Ph.D. in philosophy of science (at the Université de Montréal), I did a Ph.D. (at the Université d'Aix-Marseille) and a post-doc (at the École Normale Supérieure, Paris) in condensed matter physics. The reader is warned: in a few chapters I will speak with some confidence of the methods (and results) of physics. But this will not go beyond what presumably every physicist and philosopher of physics knows. So I will not add much to what philosophers of physics have excellently said about the methods of physics: my goal is to apply these methods of physics *and* of synthetic philosophy to concrete problems.

Switching disciplines instilled in me the habit to cross perspectives when doing research. When I studied philosophy in Montreal, I saw philosophical problems through the lens of physics, in particular from the perspective of the difficult—and still unanswered—questions raised by quantum physics. I saw connections between the two disciplines literally in every corner of philosophy. Just as I had, before, looked at physics through the lens of philosophy. Indeed, philosophy came first: in my Belgian high school, a Jesuit college, we spent months reading Plato and other philosophers. I found this exercise of crossing perspectives an invaluable source of inspiration. It will then not be a surprise that there is one more credo that I would like to transmit in this book: namely, that taking science seriously can be enormously helpful for addressing philosophical problems; *and* that there is a symmetrical benefit in using good philosophy to tackle scientific problems—of course, mainly problems of a foundational type, in say neuroscience or physics, such as about the nature of consciousness, of reasoning, of free will, problems of the interpretation of quantum mechanics, probability theory, relativity theory etc. I hope this book will corroborate at least the first part of my credo. (As said, in this philosophy book the readers will not find much evidence for the second part, philosophy helping science. But I will indicate a few lines of interdisciplinary research as examples.) Needless to say, I am not very original in my claim, many have uttered a similar expression of faith in interdisciplinary research. But, as far as I can judge, it seems not very often done, and I believe that the effectiveness of interdisciplinary research is still very much underrated.

Chapter 3 on Gettier's problem and Chapter 7 on free will are reworked and substantially extended versions of published articles³. I wish to thank here

³ Namely: L. Vervoort and T. Blusiewicz (2020), "The CMT Model of Free Will", in *Dialogue, the Canadian Philosophical Review*, Volume 59, Issue 3 pp. 415–435; and L. Vervoort and A. Shevchenko (2022). "Epistemic Relativism and the Gettier Problem: Insights from Philosophy of Science", in *Epistemology and Philosophy of Science*, Vol. 59, 1, p 58–81.

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About the Author

Louis Vervoort received a Ph.D. in philosophy of science at the Université de Montréal in 2013 and worked as a post-doc at the Université du Québec à Montréal, and since then as a professor in Russia, now at the Higher School of Economics (HSE) in Moscow. Before switching to philosophy, he obtained a Ph.D. in physics at the Université d'Aix-Marseille and did a post-doc at the École Normale Supérieure in Paris. This bi-disciplinary experience often incites him to study philosophical problems from a crossed philosophical-scientific perspective; he has the firm belief that there is a great potential of cross-fertilisation between these fields. Louis Vervoort has worked on classic problems of analytic philosophy, such as Gettier's problem, causation, the interpretation of probability, the problem of induction, free will and consciousness, and the interpretation of quantum physics—quantum nonlocality and Bell's theorem, notably. One of the key insights he would like to convey in this book is that these problems are related, and that a synthetic approach is needed to address them.

Abbreviations

CMT	Conscious-through-Monitoring-through-Theories*
def-an	definition of ambivalent neuron
DEF-C	definition of counterfactual cause
DEF-F	definition of functional cause
DEF-M	definition of manipulationist cause
def-mn	definition of multiple-neuron cause
DEF-P	definition of probabilistic cause
def-sn	definition of single-neuron cause
MCT	Maximally Coherent Theory

Chapter 1

The Method of Synthetic Philosophy. Extending the Web of Knowledge Through Problem Solving



Abstract In this chapter I describe the method of synthetic philosophy. Its key thesis is that problems are solved within the maximally coherent theory, or MCT: the theory that coherently answers the largest number of well-posed questions. This postulate provides a semi-quantitative criterion for deciding whether a philosophical problem is solved or not. I argue that the MCT can be hunted for and found by exploiting the continuity between science and philosophy. I briefly make the link with methodological credos of Quine and of Ladyman and Ross. The synthetic method can already be aided by AI, as illustrated in first case studies. Moreover, it is likely that future AI will work according to this method, in philosophy and beyond.

1.1 Introduction

Modern philosophy has a reputation of asking more questions than it answers, a tendency that has been enhanced since the rise of analytic philosophy in the twentieth century. Compared to the sciences, this situation is somewhat disheartening, since in the latter disciplines answers are found, and problems solved, at least provisionally; and the community usually *knows* when a problem is solved. In contrast, it rarely happens that professional philosophers agree that an important philosophical question is satisfyingly answered, or a problem solved. Of course, the great problems of philosophy are highly complex, for instance because they concern human beings, societies, and questions of ‘sense’. But is there really a final ground for such a profound epistemic difference between philosophy and natural science? Is there no way to curtail the explosion of questions, of questions about questions, of problems about theories, of worries about solutions?

One ground for the proliferation of unanswered questions in analytic philosophy is, I believe, the fact that ‘logical possibility’ is ever more frequently considered as a sufficient ground for deciding about ‘(ontological) reality’—where logical possibility is usually assessed by common-sense logic. If something can happen in a ‘possible world’, it is often considered a bona fide candidate for being real. But logic per se says very little about reality: it deals in the first place with correct reasoning. To

derive claims about reality—about *our* world, the one we study in this book, surely the one that many philosophers still consider worth studying—one can *use* logic and apply it to a set of hypotheses about the world (this is a well-known scheme, studied by authors as Hempel and Quine, to mention only them). The (correctly) derived statements about reality are then as certain as the hypotheses from which they start. Hence the more solid these starting assumptions are, say because they are in continuity with accepted science, the more likely it is that the derived conclusions are solid. If a minimalistic logic is used as the sole basis for evaluating ‘possible worlds’, virtually anything goes. But logical possibilities are vastly constrained by, notably, laws of nature. In sum, it seems unavoidable to use science as input if one wishes to regulate the eruption of possibilities that common sense and logic alone offer to the creative philosopher. Whenever possible, it should be science that anchors the web of knowledge into firm grounds.

I will argue in this book that philosophy is now mature enough to come to reasonably solid answers to key problems; and that there is one overarching method to do so, in essence the same method as the natural sciences use to obtain answers, in particular theoretical physics. I believe that this method is severely neglected in modern philosophy, resulting in an ever-meandering path of investigations that leads only in the best case to better-defined problems and partial answers, but rarely to solutions that allow genuine progress. This method could be called the method of ‘theory-building’ or the synthetic method, of which I will propose a pragmatic variant. Let me cast my belief in the form of a hypothesis or principle, which I call the MCT-hypothesis:

MCT-hypothesis: *Answers/solutions for philosophical questions/problems can only be given with reasonable certainty if they are coherent with, or derived within, the maximally coherent theory (MCT); and for many important philosophical problems this MCT can be constructed.*

If we understand, for a moment, the MCT simply as the ‘best possible’ theory (covering a certain problem-space), then most philosophers would presumably agree with the first part of the MCT-thesis: genuine solutions to problems agree with, follow from, the best theory about the problem-space of interest. But I suspect that many will remain sceptical as to whether this best theory can be constructed, as is claimed in the second part. In order to justify both parts of the thesis I will look in the following in some detail at how problems are solved in science. This should explain why I use the concept of ‘maximally coherent theory’; and also how such theories can be found *in a practical way*. In Appendix 1.1 examples are provided showing that current AI can already be used as a supporting tool for the synthetic method. It is to be expected that this option will become very tangible in the near future. Indeed, at the end of this chapter I will surmise that it is quite likely that, at some point, AI will be able to apply the synthetic method in its entirety and thus solve philosophical problems better than humans do. To the least, I claim we will need the proposed method to verify the solutions and decisions that the AI of the future will submit to us—if we wish to remain in control. This is far from a luxury: many of the risks associated with superhuman intelligence have already been studied (Bostrom, 2014). (Clearly,

readers who do not believe that AI will one day help to advance philosophy should not put this book aside: the synthetic method can be applied by humans without using AI—which is essentially what I did in this book.)

But first, a word on the notion of ‘theory’. ‘Theory’ should be understood here, in first instance, in the rather modest sense of a coherent set of assumptions/hypotheses/principles covering a (smaller or larger) problem-space, i.e. providing explanations in this space—not necessarily in the grander sense as one knows it for instance in physics. In physics, theories describing a whole sub-discipline, say thermodynamics or electrodynamics, are known and accepted; but this is rarely the case in philosophy, except in standard logic. Yet, if one accepts the MCT-hypothesis, there may exist, for instance, a sound ‘theory’ of probability or of free will (as investigated in Chapters 4, 6 and 7), while a full-blown theory of philosophy of science or of mind may still be lacking or debated. In sum, the method I will describe here focuses on the construction of theories/models/accounts that may refer to limited problem-spaces. Of course, as in physics, there is no *a priori* reason why ‘the’ MCT, the theory unifying all theories, could not exist *as a matter of principle*. Analogously, physicists do speculate about a future ‘Theory of Everything’—there exist even theorems about it.¹

In a scientific context, it suffices for our purposes to understand theories as hypothetical-deductive systems of propositions closed under deduction, ideally axiomatised, in any case based on a (small) number of base hypotheses, definitions, laws, principles, axioms (e.g. Hempel, 1965; Nagel, 1979; Popper, 1959). Paradigmatic examples are Newton’s mechanics and Euclid’s geometry.

1.2 The MCT-Thesis, First Round

The MCT-thesis can be considered as a heuristic rule for philosophical problem-solving. My goal now is to show that it can be transformed into a practical, workable methodology. It is a key assumption of this book, following Quine and many other philosophers, that science and philosophy are in an epistemic continuum; that their goals (roughly, the description of what is out there, of reality) and (therefore) also their methods only differ in degree (of precision), but not in kind. Thus, in order to understand the MCT-thesis it is obviously relevant to have a look at how problems are solved in science. The following analysis is well-known and has been done in detail by philosophers as Popper (1959), Hempel (1965), Nagel (1979), Quine (1951, 1960, 1969), Kitcher (1981, 1989), Schurz (1999) and several others. Here I will present a condensed and simplified synthesis, where I will pay special attention to the key role the notion of *coherence* plays in the scientific methodology, in particular in the process of theory-building.

In this book I will often take inspiration in physics. There are good reasons to do so. Physics is the most mature of the natural sciences, surely because it deals

¹ For instance, such a Theory-of-Everything should satisfy Bell’s theorem, on which more in Chapter 9.

with, in a sense, the simplest systems: inanimate systems of which the behaviour can often be reduced to that of one or a few elementary units (typically, particles). Even if this is certainly not always so, say in fluid mechanics or the quantum theory of condensed matter, which deal with aggregate systems, the corresponding theories are (historically, practically) derived from investigating simplified ‘point particles’, ‘elementary fluid elements’, ‘atoms’, etc. Philosophy often deals with incomparably more complex systems: for instance humans, which have a mind and associate in complex societies, of which the laws are only beginning to be discovered. Now, returning to our initial question, in physics and in other natural sciences it is clear when solutions to problems are accepted by the community: namely, when they are derived within, or in agreement with, the accepted theory. Mechanical questions are answered by mechanics, electromagnetic problems are solved within Maxwell’s theory of electromagnetism, etc.

It is important for my argument to note that accepted physical theories form a *coherent web*, roughly as Quine theorised it. There are concepts that occur in several sub-disciplines associated with different theories, and these concepts allow for a coherent overlap between disciplines/theories. This web is also coherent with logic and certain mathematical theories (but physical theories go beyond mathematics: they have to explain how the mathematics in the theories can be empirically understood and ‘applied’; the mathematical concepts need to be interpreted). At any rate, physicists do agree that there is no contradiction between their well-established sub-disciplines, say between optics, classical mechanics, electromagnetism, thermodynamics, statistical physics, and even quantum mechanics. Not only do these disciplines not contradict each other, there are concepts and principles that exist in more than one, or most, of them; in this sense no discipline is isolated—they form a web. For instance, notions as energy, velocity, momentum, position, time are used in most of the sub-fields, and the same holds for the principles of conservation of energy, of linear momentum, of angular momentum, etc. Therefore, if results involving these concepts and principles are empirically corroborated in one sub-field, other sub-disciplines also acquire indirect evidence. In a slogan, there is a dynamics of corroboration (or refutation!) that extends in principle over the whole coherent web, as Quine had it.²

² I will leave this notion of coherent web rather metaphorical here; we do not really need it, but it is still a handy metaphor (we do need the concept of coherence). It seems that this web can loosely be imagined as a dense network in which the nodes represent ‘knowledge units’. These knowledge units can be conceived of as sciences (including logic and mathematics); sub-disciplines of a science; or laws, principles within one sub-discipline. So the knowledge units can be imagined fine- or coarse-grained; still in other words, each of the nodes can be envisaged as a web on its own (down to the most fundamental principles). Thus, a coherent dynamics of corroboration extends over all of physics, and in principle even over all of the sciences, which all overlap with physics (As Quine has emphasised, collisions with reality, with experiment, can often be absorbed by modifying some of the nodes/hypotheses, without affecting the sub-field as a whole. The web then gets upgraded; so the web is dynamic. A dramatic collision would occur, for instance, if classical logic would appear not to adequately apply to reality anymore; then it seems that the whole web might need profound revision. But there is no convincing evidence for this possibility, also not in the light of quantum mechanics).

Of course, at the forefront of science, there are problems for which no full-blown theory yet exists, let alone a generally accepted one. Solving such problems would amount to a dynamic upgrading of the web. A case in point is the problem of describing the physics of the Big Bang and the origin of matter. In order to describe the Big Bang better than we can do now, it is widely accepted that one needs a ‘unification’, as it is called in physics, of quantum field theories and general relativity theory. In other words, *these theories need to be made coherent among each other*: at present, they are still in contradiction when applied to the Big Bang, for instance. Since there is no unified theory yet, many questions about the Big Bang cannot yet be answered with reasonable confidence. Even when there is not yet a full theory available, scientists proceed by extending existing theories by modifying only *some* of the key hypotheses (e.g. laws) of existing theories. Typically, this leads, at the forefront of science, to the proposal of several competing theories. The history and philosophy of science show that among competing frameworks, the ‘winning’ theory is the one that can unify, i.e. make coherent, the largest number of existing theories. In practice, *this is the theory that allows deriving the largest number of formerly accepted basic theories as special instances (i.e., instances valid in limiting conditions) of the new more general theory*. Returning to the previous example, one expects that the questions that are now yet unanswered about the Big Bang will be answered once quantum mechanics and general relativity are unified in a more encompassing theory of quantum gravity.

To make full sense of the MCT-hypothesis, then, the most useful idea we can learn from this little excursion in physics, is the following one. *The ‘best’ or ‘winning’ (physics) theory that is known at a given epoch of history, is the one that can adequately answer the largest number of questions, solve the most problems, in a maximally coherent way*—so in a manner that coheres, to start with, with logic and the relevant mathematics, but also with the other accepted theories. Indeed, note that if a theory can adequately answer the largest number of questions in its problem-space, then it should also cohere with a maximum number of other accepted theories (if it does not cohere with all accepted theories it must be a wrong theory and cannot correctly answer questions, old or new). So, we have found something useful; we have identified a pragmatic interpretation of what the maximally coherent theory, or MCT, is: *it is the theory that coherently answers the maximum number of (well-posed and meaningfully different) questions* in the problem-space under consideration.

Before returning to philosophy, let’s recall a few other points. Of course, the history of science has abundantly shown that the ‘best’ theory at a given time is bound to be replaced by a more powerful one at a later time; non-trivial theories of natural science are defeasible, provisional, not final (too high likelihood). We know that natural science theories, however precise, usually end up as approximations (only approximately true under certain conditions) of deeper and therefore more precise theories. But in a given period, scientists usually well know which theory is most powerful, most adequate for a given problem-space; it is always the one that

allows them to answer the largest number of questions in a coherent way.³ Even at the forefront of science, when a full theory for a complex set of interconnected problems does not yet exist, one considers the potential of hypotheses to answer questions as a sure guide for constructing the future theory.

A second point to consider is that scientific progress, understood as the construction of new efficient conceptual frameworks or, if one prefers, as the growth of the web of knowledge, does not always proceed via a discontinuous process—as when a full-blown theory is added to the web at one point in time. (Actually, the process is always gradual, but some cases appear more discontinuous than others.) The discontinuous case is illustrated by Einstein's construction of special relativity theory: the core of his theory was largely proposed 'in one piece' at a given time, anno 1905, and thus added as a finalised theory to the web. Usually, the body of scientific knowledge grows in a more continuous way, by gradually integrating concepts, laws, and models into the web, before a greater solidification is achieved by full-blown theory synthesis. A case in point is electrodynamics. Before Maxwell made the final theory (unifying optical and electromagnetic phenomena, and allowing new predictions, say the spectacular prediction of the speed of light) many local theories / sets of hypotheses had been proposed that crucially contributed to the final theory. Think of the discovery of the notions of electric current, magnetic field lines (by Faraday), the laws of Gauss, Faraday, Ampère, and many more. These more modest theories were vindicated a posteriori as sound; even before final synthesis, with them the web had greater explanatory power than without them. Although distinguishing gradual and discontinuous progress is somewhat artificial, it brings a useful idea to the fore: usually the web grows via incorporation of small-scale theories. I will argue in a moment that this is the case in philosophy, where full-blown theory synthesis seems exceptional.

Let us now return to the MCT-thesis as applied to philosophy. Of course, in philosophy 'generally accepted theories' are rare. But still, this concept admits of degree (there are semi-quantitative measures of acceptance, see further); and, especially, we have concepts that lie in a semantic continuum with 'theories', as already mentioned above. For many philosophical problems *proto-theories*, or at least *coherent sets of hypotheses* can be constructed that have explanatory power, that can solve problems, answer questions. Just as scientific explanation proceeds from laws (or theories) according to Hempel's celebrated scheme (1965), in philosophy explanation also proceeds from coherent sets of hypotheses. Now, the most valuable lesson to be drawn from the above comparison with physics is the construal of the MCT (covering a given topic) as the theory with the maximal (coherent) explanatory power, the theory that coherently solves the largest number of problems (related to the topic of interest). I claim that this construal provides us with a practical method for problem-solving in philosophy too. Sure, the 'maximal number of solved problems' is only

³ 'Coherent' here means: 1) if a generally accepted theory covering the problem-space exists: coherent within the relevant theory (i.e. logically deducible within the relevant theory, as in Hempel's 'covering law model' of explanation); 2) in any case, even if no generally accepted theory exists: coherent with logic and with mathematics if the theory in question uses mathematics; 3) in any case, coherent with other accepted theories.

a semi-quantitative notion; but it appears to be an excellent heuristic guide, as the following chapters will hopefully show. Indeed, for most important philosophical topics analytic philosophy has provided a quite well-structured cartography of the problems attached to these topics; these problems are approximately countable, at least in principle.⁴ I maintain that (synthetic) philosophy should take this analytic work by a large community seriously, and strive for constructing the synthetic theory or proto-theory that has the maximal explanatory power with respect to these often well-identified and published results; this theory is, by definition, the MCT. Synthetic philosophy solidly rests on past achievements.

In sum, it appears that *the soundness of philosophical hypotheses and theories has in principle a measurable, quantifiable character*. ‘Coherent’ in the above lines again means, a minima: 1) coherent with logic (and with mathematics if the theory / problem in question involves mathematics); and 2) coherent with established sciences, i.e. not contradicting them. As in physics, this then also means that the theory will be coherent with other maximally coherent theories for other, overlapping problem-spaces, i.e. problem-spaces that involve shared concepts and hypotheses.

At this point a note on the potential for fruitful bi-directional interaction philosophy—science seems in place. The requirement of strict coherence with natural science, and of using physics as the ultimate touchstone whenever possible, seems to imply that philosophy can never suggest fundamentally new ideas to science, ideas that would ultimately lead to fundamental revisions of it. But one should not forget that the web of knowledge is dynamic: it is, in (almost) all its nodes, provisional to all likelihood. Even theories of natural science with massive empirical support can be, and indeed were, replaced by better ones. Although this is a speculative matter, I believe that in some exceptional cases solid results of synthetic philosophy must be taken seriously as candidates not only for inspiring and grounding new lines of scientific research, but also for suggesting paradigm-changing programs; research programs that would imply, if successful, a fundamental revision of physics, for instance. This idea is briefly discussed at the end of Chapter 8. In the Epilogue, Chapter 9§1, I argue that the results obtained in Chapter 5 on causation warrant a different approach to the interpretation of quantum physics, and even to the unification of quantum mechanics and gravity. But this is speculative philosophy, i.e. philosophy that seems, at least at first sight, in tension with commonly accepted science theories and their immediate interpretations. (Therefore Chapter 9 goes beyond the main scope of this book, which is dedicated to solvable problems and tries to avoid speculation as much as possible.)

⁴ To the very least, I believe that on the basis of this theoretical number some approximate metric of explanatory efficacy should be decidable by the philosophical community.

1.3 The MCT-Thesis, Second Round. The Methodology in Somewhat More Detail

One may understand the MCT-thesis from a slightly different, corroborating perspective. We start from the (trivial) assumption that physics and philosophy share broadly the same objective: in short, to explain the world.⁵ The privileged systems studied in philosophy, humans, consist, ultimately, of physical building blocks. Alone these two facts, or assumptions, already hint to one of the essential theses of this book, namely that philosophy and science should cohere, form a coherent web of belief. Recently this idea has been promoted by Ladyman and Ross (2007) in the context of metaphysics and the ontology of the ultimate constituents of reality. I propose to apply it to philosophy as a whole. Many concepts are shared in physics and philosophy, say those of object, probability, space, time, law, cause, locality, etc. Of course, if the goals and objects of inquiry partly overlap, then it is to be expected that the methodologies also overlap. Every chapter of this book can be seen as an illustration but also as an (a posteriori) justification of this epistemic or methodological principle of the coherence between philosophy and science. Its a posteriori justification lies as usual in its problem-solving power, to be investigated in the subsequent chapters.

Now, the requirement of the coherence of the web is tantamount to the existence of strict logical criteria for synthetic philosophical theses to count as ‘sound’ or ‘(approximately) true’: to start with, they should not contradict logic and science, the most solid core of the web. But they should also not contradict other (proto-) theories, beyond science, which are gradually integrated into the web. As we saw above, the web usually grows in small steps. The first candidates to be integrated with reasonable confidence are, arguably, certain statements of philosophy of science: solid accounts of concepts as probability, which have a mathematical but necessarily also an interpretational aspect beyond the mathematics.⁶ In sum, philosophical theses can be part of the web; and must be part of it if they qualify as solid knowledge. In this book I will study problems that in one way or another resonate with physics, which makes it easier to decide about the solidity of proposed solutions. However, let me emphasise that even problems that seem a priori far from natural science, say ethical problems, may very well have a much stronger link than usually assumed. For instance, an account of free will should be part of ethics; and free will has a strong link with physics and neuroscience (Chapters 6, 7). Of course, this synthesis in a grand unified theory of philosophy seems particularly challenging and demands a much greater effort than attempted in this book. Still, for illustrative reasons I provide some very preliminary ideas, certainly not definite solutions, for how to start such a synthesis in Chapter 9, the Epilogue. I will in this book make a clear distinction between solutions that have been obtained by the synthetic method at full scale, and

⁵ Or, in the case of some philosophical disciplines, say ethics, to prescribe how a better world might look like.

⁶ Probability theory can describe, can be applied to, parts of the objective world (namely repeatable experiments or p-systems, as is argued in Chapter 4). To do so, one has to interpret the terms of the mathematical theory, which is a task of philosophy (of science).

those that remain more conjectural. The Epilogue deals with such conjectural theses that nevertheless strongly resonate with the results from the main chapters.

Thus, as many philosophers, I consider that the basic theses of scientific realism are part of the reliable web. It is an implicit assumption of natural science that there exists an objective, mind-independent physical reality; there is no evidence that the fundamental laws and properties that are described by our physical theories are the outcome of the activity of a (human or other) mind.⁷ That is why I adopt scientific realism (coupled with material realism, as usual, even if none of the claims made in this book depend on the material nature of physical reality). Anti-realist theories elicit vastly more questions than they answer—a sure indication that they are on an improbable track (more on scientific realism e.g. in Psillos, 1999). When necessary, I will say more about other background assumptions in the following chapters.

If synthetic philosophy's goal and methodology are comparable to those of natural science, then also the epistemic certainty that can be reached by it should be comparable; the difference is one in degree, not in kind. Thus, I submit that the solutions obtained by synthetic philosophy *at full scale* are comparable in certainty to solutions obtained in natural science; to the least, *they are the solutions on which further improvements can and should build*. Sure, philosophy investigates more complex systems, implying that formal theories, based on quantitative laws as in physics, are usually lacking. Hence predictions and confrontation with observation do normally not have a quantitatively precise character—but exceptions exist, e.g. many predictions derived from the interpretation of probability are quantitative. But to conclude from this that synthetic philosophy cannot be corroborated or refuted by empirical observations, is largely a fallacy, I believe. Let me start with a few examples coming from the following chapters. A sound interpretation of probability contains statements that are verifiable / falsifiable with numerical precision; it solves paradoxes of probability theory; it has empirical consequences for the interpretation of quantum mechanics and for the application of probability theory to real-world problems in general (cf. Chapter 4). A solid definition of the notion of cause can be tested in neuron diagrams (Chapter 5), and could, via the principle of causation, guide new explorative research in physics (Chapter 9). A sound definition of free will can partially be corroborated by neuroscience, and can perhaps even be used to devise new experiments (Chapter 7). Moreover, it could be integrated in a theory of ethics; which in turn could be assessed, in principle, by semi-quantitative measures probing for instance the overall wellbeing the theory 'generates' in a test group or society adopting it. True, this is a particularly ambitious research program that surely is difficult to implement, but doable in principle. In view of such examples, I would conclude that refutation of philosophical theses by empirical observations is surely

⁷ There are interesting philosophical and religious systems that counter this claim. Of course, we cannot exclude that they are right. But we can exclude that there is scientific evidence for such a creating mind. In this book we look at what science tells us, and how we can bring philosophy in harmony with it. Another caveat is this: when I say "laws are not created by a mind" I refer of course to the 'structures' or regularities of nature that are immanent in it or constitute it, not the *law statements* that are (approximate) descriptions of the objective laws (according to the scientific realism I adopt). Law statements are human creations.

much less immediate than in physics, but not impossible. In physics, in principle one falsified prediction suffices to overthrow the theory. But as Duhem, Quine and others have shown, this picture is quite simplistic: an accumulation of evidence is required. It appears that in philosophy even a larger, longer accumulation of conflicting empirical evidence is needed before the corresponding theories are revised—surely because the theories deal with much more complex systems. But the (a posteriori) explanatory efficacy of philosophical models *about real-world solvable problems* remains testable, even if it is in a subtler and more indirect manner. I hope that this thesis will be illustrated by examples given in this book.

The above comment concerns the subtle a posteriori justification of solutions via confrontation with reality or experiment. As argued above, the more practical method of a priori corroborating/justifying conclusions and solutions derived by synthetic philosophy is condensed in the MCT-hypothesis and resides in the establishment of the *theoretical* problem-solving capacity of these solutions. This is an a priori, theoretical method, in that it can be applied before one assesses the solution/theory by empirical research, i.e. by deriving and testing empirical consequences of the solution or the MCT in the problem-space of interest. (As recalled above, such a priori theoretical development of solutions also exists in physics, of course. Relativity theory is an example: it came before its experimental tests.) Note that the success of this justificatory process in philosophy hinges on the relevance and solidity of the questions and problems (usually studied in the literature) from which the synthesis starts. If our discipline makes any sense, it goes without saying that we should have a reasonable confidence that philosophers have studied meaningful problems as reported in the literature. And meaningful and well-posed problems (of synthetic philosophy) usually refer to reality, to this world, not only to logical possibility. I will often have the occasion to illustrate this claim in the following chapters, for instance those dealing with definitions, i.e. with models of what some useful concept precisely corresponds to in reality. In conclusion, both the starting problems and the solutions of synthetic philosophy should solidly be anchored in reality. There should be a way to confront synthetic theses with reality, even if this process is less straightforward, immediate, and precise than in physics.

In wrapping up, what then is, in a nutshell, synthetic philosophy? Synthetic philosophy is philosophy that aims at solving problems following the synthetic methodology, the core postulate of which is the MCT-thesis, stating that problems must be solved by the MCT in the problem-space of interest. This is a local MCT, not some MCT in a global sense, some theory that solves all problems of the book. (I did not show how such a global, unified, parsimonious MCT would look like—even if I suppose one might have an approximate idea about it by considering the various local MCTs.) In general, the local MCT is, by definition, the theory that provides the largest number of coherent answers in the problem-space of interest.

I am aware that this description of the synthetic methodology remains quite abstract. We can try to be a little more explicit about the tools one can use to find the MCT, as they follow from the above paragraphs; but the detailed process is problem-dependent and can only be made fully concrete in the following chapters. Note, then, that the definition of MCT provides two criteria, or tools, to find it: maximisation of

coherence and maximisation of explanatory power. As we saw, coherence is coherence with science and coherence among local MCTs. It is somewhat artificial to distinguish coherence and explanatory power, because both go hand in hand (from a theoretical point of view); but in the *heuristic process* of problem-solving synthetic philosophy can start with or privilege one tool rather than the other. The final criterion to judge the final result is coherent explanatory power; how one gets there is not essential and depends, ultimately, on the psychology of the inquiring synthetic philosopher. While Chapters 4, 5, 7 focus from the start on maximising explanatory power, essentially via unification, Chapters 2, 3, 6 mostly use coherency (with science and inter-MCT coherency) as the practical synthetic tool—which, in fine, should also lead to maximal coherent explanatory power. This criterion of coherency with science is sometimes expeditious for finding solutions. For instance, the outcome of Chapter 2 on the problem of induction is coherent with (the methodology of) science and can be seen as a direct application of the MCT-postulate, stating, among other things, that problems are solved within theories (hypothetical-deductive systems). Chapter 3 on Gettier's problem provides an interpretation that coheres with the MCT-postulate and with all other solutions of the book (which in turn cohere with natural science). Chapter 6 almost directly draws on physics to propose a solution.

On the other hand, Chapters 4, 5, 7 focus on maximising explanatory power from the start; coherency with science does not immediately suffice for inspiring solutions. Thus, for the problems of defining probability (Chapter 4), cause (Chapter 5), and relative free will (Chapter 7) the MCT is the model/theory that unifies the existing competing accounts for the problem considered, which then should be special (limiting) cases of the MCT; at least, that is the intended result. Maximal explanatory power can then rather straightforwardly be demonstrated. Note that for the topics of Chapters 4, 5, 7, coherency with science seems less immediate as a problem-solving tool, since they are somewhat farther away from it; but for these topics several quite solid and well-documented theories exist that clearly have explanatory power—but also a series of unsolved problems. But again, the distinction between focusing on coherence or on explanatory power as heuristic tool is somewhat artificial; both work organically at the same time. For instance, when attempting to make a synthesis between existing theories, coherence with science and logic should obviously be a permanent guiding rule, eliminating certain accounts from the start.

Perhaps the most detailed (and maybe surprising) illustration of the synthetic methodology is given in Chapter 5, where we look for a unified definition of cause. Most experts believe that an overarching definition does not exist. I will show in detail how the MCT-method allows us to gradually solve this problem, and how the most likely MCT is built in a step-by-step process. I will start by investigating one problem-space, namely mundane causal situations represented by classic neuron diagrams. These are covered in the literature by counterfactual accounts of different guises. The goal of the first step is to derive an overarching definition for this space. This is done by looking at as many problems as possible, which have handily been compiled in a reference work by Paul and Hall (2013); they all are represented by diagrams (cf. Table 1 in Chapter 5). For synthesis, it is important to consider a large number of cases. Next, I will look at an overlapping problem-space, namely the

identification of causes in physics, which is usually handled by the functional and probabilistic accounts of causation. This analysis will reveal an essential principle of causal identification, which I will term the ‘representation-relativity of causal claims’, allowing us to solve several problems of the second space. Once this principle is found, the unification of the physical and counterfactual accounts of causation will be almost immediate. The resulting higher-order account leads to a solution for a surprisingly wide array of problems encountered by classic, less synthetic counterfactual accounts, e.g. the well-studied problems related to early and late preemption, double prevention, overdetermination, transitivity, token-versus-type causation, black-box cases, etc. In the last step of unification, I will show how also the popular manipulationist account of causation can be subsumed by the final model, solving again more problems. Thus the chapter on causation offers a detailed illustration of how synthetic philosophy works, in the step-wise manner that one also finds in natural science. (Clearly, the fact that the final model applies to physics increases the likelihood that it represents the MCT; even if the bulk of the arguments leading to the model comes from philosophy.) In other chapters, e.g. Chapter 4 on probability, I will propose a synthetic account from the start, and then show how it solves problems and subsumes other accounts.

A disclaimer: I am of course well aware that the methodology for problem solving that I propose—even if based on known ideas—is not always straightforward to apply; that I have not produced here a detailed algorithm that allows one to generate answers quasi-automatically. But, then, such a thing does not exist in science either: there is no *detailed* theory for theory-building beyond the general strategy succinctly recalled above. Revolutions in science, the revolutionary theories that ultimately get accepted by the community, are made by their creative genitors in a heuristic-psychological process that remains largely mysterious. But one thing seems sure: once made, these theories unify, synthesise, and by the same token answer more questions than their competitors do.

Rather than further scrutinising the MCT-thesis and justifying it a priori, it seems clearly more instrumental to show how it works in practice. This is the goal of the next chapters. Especially the second part of the MCT-hypothesis, the claim that for many problems the maximally coherent solution and theory can be found, needs to be shown by construction.

1.4 A Note on AI and the Future of Philosophy

An essential feature of the synthetic method is that it should lend itself to fruitful interaction with AI—if not now already, then in the immediate future. Quite generally, this method starts from dressing a cartography of problems related to a philosophical topic. Certain types of AI, notably Large Language Models (LLMs), are already instrumental in making such a cartography, a tendency that should be enhanced in the future. Appendix 1.1 illustrates how a commercial LLM (a ‘chatbot’) can assist in defining an overarching definition of the concept of cause, which is the topic of

Chapter 5, by listing problems encountered by different accounts of causation. For what it is worth, I also asked the chatbot to propose a ‘best’ account, the one with the widest applicability. It chose the candidate that comes out as my winning candidate in Chapter 5. Sure, the list of problems the chatbot proposed (see Fig. 1.1 of the Appendix) is not necessarily the most complete one as the best experts would compose it, but it is nevertheless a quite solid starting point. Of course, the AI does not *yet* make human philosophical expertise redundant when it comes to synthesising *new* solutions (for instance, my version of the winning account comes with essential modifications). But exercises as these show that the prospects of AI-aided philosophical research are very real, which is also the conclusion of a deeper-digging study (Vervoort & Nikolaev, 2025).

This seems not surprising. First, AI has the capacity to store and process enormous amounts of data and texts, whereas humans are clearly limited here. Next, artificial neural nets’ core capability is to hunt for correlations, for similarities—which is the first step towards synthesis; and in the context of philosophical research, problem-solving is largely a matter of synthesis. More concretely, since LLMs work essentially as statistical text completion engines, they should be capable of finding the most probable answer to many questions—when these are discussed in enough texts and do not require overly complex step-by-step reasoning (cf. e.g. Vervoort et al., 2023). ‘Most probable answer’ should not be understood in an absolute sense, of course, but relative to the enormous corpus of (philosophical) texts the models received as training input.

It is only natural to assume that, once AI will have become a genuine problem-solving tool also at the forefront of philosophical research, it will then be capable of applying or emulating the synthetic method—somehow. Of course, that seems not what LLMs are doing for the time being (Arkoudas, 2023; Vervoort et al., 2023). One should assume that what they propose as solution to a problem is the statistically most likely (or highly likely) answer as it is extracted from published text—not a creatively generated new solution. But, then, creativity surely also works according to rules, and not haphazardly. Given the current explosive evolution of AI, I believe it is very likely that machines will implement or emulate such rules in the near future (for first ideas along these lines, see Vervoort et al., 2021; for evidence of AI-creativity in mathematics and other fields, see e.g. Bubeck et al., 2023).

Therefore, we must assume that, for better or for worse, the first artificial synthetic philosopher is soon to appear in our midst. If only to assess the output of such engines and remain in control, we will need very solid synthetic theories in all relevant problem spaces. In sum, the future of synthetic philosophy should be bright.

1.5 A Few Related Views

Several philosophers have advocated using natural science and in particular physics as a sure companion for philosophy; from the long list I will here only mention Quine (1951, 1960, 1969, 1991) and recently Ladyman and Ross (2007). Clearly, I follow

Quine's credo of the continuity between science and philosophy, and his claim that philosophy is constrained by scientific standards of clarity, evidence, and explanatory relevance (Hylton & Kemp, 2023, § 2, § 7; for a detailed analysis of Quine's developing views, see Verhaegh, 2018). Yet my approach differs from Quine's philosophy in several respects. First, Quine adopts a strongly naturalistic conception of epistemology as a chapter of natural science alone (e.g. Quine, 1969, p. 83, Verhaegh, 2018, Ch. 2). In my view such a chapter may of course be sought for and can surely be interesting, but there are important philosophical matters beyond it, as illustrated in Chapter 3 on Gettier's problem. I do not consider epistemology to be a part of natural science; both problem-spaces are meaningful and distinct even if connected. Actually, it will be seen in several of the following chapters that for many a problem it is essential to make the distinction between science and philosophy of science. *Making the distinction is the beginning of the solution* (see notably Chapters 2, 7, and 8). Next, Quine rejects the concepts of 'meaning' and 'knowledge' essentially on the basis of their being vague (Quine, 1987, pp. 108–110, Hylton & Kemp, 2023, § 2.2). I agree that these notions are vague, but again, I believe epistemology has a lot to contribute to the meaning of 'to know' and the essence of knowledge (Chapter 3). Philosophy has its reason d'être as a discipline that is clearly distinct from science, even if in coherence; it allows us to solve meaningful problems. As another example, Quine rejects the 'metaphysical' notion of necessity (Quine, 1991, p. 270). Necessity plays an important role in my philosophy: see e.g. Chapters 5 on cause and 9 on the principle of causation. Finally, there are several touchy topics in Quine's philosophy, say his view on the analytic-synthetic divide, or the worries he had regarding belief statements and counterfactual conditionals (e.g. Hylton & Kemp, 2023, §5.2): these can be analysed and dissolved by the perspectival account elaborated in Chapter 3. (I will not do the latter analyses in any detail in this book; they are a straightforward application of the account developed in Chapter 3.)

In a recent thought-provoking book, Ladyman and Ross (2007) apply the scientific paradigm to philosophy; they focus their efforts on philosophy of science and 'naturalised metaphysics' or 'metaphysics done as naturalistic philosophy of science' (ibid. p. 7). They investigate, notably, the ultimate nature of reality and the unification of the special sciences with physics. Several of their claims resonate well with some of those I will present in this book, but I will not study problems of ontology in the strict sense (even if Chapters 5 and 9 on causation are at the interface of ontology and philosophy of science), but rather from philosophy of science (Chapters 2, 4, 5, 9), epistemology (Chapter 3), philosophy of mind (Chapters 6, 7). Thus, in comparison, I conceive of the objective to base philosophy on science more generally: *I believe philosophy in its entirety should agree with science in a maximally coherent way*. Via the MCT-postulate, this is a quite pragmatic stipulation that works as a guideline for constructing a scientific philosophy for, in principle, all branches of philosophy⁸

⁸ Ladyman and Ross state (ibid., p. 28): "Metaphysics, as we will understand it here, is the enterprise of critically elucidating consilience networks across the sciences." I see an analogous role for synthetic philosophy, namely, to paraphrase, as the enterprise of elucidating coherence networks across the sciences *and the classic branches of philosophy*.

(except perhaps logic). I further wholeheartedly agree that philosophy should aim at unification and hence that “generation of arguments for disunity is not naturalistic metaphysical inquiry” (ibid. p. 5). Arguments for disunity are largely predominant in current philosophy of causation and probability, to name only these.

As said above, I will assume the usual non-sophisticated theses of scientific realism (see e.g. the introductions of Psillos, 1999; Chakravartty, 2007), as surely many philosophers of science and most scientists do. In Chapter 3 I will show that my version of scientific realism has connections with the ‘scientific perspectivism’ of authors as Giere (2006) and Massimi (2018). But I would like to emphasise that for addressing the problems studied in this book, I do not need to delve into the extraordinarily complex question of the ultimate constituents of nature or the universe. Obvious candidates are elementary particles, quantum fields, force fields or a combination of these. The appropriate reflex would be to look at what the experts of our most fundamental physics theory, quantum field theory, have to say about this problem. But at the present stage of physical theory, this problem is undecidable, as even experts admit (Kuhlmann, 2023, §5.5). Ladyman and Ross advocate a version of scientific realism called ‘ontic structural realism’, which holds that only structure and relations exist, rejecting material objects as fundamental. Now, even if this vivid debate were resolved (despite its many variants and issues, cf. Ladyman, 2023), it wouldn’t affect my answers to the classic philosophical problems explored in this book: these are independent of the debate over ultimate reality.⁹ In sum, I remain largely agnostic on the question of fundamental ontology, a matter that seems undecidable at present.¹⁰ But this local agnosticism is of course not an obstacle to formulating answers to classic problems, answers that are reasonably certain, perhaps not in some unrealistic ultimate sense, but answers that make progress and should be a safe basis for a next cycle of synthesis. Other differences with Ladyman and Ross (2007) concern the principle of causation (more in Chapter 9).

1.6 A Swift Flight Over the Remaining Chapters

Before concluding this chapter, let me present a brief overview of the remaining chapters (this section can be skipped by readers who prefer their sense of suspense not to be spoiled). Chapter 2 is a very short and straightforward illustration of how the MCT-thesis can sometimes almost immediately solve classic problems of philosophy, in this case the problem of induction. In Chapter 3 I will show that insights from epistemology intimately cohere with the MCT-hypothesis and that knowledge, the object of epistemology, must inextricably be linked to sets of hypotheses (theories).

⁹ As long as we can remain in the paradigm of scientific realism.

¹⁰ Also for this problem, the synthetic method is in principle applicable. But applying this method seems very difficult, notably because the cartography of interwoven and sometimes ill-defined problems is very complex; and, especially, because physics does not seem to privilege any of the possible answers to many or most of these problems.

The MCT-thesis and these fundamental insights from epistemology corroborate each other, form a coherent web. Logically Chapter 3 is prior to Chapter 2, because the former lends credit to my initial hypothesis, the MCT-thesis; Chapter 2 is an immediate application of that hypothesis. But I prefer to start with a short and easy chapter, to present the method in a gradual manner to the reader.

In somewhat more detail, Chapter 2 applies the MCT-hypothesis to Hume's problem of induction, sometimes called "one of the most profound philosophical challenges imaginable" (Henderson, 2020). This problem has been studied by many renowned philosophers, including Russell, Mill, Reichenbach, Carnap, Quine, and many more. I have already mentioned that the MCT-hypothesis or methodology is related to Quine's holism. The latter view can quite precisely be re-interpreted as stating that *questions can only adequately be answered, and problems solved, by whole theories*—a claim made more precise by the MCT-thesis (cf. Chapter 3). My account is also related to Aristotle's philosophy of science, at least if we interpret it somewhat charitably (for a detailed critical analysis of Aristotle's philosophy of science, cf. van Fraassen, 1980). Aristotle seems to have fore-felt that any solid knowledge, scientific or philosophical, must be based on unprovable 'first principles', as he called them, which constitute the basis of a scientific (sub-)discipline. In modern terms, these first principles are the axioms, postulates, basic laws, or basic hypotheses of theories. If we accept this, and we must accept something, we almost immediately solve the infamous problem of induction.

If solid knowledge is inextricably linked to theories, more generally sets of hypotheses, one expects that epistemology, the theory of knowledge, should cohere with the MCT-thesis. One expects a fundamental and ubiquitous manifestation of the MCT-thesis in epistemology. Now, there is indeed compelling evidence from these quarters that scientific and philosophical knowledge are embedded in theories *in a fundamental way*.¹¹ I show in Chapter 3 that one of the key problems of modern epistemology, Gettier's problem, is solved when one considers knowledge (as well as justification and truth) as being relative to sets of hypotheses, in a sense to be specified. This chapter will lead us to conclusions that can conveniently be summarised by simple and short phrases as: *one can know something only if assuming some theory*—theory in the broad sense specified above, so including, more mundanely, coherent sets of hypotheses. In a slogan: *knowing needs assuming (to start with)*. And something is true only relative to, within a set of hypotheses, a theory—a theory which can, as a whole, be a more or less adequate representation of reality. Against appearances, this position can easily be married to absolutist or objectivist views about truth and knowledge, be it in a slightly subtle way. Similarly, justification is justification within a theory, relative to a theory. Without such assumptions, pointing to a type of epistemic relativism and a mixed coherence-correspondence theory of truth, Gettier's conundrum remains with us; with these assumptions, it becomes transparent.

¹¹ Note that this claim is not in contradiction with the idea that empirical observation and experimentation are essential for the discovery and construction of theories of natural science and philosophy.

I suspect that at this point some philosophers will raise their eyebrows. Some might find this a little simplistic. Those I would remind of Russell's motto stated at the beginning of this book: we have to start with something simple, and then hopefully transform it into something less simple. All subsequent chapters heavily draw on insights gained in Chapter 3. Others might object: why would (the truth of) banal 'observation sentences', such as "The neighbour's dog just bit me", be relative to a theory? As mentioned above, it is essential to keep in mind that I will use in this book the notion of 'theory' often in a broad sense: it will, depending on the context, also refer to (coherent) sets of hypotheses, belief systems, coherent bodies of information (which may refer to sense data), etc. In this sense it comes close to the more informal 'perspective' or 'representation'. In several of the following chapters (notably in Chapters 3, 5 and 7), in order to make the distinction between this looser concept and that of scientific theory in the strict sense (T), I will often use 'theory*' or the symbol 'T*' for the broader concept. Then, it will be argued in Chapter 3 that even the truth value, and hence the knowing, of a sentence as "The neighbour's dog just bit me", depends on some—indeed rather trivial—set of hypotheses T*, such as that the sentence-utterer is not deceived, that the animal that bit him classifies as a dog according to biology (not as a wolf), etc. Philosophers of science will recognise here the well-known 'theory-ladenness' of observation sentences. Of course, in such mundane cases it is usually not necessary to specify the hypotheses T* one assumes; but in sophisticated cases it is. It is reassuring that perhaps the most paradigmatic topic of modern epistemology coheres so well with the MCT-hypothesis.

Chapters 4 to 7 concern definitions, namely of the concepts of probability, cause, and free will. Definitions of metaphysics, philosophy of science, philosophy of mind, etc. can fulfil the role of basic hypotheses of a philosophical theory, or of a coherent proto-theory. They are of the type "X is Y" (e.g. "free will is Y"), where (i) Y describes X in sufficient detail, sufficient for explanatory use, and where (ii) X is (implicitly) assumed to be something of importance, of interest, *something that exists in reality*; or to be a concept of a theory (say metaphysics or the interpretation of probability) that usefully describes or analyses reality. Such a definition can be the basis of a genuine theory about X; which, if it is a good theory, should be maximally coherent. Thus, I agree with almost all analytic philosophers that defining concepts is a highly useful task of philosophy. And indeed philosophy has done a fine job, not only in defining concepts, but also in identifying a great number of problems for existing definitions and accounts based on these definitions. This is precisely the fertile ground needed to start the next task, that of constructing definitions/theories, 'theory-definitions' if preferred, that can solve ideally all existing problems—in practice, the most relevant ones published in the literature.

This is the approach I have followed for proposing the definitions in Chapters 4, 5 and 7. Regarding probability, there is a vast literature devoted to understanding this notoriously subtle notion, beyond the relatively simple mathematics. The mathematical concept of probability can be applied, highly successfully, to real-world problems; hence one needs to interpret the mathematical symbols in order to use the calculus correctly. Here is where the problems start, as witnessed for instance by the

vast number of paradoxes of probability theory, with which even famous mathematicians have struggled.¹² These paradoxes arise because a precise definition of probability (one that interprets Kolmogorov's axioms) appears to be surprisingly subtle: there are implicit notions linked to it, and one needs to be aware of these if one wishes to solve tricky problems. In Chapter 4 I will propose a definition that makes these implicit notions explicit. This definition can be seen as an elaboration of von Mises' frequency interpretation of probability (1928/1981, 1964). Alas, the frequency interpretation seems to have become a minority position in the philosophical community (e.g. Hayek, 2019). This is regrettable, since the other interpretations, such as the classical or logical or subjective interpretations, do not comply with the MCT-thesis: they cannot solve problems in a maximally coherent manner. In Chapter 4 I propose an account that unifies the original frequency interpretation, the classical interpretation, the propensity interpretation, and certain consistent versions of the subjective interpretation. They all can be seen as special cases of the proposed model.

Regarding the notion of cause the situation is as complex as for probability. Libraries are filled with works on causality/causation; again very many interpretations are on the market. In metaphysics Lewis's counterfactual theory is popular (1973). In philosophy of science the manipulability account (e.g. Woodward, 2003) or the probabilistic theory of causation (e.g. Suppes, 1970) are favoured—and usually believed to be incompatible with other accounts, Lewis's in particular. Hence, just as for probability, it is generally believed that one overarching definition of cause is impossible. But this is a puzzling thought: in everyday speech people usually know well how to use the concept—and indeed common-sense intuition is always the touchstone for judging the adequacy of a theoretical account of cause –; but at the same time philosophy would not be able to provide an overarching definition. This sounds dubious, and indeed synthetic philosophy comes to a different conclusion (Chapter 5).

Chapters 6 and 7 are dedicated to another classic of philosophy, namely the problem of free will: what is free will, and do we have it? The debate between libertarians, determinists, compatibilists and many other schools goes on since millennia; almost all philosophers of renown have expressed their opinion on the problem. Clearly, whether something exists depends on how one defines it; once a definition is given the existence is decided by the best available theory. Hence, in principle the free will question can be studied from different perspectives. In order to understand the enormous range of questions and positions this problem elicits, I will argue that it is best studied from two perspectives: the first based on physics and its philosophy (Chapter 6), the second based on (all) other cognitive fields, notably psychology, sociology, and the attached theories of practical philosophy such as ethics, philosophy of law, etc. (Chapter 7). These two perspectives vehemently clash. The first one must be expected to be the most solid view since it derives almost immediately from physics (not from the assumption of determinism, as many believe). It leads to the staggering conclusion that we do not have free will, at least not in the absolute sense

¹² It is sometimes said that probability theory is the branch of mathematics against which it is easiest to make mistakes (Tijms 2004).

in which it is usually defined. However, I will also argue (Chapter 7) that it is useful to withhold a modified, relative notion of free will—if only to be able to engage in discussions with... most people. Having free will, in this relative sense, is a necessary condition for having moral responsibility and agency, which play such a preeminent role in ethics, legal frameworks, philosophy of wellbeing etc. Hence, to jettison the concept of free will seems risky. However, I cannot present a full-blown synthesis of both perspectives. In other words, although the definition of relative free will given in Chapter 7 solves many, and perhaps a majority, of the published problems, it surely is subject to further improvement. Needless to say, just as in science, any position in synthetic philosophy is perfectible. Until a full-blown MCT is proposed, I submit that the two basic perspectives are complementary: it depends on the context which one is best adopted.

Chapter 8 is the general conclusion, resuming some key findings. Finally, in the Epilogue, Chapter 9, I will study more conjectural theses, not obtained by the synthetic method at full scale: lines of further research that seem particularly interesting to me, and that can be based on the results derived in the main chapters. These lines of research concern practical implications of my account of free will, and the principle of causation.

1.7 Conclusion

In this chapter I have proposed a methodology leading to a criterion for judging the quality/adequacy/soundness of philosophical explanations and solutions, a criterion that is measurable, at least in theory. The rationale is clear by now: answers are adequate if they follow from the MCT (covering the relevant problem-space); and identifying the MCT is based on counting the problems that it can coherently solve. This method, which hinges on ideas that are well alive in philosophy of science, seems transparent and pragmatic—something I will try to substantiate in the following. Of course, in practice, in some cases deciding about the explanatory power of a theory is not always entirely obvious, essentially because the cartography and counting of the covered problems are not always entirely obvious; but at least we have a guideline and are not moving blindly. At the same time, I am aware that the proof of the pudding is in the eating: that I will need to illustrate and support the approach by case studies.

Notwithstanding the straightforwardness and transparency of the method, it would be naïve to believe that it will be welcomed with open arms in all quarters. I am aware that the synthetic method goes against the predominant habit of studying and discussing problems in hyper-specialised communities, often focussing on problems of purely analytic nature, and rarely attempting to make the link with other related topics (let alone society and science). This modern habit surely comes with extra-scientific benefits. But it obstructs the verification of the verisimilitude of the theses proposed—the correspondence to reality that seems often of secondary concern (a problem that does not only befall philosophy). Furthermore, the proposed method is onerous: it requires the study of large ensembles of problems and theories, and,

especially, their subsequent synthesis or unification—an activity that seems largely expelled from contemporary analytic philosophy. I believe that this lack in system-approach, in theory-based thinking, is the greatest hiatus in current philosophy. I believe it is a ground for stagnation, and potentially for a loss of credibility of the discipline in the eyes of many outsiders. And the problem is pervasive and multifarious. Many authors surely will have encountered reviewers of peer-reviewed journals rejecting a model only because one of its starting hypotheses does not match the beliefs of the reviewer—even if the model solves more problems than the competing accounts. I cannot emphasise enough: the ultimate judge of a new theory should be its explanatory power, not the *a priori* plausibility of its premises *per se*. A fine analytic job is to identify errors in reasoning starting from a theory's premises; but not to reject bona-fide premises from the start, even if these are *subjectively* counter-intuitive (an idea also stressed by Ladyman & Ross, 2007). Intuitive plausibility is in the eye of the beholder. Explanatory power is much more objective, since it is ultimately a quantitative criterion. A corollary is that it makes little sense to question and scrutinise ad nauseam the premises of a theory 'on their own', if they lead to a better theory.

Appendix 1.1. Case Study: AI as a Tool for Synthetic Philosophy

In synthetic philosophy, in order to build the solution for problem X, it is essential to keep an overview of (in principle all) the problems faced by accounts that aim at addressing problem X. Normally such an overview is the starting platform from which the creative task of synthesis proceeds. It seems clear that certain types of AI can assist us in this effort, at least for dressing the initial cartography of problems. I tested this idea using a currently popular LLM, ChatGPT, based on GPT-4 technology,¹³ for a small number of problems. I will here only present a few illustrative examples, a mini case study. A *systematic* investigation of a specific AI's problem-solving skills in philosophy will, I believe, soon become interesting, but goes beyond the examples shown here. A more detailed case study is given in Vervoort and Nikolaev (2025), focusing on causation in neuron diagrams.

For instance, I asked the chatbot to provide a list of problems faced by existing accounts of causation. Figure 1.1 displays my question (Q1) and the answer ChatGPT gave. One will note that, although neither perfect nor complete, the vast majority of listed problems are quite sensible and identify key concerns for the respective accounts of causation. I believe this is not a surprise, in view of how these

¹³ From the company OpenAI (<https://openai.com/>). ChatGPT, based on the Generative Pre-trained Transformer (GPT) language model, is available in different versions (GPT-3.5, GPT-4 and several others) (cf. OpenAI 2023). The experiment, done in December 2024, does not aim to be systematic; also, more powerful models are bound to be released. For some insights on the working of LLMs, I recommend a publication by the company DeepSeek (Guo et al., 2025). Finally, more complete lists of problems can be obtained by interacting with LLMs in a dialogue about specific accounts.

language models function: they should be excellent in summarising problems that are frequently discussed in published texts.

Q1> In philosophy, the concept of 'cause' is highly debated. Philosophers generally doubt that there is an overarching definition of the notion of cause, one that can be used in all contexts and disciplines. Instead, there are different accounts of causation, such as the counterfactual account, the manipulationist account, etc. However, all these accounts are debated and come with a series of problems. My questions is: Can you make a complete list of problems that these accounts encounter?

GPT-4> Certainly! Below is a list of problems commonly associated with various accounts of causation, organized by the account type:

1. Counterfactual Accounts of Causation

- **Problem of Preemption:** Counterfactual accounts struggle with cases of preemption, where two potential causes exist, but one preempts the other. For example, a backup cause might not occur because the primary cause takes effect, yet the backup cause still seems relevant.
- **Problem of Overdetermination:** If two independent causes are sufficient to produce an effect (e.g., two people simultaneously pushing a car), the counterfactual account has difficulty explaining which cause is primary or whether both equally qualify.
- **Problem of Non-Actual Scenarios:** Counterfactuals require evaluating non-actual, hypothetical scenarios, which can be unclear or indeterminate. Determining the "nearest possible world" can be subjective and context-dependent.
- **Fragility of Events:** Some versions assume that small changes to an event's context make it impossible to determine causation, leading to inconsistencies when the "smallest change" is unclear.
- **Asymmetry of Causation:** Counterfactuals often assume causation is asymmetric (cause precedes effect), but they sometimes fail to adequately explain why the counterfactual dependence is directional.

2. Manipulationist (or Interventionist) Accounts

- **Circularity Problem:** Manipulationist accounts often define causation in terms of interventions, but interventions themselves presuppose causal relationships, potentially leading to circular reasoning.
- **Scope Limitation:** The account works best in scientific and experimental contexts but struggles to apply to historical, metaphysical, or everyday causal claims where interventions are impossible.
- **Problem of Complex Systems:** In systems with numerous interconnected variables, distinguishing a specific intervention as the "cause" is often impractical.
- **Non-Intervenable Causes:** There are causes that cannot be intervened upon (e.g., laws of nature or events in the distant past), yet they are still causally significant.
- **Causal Relativity:** The notion of causation becomes relative to the particular variables and interventions chosen, raising concerns about whether causation is objective or merely pragmatic.

3. Regularity Accounts (Humean Causation)

- **Problem of Singular Causation:** Regularity accounts explain causation in terms of patterns or laws, but they fail to account for one-off causal relations (e.g., a meteor causing a crater).
- **Distinction Between Correlation and Causation:** Regularity accounts often conflate correlation with causation, struggling to explain why one event brings about another rather than merely accompanying it.
- **Epistemic Problem:** Regularity accounts depend on observing patterns over time, but this does not explain causation in cases where patterns cannot yet be observed.
- **Problem of Necessity:** Regularity accounts do not capture the intuitive sense of necessity that underpins causation (i.e., why the cause *must* bring about the effect).

Fig. 1.1 Dialogue with ChatGPT about causation and its problems (showing the literally transcribed text produced by ChatGPT-4)

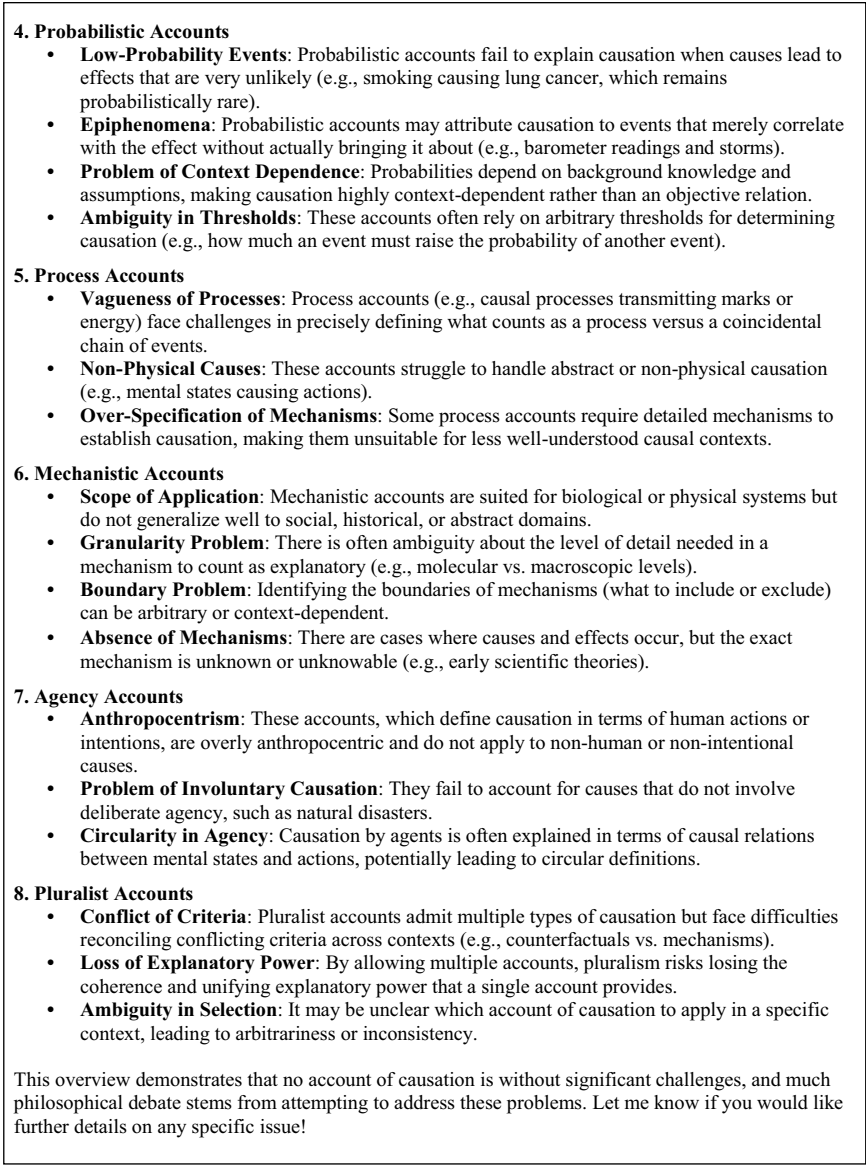


Fig. 1.1 (continued)

Immediately after ChatGPT gave its answer to **Q1** (in seconds), I asked it to pick the ‘best’ account (**Q2**): see details in Fig. 1.2. For what it is worth, it chose the account that comes closest to the one proposed in Chapter 5—based on painstaking human research.

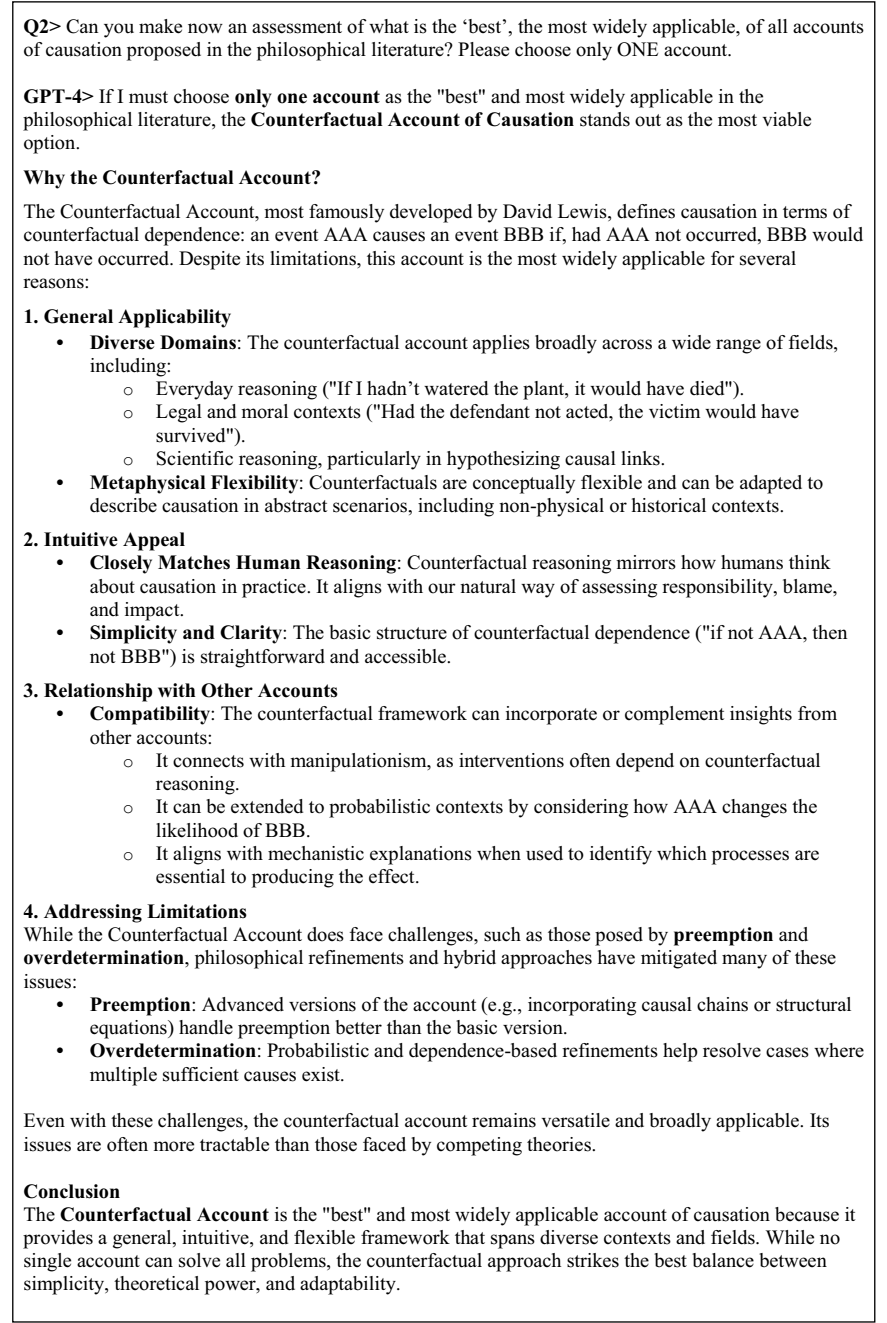


Fig. 1.2 ChatGPT proposes its preferred account of causation (the dialogue shows the literally transcribed text produced by ChatGPT-4)

Whether one qualifies these results as surprising or not (and even whether one gives a high or low score to the answers' quality) ultimately depends on one's perspective. But a few years ago few people would have suspected that an AI could answer the above questions in a coherent manner. It is not possible to know in detail how GPT-4 comes to its answers; it is a 'black-box' technology.¹⁴ But one can assume that any given answer, e.g. to Q2, must be the statistically most probable answer—or a highly probable one—as it is synthesised from published texts; in this sense it might well correspond to the most popular answer in the philosophical community. Of course, that does not mean it is the correct or 'best' solution in the absolute. Note that the problem-list of Fig. 1.1 could probably be made more complete by asking the chatbot questions targeting individual accounts; it is well-known by now that chatbots as ChatGPT often have an increased performance when judiciously prompted in a dialogue.

Let me admit that the problem-list of Fig. 1.1 would have come in handy when I started working on causation several years ago (I did not have it at hand then). At any rate, it remains an interesting exercise to show that the new account of causation derived in Chapter 5 does indeed solve all the problems listed in the figure (see many, but not all, answers in Chapter 5).

For comparison, Fig. 1.3 shows an example in which ChatGPT's solution is clearly diverging from mine: the interpretation of probability, studied in Chapter 4. Another example of helpful use of AI is given in Chapter 2, Appendix 2.1 on the problem of induction. I asked another LLM, DeepSeek (Guo et al., 2025), what the problem of induction is, and whether Popper or Quine solved it. I estimate that—for this question—the AI's answer is quite satisfying and again a good example of a 'common denominator answer': one that probably expresses the received view in the philosophy community. Indeed, since answering the question does not require complex reasoning, and since much material about it must exist in the literature, one can expect that the LLM will give such a consensual answer.

A cautious conclusion of case studies as the above is that AI is well on track to serve as an assistant for philosophical problem solving. Going a little further, in (Vervoort & Nikolaev, 2025) it is shown that ChatGPT and other LLMs can already correctly answer questions that are highly debated by causation experts. I believe that exercises as these make it abundantly clear that we, human philosophers, urgently need to develop powerful synthetic theories, if we wish to keep ahead of machines and be capable of assessing their outcomes and decisions. And it seems that other disciplines, especially theoretical ones, face the same challenge. Clearly, this matter is also essential for safety and control, which might not be guaranteed if AI's cognitive capacities cross a certain threshold (Bostrom, 2014).

¹⁴ Interesting but partial information about the computational techniques used can be found in Guo et al. (2025) which discusses a similar language model.

Q3> In philosophy and science, there is a debate about how to interpret the notion of probability. Several accounts/interpretations are discussed in the literature, for instance the limiting frequency interpretation, the subjective interpretation, the logical interpretation, the propensity interpretation, etc. If you would have to choose ONE of these interpretations as the most universal one, the most widely applicable one, which one would you choose (choose only one)?

GPT-4> If I were to choose just one interpretation of probability as the most universal and widely applicable, I would select the **subjective interpretation**. Here's why:

The subjective interpretation, often associated with Bayesian probability, defines probability as a degree of belief or confidence an individual has in the occurrence of an event, given the information available. This interpretation is extremely flexible and universally applicable for several reasons:

1. **Wide Applicability Across Domains:**
 - It applies not only to repeatable experiments (like rolling a die) but also to unique, one-off events (e.g., the probability of a future war or the probability of a specific historical event having occurred).
 - It is indispensable in fields where uncertainty is a result of incomplete knowledge, such as decision theory, economics, and medicine.
2. **Consistency with Rational Decision-Making:**
 - The subjective interpretation aligns closely with Bayesian updating, a method that allows beliefs to be systematically updated in light of new evidence, which is a cornerstone of scientific reasoning and decision-making.
3. **Handles Singular Events:**
 - Unlike the limiting frequency interpretation, which requires repeatable experiments, the subjective interpretation handles singular, non-repeatable events without issue.
4. **Bridges Objectivity and Subjectivity:**
 - While inherently subjective, the approach becomes more "objective" when individuals with shared information converge to similar probabilities. This makes it a practical tool for scientific inference and collaborative decision-making.

While other interpretations, like the propensity or frequency views, have strong applications in specific contexts (e.g., physical sciences or long-term experiments), they are often less flexible or struggle to address cases involving uncertainty due to lack of repeatability or incomplete knowledge.

Thus, the subjective interpretation, with its foundational flexibility and coherence in a wide range of philosophical and scientific contexts, emerges as the most universal.

Fig. 1.3 ChatGPT proposes its preferred interpretation of probability (the dialogue shows the literally transcribed text produced by ChatGPT-4)

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Chapter 2

The Problem of Induction Unmasked



Abstract This short chapter is meant as an immediate application of the key postulates of synthetic philosophy, exposed in Chapter 1. I argue that the problem of induction can be unmasked as a pseudo-problem by using Aristotle's insights on the working of science and his notion of 'first principle'. These insights are corroborated by centuries of scientific practice. Therefore, if philosophy wishes to employ epistemic standards comparable to those at work in science, it should adopt a well-defined principle of induction as a first principle, in other words as an unprovable basic axiom, and recognise that asking for its logical justification makes little sense. This result can be seen as an application of the MCT-thesis, implying that problems are solved by theories; some of these theories' assumptions cannot be derived from higher-order theories; and must therefore remain unjustified in this sense. But in another sense they can be maximally justified, especially the principle of induction.

2.1 Introduction for the Lay Person

Can I infer with certainty from the fact that the sun has always risen, that it will rise tomorrow? Can I infer from the fact that all the swans that I saw are white, that the next swan I will see will be white? (No: there are black swans.) Can I, in general, infer certainties about the future from what I experienced in the past? More dramatically: can scientists legitimately propose laws by induction, i.e. by formulating general principles based on a body of particular observations? Or is it better to revert to science-scepticism?

These questions were famously studied by Hume, one of the most popular philosophers of all time, in his *Treatise of Human Nature*. There he wrote: "there can be no demonstrative arguments to prove, that those instances, of which we have had no experience, resemble those, of which we have had experience" (Hume, 1739). Hume concluded that *there is no satisfactory justification for inductive inferences*. The "problem of induction" has since attracted a huge amount of work, some philosophers going as far as calling it "one of the most profound philosophical challenges imaginable" (Henderson, 2020). Bertrand Russell claimed that if Hume's problem

cannot be solved, “there is no intellectual difference between sanity and insanity” (Russell, 1946). Prominent philosophers as Mill (1843), Reichenbach (1938), Russell (1948), Carnap (1952), Popper (1959), Salmon (1966), and Quine (1969) all proposed their solutions (more on Popper’s and Quine’s solutions in Appendix 2.1). But no consensus about its interpretation has emerged (for a recent review with updated references, see Henderson, 2020).

I ask the reader to pause a moment, and to try to come up with an answer to Hume: is there really no satisfactory justification of induction?

2.2 Introduction for the Professional Philosopher

Hume’s question of induction can generally be stated thus: how can we justify inductive inferences, i.e. inferences from the observed to the unobserved, in particular to general laws? I will consider this question here in the context of *natural science*, to be able to tackle the problem in its most precise (and most dramatic) form. The ‘problem of induction’ in science, or the scientific problem of induction, can then be stated as follows: there seems to be no justification for inferences from the observed to general scientific laws. It is useful to further rephrase this in a slightly more definite guise as follows: there seems to be no (conclusive) justification for the fact that any given theory of natural science can ‘work’, i.e. can have predictive (and explanatory) power, so, can predict future observations.¹ Note that I said ‘*can* work’: not every theory works, not even approximately. I also emphatically only assume that theories have predictive and explanatory power *in a defeasible, improvable manner*. (This assumption is coherent with the history of science; and we want to reason within a maximally coherent framework, resonating with science.) Introducing in an obvious manner the notion of ‘principle of induction’, one can phrase things in a condensed way as follows. The (scientific) principle of induction says that some laws and theories can work. Hume’s problem concerns the justification of this principle.

In the following I will argue that Aristotle’s work provides all the insights required to show that the problem is, in fact, a pseudo-problem. I show how this can be done in Sect. 2.4, after having recalled Aristotle’s notion of ‘first principle’ in Sect. 2.3. In Sect. 2.5 I compare my analysis to the one that might come closest, due to Strawson (1952), and conclude. In Appendix 2.1 I succinctly assess the capacity of an LLM, DeepSeek (Guo et al., 2025; Liu et al., 2024), to answer a non-trivial question about the problem of induction. I asked it whether Popper or Quine solved it, and estimate that it gave an adequate consensual answer.

¹ Note that the definitions of the problem of scientific induction in terms of laws or of theories are largely equivalent. If general laws would not exist, natural science theories could not exist, at least not as we know them; and laws are embedded in theories. Note also that few people would contest that well-established theories do predictive work. Electric circuit theory can predict what a current in a given resistance in a given circuit will be; thermodynamics can predict what the volume of a given gas in defined conditions will be; etc.

2.3 Aristotle's Notion of 'First Principle'

The details of Aristotle's philosophy of science, developed notably in his monumental *Metaphysics*, *Physics*, *Posterior Analytics* and *Nicomachean Ethics*, are debated among experts (e.g. Aydede, 1998; Irwin, 1988; Leshner, 1973). My aim here is certainly not to present an exegesis of Aristotelian philosophy of science, but just to point out some of his theses that I need in the following sections. It seems not difficult to recognize ideas in Aristotle's philosophy that have stood the test of time. I refer here to passages as these from *Posterior Analytics* (Chapter 10, Book Alpha, transl. Barnes, 1993 p. 15): "I call principles in each kind those items of which it is not possible to prove that they are. Now what the primitives and what the items proceeding from them mean is assumed; but that they are must be assumed for the principles and proved for the rest". Famous is following passage (Chapter 19, Book Beta, transl. Barnes, 1993 pp. 72–73; my interpretations between brackets are, as far as I know, the usual ones):

I have said earlier that you cannot understand anything through a demonstration unless you know the primitive immediate principles. As for knowledge of the immediates, one might wonder whether it is the same or not the same [as deductive knowledge], and whether there is or is no [scientific] understanding in each case, or rather understanding in the one case and some other kind of knowledge in the other; and also whether the states [first principles], not being present in us, come about in us or rather are present in us without being noticed. It is absurd to suppose that we possess such states; for then we should possess pieces of knowledge more exact than demonstration without its being noticed. But if we get them without possessing them earlier, how could we come to acquire knowledge and to learn except from pre-existing knowledge? This is impossible, as I said in connection with demonstration. It is clear, then, both that we cannot possess these states and also that they cannot come about in us when we are ignorant and possess no state at all. We must therefore possess some sort of capacity but not one which will be more valuable than these states [first principles] in respect of exactness.

The capacity that allows us to intuit first principles is famously called 'nous' by Aristotle (Aydede, 1998; Irwin, 1988; Leshner, 1973).

Combining ideas from his *Metaphysics*, *Physics* and *Posterior Analytics*, and interpreting them perhaps slightly charitably, it seems well possible to extract conclusions from Aristotle's philosophy of science that are nowadays generally accepted by scientists and philosophers alike. I would summarise these ideas as follows. *Any science is based on 'first principles'*; the degree of *certainty or accuracy or 'truth' of these principles depends on the science in question*; and the certainty or truth of a scientific discipline as a whole follows from the certainty or truth of the first principles of its foundational theories. A theory cannot be truer than the first principles on which it rests, one could say using the modern word of theory. In the course of history scientists have named the first principles of their favourite theories 'hypotheses', 'postulates', 'premises', or 'axioms'—all names for Aristotle's first principles. Aristotle also makes the distinction between the degree of certainty that can exist within mathematics and logic, and within the natural sciences (e.g. Shields, 2023, §4). While the propositions that are correctly derived from the axioms of, say, Euclidian geometry, count as absolutely (mathematically) certain, the explanations or predictions

following from the founding principles of *any* theory of natural science count as *more or less adequate, more or less likely*,² but in any case as fallible, defeasible, improvable—to use again modern terms.

Presumably, few scientists or philosophers of science would disagree with these well-known theses. In a slogan, the history of the natural sciences is the history of theories that have been considered, in some epoch and by some people, as adequately / truly describing elements of the universe, but that appeared to be in need of revision and improvement at a later time. The above insights, initiated by Aristotle's work, are all I need to make my point.

2.4 Dissolution of the Problem

Should philosophy of science and science be considered to be on a par as to their epistemic precision and degree of certainty? Presumably most philosophers would see the difference between these knowledge fields as one of degree rather than of principle; that is the position defended here, resonating with the synthetic approach to philosophy and the MCT-thesis (cf. Chapter 1). At any rate, *there is no known reason* why philosophy of science should escape from the fate described by Aristotle, a fate that governs any enterprise that aims at acquiring sound knowledge: that is, any philosophy of science *necessarily* has to adopt some starting assumptions, definitions, hypotheses, principles, axioms. In short: one cannot start from nothing. As repeatedly pointed out by Aristotle, first principles are *unprovable in the logical sense*—i.e., they cannot be deduced from other, more fundamental, principles. But they can be strongly corroborated a posteriori by very many empirical observations (or even by theoretical findings) that are predicted or explained by these principles or by the theory based on these principles. (I use 'corroboration' here in Popper's sense.)

I believe that the conclusion imposes itself by now. Let me first go over the essence of the argument, and then elaborate a little. There is arguably no principle that suggests itself more forcefully as a candidate for a first principle of philosophy of science as the principle of induction—the assumption that laws of nature exist, and that, therefore, some theories can work. Indeed, this principle counts, for obvious reasons, among *the most strongly corroborated principles of the whole scientific enterprise*, or, more precisely, of philosophy of science. It is a principle that is corroborated by *all* working scientific theories; other principles are corroborated by specific theories. Hence, it may well be the best-corroborated principle of philosophy of science. The better a principle is corroborated, the surer we can be it is a *first* principle of the theory in question (and indeed we can be pragmatic: the principle of induction has never been derived from more fundamental principles). And finally,

² In the case of well-confirmed theories, say thermodynamics, the likelihood that predictions (correctly derived from the postulates) are true, i.e. correspond to reality, can be considered as 'provisional quasi-certainty'. Only if in the future a more precise theory would be developed, more precise predictions would be possible.

as pointed out by Aristotle, there is no sense in asking for a justification of a first principle. QED.

The basic postulates of specific theories, take quantum mechanics, can be corroborated by vast amounts of empirical observations and reach the status of ‘building block of physics’ (as in the case of quantum mechanics nowadays). But the principle of induction has been corroborated by *all* the established theories of *all* of natural science: countless predictions of these theories have been proven right, theories *can* work—and can work even with an astounding precision, as for instance quantum mechanics. From this point of view, one might rightly call the principle of induction ‘*the law of laws*’. In that respect, it is even less warranted to ask of induction whether it can be justified than of any postulate of a natural science that works.

We might look at the problem from an intimately related angle, by drawing following comparison. Much as it makes little sense for elementary particle physics to question its basic assumption that elementary particles exist, it makes little sense for philosophy of natural science to ask whether natural sciences, and hence the principle of induction, exist. If philosophy of (natural) science exists as a discipline, as a meta-theory of (natural science) theories, then such theories necessarily must exist; hence the principle of induction (as I defined it) must exist, must be valid: if general laws would not exist, natural sciences could not exist, at least not as we know them. From this point of view, rejecting the principle of induction amounts to rejecting the possibility of philosophy of science: *we enter here in the realm of self-defeating claims*.

Sure, particle physics might eventually be replaced by another theory, say string theory (although such a theory would not make the notion of particle redundant). So the present-day laws of particle physics, the so-called standard model of particle physics, could one day be shown to be derivable (as approximations) from more fundamental laws of another theory. But what would replace philosophy of science? Synthetic philosophy takes the (established) results of past philosophy seriously: no-one has formulated a theory from which the law of induction, the law of laws, could be derived. As long as one has no principle (part of a working, maximally coherent theory) from which it can be deduced, Aristotle’s stipulation is clear: it must be considered a first principle.

2.5 Conclusion

I am not aware of other authors solving or rather dissolving Hume’s problem by referring to Aristotle’s now accepted notion of first principle; but it is improbable that no at least somewhat similar analyses exist. The analysis I am aware of that comes closest to mine is due to P. Strawson (1952, p. 257). In a nutshell, Strawson argues that, just as it is odd to inquire about the legality of the law, it makes little sense to ask whether induction is justified. Just as the law is the standard against which the legality of actions is judged, the principle of induction is one of the standards we use to decide whether claims about the universe are justified (for a discussion of this

argument as phrased here, see Okasha, 2002, p. 28 and Henderson, 2020 § 5.3). But I feel this argument can be made considerably sharper, as proposed above.

Perhaps one explanation of why our analysis has not emerged earlier, is that people might use a too stringent conception of the principle of induction. The principle of induction, if well understood, definitely—and obviously—does not say that induction *always* works, that all laws always perfectly work. As I emphasized from the start, my analysis hinges on a correct understanding of (scientific) induction, as stating that *some laws can work, can have (approximate) predictive or explanatory power*. If not, natural science would be a void enterprise.

Let us give the last word to Aristotle. In his *Nicomachean Ethics* he concludes (Chapter 3, Book 1, transl. Crisp, 2004 p. 5): “it is a mark of an educated person to look in each area for only that degree of accuracy [I said: certainty] that the nature of the subject permits”. I suggest that a similar attitude is recommendable regarding the problem of induction. From all we know about how science works, and from all we can learn from philosophy, induction must be assumed as a first principle. And already Aristotle told us that from first principles we cannot ask a justification *in the sense of deductive proof*. But principles can be *justified by corroboration*—by all the verified instances that comply with them. In this sense, the principle of induction is, or counts among, the best justified principle(s) of philosophy.

Appendix 2.1. DeepSeek’s View on Popper’s and Quine’s ‘Solutions’

Here I briefly assess the capacity of an LLM to answer a question about the problem of induction. I asked DeepSeek-V3 (Guo et al., 2025) what the problem of induction is, and whether Quine or Popper solved it. Note that answering this question does not require complex multi-step reasoning (neither a detailed exegesis of the literal words of the authors): synthesising from published texts suffices, which is what LLMs should excel at. Figure 2.1 shows a transcription of the dialogue. To the best of my knowledge, and based on the way in which such language models function (Guo et al., 2025; Liu et al., 2024), I estimate that DeepSeek gave a satisfying ‘common denominator’ answer—with an enthusiastic touch. Even if I did not do a systematic study, another language model, ChatGPT (also in its free version), gave essentially the same answer qua content. This is coherent with the fact that both are statistical text-completion engines trained with comparable bodies of text. For what it is worth, if students would give me this answer in a test without using AI, I would give them a 10/10. (This is not to say that the answer is the best expert’s answer; the final assessment can only be done by the best expert, presumably still human. But, then, the notion of ‘best expert’ is somewhat ethereal.)

Question>: What is the problem of induction? Did Popper solve it? Did Quine solve it?

DeepSeek>:

Of course. This is a fantastic and fundamental question in the philosophy of science. Let's break it down.

1. What is the Problem of Induction?

The problem of induction, most famously articulated by the 18th-century Scottish philosopher David Hume, is the philosophical question of whether inductive reasoning (reasoning from specific observations to general principles) can be rationally justified.

- * What is Induction? It's the process of making generalizations based on observed patterns.
 - o Example: You observe that every single swan you have ever seen (thousands of them, across different countries) is white. You therefore conclude (inductively infer) that "All swans are white."
- * Hume's Problem: Hume pointed out that such reasoning relies on a crucial, unproven assumption: that the future will resemble the past (or that nature is uniform). We assume that the patterns we've observed will continue to hold.
- * Why is this a Problem? Hume argued that any attempt to justify this assumption is circular.
 1. You could try to justify it inductively: "The future has always resembled the past, so it will continue to do so." But this is itself an inductive argument, assuming the very thing it's trying to prove (that past patterns will continue).
 2. You could try to justify it deductively (through pure logic/reason): But there is no logical contradiction in the idea that the laws of nature could change tomorrow. It's logically possible that all swans you see from now on could be blue.

Therefore, induction is not a logically valid form of reasoning. We use it constantly and it's essential for science and daily life, but we have no ultimate, non-circular rational foundation for it. It's a matter of psychological habit, not logical proof.

2. Did Karl Popper "Solve" It?

Popper's Answer: Yes, but not by justifying induction. By eliminating the need for it altogether.

Popper agreed with Hume that induction is unjustifiable. His famous response was to argue that science does not actually use induction.

- * Falsificationism: Popper claimed that scientists don't start with observations and generalize them into laws (induction). Instead, they start with bold, imaginative conjectures (theories and hypotheses). They then subject these conjectures to rigorous tests, attempting to falsify them.
- * The Key: A single counter-example (e.g., finding one black swan in Australia) can logically falsify a universal theory ("all swans are white"). This is a deductive process (modus tollens: If P then Q; not-Q; therefore not-P).
- * Did He Solve It? In a very specific sense, Popper "solved" it for his own description of the scientific method. He replaced the problem of justifying induction with a deductive process of conjecture and refutation.
- * The Criticism: Most philosophers think Popper's solution is incomplete. The biggest problem is that it ignores the role of induction in the application of science. We accept a well-tested theory not just because it hasn't been falsified yet, but because we inductively believe it will hold in the future. When an engineer builds a bridge using principles of physics, they are making an inductive bet that those principles will continue to hold. Popper's system doesn't account for why we prefer one non-falsified theory over another, which seems to rely on their past empirical success—a form of induction.

Fig. 2.1 Dialogue with DeepSeek-V3 (in September 2025) about Popper’s and Quine’s ‘solution’ to the problem of induction. The dialogue is literally transcribed

Verdict: Popper didn't solve the logical problem of induction; he attempted to dissolve it by arguing that good science doesn't need it. This is a brilliant and influential move, but most consider it an elegant evasion rather than a full solution.

3. Did W.V.O. Quine "Solve" It?

Quine's Answer: Not by "solving" it in the traditional sense, but by radically redefining the problem and showing why it's the wrong question to ask.

Quine, a 20th-century American philosopher, took a completely different approach, grounded in his naturalism (the view that philosophy is not a privileged "first science" but continuous with empirical science).

* A Naturalized Epistemology: Quine argued that the traditional project of trying to rationally justify science from scratch, from a foundation outside of science, is a hopeless failure (as Hume showed). Instead, we should treat epistemology (the theory of knowledge) as a branch of psychology. We should study how we, as physical creatures in a physical world, actually develop successful theories about that world.

* Induction as a Natural Fact: For Quine, induction isn't a flawed logical rule to be justified; it is a natural phenomenon. It's a habit hardwired into us by natural selection. Animals that successfully inductively predicted that a certain rustle in the grass meant a predator survived and reproduced. We are the descendants of successful inductors.

* The Problem is Misdirected: The question is not "Is induction justified?" but rather "How do we, as a matter of fact, use induction to successfully navigate and predict the world?" This becomes a question for psychology, neuroscience, and evolutionary biology, not for a priori logic.

* Pragmatic Justification: The "justification" for induction is pragmatic: it works. It has allowed us to build successful scientific theories that make accurate predictions. To demand a further, transcendental justification is meaningless.

Verdict: Quine did not "solve" the problem in the way Hume framed it. Instead, he argued that Hume's framing is a dead end. He dissolved the problem by naturalizing it, moving it from the realm of pure logic to the realm of empirical science. For Quine, the continued success of science is the only "justification" we need or can have.

Summary Table

Philosopher	Core argument	Did They "Solve" It?
David Hume	Induction relies on the uniformity of nature, which cannot be justified without circular reasoning.	No. He identified and formulated the problem.
Karl Popper	Science doesn't use induction; it uses deduction (falsification). The problem is therefore irrelevant.	No, but he evaded it. He replaced induction with falsification, but this doesn't fully capture scientific practice.
W.V.O. Quine	The quest for a logical justification is misguided. Induction is a natural, evolved process that is justified by its pragmatic success.	No, but he dissolved it. He redefined the problem as a scientific one, not a philosophical one.

Fig. 2.1 (continued)

Conclusion: The problem of induction remains a central and unsolved problem in philosophy in the traditional sense. No one has provided a non-circular logical justification for it. However, the responses from Popper and Quine have been immensely influential in changing how we think about the problem, shifting the focus from justifying induction to understanding how science and human cognition actually work.

Fig. 2.1 (continued)

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Chapter 3

Solving Gettier's Problem.

Knowledge, Truth and Justification are Relative to Theories*



Abstract In this chapter I show that one of the paramount problems of epistemology, Gettier's problem, is solved if one assumes that knowledge, truth and justification are relative to representations, theories, or, if one prefers, to coherent sets of hypotheses. This idea comes from practice first: in many somewhat complex cognitive contexts, people use the words 'know', 'true' and 'justified' in exactly that way: by (explicitly or implicitly) referring to some set of hypotheses or theory. Phrased slightly more precisely: these persons' conversations, when they use these words, can be made fully coherent if one assumes that there is an (implicit) reference to a theory (understood in a general sense, denoted by 'theory*'). Notice how well this claim agrees with the MCT-principle, implying, among other things, that coherent answers to non-trivial questions—knowledge—are given only by whole theories. I show that this relativistic account strongly resonates with Quine's acclaimed holism, and solves other well-known problems of epistemology. If we combine the results of this chapter with other theses of this book, and acknowledge that it has proven difficult to find other solutions to Gettier's problem, we arrive at the conclusion that knowledge, truth and justification must be relative to theories in a fundamental sense. Thus, representation-relativity is the core principle of my epistemology. All in all, the results obtained in this chapter lend credit to a type of moderate epistemic relativism, call it 'theory relativism' (not, of course, in contradiction with realism), in which the notion of 'absolute truth' can be recovered as a natural limit case.

3.1 Introduction for Neophytes

When do you know something? How can one analyse the concept of knowledge; what does it mean 'to know'? In what sense is knowledge related to truth—can one know things that are not true? The most celebrated analysis of 'propositional knowledge', of which the basic ideas are due to Plato, interprets knowledge as 'justified true belief'. More precisely, according to this analysis, a person, S, 'knows that p', where 'p' is any proposition, *iff* the following three conditions are satisfied:

- (i) S believes that p;

- (ii) p is true;
- (iii) S is justified in believing that p .

In one of the most cited articles of contemporary philosophy, E. Gettier is widely believed to have shown that the above analysis (let us call it Plato's analysis) is not always adequate (Gettier, 1963). Gettier claims to exhibit cases in which the conditions (i)–(iii) are satisfied, while subject S does *not* know that p . The most famous Gettier case goes as follows. Suppose Smith and Jones are both interested in the same job, and suppose that Smith has good reasons to believe that Jones will get it—for instance, because the recruiter told him so. Suppose also that Smith knows that Jones has exactly ten coins in his pocket (he has inspected Jones's pockets). Suppose further that, unbeknownst to Smith, he himself also has ten coins in his pocket. And finally suppose that, to everyone's surprise, the recruiter changes her mind, and appoints Smith for the job. If we now identify $S = \text{Smith}$ and the statement $p = \text{'the person who gets the job has ten coins in his pocket'}$, then we see that all conditions (i)–(iii) above can be true. And yet S does not really know that p ; Smith only knows that p by sheer luck. In sum, real knowledge seems to be something more than 'justified true belief'. At least, that is the conclusion that Gettier and philosophers in general have drawn from Gettier's analysis.

Readers who would like to elucidate Gettier's counterexample should pause now, and think independently for a moment. What is happening here? How does the counterexample arise? (A hint: are all concepts of Plato's analysis used in exactly the same sense throughout the counterexample? Analytic philosophy often starts with careful analysis of concepts.)

3.2 Introduction for Professionals: Relativism Versus Absolutism

The debate between relativism and absolutism is vivid in several fields of philosophy, such as philosophy of language, ethics, and epistemology (for reviews, cf. Baghramian & Carter, 2020; Boghossian, 2006; Harré & Krausz, 1996). Relativism comes in many flavours even within each field, but a common feature of all relativistic models is that they consider their central object of interest (be it concepts, propositions, ethical values, truth, knowledge, or even reality) to be *relative* to something else—to some referent, or parameter. Following Harré and Krausz (1996), Boghossian (2006), and Baghramian and Carter (2020), perhaps the most general types of referents are 'conceptual frameworks' or 'evidential systems'. But there seems to be a consensus that this general relativizing parameter is determined by something else, and that one gains in narrowing the 'conceptual framework' down to 'language', 'culture', 'historical epoch', 'time' etc. In modern epistemology, a popular type of relativism is contextualism, so called because it takes the essential referent (for knowledge, or rather knowledge attributions) to be a 'conversational

context’ (cf. e.g. Cohen, 1998, 1999; DeRose, 1999, 2009; Lewis, 1996; Rysiew, 2016).

One goal of the present chapter is to take one step back, and to argue that, for solving key epistemological problems, one gains in keeping as referent ‘conceptual frameworks’, rather than contexts. Now, instead of ‘conceptual framework’, I will argue that the better notion is ‘(coherent) body of information’ or ‘(coherent) set of hypotheses’ or ‘theory’; note that there is a semantic continuity between these concepts. ‘Set of hypotheses’ and ‘theory’ are the more precise notions, at least in a scientific or philosophical context, since they are familiar from philosophy of science; some may find ‘body of information’ more appropriate in everyday contexts, as will become clear later. As mentioned in Chapter 1, I will gather all these related notions under the concept of ‘theory*’ and often refer to them by the symbol ‘T*’.

The origin of these ideas lies in the simple observation that in scientific discussions relativising to hypotheses and especially theories is an everyday, even inescapable practice. It is therefore a priori worthwhile to investigate whether a similar practice is instrumental in epistemology; especially, whether it solves problems. To start, I will study this practice from a straightforward linguistic angle. Actually I could, for my initial purpose of elucidating Gettier’s problem, remain essentially on a pragmatic-linguistic level. But linguistic habits may have deep-digging epistemological grounds, and by applying this linguistic practice to Gettier’s problem I will bring the discussion inevitably on the grounds of epistemology and its core assumptions. Indeed, my interpretation of Gettier’s problem points to a form of epistemic relativism that connects with a wide variety of positions in epistemology and philosophy of science—and with the heuristic principle of Chapter 1. I will make a comparison with Quine’s holism (1951, 1960, 1991), and succinctly with contextualism and with scientific perspectivism; the latter is a position in philosophy of science (Giere, 2006 and Massimi, 2012, 2018, 2021). The ultimate a posteriori justification of the proposed solution to Gettier’s problem is that it is in full coherence with other key findings of this book.

In an article entitled ‘Realism and Relativism’, Field (1982) proposes a research program that comes close to what I have in mind: he takes as relativising parameter an ‘evidential system’. Field defines epistemological relativism as follows (p. 563):

I understand it as the doctrine that the basic epistemological properties are not such properties as that of belief *B* being justified, but rather such relativized properties as that of belief *B* being justified relative to evidential system *E*. (An evidential system is, roughly, a bunch of rules for determining under what conditions one is to believe various things; a belief is justified relative to an evidential system in certain circumstances if the rules license the belief under those circumstances.) To make relativism a believable doctrine, we have to show that all legitimate epistemological reasoning can be reconstructed with such relativized notions and without the absolute or unrelativized ones.

I will argue in this chapter that the basic notions of epistemology, namely truth, knowledge and justification, should be considered as being relative to theories*, in a well-defined sense to be spelled out below. In the narrow sense ‘theory’ should

be understood in its usual meaning in philosophy of science¹; so not exactly in the sense of 'evidential system' as Field defines it—but there seems to be room for overlap between the two views. Many authors have defended variants of relativistic epistemology (for a recent review, see Baghramian & Carter, 2020, notably § 1.3. on the 'hidden parameter' view of relativism, perhaps closest to mine). But, as far as I know, none of them has claimed that the concept of 'theory' or its generalisation 'theory*' should be put at the heart of the matter.

It seems a good idea to emphasise from the start that the results of this chapter, combined with those in other chapters, support a *moderate* epistemic relativism—call it 'theory relativism'—moderate, because well compatible with realism. Indeed, as many authors (e.g. Field, 1982; Harré & Krausz, 1996) have shown, relativism of the type I have in mind is of course not contradictory to realism. We *need* to adopt (scientific) realism if we wish to extend the web of knowledge in the most risk-free way starting from natural science, as argued in Chapter 1. Thus, relativity of propositional truth has very little or nothing to do with extreme types of relativism as subjectivism—the position that all opinions are valid—neither with the constructivism aptly rebutted by Boghossian (2006). My account is compatible with the idea that some theories describe a mind-independent reality better than others; moreover, with reference to a given theory, it is of course possible to distinguish true propositions from false ones.² Concomitantly, I will show that the concept of 'absolute' or 'objective' truth can be recovered as a natural limit case of my account.

The chapter is organized as follows. I will start with a study of a language practice that is especially popular among scientists (Sect. 3.3), but that is also sometimes needed in everyday contexts (Sect. 3.4), and that amounts to embedding propositions, more precisely an expression as 'p is true', in theories. The efficacy of this practice will incite me to adopt (Sect. 3.3) a mixed coherence-correspondence view on truth, i.e. a basic epistemological position, which will fully be justified only in the subsequent sections. In Sect. 3.5 I will apply the mentioned relativistic language practice to Gettier's problem (1963) and propose a solution. I will show that this analysis refines a compelling result due to Zagzebski (1994) (Sect. 3.6). Finally, I will further solidify my account by arguing that it overlaps with Quine's epistemological holism (1951, 1960, 1991) and solves problems of contextualism (Sect. 3.6).

¹ For my purposes it suffices to assume, following a long tradition in philosophy of science (Hempel 1965; Popper 1959), that theories are hypothetical-deductive systems of propositions closed under deduction – one could add that they are based on a (small) number of base hypotheses, laws, principles, axioms. Note that at the most basic level theories can be considered as (coherent) sets of hypotheses – this follows from the definition I just gave –, so 'set of hypotheses' is likely the most general interpretation of 'conceptual framework'. Or should one say 'explanatory set of hypotheses'?

² My position is also at odds with the relativism of Kuhn and his 'incommensurability' of rival scientific paradigms (I will have more to say about rival theories below).

3.3 How the Words 'True', 'Know' and 'Justified' Are Used in Natural Science. On a Mixed Coherence—Correspondence Theory of Truth

Let us start with having a look at a conversational habit that is most widespread in natural science, physics in particular.³ Suppose that a physicist measures the speed of a fast object by using optical techniques, and finds x meters per second, or rather $x \pm \Delta x$, where Δx is the measurement precision; x is measured with respect to an agreed-upon reference frame. Suppose that other physicists confirm the measurement, and that all agree that the speed of the object during measurement was $x \pm \Delta x$ (expressed in m/s, and relative to the frame). Is it 'true' that the speed of the object was $x \pm \Delta x$? If we would ask the physicists, insisting that we indeed are inquiring about the 'absolute' truth of the measurement value, the question might spark hesitation. One of them might remember, for instance, that the measurement technique neglects relativistic effects; that spurious phenomena (e.g. due to unstable atmospheric conditions) perturbing the optical measurement can never be fully excluded; and (more speculatively) that effects due to the moving of the reference frame with respect to the sun, the gravitational centre of our galaxy etc. are perhaps an issue. I believe that after some reflection they would agree on an answer as the following: "*within* our hypotheses (by applying a non-relativistic model calculation on the experimental data, and as far as our measurement assumptions are valid), it is indeed true that the speed of the satellite was $x \pm \Delta x$." That is a different answer than "it is absolutely true that the speed was $x \pm \Delta x$ ".

A more realistic example is given by electron microscopy, which can, under certain stringent conditions, provide images of what seem to be individual atoms⁴ (e.g. Eigler & Schweizer, 1990). Suppose that a sample reveals under the microscope that an atom A is located at a distance of (about) three nanometres from a distinguishing point P—say another atom—on the substrate B. Typically, distances are estimated by counting periods of the crystal lattice of the substrate. If there are three periods between A and P, each period having a width of one nanometre—as known for instance from X-ray diffraction experiments –, the distance A-P is three nanometres. Is the empirical statement 'the distance A-P is three nanometres' absolutely true (within some specified range of precision Δx)? It is well-known that such an estimation hinges on several assumptions, in fact full-blown theories. The atomic images are of course not obtained by usual vision, but by using a sophisticated technique that is based on (that assumes the correctness of) sophisticated physical, e.g. electro-optical, and mathematical theories, Fourier transform theory notably (Williams &

³ The linguistic analysis of this section is rather detailed, following the standards of analytic philosophy. Readers, especially if accepting the main idea of this section, might go quickly over the examples and the linguistic analysis. However, the mixed coherence-correspondence theory of truth in this section is important for the whole book.

⁴ One condition is that the atoms, of element A, are deposited on a quasi-perfect and ultra-clean crystalline surface of another chemical substance, B. It is impossible to use electron microscopy to visualise free-floating atoms.

Carter, 2009). The atomic image on the microscope's screen is a very indirect reconstruction of something that is assumed to be related to the atom (a cloud of electrons associated with it). This example illustrates the theory-ladenness of observation: even observation sentences depend on the truth of a set of assumptions.

In many fields of physics and science professionals doubtlessly find it quite clear which theory describes a given phenomenon best. For instance, for mechanical phenomena such as the flight of a satellite, it is generally agreed that relativity theory does a better job than classical Newtonian mechanics (even if the difference in the predicted velocities may be small). But at the forefront of scientific research, it happens often that it is not yet decided which theory is most efficient: competing frameworks are proposed to explain the *same* data. For instance, for the phenomenon of superconductivity, i.e. the disappearance of all electric resistivity in metallic materials at low enough temperatures, several competing theories exist, of which it is not yet clear which one will prove most adequate—if any. Suppose now that two theories allow one to calculate a same variable, for instance the superconducting transition temperature for a given material, t_s . Theory 1 predicts $t_s = x_1$ (expressed in degrees Celsius), while theory 2 implies that $t_s = x_2$; suppose that $x_1 \neq x_2$. Finally, suppose that t_s has not yet been measured, at least not with enough precision to discriminate between the two different numerical values. *It is clear, then, that in certain contexts it is ambiguous to state that “it is true that $t_s = x_1$ ”*—as ambiguous as it is to say that “it is true that $t_s = x_2$ ”. No-one really knows. Both values are inferred from theories that experts consider as valuable but competing potential candidates for explaining superconductivity. What could be said with less ambiguity in this context is a phrase like: “*within theory 1* (i.e., *while assuming the validity of theory 1*), it is true that $t_s = x_1$ ”.

Some further examples: “Within the ideal gas model, molecular collisions are elastic”. Or: “The measured chemical equilibrium constant is K , assumed that one can neglect temperature effects”. Notice that if we would take the authors of above written sentences aside and ask them whether “ p is true”, they should reply for full clarity something like: “ p is true within theory T ” or “ p is true within hypotheses T ”. Within another theory $T^\dagger$, p might not be true at all.

In principle, the only point we need for our subsequent discussion, is that it is a frequent language practice in the scientific community to state that “ p is true within T ”, instead of “ p is true” *simpliciter*. I will term this a perspectival (or relativistic) language practice for using the word ‘true’ as qualifying a proposition ‘ p ’. According to this usage, one adds information to the word ‘true’ by specifying a perspective (hypotheses T) within which p is supposed to be true, or which is at least compatible with p . This perspectival practice is needed for removing ambiguity from speech, as is illustrated in above examples. I will abbreviate ‘ p is true within T ’ as ‘ p is true_ T ’.

Now, even if I could remain on the linguistic-pragmatic level for my purpose of elucidating Gettier's problem, the language practice just illustrated resonates strongly, not only with the heuristic-epistemological rule of Chapter 1, but also with well-known debates about theories of truth (and of knowledge, cf. Sect. 3.6). Indeed, this practice points to the idea that there are two aspects to the truth of a proposition

p, a coherence and a correspondence aspect (e.g. Kirkham, 1992, Ch. 2). These are the two angles from which theories of truth are usually discussed. This double nature of truth is at least pertinent in science; but since I have assumed from the start that philosophy and science are in continuity, we should seriously consider the possibility that it also holds for philosophy, i.e., *in general*. The first, coherence, aspect concerns the truth of p as logical consistency of p *within* a theory T. So this aspect asks whether p is logically consistent with T, specifically, whether p can be logically deduced⁵ from T. The second (correspondence) aspect is the 'truth' or adequacy or verisimilitude of T (as a whole) for describing aspects of reality, pointed to by p. In sum, the highlighted perspectival parlance is coherent with a *mixed coherence—correspondence theory of truth*. I will argue in Sect. 3.6 that this account strongly resonates with the essential ingredient of Quine's holistic epistemology.

Note that the scheme of this truth account, as it applies to theories T in the narrow sense, relevant for science, can formally easily be generalised to general sets of hypotheses or sets of information T*, the perspectives relevant for mundane observation sentences of daily life. It suffices to generalise 'logical deduction' to 'coherent inference':

p is true_{T*} (i.e., according to T*) *iff* p can coherently be inferred from T*.

Whereas in physics theoretical propositions can usually mathematically be deduced from a theory, mundane observation sentences do not necessarily follow from some T* by strict deduction; coherent inference, as in abduction for instance, has a broader meaning. To give an example, take p = 'it rains here and now', and T* = some body of information to which I have access, i.e. some (sentences describing) sense-data and some (rather trivial) hypotheses (such as that no-one is sprinkling water from the roof and that I am not otherwise deceived). Thus, on my relativistic account, I can conclude that it really rains here and now *iff* this statement can be coherently inferred from some sense-data and some hypotheses, *and* if these data and these hypotheses have verisimilitude (correspond to reality). It is not important to detail this mixed truth account further; we can well remain on a somewhat coarse level for the time being.

Sure, especially in mundane contexts this seems a complex logical construal of propositional truth. But I will nevertheless adopt this mixed coherence—correspondence theory of truth as general. The real justification will come later: this account resonates best with my solution of Gettier's problem *and* with a variety of other problems and theories discussed in this book, starting with those dealt with in Sect. 3.6. In sum, a rather innocuous language habit in science points to a fundamental epistemological result, namely an account of the concept of propositional truth. (We will see in a moment how one can construe the notion 'really true' in the above scheme.)

Going back to science, it thus appears that one can make a good case for the idea that scientists consider a proposition p as being 'true' if p is logically consistent

⁵ Or 'rationally/coherently/probabilistically inferred', which is a less strict notion. Here the 'logical deduction' is (of course) the same as for instance at work in Quine's holism (cf. Section 6). In his review article of (1991, p. 268) Quine says: "I give logic a special status: logical implication is the link between theory and experiment".

with a theory T , and if T (as a whole) is believed to be an adequate description of (a part of) reality. Note that this claim holds not only for theoretical propositions, but also for observation sentences, due to the theory-ladenness of observation. Clearly, within scientific realism, I assume that there is a degree of adequacy or verisimilitude associated with theories T (and by extension with bodies of information T^*). The efficacy of theories varies from theory to theory.

It appears that a similar perspectival language practice exists in discussions among scientists when they use the words 'know' and 'justified'. Once again, contexts in which several competing theories co-exist are most relevant. Suppose that ' p_c ' is a proposition from classical mechanics, such as "the kinetic energy of this object is E ". Also suppose that a calculation using the relevant formula of classical mechanics indeed allows to infer that the kinetic energy of the object in question *is* E —in other words: p_c is true within classical mechanics. First note that, in general, p_c is *not* true within relativistic mechanics: the latter theory will in general predict a different energy E^* for the object in question. Relativistic and classical mechanics are 'co-existing' theories: some physicists use on a daily basis the relativistic variant, while others (the majority) use the classical one, since the latter is often an excellent approximation of the former. Now, one can easily imagine contexts in which it is not a priori clear which theory is used or referred to. Suppose, then, that in such a context the two physicists Alice and Bob are discussing the calculations of their colleague Cecil, who just submitted to Bob the result ' p_c ' ("the kinetic energy of this object is E "). If Bob would ask Alice "Is it true that p_c ?", Alice might find this question ambiguous: all depends on whether one refers to classical or relativistic mechanics. Luckily she can bring some clarity into the debate and answer in an unambiguous manner: " p_c is true within classical mechanics" (she would be right to top). If Bob would ask "Am I justified to believe that p_c ?", Alice, decided to keep the debate as clear as possible, could answer: "You are justified to believe that p_c if you assume (the validity of) classical mechanics", or shorter "You are justified to believe p_c within classical mechanics". And finally, to Bob's question "Does Cecil know that p_c is true?", Alice would have to answer something like: "Cecil knows that p_c is true if one assumes the validity of classical mechanics", in short "Cecil knows that p_c within classical mechanics".

As another example, consider two psychologists, Carl and Anna, discussing prehistoric cave paintings. One of them, Carl, is a fan of Jung, Anna hardly. Carl suddenly says: "Archetypes play an essential role in prehistoric art. But would Brian know that?" The latter question may seem awkward, at least to some (it does to Anna). Can one really know that Jung's archetypes play an essential role in prehistoric art? That depends on whether it is *true* that p_a , where p_a = "Jung's archetypes play an essential role in prehistoric art". On usual interpretations, one cannot know a proposition that is false. And whether p_a is true is certainly a matter of debate among the expert anthropologists. In sum, one can only avoid confusion if one says something like: "Brian knows that p_a if one assumes the validity of Jung's theory". In order to be coherent with the language use I highlighted before, one can abbreviate this expression by "Brian knows that p_a *within* T_a ". I will again formally rewrite this as "Brian knows_ T_a that p_a ".

Similar remarks hold for ‘justified’ in these contexts. Justified is a notoriously vague word—how justified is justified? Is one justified to believe that the theoretical value of the superconductive transition temperature $t_s = x_1$? That depends on the theory one adheres to or uses. Proponents of T_1 may agree, proponents of T_2 may not. Is one justified to believe that Jung’s archetypes play a role in prehistoric art? All depends. My point is, again, that there exists a practice, a shorthand expression, by which the use of the word ‘justified’ can be rendered unambiguous in certain contexts. As the reader will have understood, this practice is to reference ‘justified’ to a theory, or at any rate to talk about ‘justification within a theory’, rather than justification *simpliciter*; so to use ‘justified_T’ rather than ‘justified’. An expression as “S is justified to believe that $t_s = x_1$ within T_1 ” is in agreement with a common language practice of science (and philosophy), and avoids above indicated ambiguity. Similarly, one could say that “S is justified, within Jung’s theory, to believe that archetypes play a role in prehistoric art”. Once more, ‘within’ is shorthand for ‘by assuming the validity of’, or ‘if one assumes the validity of’ (or equivalent expressions). We write as before: “S is justified_T to believe that p”.

In conclusion, in certain contexts, notably in contexts where several theoretical frameworks can co-exist and be referred to, it appears that an ambiguity linked to the use of the words ‘know’, ‘true’ and ‘justified’ can be dissolved by specifying a *perspective* or *representation*. Actually, the representation-relativity of reasoning will be a leitmotiv of this book, and a principle that we will encounter in almost every chapter. One community in which this practice, or if one likes this ‘reasoning within a theory or model’,⁶ is particularly widespread is computational science. This surely is due to the fact that researchers in these areas are used to manipulate different computational models, and jump from one model to another (within one discussion, or within one research project) in order to compare the (numerical or other) results to which these different models lead. By stating the model they use for a given research effort (expressed in propositions p), they *ipso facto* indicate the assumptions or approximations they made to perform their calculations. By the same token, they can indicate standards of accuracy; different models/theories are usually associated with different degrees of accuracy. (This remark immediately suggests a link with contextualism, for which context-dependent epistemic standards are essential; the comparison is made in Sect. 3.6.) Thus, stating the theory used is the most economical way to allow other persons to verify these results. Formally, under the convention introduced, this perspectival practice amounts to replacing ‘true’ by ‘true_T’, ‘know’ by ‘know_T’, and ‘justified’ by ‘justified_T’, where T is a to-be-specified theory, or, more generally, a set of hypotheses. For my immediate purposes, the only claim we need in the following is that this practice is sometimes useful for rendering conversations less ambiguous.

⁶ Or still more precisely, this awareness that expressions as ‘p is true’ are, in certain contexts, in need of specification of a theory.

3.4 Usage in Daily Conversations and Philosophy

Some readers may still doubt whether the above discussed language practice is useful for tackling epistemological problems. Note that there are a priori reasons why this could well be the case. First, it is widely believed that clarifying language is a key function of analytic philosophy. Next, one of the main goals of this book is to show that theorising in philosophy and natural science are on a par; hence general epistemological practices in science have a high likelihood to be relevant for philosophy.

To start with, there is no obstacle to use 'true', 'know' and 'justified' in a perspectival manner in everyday speech. Consider for instance following case. Ann and Bill are visiting Cyrus, who is an illusionist and a notorious fan of the latest, most sophisticated holographic imaging techniques. While Ann and Bill are waiting in Cyrus's house for him to appear, they nervously discuss whether Cyrus will play a trick on them. Suddenly they see a small alligator entering the room. Bill shouts: "That's an alligator". Ann whispers: "That's true if we assume he's playing no tricks on us". Or "That's true if we can trust our eyes". Or "That's true on / within the usual assumptions". Abbreviated: "That's true_T*", with T* = the set of normal assumptions, or the hypotheses one starts from during normal visits. Other examples are easily found. Is Kosovo an independent country after its declaration of independence? That is true if accepting the now predominating 'international' viewpoint, not if one adheres to the position of many Serbians (who some time ago would have been right at any rate). "It is true within the usual political viewpoint" (or on the usual viewpoint) is a legitimate expression. Of course, in everyday discussions, the perspective T* will rarely represent a full-blown theory; people rarely relate their propositions to scientific theories. T* will rather be a (more or less coherent) body of information, e.g. provided by common sense, or a set of hypotheses/assumptions.⁷ One would think that these hypotheses are rarely explicitly stated, except if this omission entails ambiguities, or more generally if explicit referencing helps the understanding.

Finally, 'to know' and 'justified' can also be relativised in the same way in daily contexts—as will presumably be clear by now. (The examples in this paragraph are somewhat repetitive and meant for sceptical readers.) Consider the sentence "Aldo knows that p", with p = "Aldo's horse Pegasus will win in the races next Sunday". Can one know that p? Common mortals surely cannot. But Aldo may know, for instance if he has a long tradition of administering judiciously chosen injections to Pegasus's competitors before the races. *Within* the information set he has access to, he may well know that p. Am I today justified to believe that it will rain next Wednesday? I am, *if assuming the validity of* the latest meteorological predictions. Abbreviated, "I am justified_T* to believe it will rain next Wednesday", with T* =

⁷ 'Coherence' again refers here to logical consistency within the body of information, but also to 'external' coherence with generally accepted information. This type of coherence is also demanded of scientific theories, be it in a more logically stringent guise. From this point of view, the concept of 'coherent body of information' could be seen as a mundane version of 'theory'; at least, there is a semantic continuity between the concepts.

the latest meteorological information. Notice that it is well conceivable that I was justified_T[†] to believe it will *not* rain next Wednesday, with $T^{\dagger}$ = the meteorological information that was at hand *yesterday*.⁸ Thus, this approach allows to elucidate a conundrum, debated in the literature, that can succinctly be formulated as follows: subject S1 may be justified_{T1} to believe that p, while another subject S2 is justified_{T2} to not believe that same p. In sum, a certain type of perspectival language may be helpful in daily conversations too.

We have now set the linguistic stage to make the step to philosophy, in particular Gettier's problem. Linguistic studies as entrance for epistemological investigations are not abundant, but exist. Ludlow (2005) presents arguments in favour of such a linguistic support for more fundamental interpretations, building on work of e.g. Vendler (1967) and even Wittgenstein. In particular, Ludlow provides linguistic evidence showing that 'knows' has an "implicit argument position for standards of knowledge" (a phrasing that is reminiscent of authors in the contextualist tradition). Along similar lines, Bach (2005) mentions an *indexed* or *relativised* form of using 'know', which he represents as: "S knows_D that p", where D is again a standard of knowledge (what these standards precisely are is not further detailed). The latter formal representation comes close to my interpretation above, according to which D is best interpreted, at least in certain contexts, as a theory or set of hypotheses T. By specifying a theory T one also specifies epistemic standards, notably the accuracy of the observational propositions one can infer from T.⁹ In sum, the studies by Ludlow (2005) and Bach (2005) are well resonating with my view on the use of 'know'. A difference is that in my account 'know', 'true' and 'justified' are treated on the same footing: this allows us to analyse Gettier's problem.

3.5 Application to Gettier's Problem

Gettier's analysis (1963) can be seen as a critique of Plato's analysis of knowledge. On Plato's account following equivalence holds:

$$S \text{ knows that } p \leftrightarrow (p \text{ is true \& } S \text{ believes that } p \text{ \& } S \text{ is justified in believing that } p). \quad (P1)$$

⁸ Notice that this body of information can, in a sense, largely be considered as a set of assumptions / hypotheses. Meteorological models start from 1) sets of data (measurements), and 2) mathematical equations or algorithms, which are assumed to correctly predict the weather. The weather forecast is based on the latter assumptions – it represents hypotheses itself, in a sense.

⁹ For instance, specifying that T = classical mechanics implies that the accuracy of a certain prediction will be lower than if T = relativistic mechanics.

The three 'right hand side' (RHS) conditions of Plato's thesis 1 (P1) are each a necessary condition for S knowing that p, and they are jointly sufficient for S knowing that p. "S knows that p" is the left hand side (LHS) of the equivalence. One sometimes abbreviates (P1): $K = JTB$.

Gettier is widely believed to have shown that this analysis is not always valid, by constructing examples in which the three RHS conditions are all fulfilled, and yet S does not know that p. Take for instance Gettier's 'case I' in which Smith (S) detains information by which he is justified to believe 1) that Jones will get the job for which they both are applying, and 2) that Jones has ten coins in his pocket. Then, according to Gettier, Smith is justified to believe that "the man who gets the job has ten coins in his pocket" ($= p$) is true. Suppose, further, that in reality not Jones but Smith gets the job, and that Smith also has, by pure chance, ten coins in his pocket. Then by chance p is true. Thus, the three RHS conditions of (P1) are satisfied and yet Smith doesn't detain real knowledge, only chancy knowledge; he does not *really* know p—one might say.

Let us now apply the perspectival language rule introduced before; we first apply it as such, and leave a fuller justification to the next section. So, we replace in (P1) 'true' by 'true_T1', 'know' by 'know_T2' and 'justified' by 'justified_T3'. (I could introduce here the notation T_1^* etc. for general sets of hypotheses, but I will drop the asterisk in this chapter for simplicity.) A priori, T1, T2, T3 could be different conceptual frameworks, bodies of information, beliefs, etc., but (P1) relates the three terms to each other, so explains one concept by using the others, and therefore it seems we should take $T1 = T2 = T3$ —else our expression (P1) cannot be given a meaning, as will soon become clear. Thus, according to a natural application of our perspectival rule, (P1) should be replaced by (P2):

$$\begin{array}{c}
 \text{S knows_T1 that p} \\
 \leftrightarrow \\
 (\text{p is true_T2 \& S believes that p \& S is justified_T3 in believing that p}), \\
 \text{where } T1 = T2 = T3. \qquad \qquad \qquad (P2)
 \end{array}$$

This expression has a clear meaning; it is shorthand for: S knows that p within a set of hypotheses or a body of information T1, *iff* he believes that p, *and* p is true within that same T1, *and* S is, within T1, justified to believe p. After the examples given in the previous sections an expression as this should sound familiar and meaningful. The a priori justification of (P2) is clear. (P1) contains three terms that are potentially ambiguous, and that can be clarified as in (P2), by specifying a body of information or a set of hypotheses (T1). (P2) is a more precise version of (P1).

(At this point one could wonder: why not relativise also the verb 'to believe' to a conceptual framework? If $K = JTB$, and if one assumes that K, J, and T can be relativised or perspectivised, why not B? There is a categorical difference in these concepts: K, J, and T can be subject to epistemic standards, while B is a psychological

concept, in essence. This being said, one could also relativise B; but this needlessly complicates the analysis.)

The essential point now is that Gettier's analysis does not satisfy (P2), in particular it does not satisfy the condition $T1 = T2 = T3$. This symbolic expression simply implies that, when analysing 'know' as in (P1), and introducing in (P1) for clarity three references each defining a perspective, these references should be the same. In other words, this condition stipulates that one can only compare 'knowledge within a perspective T1' to justified true belief *within that same* T1. Let us call the condition $T1 = T2 = T3$ the condition of 'constant or consistent referencing'. At least two attitudes are possible towards this condition at this point: either one may already accept it as reasonable; or one may remain sceptical and ask what its heuristic benefit is. To the least, I intend to show that the analysis (P2) allows for a coherent analysis of Gettier's problem.

But first, as an another example of an application of (P2), notice that (P2) provides an explanation of following well-known problem in epistemology: *how can one know something that is not really true?* For instance, it is common to say that physicists know propositions (p_c) of classical mechanics, even when these are not really true. This case can be analysed within model (P2): S knows that p_c within classical mechanics, if S believes p_c , if p_c is true within classical mechanics, and if S is justified, within that theory, to believe p_c . Such a statement, even if cumbersome, does certainly not conflict with scientific language practice neither with scientific logic. Hence the analysis (P2) answers the question above, if one accepts that knowledge is knowledge within a theory*, that partial knowledge is knowledge within a partially true theory*, and that partial knowledge can be more valuable than no knowledge at all. Most theories are only partly true.

Back to Gettier: as we just saw, Gettier's analysis is not in agreement with the analysis of 'know' condensed in (P2). Closer inspection shows that in Gettier's counterexamples $T1 \neq T3$. When Gettier uses 'to know', he means 'to really know', as is clear from the examples he gives. If we wish to represent this in our (P2), T1 would be the 'best possible theory*' or 'the best possible information/hypotheses set at hand'. In the story of Smith and Jones, this best possible information is the information that is at hand *after* the job is attributed—when one indeed really can know who got the job. But the 'justification' of Gettier's examples is only justification within a truncated perspective or information set T3. Indeed, Smith was justified to believe that Jones would get the job *based on preliminary information (T3)* about the boss's future decision¹⁰—but information that turned out to be inadequate: the boss changed his mind. T3 is in any case not the same information or set of hypotheses as T1; T1 is a revised and improved body of information with respect to T3 (one could write $T1 > T3$). In short, $T1 \neq T3$, in violation of the condition of consistent referencing. (Let me remind the reader that it is a key assumption of scientific realism

¹⁰ Thus, I understand T3 here as a body of information that includes, for instance, a promise of the boss regarding his future decision. Or, one could consider T3 as a set of hypotheses including, for instance, the hypotheses "The boss said that Jones would get the job" and "The boss is usually trustworthy".

that different theories usually have a different degree of adequacy or verisimilitude. Therefore one can in principle introduce a metric ($<$) to compare theories in the same problem space.)

An identical analysis is valid for Gettier's 'case II'. Now p = "Either Jones owns a Ford, or Brown is in Barcelona". If Smith has no clue where Brown is, but has evidence that Jones owns a Ford (he always had one in the past), he is justified to believe that p , based on the hypothesis that there is constancy in Jones's car possession. If in reality Jones doesn't own a Ford any longer, but Brown is by chance in Barcelona, p would be true and yet Smith would not really know that p . In our formalism: Smith is justified_T3 to believe that p , with $T3$ = the body of information that includes the hypothesis of constancy in Jones's car possession. At the same time p is true_T2 with $T2$ = the best possible body of information (including the information that Jones has no Ford and that Brown is in Barcelona). Smith does not know_T1 that p , with $T1 = T2$ = the best possible theory*. But the latter conclusion is only a counterexample to (P1), not to (P2), since, again, $T1 \neq T3$.

3.6 Interpretation. Comparison with Zagzebski, Quine, and Contextualism

In this section I will further elaborate and justify my above claims, notably by comparing them to Plato's theory of ideas, to Zagzebski's analysis of the inescapability of Gettier cases (1994), to Quine's holistic theory of meaning, and to epistemic contextualism.

To recapitulate, in the preceding sections I have shown that in certain conversations it is useful to specify the body of information or theory or perspective one relates to. Obviously, *explicit* referencing towards a theory or information set is an uncommon practice in daily conversations: it is not only cumbersome, but often also unnecessary. Explicit referencing is often redundant because in a non-ambiguous context the 'background information' or 'background set of assumptions' is broadly shared by all parties of the conversation: it is the body of information and assumptions that is available to common sense, or that is shared by people engaging in a 'usual' or 'normal' conversation—a conversation that is not in need of special epistemic precautions. At the same time, it seems at least worthwhile to consider the possibility that explicit referencing is as natural and helpful in epistemology (for solving problems) as it is in other contexts. I submit that this is the case for Plato's and Gettier's analyses of knowledge. On this view, then, (P2) is the more precise variant of (P1). This assumption allows for a transparent interpretation of Gettier's counterexamples to (P1): these counterexamples are based on a shift in the reference or perspective. If to the left and right of the *iff*-arrow in (P2) one uses a different measure or reference of knowledge and justification (i.e. if $T1 \neq T3$), it is not surprising anymore that the equivalence does not hold. (P2) contains in particular the condition of consistent referencing ($T1 = T2 = T3$), which emerges as a direct

consequence of perspectival language use and the fact that (P2) relates ‘know’, ‘true’ and ‘justified’. If one uses a reference for these words, they should be the same if one defines one concept by the others; without this condition (P2) would make no sense.

In short, what Gettier has shown is that *real* knowledge is not the same as *partially* justified belief (see Cabbolet (2024) for a similar conclusion). (P2) also suggests an answer to the question: under which conditions *is* justified true belief equal to knowledge? Formally: if $T1 = T2 = T3$ in (P2)—so if the condition of consistent referencing is satisfied. For several reasons, all related to the desideratum of maximising explanatory power, I said above that I would jump to conclusions and take this perspectival solution to Gettier’s problem as the basis for a perspectival epistemology—which aims at telling us something fundamental about the theory-relative nature of truth and knowledge. In the following I will corroborate this position by arguing that it can subsume other well-known accounts.

To start, Plato’s wider epistemology can be understood as a special case of (P2). Indeed, on Plato’s classic view, real knowledge (*episteme*) is knowledge of the eternal truths, the Ideas; an important part of Plato’s philosophy was aimed at distinguishing it from ‘common belief’, or ‘opinion’ (*doxa*). Now, it is natural to associate Plato’s *episteme* to (fully) justified belief in (absolutely) true propositions. Thus, Plato’s thesis (P1) can be seen as a natural limit case of (P2), with $T1 = T2 = T3$ = the best possible theory or information set, or the final or ‘divine’ theory if one likes.

Interestingly, it seems that when one uses ‘to know’ and ‘true’ in everyday speech, one uses them in general in this platonic sense: one means ‘to really know’, and ‘really true’ (recall also how Gettier interprets his examples). Not only people in general but also philosophers and scientists seem to find this concept of absolute knowledge natural, if we can rely on the frequency with which it appears in theology, philosophy, science (think of the related notion of ‘Theory of Everything’ used by physicists). But on my account this unrelativised or natural or ‘purely factual’ interpretation of ‘know’ should be seen *as an idealisation, in other words as an oversimplification*: it throws philosophy into unsolvable conundrums as Gettier’s problem if not properly re-constructed. Natural language is full of such idealised concepts; cases in point are the notions studied in the next chapters: free will (Chapters 6 and 7), cause (Chapter 5), and probability (chapter 4). Philosophers having studied these concepts professionally will presumably agree that there is a hidden part in their meaning. On the other hand, it is reassuring that the notion of objective or absolute truth can be given a clear meaning on our account: it is the truth as described or covered by the best possible theory/set of assumptions. In sum, our type of epistemic relativism is well compatible with a belief in objective truth, even if in a slightly subtle way.

Of the many investigations that have contributed to elucidate Gettier’s analysis, the well-known account of L. Zagzebski (1994) seems particularly pertinent. In ‘The inescapability of Gettier problems’, Zagzebski proposes a condition that is imposed on all usual accounts of knowledge, whether of Plato’s type (P1) or involving a fourth, fifth etc. RHS condition. The conclusion of the article is that all counterexamples to (P1) *à la* Gettier arise when “an accident of bad luck is cancelled out by an accident of good luck” (Zagzebski, 1994, p. 66). And such a cancelling-out can only

occur when there is a type of 'difference' or 'independence' between knowledge and justification. In the author's words (Zagzebski, 1994, p. 73):

The notion of knowledge requires success, both in reaching the goal of truth, and in reaching it via the right cognitive path. The notion of justification or warrant is less stringent, requiring only that the right path is one that is *usually* successful at getting the truth. It is this difference between the notion of knowledge and the notion of justification that is responsible for Gettier problems.

Zagzebski concludes (*ibidem*, p. 71):

It appears, then, that no account of knowledge as true belief plus something else can withstand Gettier objections as long as there is a small degree of independence between truth and the other conditions of knowledge.

Zagzebski does not explain in more detail what this difference or independence amounts to. Now, it seems clear that her claim can straightforwardly be recast in terms of a perspectival interpretation. Within my model, the counterexamples arise when T3, the reference for justification, is different from, less adequate than T1, the reference or perspective of knowledge (and truth); I wrote $T3 < T1$. This nicely fits to the above quotes. I also argued that in order to be 'Gettier-proof', one should 'consistently reference': T1 must be equal to T2 and equal to T3, where equality is measured in degree of verisimilitude. Clearly, this condition of consistent referencing corresponds to the condition of absence of independence Zagzebski refers to. In sum, both analyses overlap: even if there is only a small degree of independence, *even if the (implicit) references of the words 'know', 'true' and 'justified' are almost the same* ($T1 \approx T2 \approx T3$), "justified true belief will never be sufficient for knowledge" (Zagzebski, 1994, p. 69). Thus, the analysis in Sect. 3.5 provides an interpretation of Zagzebski's difference or independence between knowledge and justification. The latter is a difference in the information / hypotheses (the T's) that are implicitly referred to, in particular a difference in adequacy or verisimilitude of the latter.

It seems important to also look beyond the horizon of Gettier's analysis. I find it particularly relevant to make the link with W. Quine's acclaimed epistemological holism, which he developed, notably, in his seminal article 'Two Dogmas of Empiricism' (1951). (Quine withheld his view on holism in his later review paper (1991), cf. Verhaegh, 2018 §6.3; see also similar ideas by Hempel (1950)). Quine famously discusses the meaning and verifiability of scientific statements, criticising the analytic-synthetic distinction and reductionism—"the belief that each meaningful statement is equivalent to some logical construct upon terms which refer to immediate experience" (1951, p. 20). He says:

The dogma of reductionism survives in the supposition that each statement, taken in isolation from its fellows, can admit of confirmation or infirmation at all. My countersuggestion, issuing essentially from Carnap's doctrine of the physical world in the *Aufbau*, is that our statements about the external world face the tribunal of sense experience not individually but only as a corporate body. (1951, p. 38)

And:

The unit of empirical significance is the whole of science (1951, p. 39).

Thus, Quine argues that propositions cannot be verified or tested *individually* by direct empirical observation; what one tests or verifies is a whole web of hypotheses, a whole theory one could say, or perhaps even science as whole (a generalisation I would not make¹¹). Clearly, this epistemological/verificationist holism resonates remarkably well with the claims presented in this chapter. Quine's theory of verification could be seen as part of a truth theory, providing information about when (observation) statements (in science) are true. My main point is therefore the following: I believe that Quine's verificationist holism can be subsumed under the mixed coherence—correspondence theory of truth I adopted in Sect. 3.3 (coming from a wholly different linguistic angle); or, perhaps both are overlapping construals of the same idea.

Indeed, Quine's thesis can be restated, perhaps slightly more clearly and generally, in theory-relativised parlance, not only in a scientific but in a general context. Namely: the truth-value of propositions is attributed via the truth of a whole theory (in the broad sense); propositions are solidly embedded in theories T*/sets of hypotheses T*. More explicitly, a statement (whether an observation sentence or a theoretical statement) is true if it can (logically/rationally) be inferred from T* *and* if T* as a whole is an adequate theory* (i.e. if T* corresponds to reality, if it describes reality well, notions that clearly come with a degree). This is the mixed coherence-correspondence account of truth of Sect. 3.3, resonating with my solution to Gettier's problem.

Thus, there seems to be a strong overlap with some of Quine's celebrated views. I believe it would be interesting to go deeper into the similarities and differences in my view and Quine's, but this would go beyond the scope of this book (some topics of tension were mentioned in Chapter 1). This being said, I do not know how Quine's philosophy would solve Gettier's problem. For addressing this problem one needs, I believe, the relativity of statements with respect to theories*, not to science as a whole. Also, as shown above, several other epistemological riddles are solved by making the distinction between true_T1 and true_T2. On the other hand, just as for the problem of induction (cf. Chapter 2), Quine could perhaps deny that there is an interesting problem. In his (1987, pp. 108–110) he claims that the word 'knowledge' is imprecise and unscientific. He might therefore dismiss Gettier's problem as not being a real (scientific) problem.¹² I agree that 'to know' is vague, but not that there isn't a relevant philosophical problem. It seems to me that Quine

¹¹ True, in a paper written forty years later, Quine invokes not science as a whole, but rather "chunks of [science], clusters of sentences just inclusive enough to have critical semantic mass. By this I mean a cluster sufficient to imply an observable effect of an observable experimental condition" (1991, p. 268). Nevertheless, Verhaegh argues that Quine remained faithful to his radical holism (Verhaegh 2018, §6.3). On my view 'theory' is the better reference unit: if a cluster of sentences of a theory is definitely shown to be in contradiction with experiments, then the whole theory will ultimately be rejected (at least under certain conditions). Next, in the history of science as we know it, disciplines centred on theories have changed *independently* on countless occasions, leaving the rest of the network untouched. Finally, taking 'theory' as the reference unit is most instrumental for solving a series of other problems studied in this book.

¹² I thank Sander Verhaegh for having drawn my attention to (Quine 1987) and Quine's likely interpretation of Gettier's problem.

is caught in an overly restrictive canvas of seeing epistemology only as a chapter of natural science—a position that tends to reject philosophy as a genuine, theory-based discipline, and that has a self-contradictory touch to it. My account allows one to see in what sense ‘to know’ is vague, and how to remediate this problem.

It seems one can now make essential generalisations: once we have adopted the above relativistic coherence-correspondence account of truth, it is then also quite compelling to adopt following key thesis: namely, that the *meaning* of propositions *and* concepts derives from their meaning within a T^* . We will encounter essential illustrations of this thesis throughout this book (cf. the general conclusion, Chapter 8, also Chapter 6). This claim agrees again with a view that can be retraced to Quine, it seems (1960, 1969): *epistemological/verificationist holism and semantic holism are inextricably related—ultimately one and the same thing*.¹³ I will not elaborate on this idea here any further; but I hope that at the end of this book the reader will agree that many of my conclusions in the subsequent chapters corroborate the perspectival claims presented above. I believe they find their clearest expression in the theory-relativity or representation-relativity of truth, knowledge, and justification.

(If, by the end of this comparison with Quine's ideas, some readers conclude that I have merely re-derived some of his theses, I would gladly accept that verdict. Let us then hope that I have elucidated some lesser-known connections, and that this effort serves to illustrate the very Quine-compatible idea that one can reach the same nodes in the web of belief by different routes.)

Let us finally compare the above account to a well-known position in contemporary epistemology, namely contextualism, a type of epistemic relativism. As mentioned in the introduction, my initial goal was to follow Field's injunction to inquire whether key concepts of epistemology can be relativised to ‘evidential systems’. I will only compare my model here to some of the most salient claims of contextualist epistemology. Note that one alleged solution to Gettier's problem within contextualism was offered by D. Lewis (1996), but his solution relies on a complex set of unintuitive rules; it is now considered as controversial (e.g. Cohen, 1998; Heller, 1999).

In its paradigmatic form epistemic contextualism states that the (truth value of) propositions expressed by ‘knowledge sentences’ (such as ‘S knows that p’) depend(s) upon the conversational context in which they are uttered (e. g. Cohen, 1999; DeRose, 1999; Lewis, 1996; Rysiew, 2016). Now, it is trivial that the key ideas of contextualism and of the relativistic view based on (P2) can be translated into each other. In short, if knowledge attribution depends on the context, it depends on a given body of information: a conversational context is trivially linked to a body of information—without which the conversation would not be possible. Conversely, if the verb ‘know’ depends on a given body of information T^* , knowledge sentences (KS) indeed depend upon context, because the information set T^* in which one embeds a knowledge type depends on the context of conversation.

¹³ I will justify it this way: on the logical-epistemological level, the meaning of something, say X, can be construed, it seems, as a body of essential statements about X that are (supposed to be) true. If truth-about-X is ultimately relative to a T^* , then meaning-of-X also is.

Let us make a slightly more detailed comparison. Since there are infinitely many theories*, and since in most contexts several of these are likely to have some pertinence, the attributor of a KS can, a priori, refer to a few of these theories when she utters the (often ambiguous) phrase ‘S knows that p’. In context 1 she may mean ‘S knows_(T1) that p’, and in context 2 she may want to say ‘S knows_(T2) that p’ (or ‘S does *not* know_(T2) that p’).¹⁴ These and other epistemological riddles can straightforwardly be handled by my model (see one more in Appendix 3.1). Here, then, is one argument in favour of theory-relativism as compared to contextualism: *in one and the same context* several theories* can be discussed or referred to, and in that case the expression ‘to know that p’ must be accompanied, if full clarity is required, by a specification of the theory*, not the context.

In sum, there seems little doubt that context-relativism and theory-relativism at least broadly overlap in several respects. I submit that the added value of my model lies in the following: (i) its consistency with (language) practices in science and philosophy at large; (ii) the fact that it can offer a precise and straightforward interpretation of Gettier’s problem, as well as of the other epistemological problems and theories dealt with above; (iii) especially, the fact that it resonates favourably with the solutions to other key problems studied in this book.

3.7 Conclusion

In order to solve Gettier’s problem, we could have remained essentially on the level of linguistic practice, if we wished to play it safe, but we wanted to get to the core of things, and investigate what is to be learned from this particular problem for general theories of truth and knowledge—the core of epistemology. I proposed that the key principle to be derived is that of a fundamental theory-relativity or representation-relativity of knowledge, truth, and justification—in any context, scientific or mundane. My main arguments are based on 1) the fact that Gettier’s problem is a key problem of contemporary epistemology (so a coherent solution should be considered with particular attention); 2) the fact that the relativistic account of truth (a mixed coherence—correspondence view) that is most naturally compatible with the above solution to Gettier’s problem strongly resonates with Quine’s acclaimed holism; 3) especially, the fact that representation-relativity will turn out to be the

¹⁴ See the examples given above, and see Cohen (1999, p.58) for a well-known example, which can easily be analysed in relativised parlance. Following the example of Cohen, Smith may know that Mary’s plane stops in Chicago based on common sense and the assumption that official schedules are usually reliable. So Mary might say (or mean) that Smith knows the fact *within* the body of information offered by common sense and that assumption. But the same Mary might also say (or mean) that Smith doesn’t know the fact for sure if one takes potentially more reliable data into account (the same body of information plus the information gathered by checking with an airport manager). Relative to a potentially more complete body of information, Smith *doesn’t* know for sure.

main tool for problem-solving for a majority of problems studied in this book (see notably Chapters 5–8).

Some particular results were the following. The truth-value that one is (or should be) willing to attribute to any utterance depends on the (truth-value or the degree of truth of the) hypotheses/theories one necessarily must assume¹⁵ when making the utterance. Even (the truth-value of) a banal statement as ‘this is a tree’, or ‘the dog Droopy just bit me’ depends, in a sense, on (the truth-value of) a set of hypotheses—including for instance the assumption that one’s senses are not manipulated, that a certain dog really is called Droopy, etc. Needless to say, being aware of theory-relativity is mostly useful for solving complex problems: one does not need to be constantly aware of it in daily life! An inextricably and obviously related conclusion is then that *scientific knowledge is solidly embedded in theories alone* (on the logical-epistemological level); but more generally, that *any* knowledge, even of everyday things, is embedded in sets of hypotheses. Thus, our analysis of Gettier’s problem does not only suggest an account of truth, but also of knowledge; I will not elaborate, but it is clear that it corroborates ancient wisdoms about knowledge-gathering. These wisdoms can pleasantly be condensed in slogans, perhaps platitudes, such as: *(advanced) knowledge is all about theory building*—which is a variant of the MCT-hypothesis of Chapter 1. And: *the atoms of knowledge are theories*. I hope that such claims are self-explaining by now.¹⁶

To repeat, and to defuse once more charges of radical rationalism, theory-relativism can well be brought in harmony with the idea that many theories and sets of hypotheses are, *in the process of discovery*, abstracted by the human mind *from empirical observations*—and therefore that observations and experiments play a primordial role for knowledge gathering too, besides theory building. But theories* are the key ingredients at the level of a logical-epistemological analysis of what knowledge is. As emphasised above, theory-relativism can also well accommodate the notion of absolute or objective truth—as a limit concept.

My account is a moderate form of epistemic relativism, and is related to a form of perspectivism in philosophy of science. In philosophy of science, theory relativism could be compared to, for instance, the scientific perspectivism of authors as Giere (2006) and Massimi (2012, 2018, 2021), which are perspectival realisms (see discussions of perspectivism also in, e.g., Chakravartty, 2010 and Callebaut, 2012). Indeed, the following view is a natural companion to my epistemological account: what we can scientifically know are (ultimately) not absolute objective truths but theories, which offer perspectives to the objective universe—hence a connection with Giere’s

¹⁵ On a logical level, the necessity of the hypotheses/theories one assumes reveals itself in the following way: these hypotheses/theories must necessarily be announced or stated when someone asks ‘why?’, when someone asks ‘why is p true?’. This is the logical / rational way to justify the assumed truth of p.

¹⁶ To be explicit, once more: the truth (value) of statements depends on, is decided by, the theory one adopts. More obviously, the justification of statements is done via theories, and therefore hinges on the latter: propositions are justified if they derive from a justified theory (and we know the criteria for theory justification, based on prediction, explanation, problem solving). It is then also natural to say that the building blocks of knowledge are theories: not concepts or propositions in isolation.

and Massimi's views. Again, this position is not in conflict with the concept of objective truth; on the contrary, it is consistent with it. I will not study here in detail how deep the connection with Giere's and Massimi's accounts goes, but just note that many types of (scientific) perspectivism exist. To the best of my knowledge, none of them has put 'theories' or 'theories*' at the centre of the vantage point.

Appendix 3.1. On the Factuality of Knowledge

This appendix is intended for experts in epistemology, and can safely be skipped by other readers. At the risk of being redundant, I would like to respond to a worry I often heard, and that was phrased by an anonymous reviewer of (Vervoort & Shevchenko, 2022). Some readers keep finding it problematic that the above account asserts the following. If it is the case that *p*, according to one theory, call it *T*₁, and it is the case that not-*p* according to another theory, call it *T*₂, then one could truthfully assert (S1) 'John knows that *p* and Mary knows that not-*p*', if (S1) is understood as (S2) 'John knows_*T*₁ that *p* and Mary knows_*T*₂ that not-*p*'. However, so goes the objection, given the factuality of the verb 'know', no-one can assert (S1) without contradicting themselves. It is added that our proposal boils down to an attempt to provide the verb 'know' with an unnatural interpretation, and that we need to show how this verb's factual component can be accounted for in terms of our account.

I would answer the following. Natural language is rife with imprecisely defined concepts, and 'know' is one of them; it is precisely the task of philosophy to elucidate them. In a nutshell, statement (S2) is more precise than (S1), and avoids Gettier's problem *and* the problem just raised. As explained in more detail in Sect. 3.3 by my mixed coherence—correspondence view of truth, the factuality of knowledge can well be retrieved in the proposed model: factual knowledge is knowledge within a *T* that adequately describes the facts. "I know that *p*" [where *p* corresponds to a fact] means, really, "I know_*T* that *p* where *T* describes the facts adequately", in full, "I know that *p* relying on hypotheses *T* that describe the facts adequately". I agree, this may sound contrived; but it is a way out of several conundrums—and note, I do nothing more than generalising insights that are very alive in philosophical and scientific practice. Can epistemology afford to close its eyes on an epistemic reality in a realm that is reputed to employ high standards of knowledge, and therefore presumably language practices doing justice to these standards? Finally, in Sect. 3.6 I have argued that in everyday / natural language 'know' usually corresponds to 'really know' or 'know within the absolute *T*', which is a natural, but often imprecise, limit case of my model. In conclusion: knowing implies assuming; and sometimes one has to declare what one assumes.

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Chapter 4

The Scientific Concept of Probability



Abstract In this chapter I revisit the frequency interpretation of probability of von Mises, and argue that it contains the key elements of any coherent interpretation of scientific probability—notwithstanding the heavy criticisms von Mises collected from mathematicians and philosophers alike. I propose a modified definition that is not based on von Mises’ ‘infinite collectives’ but retains an essential ingredient of his interpretation, namely that probability can only quantitatively be defined for events that are, or can be, repeated in similar conditions and that exhibit frequency stabilisation. New is that the mentioned ‘conditions’ should be ‘partitioned’. Thus, I will partition probabilistic systems into object and environment, or in object, initiating, and probing subsystem, and show that such partitioning solves a series of classic problems. Other consistent interpretations can be unified with frequentism, e.g. Bayesianism à la Jaynes. I explore in detail the tantalising subjective or operationalist touch of probability, and argue that it can be put on objective grounds. One further upshot of the model is that several quantum puzzles can be seen as deriving from an imprecise interpretation of probability.

4.1 General Introduction

Philosophers are since a long time interested in the meaning, or interpretation, of the concept of probability, as witnessed by the enormous literature that exists on the topic. When people with a scientific training hear about this ‘problem of interpretation’, their first reaction is often one of surprise (as was my case). *Why is there a problem?* Are all problems of probability not adequately solved by the well-understood calculus of probability, based on Kolmogorov’s axioms? That there *is* an interesting problem-space can be recognised by following two considerations—to which usually not much attention is paid in science curricula (regrettably, I now believe). First, in textbooks of applied probability, the science student learns that two or more interpretations of probability exist, notably the ‘classic’ interpretation of Laplace, applicable to chance games (using dice, cards, urns with balls, etc.), and the ‘frequency’ interpretation, applicable to basically anything else. Especially modern textbooks add other

interpretations, notably the subjective interpretation. Rough definitions of these are the following. According to the classic Laplacian view, the probability of event E occurring in a probabilistic experiment is the ratio N_E/N , where N_E is the number of outcomes in which E occurs, and N the total number of different outcomes. (This is a rough definition; it presupposes for instance that the outcomes are equally likely and mutually exclusive.) According to the frequency interpretation, the probability of E is the relative frequency with which E arises in long (ideally infinite) series of repeated experiments. Both these interpretations are ‘objective’. According to the ‘subjective’ or Bayesian interpretation of probability, probability values are subjective measures of belief; different persons may attribute different probabilities to the same event. Several other interpretations exist, and each in several variants. As a consequence, a philosophically-inclined student, consciously or unconsciously driven by an urge for unification, might wonder: *Why there isn’t one overarching interpretation?*

I will argue, following von Mises, that probability can be seen not only as a mathematical theory, but also as a physical one. In a sense, probability theory is the most encompassing physical theory we have. All physical systems comply with it: obviously all probabilistic systems comply with it¹; and deterministic systems are a special case of probabilistic systems characterised by probabilities 0 or 1. This fact already seems to suggest that there should be *one* fundamental interpretation of (scientific) probability *as a basic physical property or category*. But can one really assert this intuition in view of the plethora of counterarguments and pluralistic positions that thrive in the literature? Isn’t that a thought too bold to entertain? Possibly, but according to the MCT-thesis we should take it seriously.

A second eye-opening consideration as to whether there is a real problem, stems from the intriguing chapter of the paradoxes of probability theory. Applied probability is rife with such paradoxes: problems the solutions of which were disputed even among famous mathematicians, including the fathers of probability theory. Many mathematics books on probability contain at least a few: Bertrand’s paradox, the Borel–Kolmogorov paradox, the Monty Hall problem, etc. Together with the first consideration, these paradoxes show that the *application* of measure-theoretic probability theory (Kolmogorov’s axiomatic theory) to *real-world* problems is non-trivial. But it *can* of course be applied to real-world problems, for instance physical problems, say die throwing, chance games, or countless problems of statistical physics. In sum, these two considerations show that the interpretation of probability—What is probability beyond the mathematics? How to apply the theory?—is a genuine problem at the interface of mathematics, physics and philosophy. I now believe it is a particularly fascinating one, and a uniquely subtle one to boot. People who have delved into this question will not need to be convinced. Novices may now pause, and try to come up with their own answer to the above questions.

Following the heuristic rule of Chapter 1, I believe the approach to tackle the problem of interpreting probability should be one of unification, as always: one should try to unify existing interpretations (as far as it is possible), and show that

¹ I mean that the probabilities predicted by probabilistic physical theories, say quantum mechanics, obviously comply with probability theory.

the unified theory can solve more problems, including the mentioned paradoxes and the very many problems discussed in the literature. I found it instrumental to guide this quest by following additional questions: What is a probabilistic *system* (in the physical sense)?, and: How does one extract probability values from empirical data? These questions put the search on firm empirical grounds, which seems not a luxury. Indeed, I believe it is not enough emphasised in the philosophical literature that in science *any* experimental probability is determined, measured, verified *as a relative frequency*. Recalling the verificationist principle of meaning of the logical empiricists, *and* the credo of synthetic philosophy that science is the ultimate judge of philosophy (at least for problems that have such an immediate link with science), this seems to give a head start to the frequency interpretation. But again, the real question is: does the latter allow one to unify and concomitantly to answer the largest number of well-posed questions? That is what will be studied in this chapter.

Some people immediately object that theories such as quantum mechanics may predict probabilities as real numbers that are not obviously ratios or frequencies (think of the square of a modulus of a quantum wave function, a transition amplitude, etc.). But to *verify* these numbers one always determines relative frequencies. And as most empirically determined quantities (say the magnitude of speed, energy, magnetic field, etc.), these relative frequencies necessarily have an experimental error or precision; they lie within a numerical range ΔQ . As emphasised in the next section, it is the key assumption of my von Mises-style model, *and one of the best corroborated facts of science*, that ΔQ can be made as small as practically needed by increasing the number of trials. In other words, the measured probability $Q = P \pm \Delta Q$, where P is the ‘real’ or theoretical probability value and ΔQ the experimental precision or uncertainty. Any real number P between 0 and 1 (as predicted say by quantum mechanics) can be written under the form $P = Q \pm \Delta Q$ with Q rational; and ΔQ can be made arbitrarily small by increasing the trials. In sum, one can determine ratios that arbitrarily closely approximate² the theoretical P .

Several authors have scrutinised the frequency interpretation (for reviews on this and other interpretations, see e.g. the excellent Fine (1973), von Plato (1994), Gillies (2000), Khrennikov (2008)). K. Popper and B. van Fraassen were at some point proponents of the frequency interpretation—before becoming adepts of other positions. I will have a closer look at Popper’s arguments in favour of the so-called propensity interpretation in Sect. 5. In his (1980, pp. 190–194), van Fraassen analyses how to link physical experiments to probability functions, and proposes following definition (1980, p. 194):

The probability of event A equals the relative frequency with which it would occur, were a suitably designed experiment performed often enough under suitable conditions.

My definition will be seen to be broadly in agreement with the above one. But it appears that the notion of probability has so many hidden dimensions that a more elaborate model is needed in order to address the numerous extant problems. To

² Note that if, in this verification process, the predicted probabilities do consistently not correspond to the measured frequencies (within the experimental error), the theory is at risk of being rejected.

mention only two of these implicit notions: I will give in particular a further analysis of what the ‘suitable conditions’ in van Fraassen’s definition are, and what ‘often enough’ might mean. This being said, it will be seen that van Fraassen’s condensed phrasing can function as an excellent mnemonic device.

Besides philosophers, the fathers of probability theory, from Pascal, Fermat, De Moivre, Laplace to von Mises and Kolmogorov, were often interested in the emerging philosophical questions. The analytic definition of probability I will propose in Sect. 4, the core of this chapter, can be seen as a variant of von Mises’ definition (1928/1981, 1964). When trying to construct my interpretational model, I found Kolmogorov’s and his pupil Gnedenko’s standard references on the mathematical theory highly instrumental³ (Kolmogorov, 1933/1956, Gnedenko, 1967). Actually, Kolmogorov adopts an interpretation of probability that is frequentist (1933/1956, p. 3, see below); he explicitly refers to von Mises as his source of inspiration.

A last preliminary remark concerns the use of the word ‘probability’ in daily contexts. Even if there may be other uses of the notion of probability in general philosophy, psychology, and certainly everyday discourse,⁴ and even if some of these may be coherent, I will focus here on the notion as used in natural science. I will find as usual initial inspiration in physics, but the aim is to encompass natural science as a whole.

The chapter is organised as follows. In Sect. 2, I will anticipate some of the modifications I will introduce to von Mises’ interpretation. In Sect. 3 I will analyse some simple probabilistic systems in order to introduce the main concepts of my definition, which will be presented in final form in Sect. 4. Problem-solving and comparison with other interpretations are done in the Sects. 4 and 5. Section 6 concludes and proposes lines of further research.

4.2 Introduction for the Professional. Preliminary Remarks on Similarities and Differences with von Mises’ Model

It was A. Kolmogorov who published the first and simplest axiomatic system for probability in 1933 (1933/1956). This theory is generally believed to cover all probabilistic systems of the natural, applied and social sciences. At the same time Kolmogorov’s

³ I benefited in particular from Boris Gnedenko’s book on probability calculus. Gnedenko did not only contribute to the calculus (e.g. by elaborations of the Central Limit Theorem and in statistics); his work reflects his keen interest in the foundational issues. Also Kolmogorov insisted on the need of an interpretation of probability (cf. his (1933/1956) p. 9). Interestingly enough, the axiomatisation of probability theory and the discovery of quantum mechanics overlapped in time. Von Mises but also Kolmogorov and Gnedenko followed the development of quantum mechanics with interest, surely realising that probability theory governs an ever wider part of the natural phenomena.

⁴ In daily conversations, for instance when discussing “the probability that person X will do Y” it is more appropriate to use the word ‘likelihood’ rather than ‘probability’; the former concept is not necessarily related to the strict rules of probability calculus. Thus, in principle, it is preferable to reserve ‘probability’ for the concept that complies with the mathematical theory.

theory—the calculus of probabilistic or random events—does not define what a ‘probabilistic/random event’ is and does not provide an interpretation of probability. For a deeper understanding of what probability is, and for applying the calculus to real world situations, one needs to resort to the ideas of other fathers of probability theory, such as Laplace, Fermat, Venn, von Mises, and to the philosophers having investigated this matter (for reviews, see notably Fine (1973), von Plato (1994), Gillies (2000), Khrennikov (2008)).

In the broader physics community the most popular interpretation surely is the frequency model—especially the limiting frequency version, usually attributed to Richard von Mises (1928/1981, 1964). But in philosophy and the foundations of quantum mechanics other interpretations, in particular the subjective interpretation, are increasingly popular. Suffices to consult contemporary works on the interpretation of probability in quantum physics⁵ (and, to a lesser extent, in statistical mechanics) to realise how vivid and wide-ranging the debate is, also in the physics community. Thus, the objective versus subjective controversy is far from being settled (see e.g. Gillies (2000) and for an overview in physics, Beisbart and Hartmann (2011), Ch. 1). I will try to come to grips with this remarkably subtle debate in the following sections.

Now, the dramatic underrepresentation of the frequency interpretation à la von Mises in the contemporary philosophical debate is very striking. This state of affairs is partly due to the fact that naïve frequency interpretations have extensively been analysed and found wanting (cf. e.g. Fine (1973), von Plato (1994), Gillies (2000), Hájek (2009, 2023) and references therein). More importantly, von Mises’ calculus, based on the notion of ‘collective’, acquired both among mathematicians and philosophers the reputation of being complex and possibly incoherent. During the last century several alleged mathematical flaws were discussed in publications, but today most commentators seem to agree that von Mises’ calculus can well be made consistent if amended and interpreted adequately (e.g. Gillies, 2000; p. 104; von Plato, 1994, p. 197; Khrennikov, 2008, p. 27). Still, the damage was done.

Indeed, I believe that these numerous criticisms have unduly discredited von Mises’ work. I will go further: I believe that neglecting the key ideas that are explicit or implicit in this work simply amounts to being blind for the most fundamental properties of scientific probability (beyond the mathematics). One could say that where Kolmogorov describes probability, von Mises explains it.⁶ Von Mises’ theory not only recovers Kolmogorov’s axioms, it also explains what probability values or measures ‘are’, where probability comes from—and *it is the only one to achieve this in detail*. The subtlety of it, and hence the many uncharitable criticisms, are, I submit, in the first place due to the fact that the theory does not deal with objects (or events) per se, as usual physical theories, but with *collections* of systems or events (“mass phenomena and repetitive events”, in von Mises’ words ((1928/1981) p. v)). I conjecture that one needs a genuine change of mind-set to come to grips with this particularity, as we will study in this chapter.

⁵ See e.g. Caves et al., (2002, 2007), and Bub (2007).

⁶ I thank Richard Gill for this apt phrase.

In *rough* essence von Mises interprets the probability $P(R_j)$ as $\lim_{n \rightarrow \infty} n(R_j)/n$, where $n(R_j)$ is the number of events or trials that have result R_j in a series of n trials ($j \in \{1, \dots, N\}$). This concept of *limiting relative frequency*, for the limit of n tending to infinity, is considered problematic by several authors (see e.g. Neapolitan (1992), who disagrees). This is a remarkable state of affairs. By definition of limit,

$$P(R_j) = \lim_{n \rightarrow \infty} \frac{n(R_j)}{n} \Leftrightarrow \forall \varepsilon > 0, \exists N : \forall n > N, \left| \frac{n(R_j)}{n} - P(R_j) \right| < \varepsilon. \quad (4.1)$$

Mathematicians as Richter (1978, p. 426) and others (Neapolitan, 1992) complain that the last inequality in (1) may not be satisfied. But this is in overt contradiction with empirical evidence: an overwhelming amount of experimental data has shown that in long (of course not infinite) experimental series the values $n(R_j)/n$ do tend towards a fixed number *in a way that complies with (1)—at least under conditions that I will specify in detail in the next sections*. (Let me warn in advance: convincing oneself that (1) is correct may demand quite some attention. A clear graphical illustration of the limiting behaviour of typical random variables is given in Gillies (2000), p. 93, Fig. 5.2.) A first somewhat trivial confusion regarding (1) is related to the observation that surely for some chosen n and small ε the difference $\left| \frac{n(R_j)}{n} - P(R_j) \right|$ might be larger than ε . But experience has shown that one should be able to find a number N large enough such that for all $n > N$ the difference becomes smaller than any ε . This well-corroborated scientific hypothesis of frequency stabilisation—the ‘Ur-phenomenon’ of probability theory as von Mises calls it—is well captured by (1) *if one understands it correctly*.

A less obvious and frequent confusion is linked to the following. Note first that the existing empirical evidence corroborates the initial hypothesis (namely of frequency stabilisation as a real-world phenomenon) as strongly as any hypothesis of the natural sciences can be confirmed: hypotheses of natural science are never confirmed with absolute certainty. Now, of course, in some series of trials aiming at determining probabilities, it may be that there are changes in the limiting value. For instance, it may be that after very many repeated measurements of a magnetic field strength B in a given point X , leading to a constant result of say 0.5 Tesla (which would allow one to infer that $P(B_X = 0.5) \approx 1$), the next measurements in the series unexpectedly jump to 0.7 Tesla. *This in itself would not be enough ground to reject the hypothesis that, in the conditions in which the first series of measurements were done, there is a fixed field $B_X \approx 0.5$ Tesla in X* . If the next trials constantly indicate that now $B_X \approx 0.7$ Tesla, physicists would look for causes of the jump. The conditions (environment) of the experiment must have changed, or the magnetometer must have started to malfunction; 0.7 Tesla is the field strength in the new (stable) experimental conditions—which arose due to some known or unknown systematic cause. If 0.7 Tesla is only measured for extremely rare trials in the series (with probability measure 0), this jump would be termed ‘erratic’; it would be due to an unknown, unsystematic cause and it would *not* alter the conclusion $P(B_X = 0.5) \approx 1$. Similarly, if in a given experiment that is known to be genuinely probabilistic a limiting frequency could

not be identified, one would not reject the law of frequency stabilisation, but look for causes to explain the erratic behaviour. We will look at all concepts involved here in more detail in the next sections.

Thus, at the very basis of the frequency interpretation lies the scientific hypothesis that if certain well-defined experiments (trials) would be repeated an infinity of times, certain ratio's (involving outcomes of these trials) would converge to a limit.⁷ Some read this as a strongly counterfactual statement ("infinite series do not exist"); *but note that that is not the only possible reading*. Definition (1) can also be read in a very pragmatic and factual way: the longer one makes the trial series, the closer one comes to a certain number (in a way compatible with (1))—as confirmed by countless experiments in all areas of natural science. I therefore emphatically disagree with authors as A. Hájek, who comments: "Note that at this point we have left empiricism behind. A modal element has been injected into frequentism with this invocation of a counterfactual; moreover, the counterfactual may involve a radical departure from the way things actually are, one that may even require the breaking of laws of nature" (Hájek, 2023, Sect. 3.4). Much to the contrary: frequentism, if well defined, is likely the only interpretation that takes the laws of nature seriously. I will look in quite some detail into the allegedly counterfactual touch in von Mises-style theories in the next sections.

Indeed, this stabilisation hypothesis suggests one of the most fascinating questions of philosophy: *Why does frequency stabilisation happen? Why is frequency stabilisation a ubiquitous property of nature?* To my personal taste, these (and related) questions point to some of the most intriguing problems of science and philosophy. I will return to them in the Epilogue.

For the sake of completeness, I very succinctly analyse some of the criticisms of von Mises' theory (mainly related to his use of the notion of infinity) in Appendix 4.1. It should be remarked, though, that von Mises dealt in detail with many of them (see in particular the introductions of his 1928/1981 and 1964). If probability theory is a physical theory, then it is useful to remember that the concept of infinity is ubiquitous in theoretical physics, even if it is an idealisation in calculations—a *useful one when the formulas correctly predict what happens in reality*.

Nevertheless, in the following I will argue that it is not necessary to define frequency stabilisation in terms of convergence in the limit $n \rightarrow \infty$. One can define 'probability-as-measure-with-precision- δ ' in a pragmatic way without any counterfactual notion of infinity (cf. Sect. 4). Although I have never encountered a final argument against von Mises' use of infinity, introducing the notion of precision has the advantage (besides of avoiding putative subtleties) of allowing to define probability in a more inclusive way, as it is used in cosmology, meteorology, and other areas.

My frequency interpretation will be seen to be much simpler than von Mises' theory (1928/1981, 1964). A *real-world* (or *physical*) theory (T) for probability is the union of a calculus (C) and an interpretation (I): $T = C \cup I$. The theory 'I' may contain e.g. the philosophical hypothesis that the theory represents things or events

⁷ More on the modal aspect of probability in van Fraassen (1980).

‘out there’; more generally, it contains rules stipulating to which objects to apply the mathematics, and how to do this. Von Mises proposed both an interpretation I and a calculus C—a physical calculus one might say; not an abstract, purely mathematical theory as Kolmogorov’s. Both his ‘I’ and ‘C’ are based on the concept of collective, an in principle infinite series of experimental results. His calculus strikes however by its complexity, and nowadays probably no-one would consider using it on a regular basis (see e.g. his treatment of de Méré’s problem, von Mises (1928/1981) p. 58ff). Therefore, I will not use von Mises’ calculus of collectives; my calculus is Kolmogorov’s; both lead to the same results.

In particular, I believe that it is not necessary to resort to von Mises’ notion of ‘invariance under place selection’ to characterise randomness. Von Mises’ ‘randomness condition’ is, in essence, the assumption or postulate that the limiting frequencies of a collective remain invariant in sub-sequences obtained under ‘place selections’, i.e. certain functions defined on the original collective (von Mises, 1964). This characterisation triggered numerous mathematical questions and criticisms—to which von Mises responded. Now, even if von Mises’ work on invariance under place selection is mathematically highly interesting (cf. Appendix 4.1), *I believe it is superfluous for its goal of defining randomness*. Indeed, it is enough to define randomness via the notion of frequency stabilisation. To the best of my knowledge, the most relevant argument is due to R. Gillies, who proved that the assumption of frequency stabilisation *without* the randomness condition suffices to imply Kolmogorov’s axioms (Gillies (2000), p. 112). This clearly points to the conclusion that this condition is indeed redundant, and that frequency stabilisation is the only relevant criterion for probabilistic randomness (besides unpredictability). I will elaborate this idea in Sects. 3 and 4.

Besides a simplification, my model is also intended as a clarification of the frequency interpretation. It helps in avoiding the paradoxes to which probability theory is so sensitive. I will show that when the application of probability theory becomes subtle, one gains in *partitioning* probabilistic systems into subsystems: namely test object; initiating; and probing subsystem (or environment, in some contexts of inquiry). This partitioning allows solving a series of classic problems (Sect. 5): it solves Bertrand’s paradox; it offers an explanation of the ‘subjective’ temptation and of how it can be avoided; and it allows unifying the frequency interpretation with other interpretations. Finally, and perhaps surprisingly, my frequency model allows to re-interpret many puzzles of quantum philosophy as direct consequences of an inadequate interpretation of probability.

A last preliminary remark concerns the tantalising subjective or epistemic or operationalist dimension of probability theory, which we will encounter at different occasions throughout this chapter. I will only come to a final conclusion in the last section. The ‘subjective versus objective’ debate manifests itself already in the question of what the subject matter of probability theory is—to what exactly to apply it? According to von Mises probability theory treats “problems in which either the same event repeats itself again and again, or a great number of uniform elements are involved at the same time” (von Mises (1928/1981) p. 11). I will push his answer a little further and arrive at the conclusion that probability theory applies to a special

type of random events or systems, which I will term ‘p-random’ or ‘ ρ -random’—hence ‘ ρ -systems’. Since I attribute probabilities to systems, my objectivist ambitions are clear. Indeed, my definition will be seen to be applicable to both ‘chance games’ and natural probabilistic phenomena: both happen in real-world systems.⁸ (This is the first and simplest synthesis of this chapter.) In this objectivist perspective, I assume, likely as most scientists, that these natural random phenomena, such as quantum phenomena, spontaneously happen in nature *in agreement with the laws of probability theory—also when nobody looks*.⁹ Of course, this assumption is in agreement with scientific realism, which I have to adopt in order to avoid the need to conjure up contrived answers to such questions as: “why can all people who use the same experimental protocol measure the same probability values for probabilistic events?” For the realist the answer seems straightforward and coherent with all of science: because probability is an objective, ontological category of nature; it characterizes *objective chance* (more in §4.2, **C2** below). (For a well-developed and different view, see Gillies (2000, Ch. 8) and, in the case of statistical mechanics, Uffink (2011).)

And yet I believe that this is not all that must be said about this unusually subtle matter. *There is an epistemic aspect to probability too*; more precisely: a somewhat complete physical/interpretational theory of probability will have to highlight epistemic aspects of the application of the theory. According to a typical definition “Objective interpretations of probability [...] take probability to be a feature of the objective material world, which has nothing to do with human knowledge or belief. Clearly the frequency and propensity interpretations are objective” (Gillies, 2000, p. 2). I agree, and yet I will argue in Sect. 4 that there is an epistemic dimension in the application of probability theory, in this precise sense: the ascription of whether a system or event is probabilistic (indeterministic) or deterministic, or neither of both, does depend on the knowledge state of the attributor. But let me reserve my final analysis to the conclusion (Sect. 6).

Let us now enter into the heart of the matter. Readers familiar with the frequency interpretation might go quickly over Sect. 3 (but halt to read Kolmogorov’s interpretation and his reliance on the notion of ‘condition’) and focus on Sect. 4.

⁸ What I will term, following von Mises, chance games are games or procedures based on man-made probabilistic systems, such as dice, urns containing balls, card decks, roulettes, etc. ‘Natural’ probabilistic phenomena are ubiquitous in nature and can be found in diffusion, population dynamics, fluid dynamics, quantum mechanics and, actually, in any branch of natural science.

⁹ It seems to me that one more convincing argument in favour of this claim resides in the following. One can measure averaged quantities (such as average kinetic energy) on ensembles of systems (say molecules in a gas), quantities that can also be calculated by integrating over probability distributions. The favourable comparison between measurement and theory—demonstrated many times by physicists—is indirect but convincing evidence in favour of the probabilistic assumptions made in the calculations, *and* of the claim that probabilities are objective.

4.3 Examples. Frequency Stabilisation and Partitioned Conditions

Let us have a look at a typical probabilistic system, namely a die. The key question to start with is: when or why is a die ‘probabilistic’, or ‘random’ (or rather the throwing events or outcomes)? Simply stating in non-anthropocentric terms what a die throw is, brings already to the fore a few essential notions. In physical terms, a ‘die throwing event’ or ‘die throw’ consists of the 3-dimensional movement of a (regular) die, that is 1) characterised by the time evolution of its centre of mass and its three Euler angles, 2) *caused* or *initiated* by an ‘initiating/initialising system’ (e.g. a randomising and throwing hand, or automat), and 3) *probed* by an ‘observing/probing system’ (e.g. a table) that allows to observe (in general ‘measure’) an outcome or result *R* (one up, two up, etc.). In the case of die throwing, it is easy to realise that if we want to use the die as it should, i.e. if we want to be able to observe or measure the usual probability for the different results of the throws, we have to repeat the experimental series (i.e. the throws) not only by using the same regular die, or similar regular dies, but also by applying the usual ‘boundary conditions’. Here I obviously do not refer to the detailed initial conditions of each individual throw, which are unknown, but to the conditions of our random experiment *as they can be stated in an experimental protocol*, and thus be communicated to other experimenters (“throw roughly in this manner, probe on a table”, etc.). Countless experiments have shown that if the protocol is well-defined and well-repeated, the same probabilities will be generated. In a slogan: same experiment, same probabilities. (If this notion of protocol rings operationalist bells to some readers, I can only agree. But, as said, I believe this aspect can be put on an objective basis.)

These boundary conditions, then, are related to our throwing, the table, and the environment in general. *Irregular conditions in any one of these three elements may alter the probability distribution.* We can for instance not substantially alter our hand movement, say by putting the die systematically ace up on the table: this would change the probability distribution for the six outcomes from the usual (1/6, 1/6, 1/6, 1/6, 1/6, 1/6) to (1, 0, 0, 0, 0, 0). Nor can we put glue on the table, position it close to our hand, and gently overturn the die on the table while always starting ace up—one of the outcomes could occur more often than in a ratio of 1/6. Nor can we do the experiments in water instead of air: again one can imagine situations in which the environment alters the usual probabilities.

It will make sense to isolate the mentioned elements, and to consider a random event as involving a *random system* containing three subsystems, namely 1) the (random) *test object* itself (the die), 2) the *initiating system* (the throwing hand), and 3) the *probing* or *observing system* (the table, and let’s include the human eye). I will call such a composed random system a ‘*p*-system’ or probabilistic system (‘*p*’ stands for the probability distribution generated by the system, see the general definition in **C3** below). Just as one can associate subsystems to the random event, one can associate (composed) *conditions* to it, so conditions under which the random experiment occurs: *initiating/initialising* and *probing conditions*.

Here is another example, from darts. According to the frequency interpretation, if we want to determine the probability that a given robot, equipped with a throwing arm, hits a given section of a dartboard (suppose there are 10 sections S_1 to S_{10}), we let it throw say 1000 times, and count relative frequencies: n_1 hits in S_1 , so P_1 (the probability the robot hits S_1 in the given conditions) is approximately $n_1/1000$, etc. Again, if we want to define probabilities for the experiment, the robot, the darts *and* the board need to ‘behave well’ and operate in well-defined constant conditions. The probabilities are defined *relative to these conditions*, referring to the test object (the darts), the initiating system (the throwing robot), and the probing or observing system (the board, and we can again include some optical detection system). For instance, if during the experiment the robot’s output force decreases due to a failing component in its circuit, P_1 might be lower than $n_1/1000$, the value obtained under normal conditions.¹⁰ All this may look almost trivial; and yet it will prove useful to explicitly include these observations in the definition of probability.

At this point an essential remark is in order. Instead of ‘initiating system’ and ‘probing system’, it can be more appropriate to speak of ‘environment’, namely in the case of spontaneous or natural probabilistic events (versus artificial ones, as the outcomes of die throws and chance games created by human intervention). Many random events occur spontaneously, without any known cause. A spontaneously disintegrating nucleus has probabilistic properties, for instance the lifetime at which it disintegrates. Such properties are, on usual interpretations, not ‘initiated’ by a person or a robot, as in die throwing. Neither are they probed, except when subject to lab experiments. But the nuclei disintegrate spontaneously according to a probabilistic pattern that is numerically well-defined, *only if the environment is well-defined and stable*. Changing the environment, e.g. by irradiating the nuclei or by strongly heating them, may very well change the probability distribution in question. In other words, if we want to experimentally determine the half-life of the nuclei, i.e. the probability of disintegration, we have to put them in the lab in well-defined (initialising) conditions of temperature, pressure, etc. and measure their properties in well-defined (probing) conditions, which scrupulously correspond to their natural environment—if we want to know their probabilistic behaviour ‘in nature’. So, also in the case of natural stochastic phenomena the initial and final conditions re-appear, be it under a different guise: as the conditions describing the environment.

By the above partitioning in subsystems we have clarified the concept of ‘conditions’. This concept is not explicit in von Mises’ notion of collective, but it is explicitly mentioned by Kolmogorov (1933/1956, pp. 3–4), in the brief paragraph where he discusses how to apply his axiomatic theory to real-world *experiments* (sic). In this paragraph Kolmogorov explicitly acknowledges von Mises as his main source of inspiration (1933/1956, p. 3); and it is clear that Kolmogorov’s interpretation comes very close to von Mises’. Here is the key passage:

“There is assumed a complex of conditions, [C], which allows of any number of repetitions [...]. Under certain conditions, which we shall not discuss here, we may assume that to an

¹⁰ Notice there is some arbitrariness in where we draw the line between the different subsystems. But that doesn’t play any role in the following.

event A which may or may not occur under conditions [C], is assigned a real number $P(A)$ which has the following characteristics:

- (a) One can be practically certain that if the complex of conditions [C] is repeated a large number of times, n , then if m be the number of occurrences of event A, the ratio m/n will differ very slightly from $P(A)$.
- (b) If $P(A)$ is very small, one can be practically certain that when the conditions [C] are realized only once, the event A would not occur at all."

(Notice that 'conditions' is used in two different meanings. With the 'conditions that are not discussed here', Kolmogorov presumably refers to von Mises' conditions for applicability of probability theory, namely his conditions of frequency stabilisation and 'randomness'. I will argue below that the former suffices; it encompasses the latter.)

The notion of condition also appears in Gnedenko's (1967, p. 21):

On the basis of observation and experiment science arrives at the formulation of the natural laws that govern the phenomena it studies. The simplest and most widely used scheme of such laws is the following: *Whenever a certain set of conditions C is realized, the event A occurs.*

And a little further (1967, p. 21):

An event that may or may not occur when the set of conditions C is realized, is called *random*.

By partitioning, we characterised these conditions in some more detail than in the mentioned works.

Interestingly, this interpretational notion of condition can immediately be linked to the calculus—which shows how essential it is. Indeed, even unconditional probabilities ($P(R)$) can be regarded as conditional *on the conditions or circumstances of realisation*, as is implicit in Kolmogorov (1933/1956, p. 3) and explicitly stated by Gnedenko (1967, p. 67). In a scientific context these circumstances can be described as variables assuming a certain value (in that sense they correspond to events); in the simplest case such variables C have dichotomic values, i.e. $C = c \in \{\text{yes, no}\}$ or $\in \{\pm 1\}$, say. Thus, the natural generalisation in the model I will propose below is to replace, if helpful, $P(R)$ by $P(R|C)$ where C is the set or n-tuple of all relevant variables that describe the initiating, probing, and/or 'environmental' conditions in which the event R occurs. (Caution: C refers here to variables that are *constant* for all considered trials of the experiment; C are *not* random variables within the experiment considered. If C represents random variables then in general $P(R) \neq P(R|C)$, of course, and one cannot replace $P(R)$ by $P(R|C)$.) For instance, in the example of the disintegrating nucleus, the probability that the disintegration time $t = t_1$ can be formalized as $P(t = t_1|T, p)$, where T and p fix the temperature and pressure¹¹ (in certain intervals). *Note that if these conditions are not made explicit as in $P(R|C)$,*

¹¹ In principle it is more precise to write not $P(t = t_1|T, p)$ but rather $P(t = t_1|T = \tau, p = \pi)$ with τ and π numerical values (or intervals); so to explicitly relate the probabilities to events instead of variables. This is understood in the widespread abbreviated notation I often use.

then they have to be considered as implicit (some authors use the notation ' $P_C(R)$ '). Again, at face value all this is quite straightforward.

With this in mind, it seems not difficult now to deduce the *conditions of randomness* of a physical event as a die throw—the criterion to which von Mises spent so much mathematical effort. Suppose we throw a die 20, 30, in general n times ($n \gg 6$), and that we note how often the 6 results R_j ($j = 1, \dots, 6$; $R_j = 1, \dots, 6$) occur in these n throws. We can then determine the relative frequencies of the results R_j , namely the 6 ratios n_j/n where n_j is the number of throws that have result R_j and n the total number of throws. Now, according to the frequency interpretation (and, I submit, according to the practice of any scientist trained in experimental statistics), in order that the die throws can be termed 'probabilistic', it is a *necessary condition* that the relative frequencies of the R_j (the ratios n_j/n) converge towards a constant number when n grows. For a regular die thrown in regular conditions it is an empirical matter of fact that the six ratios n_j/n converge towards $1/6$. *If such frequency stabilisation would not occur, one cannot speak of (scientific) probability.* If for instance the die would subliminate during the experiment in an asymmetric manner, thus gradually losing weight on one side, or if we modify the usual die throwing in a manner that does not lead to frequency stabilisation, we simply cannot attribute a probability to the outcomes. In such irregular conditions, if we repeat the throws, the above-mentioned ratios will not converge but erratically jump from one value to another without showing the typical limiting behaviour of probabilistic systems. In von Mises' words (1928/1981, p. 12):

It is essential for the theory of probability that experience has shown that in the game of dice, as in all the other mass phenomena which we have mentioned, the relative frequencies of certain attributes become more and more stable as the number of observations is increased.

Thus, frequency stabilisation gets the status of a principle, or postulate, expressing something quite akin to a law of nature—and yet on the surface different from the usual laws, since now human agency (repeated observations) is explicitly referred to. Probability-theory-as-a-theory-of-the-universe, at least in von Mises' perspective, has to assume this principle, as backed-up by empirical evidence. Of course, any real-world theory has to assume postulates, laws, ... (recall Chapter 2 on the problem of induction).

As already mentioned, frequency stabilisation and empirical probability¹² are characterised by a precision (' δ '), which will be integral part of the general definitions (DEF1&2) I propose below. The experimental series n_j/n stabilise as a function of n towards the constant number P *within an experimental precision* δ (typically in an irregular, oscillating manner, but with on average ever smaller amplitudes, in agreement with (1)). This means that from some value n on, n_j/n will lie in the interval $[P - \delta, P + \delta]$. Typically, this precision is of the order $1/\sqrt{n}$, in agreement

¹² Experimental or empirical probability is the object of interpretational/physical theories as the one we are interested in; to be compared to the purely mathematical probability of Kolmogorov's theory. Ultimately they should refer to the same thing. Analogously, physical properties usually are represented in physical theories as error-free variables with an exact value, which almost never coincides *exactly* with the experimental value.

with the Central Limit Theorem (Feller, 1991; see examples in Gillies, 2000, Ch. 5 and p. 94).

Since both these notions of precision and repeated observations are part of my definitions **DEF1&2** of probability, one could be worried that they have an obtrusive operational or anthropocentric or subjective dimension. Now, on closer inspection, one will see that *this is just an enhancement of a dimension that is present in any physical theory*. Take for instance Maxwell's electromagnetism. Maxwell's theory (as von Mises') is partly a mathematical theory: the electromagnetic fields satisfy equations, which are considered exact. These equations allow to extract, in principle, perfectly specified values for the fields, having $\delta = 0$. But the values measured in experiments only approximately correspond to the theoretical values, because of non-ideal *conditions*, such as non-zero background fields, or background charges that are not infinitely far away, etc. The precise value of the measured property depends on the specific conditions in which the experiment is done. So there is an operational/anthropocentric/subjective/epistemic dimension in the *application* of usual physical theories too; experimental conditions and precision are fundamental ingredients of the application of the theory. As a consequence, there is also a counterfactual dimension in the use of Maxwell's theory: the exact, 0-error value of the fields could only be measured if other background fields would exactly be equal to 0. But 'exactly 0' is a counterfactual notion in physics. Perhaps an even more striking example comes from the definition of 'electric field' in electromagnetism. Most textbooks on electromagnetism define the electric field (at an arbitrary point R) along these lines: the field is the electric force per unit charge that *would* be experienced by a tiny positive test charge if it *would* be placed at R. In sum, counterfactuality is ubiquitous and harmless in science, as in probability theory—a fact that seems not enough acknowledged (see e.g. Hájek, 2023, Section 3.4). (More on this intriguing issue in **C4** in Sect. 4 and in Sect. 6.)

It is just that in probability theory the operational, subjective dimension is a bit more prominent and confusing, apparently. More precisely, for probability the experimental conditions have a more decisive, and yet less easily graspable, influence on the experimentally determined values. I submit that this is ultimately due to the fact that this theory is all about partitioned systems and their repetitions, not objects per se, something we are not used to. What adds to the mystery is this: probability theory is, I believe, a more fundamental theory than usual physical theories. It is so general that it can be abstracted in an axiomatised mathematical theory. That is not the usual fate of physical theories (more in the Conclusion).

Interestingly enough, the idea that empirical probability 'needs' the notion of precision on a fundamental level, seems also suggested by Kolmogorov's insights condensed in point (a) in the above quote of his (1933/1956, p. 3–4).

It may be useful to further illustrate the ideas of partitioned conditions and frequency stabilisation by a realistic example from the physics lab, namely recent experimental findings in fluid mechanics and nonlinear physics, which have sparked considerable interest in the physics community in the last decades (e.g. Bush et al., 2024; Papatryfonos et al., 2024). A group of experimentalists have discovered that oil droplets can be made to hover over an oil film (Bush et al., 2024; Couder & Fort,

2006; Couder et al., 2005; Eddi et al., 2011). To that end, the oil film is made to vertically vibrate using an external motor. If small oil droplets are deposited on such a vibrating film they sometimes begin to horizontally walk over the surface, for indefinite time; actually they bounce so fast on the film that they seem to hover.¹³ However, this stable walking regime only occurs in well-defined experimental conditions, i.e. for precise values of the physical parameters of the system, essentially the frequency and amplitude of the external vibration, the size of the droplet, the geometry of the oil film and bath, and the viscosities of film and droplet. If these parameters are fine-tuned and lie within precise ranges of values, well-documented by the researchers, the droplets walk horizontally; outside these value ranges the movement becomes erratic and/or the droplet is captured by the film. In the walking regime it is possible to experimentally determine certain probabilities, notably the probabilities P_R that the walking droplet or ‘walker’ passes through a certain space region R . To that end, the physicists have counted relative frequencies (cf. e.g. Couder & Fort., 2006, Fig. 2–3) much as I illustrated above in the case of a dartboard: they determine n_R/n , where n_R is the number of trials in which the droplet passes through R , and n the total number of trials. Now, before determining numerical values of probabilities as P_R , the physicists have spent weeks, months, and possibly years to identify the exact conditions in which a stable probabilistic pattern (i.e. probabilities P_R) occurs. To *prove* that the system is probabilistic (and not erratic) there is only one way: namely to prove that frequency stabilisation occurs. When experimentalists state numerical values of probabilities in publications, they (should) have ascertained that such a frequency stabilisation has occurred in their experiments (if ratios/frequencies would be presented as ‘probabilities’ without anterior verification that these ratios lie in a stabilising regime, scientific deontology would be violated). In physics, this typically manifests itself by the fact that a probability histogram (e.g. n_R/n as a function of discrete values of R) is better and better approximated by a smooth curve when n is increased (see e.g. Couder and Fort (2006), Fig. 2–3 where both histogram and curve are shown); often one publishes just the curve. This curve, in other words these limiting values of n_R/n , are supposed to be ‘the’ probabilities, or good approximations of it.¹⁴

A key point highlighted by this hydrodynamic system is that frequency stabilisation, i.e. a probabilistic pattern, *only occurs under well-defined conditions, and by*

¹³ These walking droplets exhibit moreover a behaviour that strikingly imitates quantum behaviour, including double-slit interference, tunneling and quantisation of angular momentum. Couder et al. have quite convincingly shown that the origin of such a quantum-like behaviour lies in the wave-front that accompanies the hopping droplets (the wave is created by the external vibration and by the back-reaction of the bouncing droplets on the oil film). Thus, these researchers could claim that these walking droplets are the first realisation of a “particle + wave” system (Couder et al., 2005, 2006). Here we do not focus on these intriguing quantum-like features, but on the classical probabilistic features, which were also extensively investigated by these researchers.

¹⁴ For other examples of experimental determination of probability values, cf. Papatryfonos et al. (2024) and Nikolaev and Vervoort (2023).

no means always. Very small variations¹⁵ in some variable (e.g. in the external vibration frequency, the droplet size, slight air movements, etc.) may lead to completely different probabilities, or to erratic behaviour without stable relative frequencies, or to absorbance of the droplet by the film. One has not necessarily control over these variations. Therefore, if a physicist who is novice in the dynamics of oil droplets on films—almost anyone—is asked to determine the probabilities P_R by experiments, she may well come to the conclusion, even after weeks of experimentation, that the system is *not* probabilistic; or, if she is more cautious, that she did not identify conditions in which the system is probabilistic.

Before generalising these illustrations in definitions, in the above jargon probabilistic/random systems include deterministic systems. Indeed, in deterministic systems frequency stabilisation will also occur. If we move a die by placing it systematically—deterministically—ace up on the table, the six $P(R_j)$ will trivially converge to a value (1, 0, 0, 0, 0, 0). We could exclude deterministic cases from the definition of random events/systems by demanding that, besides exhibiting frequency stabilisation, the events/systems should be unpredictable. Thus ‘genuinely’ random events show frequency stabilisation *and* are unpredictable—or perhaps one could replace the latter by another predicate that qualifies the ‘unstructured’, disordered nature of randomness. (In a sense the unstructured-ness is only partial, since probabilistic systems are still subject to the mathematical structure of probability theory.) Deterministic systems could be called ‘trivially probabilistic’: their probabilities have the trivial values 0 or 1.

4.4 Generalisation, Definitions

4.4.1 To What to Apply Probability Theory?

If the essential feature qualifying physical events or systems as random-in-the-sense-of-probabilistic is frequency stabilisation, then following (neglected) question pops up: are all so-called random physical systems genuinely random in the sense of probabilistic? In the case of systems *studied in physics*, one usually declares that these are either probabilistic (indeterministic) or deterministic; so here random can be considered identical with non-trivially probabilistic. However, the above question becomes more interesting in the case of ‘physical systems’ in a larger sense, especially if the systems in question contain humans or are influenced by humans. In this case it is clear that *not* all so-called random systems are random-in-the-sense-of-probabilistic. *Frequency stabilisation is by no means guaranteed for an arbitrary disordered or spontaneous event or system.* Consider the number N of patients in a certain hospital

¹⁵ The extreme dependence of physical properties on certain variables is typical for nonlinear systems, such as those investigated by the teams of Couder and Bush. The oil droplet bouncing on the film is governed by a nonlinear equation of movement, e.g. due to the so-called ‘viscous friction’ between droplet and film.

division on a given day of the week, or the number N of cars parked in a street; and consider the probabilities that these variables assume a certain value ($P(N = N_1)$). The precise value of N varies unpredictably, randomly over time. Perhaps it is possible to define conditions (including e.g. the month of observation, etc.) for which the corresponding probabilities do stabilise when long series of data are gathered; but the notion of ‘long series of data’ becomes ambiguous here. Indeed, there may be *systematic causes* that change the limiting frequencies when these are identified in real experiments: a new vaccine may be developed that changes $P(N = N_1)$ over the duration of the experiment, or a new shop is opened in the street increasing $P(N = N_1)$ if N_1 is low. For (simple) physical events as a die throw one has a clear idea of what the conditions are that need to be repeated (one can repeat them ‘as often as one wants’); but for complex systems, in particular human-induced ones, the conditions leading to frequency stabilisation may be so complex (or the set of conditions may be so large) that *they cannot be defined in an experimental protocol—they are not known in detail*. In such cases, experimental ratios $\frac{n_{N1}}{n}$ can be computed and are often still called probabilities; but this ratio is not a genuine probability: probability theory and statistics are not reliably applicable to it. It may make sense to call such numbers ‘quasi-probabilities’.¹⁶

As a realistic example, temperatures and rainy days in a given city look random, they even look statistically random, but they don’t show frequency stabilisation over long periods, e.g. due to climate change. Meteorological probabilities may be seen as an interesting limiting case. If a frequency ‘at infinity’ cannot be determined, one may wonder whether the notion of probability is still sound here. I believe it is, *if one introduces the notion of precision in the definition*, as is done below. Indeed, contemporary meteorologists heavily use statistical talk: they assume that the conditions at a given time in a given place are sufficiently similar to those of which they have a large data set, allowing them to infer approximate probabilities. Since these data sets are very large, and climate conditions change rather slowly—at least in the previous decades!—*it is likely that the key requirement is satisfied, namely that frequency stabilisation occurs within a reasonable precision interval δ* . What is reasonable is an (inter-)subjective matter, depending on the standards of precision of the field of application; it depends on which use one wishes to make of the inferred probability. In any case, one has here not the precision one can attain in more simple physical systems, for which the sampling size—of data corresponding to conditions that can be specified in an experimental protocol—is virtually infinite. But this problem can be handled if (empirical) probability is defined relative to a precision as is done in the definitions **DEF1** and **DEF2** below.

In the former section we saw a concrete example from fluid dynamics that illustrates the daunting practical difficulty for finding conditions of probabilistic stabilisation. Arguably a majority of physical systems are so complex and non-linear that they

¹⁶ Such quasi-probabilities may still have some practical value. For instance, hospital managers may use them to estimate the number of medical doctors to appoint in a given week. But this estimation is likely imprecise and unstable.

are extremely sensitive to variations in the parameters that describe them. As illustrated above, this leads often or ‘normally’ to the impossibility of determining conditions of frequency stabilisation. Interestingly, this is again a statement of pragmatic nature—one can argue that an omniscient being could know such conditions.

In conclusion, frequency stabilisation happens often, but certainly not always. (A particularly beautiful mathematical argument for the ubiquity of probabilistic randomness is provided by the Central Limit Theorem—sometimes called the unofficial queen of probability theory.) So not all randomness is the randomness that is the object of probability theory. This justifies the introduction of the concept of ‘ ρ -randomness’, to be contrasted with ‘randomness’ and ‘non- ρ -randomness’: ρ -randomness, or probabilistic or structured randomness, is randomness (chance) that is characterised by frequency stabilisation ‘for large n ’. Before one applies probability calculus to an event, one conjectures or has evidence that it is ρ -random—a feature that can ultimately only be confirmed by experiment. I use the symbol ‘ ρ ’: in physics the familiar way to characterise a variable R as random, is to state that it has a ‘probability density function’, a parameter often symbolised by ρ ($\rho = \rho(R)$). In physics variables usually are continuous, so R can take an infinity of values (more on this point in **C5** below).

4.4.2 Definitions

The stage is now finally set for introducing the two essential definitions of my philosophy of probability, **DEF1** and **DEF2**. In the comments **C1–C6** I elaborate on the auxiliary terms and assumptions of these definitions, and mention the most immediate results to be inferred from them—so the most immediate justifications of the definitions. Von Mises starts by defining a collective as a series of random results and defines probability with respect to a given collective. But I have just argued that probability cannot be attributed to any random series, but only to ρ -random series or ρ -systems. Let us therefore first define the concept of frequency stabilisation: it is logically prior to that of probability.

In accordance with the discussions above, I define probability (with precision δ) relative to an ordered 5-tuple $\langle C, T, S_C, \{R_j\}, \delta \rangle$ or, in a more condensed way, to the 4-tuple $\langle C, T, \{R_j\}, \delta \rangle$ (S_C can in principle be derived from T and C). Here C refers to the repeated experiment done on the test object T , more precisely to the partitioned experimental conditions mentioned above. S_C is the larger system that consists of T in the conditions C (it is the ρ -system already mentioned). $\{R_j\}$ represents the set of possible results/outcomes/events R_j of the experiment ($j = 1, \dots, N$). (The R_j can always be chosen as discrete numerical values or labels of continuous intervals of the real numbers.)

DEF1 (of frequency stabilisation). A system S_C , or an outcome/event R_j occurring in the system, possesses the property of frequency stabilisation (characterised by precision δ) *iff*

- (i) it is possible to repeat n ‘identical’ experiments (with n a number sufficiently large for the precision δ) on the test object T of S_C by observing or measuring T n times under ‘identical’ conditions C (materialised by the initiating and probing subsystems, or more generally, by the environment); and
- (ii) in this experimental series the relative frequencies of the outcomes R_j ($j = 1, \dots, J$) of the experiments on T (detected by the probing subsystem) converge towards constant numbers when n grows; more precisely, when n grows the ratio $\{(\text{number of times that outcome } R_j \text{ occurs in the experimental series})/n\}$ converges (within δ) towards a constant number, for all j .

DEF2 (of probability). Only for an outcome/event R_j , or system S_C , that shows frequency stabilisation according to **DEF1**, as can be tested by an experimental series in well-defined conditions C , the probability of the outcome/event R_j ($j = 1, \dots, J$) is defined (within precision δ , and for all j), and given by:
 $P(R_j) = (\text{number of times that outcome/event } R_j \text{ occurs in the series})/n$ (for n large enough to attain frequency stabilisation within δ).

These are, in principle, definitions of ‘frequency stabilisation within the precision δ ’ and of ‘(empirical) probability-cum-precision- δ ’. In **DEF1**, “a ratio converges within δ to a constant number (P)” means that the ratio varies only in the interval $[P - \delta, P + \delta]$. I believe that the notion of precision is an intrinsic and unavoidable ingredient of probability as empirical notion, as elaborated in **C4** below. However, if one wishes, this model allows one to define ‘the’ probability, in an absolute sense: it is the number in **DEF2** corresponding to $(\delta \rightarrow 0, n \rightarrow \infty)$, the theoretical number that can *sometimes* be predicted exactly for idealised systems, e.g. chance games and quantum systems. Hence, one can recover von Mises’ limiting frequency definition as a limit case of ours.

Regarding terminology, I will say that the outcome R_j ‘has’ the probability $P(R_j)$ with or within the precision δ . Note that the outcomes $\{R_j\}$ are the values of a property (variable) X of T probed in the experiment C . Therefore probability can also be associated with (physical) properties/variables (as done in physics, and as can trivially be done for any probabilistic system¹⁷). Note also that these conventions encompass the usual habit of associating probabilities with ‘events’ (E_j): the event E_j consists in the variable X assuming a value R_j in the experiment C ; hence, $P(E_j) = P(R_j)$. **DEF2** implies that probabilities only make sense, are only defined, for systems S_C that exhibit frequency stabilisation in some property X according to **DEF1**. One can also say that the experimental outcome R_j (or even the object T in conditions C) exhibits frequency stabilisation.

¹⁷ For instance, for usual die throws the property of the die that is tested/observed is $X = \text{number of eyes on the upper face (at rest)}$.

I conjecture that the above definitions cover all events or systems that are accurately, numerically described by probability calculus. My main argument lies in the unifying and therefore problem-solving capacity of the model based on **DEF1&2**, as exposed in the next section (results **R1–R4**) and in auxiliary comments **C1–C6** below. I cannot review here all probabilistic systems and show how to apply the definitions to them, but I will give examples for several more (classes of) systems, besides the ones already treated; and one can find many more in the works of von Mises (e.g. 1928/1981, 1964).

C1. Compliance with Kolmogorov’s axioms. It is well-known that the frequencies or ratios defining probability in a model like the above obey the axioms of Kolmogorov’s probability theory (e.g. von Mises, 1964; van Fraassen, 1980; Gillies, 2000; Khrennikov (2008)). Thus, in von Mises-style theories probability is defined in terms of a measurable quantity; and the axioms of probability theory become theorems in his theory. Most other interpretations of probability do *not* explain why probabilities satisfy these axioms—so this is a genuine argument in favour of von Mises-style interpretations.

C2. The gist of the interpretation. On the ontic (objective) and epistemic (subjective) aspects of probability theory. If one reduces the definitions to their essence, they state that probability *only* exists for p -random systems that *can* be subjected to massively repeated experiments or observations, occurring in scientifically well-defined conditions; and that in that case probability is given by a simple ratio or frequency. Under the scientific realism I assume, probabilities also make sense (‘exist’)—more precisely: describe something real and objective pertaining to systems—even if no experiments have been or will be done on them; *but it must be possible that such an experiment be done*, at least in principle, as stipulated in **DEF1**. This is the gist of the interpretation. The ‘in principle’ proviso means: empirical verification of the probability value must be possible, but it can be an *indirect* verification (via a scientific theory): see the discussion below.

Hence, according to this interpretation probability is not a property of individual objects or events per se; rather, it is a property of *repeatable experiments* (C above), or of repeatable events, even better of (*collections of*) *repeatable systems-in-context* (p -systems)—a phrasing that resonates with realism. As a side-remark: within scientific realism, the distinction ‘object versus composed object (system)’ is not utterly fundamental, once one is aware of it.

In the case of physics, the idea that probabilities refer to experiments has been highlighted by several philosophers, including Popper¹⁸ (1957, p. 67) and van Fraassen (1980, Ch. 6); it is implicit in von Mises’ theory and Kolmogorov explicitly mentions the experimental conditions, as recalled above (1933/1956, p. 3). I believe this conclusion, that probability is a *property of systems-in-context that can repeatedly be tested in experiments*, should be generalised to any scientific notion of probability, also beyond physics. The general wisdom here is clear: untestable notions are unscientific.

¹⁸ Popper (1957, p. 67) famously interprets probability as the *propensity* that a certain experimental set-up has to generate certain frequencies.

Notwithstanding the operationalist concepts used in the definitions, under scientific realism any sound scientific model and definition should say something about the objective world; and indeed probability values can objectively (or inter-subjectively) be verified and agreed-upon, as all scientific properties can. So each probability (value) must correspond to something real, ontic; if one wishes to give a new name to this objective property, only to emphasise that it is an ontic category, one could call it ‘objective chance’. The key idea is: *probability values also govern probabilistic systems when nobody looks*, e.g. molecules in a gas, biological populations, quantum systems, etc. The temperature of a gas, for instance, is a verifiable property of a massive collection of molecules. This temperature is determined by the probability distribution of the molecules’ velocity magnitudes, a probability distribution that is an objective feature characterising the movement of the molecules (cf. von Mises (1928/1981 p. 20) and Appendix 4.2). It makes sense to call this feature the objective chance (some prefer ‘propensity’) to assume given velocity values in given conditions. A sure witness of the objectivity of such a probability distribution is the fact that everyone can measure the same temperature of the gas; neither temperature nor probabilities are subjective on this very science-compatible interpretation.

Indeed, a merit of the definitions is that they apply both to artificial systems (chance games) and natural ones: the latter behave in a p -random manner without human intervention. In the case of chance games the initialising and probing subsystems can be identified as truly separate systems acted upon, or made, by humans (more in **C6**: Laplace’s interpretation for chance games resorts under **DEF1&2**). In the case of natural random phenomena the initialising and probing systems coincide with—are—the environment, which materialises the experimental conditions mentioned in the definitions. Natural random events are normally not initiated nor measured by humans in repeated experiments, but according to the above model they only have a well-defined probability if they occur, or can occur, on a massive scale as could be imitated in the lab, or studied ‘out there’ under reproducible conditions. Again, the probability values of such natural events can only be revealed, in any case ‘proven’, by experiments in their environment or imitating their environment; they can sometimes be hypothesised by calculations based on experimental data or theoretical assumptions¹⁹ (for some examples from physics, cf. Appendix 4.2).

Elaborating on this last remark on ‘theoretical’ probabilities: in view of their primordial link with experiment, some might worry that **DEF1&2** are not applicable to certain probabilities that are predicted by theories and seemingly not directly

¹⁹ Von Mises found it important to emphasize that when one applies probability theory to a concrete problem, one always starts from assuming a certain probability distribution. One of his well-known maxims is “Probability in, probability out”. E.g., when one calculates the chances in urn picking by combinatorics, one typically (and often implicitly) starts from assuming that the chance that a given ball is picked is equal for all balls. Then one proceeds to the problem posed (such as: what is the probability that two successively picked balls are white?). In theoretical physics too, one often guesses or has evidence or derives by using physical theory that the probability distribution of a certain property is of a certain type (say a Gaussian). Based on this hypothesis one then uses probability calculus or statistics to derive other probabilistic predictions—where it is often only the latter that can be tested. This is an example of *indirect* verification of probabilities.

measurable by experiments. An example is again the probability of the velocity of gas molecules. Since von Mises has treated this case (1928/1981 p. 20), I will only briefly summarise it in Appendix 4.2. The key idea is this: even if experiments for measuring molecular velocities are not available (yet), they are a conceptually immediate extrapolation of possible experiments (they are valid thought experiments), and, most importantly, *the inferences of such thought experiments can be tested by experiments in agreement with DEF1&2*, as explained in the Appendix. In sum, when I stipulated above that “it must in principle be possible that a repeated experiment be done”, the ‘in principle’ means that the assumption that the property in question is characterised by a probability must be an assumption within a theory that, as a whole, has empirically verifiable consequences. (We know at least since Quine that statements, e.g. attributing probability values, are always tested as part of theories.)

Also, note that the ‘repeatable experiment’ in DEF1&2 can be rudimentary: it can consist of simply repeatedly ‘observing’ a property on an ensemble of test objects ‘out there’, in their natural environment—a simple type of measurement, simpler than for instance quantum measurements done in sophisticated lab conditions. This possibility is explicitly added in DEF1&2 to include probabilities extracted from statistical data, describing for instance properties of biological populations, meteorological events, or cosmological systems. Such data can represent the outcomes of tests, observations, done in homogeneous-enough conditions on the elements of the population. From such experimental data probabilities can be extracted—but *only if the data are characterised by frequency stabilisation within a certain acceptable precision*. So, for instance cosmological probabilities studied by Beisbart (Beisbart & Hartmann, 2011, Ch. 6) are covered by DEF1&2, thanks to the notion of precision (cf. Appendix 4.2 and C4 below).

A recurrent question that haunts this chapter is to what extent the operationalist concepts in the definitions point to subjectivism or operationalism—are *these fundamental*? It is very intriguing that the model overtly posits an intimate link between probability measures and experiments exhibiting p -randomness, or if one prefers between probability and experimental conditions (C). I believe one cannot avoid this operationalist or anthropocentric move; and one wonders whether it has essential philosophical implications, pointing to the idea that (the application of) probability theory has an epistemic, conventional aspect to it. The notion of precision in DEF1&2 seems to corroborate this idea. I have already argued in Sect. 3 that such an operationalist/epistemic/subjective element should not frighten us too much; it is ubiquitous in physics.

Indeed, consider again the idea that the probability of X can only be well-defined if well-defined experiments for X can be devised—described by a protocol usable by trained experimenters. This claim seems to suggest that the concept of probability points to our human cognitive limitations. The phrasing in clause (i) of DEF1 “it is possible to repeat n identical experiments” might trigger a naïve question: *possible for who*? A more skilled experimenter might perform experiments and write protocols disclosing probabilities that remain unknown to others—recall e.g. the elusive hydrodynamic probabilities mentioned in Sect. 3. Do probabilities only exist for these experienced experimentalists? In a trivial sense, yes, but in a more profound

sense, no, of course. “Possible” means “possible for anyone who can”, “possible for the most capable experimenter”, and ultimately “possible in the absolute”. We get here very close to Laplace’s famous conjecture that we only need probability because of our cognitive limitations. An all-knowing being would know the course of events with certainty—attributing only zeros and ones to probabilities, as is fitting for a fully deterministic universe. While *we* can only fix and know the (initiating) conditions of a die-throwing experiment up to a certain point, allowing us to extract non-trivial probabilities, one can perhaps legitimately imagine an all-knowing being who knows and fixes the experimental conditions of another, more fine-grained die-throwing experiment, leading to trivial probabilities. Yes, it seems so, *but in any case that would be another experiment, characterised by different experimental conditions*. Which probabilities are correct? Both—*both refer to different p -systems*. In sum, whether one considers the ultimate nature of an event/system as genuinely random as opposed to deterministic, depends on one’s knowledge state—on whether one factually knows (or assumes the existence of) an experiment that leads to trivial probabilities (0 or 1) or not (recall that deterministic and genuinely random systems are two sides of the same coin: both are p -random). *I would consider this as the most relevant epistemic, subjective dimension of the application of probability theory*. An example of how this relativity of (in)deterministic ascription manifests itself is given in Appendix 4.3.

Thus, from the point of view of the philosophy of physics, the mentioned link probability—experiment and the said conventional dimension lead to following fascinating metaphysical question: Are indeterministic (i.e. non-trivially probabilistic) events *ultimately* non-trivially p -random, or is there *ultimately* a hidden determinism in nature? Do indeterministic properties, including quantum ones, all emerge from deterministic properties—as Laplace conjectured? This question goes much beyond the scope of this chapter, but, of course, the orthodox interpretation of quantum mechanics claims that indeterminism is ontologically fundamental, and hidden determinism a fantasy. However, I believe this question of the ultimate nature of indeterministic phenomena is an unsolved problem. Against the orthodox position in the quantum physics community, there are cogent and rarely discussed arguments in favour of hidden determinism (Nikolaev & Vervoort, 2023; Vervoort, 2019, 2020; Wuethrich, 2011). I repeat that this is not the same claim as the main contention of the subjective interpretation of probability, which makes probability an epistemic, subjective category. Probability values are objective (cf. **R3** below); but at the same time the question of whether a system is considered indeterministic (non-trivially probabilistic) or deterministic does depend on the knowledge state of the

agent attributing probability values. So it is this ascription of a deterministic or indeterministic nature to a system that could be termed epistemic²⁰ (an idea illustrated in Appendix 4.3).

Now, this epistemic claim can, in fact, easily be grounded on an objective basis; it appears to be not fundamental. As already suggested above, it suffices to realise that probability does not belong to systems per se, but to p -systems. *It may be undecided whether ‘a system’ is deterministic or indeterministic, but whether a p -system is deterministic or not is an entirely objective matter.*²¹ Interestingly, as the example in the appendix illustrates, one always comes to this conclusion: the more knowledge observers have about a system, the surer they will consider it deterministic. The all-knowing observer is a determinist, so to speak. Does this hint to the heretic conclusion that Laplace and Einstein win against Bohr (Nikolaev & Vervoort, 2023; Vervoort, 2019, 2020)?

But let us not get carried away. These matters of the philosophy and foundations of quantum physics, however interesting, should not divert us from our primary goal, which is to define probability. Further conclusions about the subjective–objective debate are given in the next section (R3 on the subjective interpretation of probability) and in Sect. 6, the Conclusion.

C3. On ‘identical’ or ‘similar enough’ conditions. DEF1 of p -randomness relies on the notion of ‘identical’ or ‘similar-enough’ conditions and objects. It may thus look suspicious in philosophers’ eyes. But it is not difficult to see that the definition is sound in the scientific sense: it allows to define p -randomness and probability in a verifiable manner, a manner that allows for objective or if one prefers inter-subjective agreement. Indeed, the conditions for doing the ‘frequency stabilisation test’ can be put down in a protocol and communicated for any particular system: “do

²⁰ One easily gets in a Babylonian confusion of tongues in this debate, so I will be explicit, at the risk of repeating myself. In contrast to the general literature on the philosophy of probability, in the philosophy of physics literature the notion of ‘subjective’ probability is often associated with determinism, and ‘objective’ probability with indeterminism (e.g. Wuethrich, 2011). The underlying thought is that ‘objective’ probability is the only possibility we are left with in a fundamentally indeterministic (say quantum) world; while conceiving probability ‘subjectively’, as an expression of our ignorance of underlying deterministic processes, is appropriate in a deterministic world. (In this problem of the ultimate (in)deterministic nature of elementary particles, instead of ‘objective’ probability I prefer ‘ontic’; and instead of ‘subjective’ I prefer ‘epistemic’.) Now, according to my model, even in the latter subjective or rather deterministic or epistemic view of probability, probability values are still objective measures in the strict sense (as used in the philosophy of probability)—namely relative frequencies. So it is more precise to consider not the probability value itself as subjective, but rather the “ascription of a deterministic or indeterministic character” to the event under scrutiny (cf. Appendix 4.3, based on Vervoort, 2020). For a related view, see Bacciagaluppi (2020).

²¹ Similarly, the question of whether a variable/property X of system T in conditions C is deterministic or indeterministic has an objective answer, in principle: it depends on whether a repeatable experiment (measuring X and representing C) on T exists that always leads to trivial probabilities for the values of X . If so, X (and T in conditions C) are deterministic. For instance, the spin σ_z of an electron T in a state $\psi = (u_z + u_{-z})/\sqrt{2}$ measured along the axis z is *indeterministic* (the context C is here described by the electron’s initial state ψ): the spin has two possible values ($+1$ and -1) and both have a probability $1/2$.

such-and-such ('identical') initial and final actions on such-and-such ('identical') systems (or put or observe these 'identical' systems in these 'identical' environmental conditions)—and the probabilities P_j will emerge. I found frequency stabilisation and the probabilities P_j , so if you do the experiment in the 'right, identical' conditions, you should find them too." The term 'identical/similar', which is ubiquitous in the philosophy of probability theory, can thus be defined in an operationally consistent manner: *'identical' is 'identical' insofar a probability distribution p emerges.*

Thus, it is helpful to introduce the notion of 'p-identical', or rather 'p-identical', events/systems. To identify the probability of an event (R), one needs to perform a series of experiments on p-identical systems (including test object, initiating and probing subsystem, or environment)—systems that lead to the probability distribution $p(R)$ of the results R generated by the system. Vitrally, all subsystems *combined* lead to a stable p ; p is only well-defined with respect to such a partitioned system in which all parts 'work together' to lead to precisely p . Changing one element of the system will lead, in general, to a different p . Therefore I define the concept of p-system = a system {test object, environment, initiating/probing subsystem} that is characterised by frequency stabilisation (in at least one of its properties). Many problems can be avoided by using the notion of p-system, or p-identical system, for instance the 'reference class problem' treated in C4 below and other problems addressed in Sect. 5. Again, in a slogan: probability belongs to p-systems repeated/repeatable in p-identical conditions.

C4. On the meaning of 'n' and the role of 'δ'. A difference with von Mises?

Frequency stabilisation and probability are defined by means of convergence of a certain ratio when n grows. Here n is a variable or index that counts the number of trials or repetitions of an experiment; more precisely, it represents the cardinality of a set of outcomes of a real experiment that are subsequent in time (or of a set of outcomes randomly drawn from the experimental series, cf. von Mises, 1928/1981, 1964). As most experimentally determined quantities, an experimental probability measure has a precision δ ; and scientific practice, especially in physics, has provided overwhelming evidence that this precision δ is a function of n , as noted above. The link $\delta \leftrightarrow n$ is ultimately a (hypothesised) fact of nature, and an entrenched tool of science and statistics.

Clearly, the phrasing in DEF1&2 is close to von Mises' original definition of probability as $P(R_j) = \lim_{n \rightarrow \infty} n(R_j)/n$. So close that one may wonder whether there is a real difference. Sure, the definition " $P(R_j) = n(R_j)/n$ for a number of trials n that is large enough for the precision δ " avoids the notion of infinity which some find confusing (recall Appendix 4.1 and Sect. 2). But note that from a pragmatic point of view, if one would use DEF2 to experimentally determine a probability *and if the p-system is repeatable 'at will'*, my definition leads to numbers that are equal, *to any desired precision*, to those identified via von Mises' definition. At least operationally there is then no difference in the definitions: they can lead to the same results, the same numbers. Scientifically speaking they are identical, one might say.

But at least on the surface my definition is more inclusive: it allows to determine ‘probabilities with a precision δ ’, or δ -probabilities, even when one cannot determine $n(R_j)/n$ for n so large that $\delta \rightarrow 0$, which happens in many somewhat complex systems, e.g. from meteorology, biology, etc. For ‘simple’ physical systems as a die or as certain quantum systems, one knows the conditions for repeating the series (quasi-)perfectly; one knows the p -system (quasi-)perfectly; and one can determine probabilities with practically $n \rightarrow \infty$ and $\delta \rightarrow 0$. For other systems one has no access to massive data, for instance because one cannot repeat the conditions C (the p -system) very many times.

However, I claim that even in that case von Mises’ definition as limit is entirely sound as a construct of a natural science theory: it refers to the mathematical object one *would* identify if one *could* repeat C infinitely many times. I have already argued in Sect. 3 that such an expression is not different-in-kind from other counterfactuals met in science; in natural science mathematical objects, say a field-value from electrodynamics, are always compared to their experimental real-world referents under counterfactually idealising conditions (“this mathematical object, this number, would be found/measured if these idealised, counterfactual conditions would be met”). In other words, mathematics-based theories in physics refer to quantities that are always only approximately measurable, quantities of which the real-world referents always depend on experimental conditions C , and that are (almost) always characterised by a precision δ . So one should not expect that a real-world (experimental) probability, the object of von Mises’ theory, *exactly* corresponds to a theoretical probability, as they can *in rare cases* be calculated with infinite precision ($\delta = 0$) within certain assumptions, say by Laplace’s formula or by quantum mechanics. Demanding that von Mises’ probability should be verifiable with infinite precision is as unreasonable as asking that any physical property be measurable with infinite precision.

The often-heard complaint “frequencies do not always stabilise” reveals a confusion of the same kind: only in idealised cases can one repeat *exactly identical* conditions C at will; often these conditions (describable by a protocol) are only approximately known and controllable—except in simple cases. It is again here that my definition allows to determine approximate probabilities corresponding to frequencies that only approximately converge (within δ) *due to only approximately fixed conditions*. Indeed, if some frequency does not converge with $\delta \rightarrow 0$, even on sufficient repetitions, my model suggests that this is due to some insufficiently fixed/stabilised condition-variable(s), i.e. some of the variables in the sets C_i , C_f , C_e in $P(R|C_i, C_f, C_e)$, with C_i , C_f , C_e representing respectively the initiating, probing and environmental conditions. (As an example, recall the case of the variation in magnetic field given in Sect. 3, due to a varying systematic cause in C_e .) Indeed, as we will elaborate in the next chapter, these conditional variables are in *causal relationship* with R (in agreement with our intuition that these conditions co-determine R). Therefore, fluctuations in some of these variables should lead to fluctuations in R and hence in $P(R)$ ($= P(R|C_i, C_f, C_e)$). In the last discussion I have assumed that, even if δ does not go to zero, it is sufficiently small to speak of the probability of the variable at stake (R). This means that the most important causal factors (conditions) are fixed/constant; the fluctuating variables have a much weaker influence on R . I

have discussed these issues here without going into the mathematics; I conjecture that such an exercise would be interesting and could, perhaps, lead to quantitative insights.

In conclusion, one can defuse criticisms against the limit-definition if one understands it correctly as referring to the mathematical object one *would* determine if one *could* repeat *exactly* the conditions C infinitely many times. My definition boils down to a more explicit variant.

In his (2009) Hájek considers an experiment of coin-tossing in a train that moves periodically back and forth on its track pointing west–east (see details in (2009), p. 218). The outcomes are recorded at well-defined space–time points, and can therefore be represented in a space–time diagram. In the case considered they form the following periodic series as a function of time: HHTHHTHHT..., so that the probability of obtaining H, $P(H) = 2/3$. The author claims that one could also ‘look at the data’ on the space–time diagram from west to east, and then one ‘sees’ the series HTHHTHHT... with limiting frequency $P(H) = 1/2$. The author concludes: “Now, why should one answer have priority over the other? In other words, we have more than one limiting relative frequency, depending on which spatio-temporal dimension we privilege.” But this objection does obviously not apply to von Mises’ frequentism, which explicitly attributes probabilities to real experimental series, not of course to deterministically selected partitions of outcomes, for example by projecting them on the spatial axis in a space–time diagram! So, ‘n’ in the definitions **DEF1&2** is a number that represents the cardinality of a series of results that follow each other in time in the experimental series (or of a series of outcomes randomly drawn from the experimental series, cf. von Mises, 1928/1981, 1964). [Hájek’s example is contrived also because the experimental series he considers is perfectly periodic, i.e. deterministic: there must be a mechanism ensuring such a perfect periodicity. A toss in the forward moving train always gives H, in a backwards move always T; there are two times more forward moves. So the ‘probability’ $2/3$ only makes sense for a fully unaware observer (a robot), so a ρ -system that *only* includes an observing system registering the outcomes, not a ρ -system that includes a movement detector—which would allow one to define and measure probabilities $P(H|n) = 1$ for n indicating that the die toss is done in a forward move.]

In sum, objections as to an alleged problem of undefined limiting values (Hájek, 2009, 2023) are dissolved by realising that sets of real experimental series must be considered. It should be noted that several criticisms against the frequency interpretation neglect the real-world basis of experimental series of frequencies: these are physical or real-world series, not logical nor mathematical ones (as another manifestation of this confusion, see the criticism by Fine (1973), p. 93, and my objection in Appendix 4.1). The ‘reference class problem’ (Hájek, 2009, 2023) is solved immediately in a similar way: probabilities are relative to a well-defined experiment, or if one prefers to a ρ -system, or if one prefers to the outcomes—ordered by increasing n —of well-defined experiments involving ρ -systems.

(I hope that the reader, by the end of this chapter, will agree that the operationalist/epistemic elements in probability theory are innocuous, but for some philosophers

a definition closer to von Mises', not relying on precision, might still be more satisfactory, *metaphysically speaking*: one could say that 'the real' probability is the limiting frequency 'at infinity'—if one is aware of the surprisingly many implicit notions discussed above. An ideal, theoretical number, corresponding to the theoretical precision interval $\delta = 0$, a number that one can rarely know with infinite precision but that one can determine with increasing precision when n grows in stable conditions.)

C5. Applicability to continuous variables. In physics most p -random phenomena are characterised by continuous variables rather than discrete ones. Kolmogorov was the first to rigorously show that his calculus applies to continuous variables too (Kolmogorov, 1933/1956). Thus, in this case the variable R will range over an interval of the real numbers. A notorious exception are certain quantum systems, notably when they are carefully 'prepared'—I said 'initiated' above. Still, many or most quantum properties have a continuous spectrum. Note that the origin of the discreteness of outcomes of chance games is obvious: the highly symmetrical objects used are constructed on purpose by humans to generate a limited number of outcomes.

In the case of a continuous p -random variable R one defines in probability calculus the probability density function $\rho(R)$, the meaning of which is however defined via the concept of probability. Indeed, in the continuous case the probability $P(R \text{ is in } dR \text{ around the value } r) = \rho(R = r) \cdot dR$. This equality defines ρ via P as characterized in **DEF2** (here $P(R \text{ is in } dR \text{ around } r)$ is the probability that the variable R assumes a value in the infinitesimal interval dR centred on $R = r$). One can thus safely treat probability density functions and probability on the same foot; a model that interprets probability for discrete variables also does the job for continuous variables. Note that one can formally describe discrete cases by a density function $\rho(R) = \sum_j P_j \cdot \delta(R - R_j)$, where $\delta(-)$ is the Dirac delta-function: the discrete case is a special case of the more general continuous case.

C6. Unification with the classical interpretation. Frequentist models in general cover the classical interpretation of probability, due to Laplace and others, which is used to tackle chance games. Since von Mises (1928/1981, p. 66 ff.) and many modern texts have proven this result, I will briefly present an argument as applied to **DEF1&2**. Notice, first, that **DEF1&2** can be applied to such chance games. The initiating/initialising subsystem is most of the time a randomizing hand (tossing a coin or a die, pulling a card or a ball from an urn, etc.); the probing subsystem is often simply a table (observed by a human).

In Laplace's well-known terminology, for calculating a probability of a certain outcome, one should consider "events of the same kind" one is "equally undecided about"²²; within this set, the probability of an outcome is the ratio of "favourable cases to all possible cases". Notice now that dice, card decks, urns containing balls, roulette wheels, etc. are constructed so that they can be used to produce *equiprobable* (and mutually exclusive and discrete) *basic outcomes*, i.e. *having all P_j equal*,

²² The events thus fulfil the 'principle of indifference' introduced by Keynes.

and given by $1/J$ (J = the number of basic²³ outcomes). Equiprobability is at any rate the assumption one starts from for making mathematical predictions, and for playing and betting (recall that one has to make *some* probabilistic assumption to start from, according to von Mises' maxim "probability in, probability out"); and indeed Laplace's "events of the same kind one is equally undecided about" would now be termed equiprobable events. Along the lines exposed above, a chance game can thus be seen to correspond to a p -system with $p(R) = \sum_j (1/J) \delta(R - R_j)$.

In the special case of equiprobable and mutually exclusive events, the frequentist and classical interpretation indeed lead to the same numerical values of probability. Within the frequentist model, $P(R_j) \rightarrow \frac{n_j}{n} = 1/J$ (the numerator n_j = the number of R_j -events among n ($\gg J$) exclusive and equiprobable events each having a probability $1/J$). The result, $1/J$, is equal to the prediction given by Laplace's formula (1 favourable case over J possible cases).

Note, however, that Laplace's formulation is not superfluous. It allows in the special case of chance games for calculation, i.e. theoretical prediction: in complex examples 'favourable cases' and 'possible cases' can conveniently be calculated by the mathematical branch of combinatorics—a theory of counting, initiated by the fathers of probability theory. In sum, it appears that the classical interpretation is only applicable to a small subset of all probabilistic systems, namely chance games, in general artefacts having high degrees of symmetry leading to easily identifiable, discrete basic outcomes. For this subset Laplace's interpretation can be seen as a *formula*, handy for calculation, rather than an interpretation of probability.

These notes conclude the basic description of my interpretation of probability. As always, the way to validate and strengthen a model is to show that it applies to non-controversial cases, and that it solves problems left open by other accounts. Besides in the examples given, it is useful to explicitly verify the application of the definitions in the somewhat subtler case of probabilities hypothesised in physics (Beisbart & Hartmann, 2011, pp. 143–167). A few more examples are given in Appendix 4.2; the examples given throughout this chapter hopefully illustrate how to apply the definitions also to other cases. The exercise in the appendix illustrates, notably, the essential role of the 'environment' (say temperature), part of the p -system, in the case of natural random phenomena.

4.5 More Problems Solved

In this section I show that the account based on **DEF1&2** allows us to interpret controversies and to solve a variety of problems of the interpretation of probability. Some conundrums have already been addressed in the previous section. Here I will

²³ The 'basic events' or 'basic outcomes' of coin tossing are: {heads, tails}, of die throwing: {0, 1, ..., 6}, etc. The probability of 'non-basic', or rather composed, events (an even number of eyes; a six or an ace; etc.) can be calculated by using probability calculus and combinatorics.

propose my solutions to problems **R1–R4**, on single-case probabilities and the unification with the propensity interpretation (**R1**); on Bertrand's paradox (**R2**); on the unification with certain subjective interpretations and the origin of the subjective shift (**R3**); and on the probabilistic origin of many quantum puzzles (**R4**).

R1. On single-case probabilities and the unification with the propensity interpretation. An immediate consequence of the model based on **DEF1&2** is that it dissolves the intensely debated problem of 'single-case probabilities' (see references in Beisbart & Hartmann, 2011, Ch. 1 and a detailed discussion in Gillies, 2000, Ch. 6–7). According to von Mises-type interpretations the solution is clear: they imply that it makes no sense to talk about the probability (as a quantitative measure) of an event that cannot be repeated in experiments. For instance, it makes no sense to speak about the 'probability' of a dictator starting a world war and the like: no experiments can be repeated here, and even less experiments in well-defined conditions. One might of course attribute a subjective *likelihood* to such events or hypotheses; but we are interested here in the scientific notion of probability. No experiments, no natural science. (As already hinted to above, one might perhaps still speculatively use the notion of probability *in philosophy* to refer to hypothetical properties of classes of dictators that would share enough similarities to envisage stable statistics. Perhaps this speculation has some use in certain philosophical contexts, cf. § 6.2, but surely no quantitative information could be extracted from it.)

Popper's propensity interpretation of probability was an attempt to accommodate such probabilities of single events—quantum events were his inspiration, such as the disintegration of one particular atom (Gillies, 2000, Ch. 6). Intriguingly, previously Popper had done extensive work on the frequency interpretation, but he ultimately rejected the latter in order to make sense of single-event probabilities (Popper, 1959). Now, this inspiration from quantum mechanics is quite puzzling: what is the difference with classical systems? What is the relevant difference between the probability of this particular die showing a 6, and the probability of this particular atom disintegrating in the next hour? I do not know of any relevant difference. The practice of physics has shown that *any probabilistic quantum property of any single quantum system* can only be attributed a probability value if the property can be measured on an ensemble of such systems—as in the case of macroscopic systems. Sure, one can *metaphorically* say that *this* atom has a certain disintegration probability, just as one can metaphorically say that this die has a certain probability to show a six. But this has no immediate empirical meaning other than the meaning conferred by the frequentist account. And indeed, propensity interpretations typically state something along these lines: a probability P of an outcome/event is the propensity of conditions C , or of repeatable experiments, to produce outcomes/events characterised by a frequency P . Such a definition is still based on a frequency interpretation. Therefore, the added value of the propensity interpretation seems far from obvious. Popper has argued that frequentism associated with conditions, as they are an explicit part of **DEF1&2**, amounts to a propensity interpretation. But since a well-understood frequentism answers so many more questions, as argued in this chapter, this is turning the logic

upside-down: it is the propensity interpretation that could be seen as a trimmed-down variant of a detailed frequency interpretation.

Perhaps proponents of the propensity interpretation prefer it because the expression ‘probability is a propensity of conditions’ has a more evident objective touch to it; it confers the impression that there is *something in nature* that creates probabilities. Popper compares propensities to physical forces (Popper, 1959). That is fine in itself; but I fear we enter here in the realm of semantic fuzziness and that there is nothing substantial to be gained, that there are no new important questions answered. I argued above in some detail that my frequency interpretation also comes to the conclusion that probability is an objective category (notably for the obvious reason that frequencies emergent from ρ -systems are objective); it is in any case a basic assumption of my model; and see **R3** below. Also, it is no more objective to attribute probabilities to single atoms than to ρ -systems.

R2. Solving Bertrand’s paradox. A classic paradox of probability theory, debated by the fathers of probability calculus and many mathematicians since, is Bertrand’s paradox. It goes as follows: “A chord is drawn randomly in a circle. What is the probability that it is shorter than the side of the inscribed equilateral triangle ?” (see e.g. Marinoff, 1994). Bertrand showed in 1888 that apparently three valid answers can be given—so which is the right one?

Bertrand’s problem is an archetypical case of a paradox that originates from neglecting the fact that probability is physical, not only mathematical. Bertrand’s puzzle refers to a real-world situation; it is not only a mathematical problem. In the real world there are many ways to ‘randomly draw a chord’ (which may not be obvious upon first reading of the problem—hence the confusion). One can for instance randomly chose two points (homogeneously distributed) on the circle by using a spinner²⁴; a procedure that leads to the probability 1/3, as can be measured and calculated. But two other (classes of) randomisation—or *initiating*—procedures are possible, leading to two different outcomes (1/2 and 1/4). In sum, *Bertrand’s problem is not well posed*, a conclusion that is now quite widely accepted (see e.g. Marinoff (1994)).²⁵ Note that this conclusion immediately follows from **DEF1&2**, according to which probability is only defined for experiments in well-defined conditions, *including initiating conditions*—it is these that are not defined in the phrasing of the paradox. Probabilities need to be associated with a ρ -system, which includes, in the case of chance games, an initiating subsystem. It is not surprising that in a situation that is experimentally ambiguous no unique probability values can be defined.

It will be seen that the literature abounds with paradoxes which stem from formulations that are experimentally complex or ambiguous. The infamous ‘Monty Hall’

²⁴ More precisely, by fixing a spinner at the centre of the circle; a pointer on the spinner and two independent spins generate two such independent random points, through which then a ‘random chord’ can be drawn.

²⁵ The latter reference also reviews von Mises’ and Gnedenko’s treatment of the problem, cohering with the one here presented (p. 23). Keynes, for unclear reasons, holds fast to the principle of indifference (p. 23).

problem, popularised by Marilyn vos Savant, will be seen to become equally straightforward if one remembers that, here to, one looks for a relative frequency of an outcome R of a well-defined experiment.²⁶

R3. Unification with certain subjective interpretations. Origin of the subjective interpretation. The subjective or Bayesian interpretation of probability, due to authors as De Finetti, Ramsey, Jeffreys, Jaynes and several others, comes in as many variants as there are authors (see excellent reviews in von Plato (1994), Gillies (2000)). These range from full-fledged subjective versions such as De Finetti's, interpreting probability as a measure of strictly individual belief, to variants that seem easier to adopt in scientific contexts, such as Jaynes' theory. According to subjective interpretations, depending on the variant, probabilities can be attributed to events, hypotheses, propositions, etc. Neutral assessments often come to the conclusion that the hard-boiled subjectivist versions 'become empty' after detailed scrutiny (Gillies, 2000, p. 84).

In the case of Bayesianism à la Jaynes, the situation is a priori less clear. How does such a theory compare to frequentism? This is an important question, because Bayesianism has become popular in several scientific fields, including in quantum foundations and quantum information theory (e.g. Caves et al., 2002, 2007; Bub, 2007). It has already inspired valid scientific results. So, if frequentism is the mother of all interpretations, following question is pressing: can these results also be obtained within the frequency interpretation? I conjecture that they can. Although I cannot review here this broad Bayesian research program, I will show that some variants of Bayesianism can be absorbed in the frequency interpretation. I will focus on Jaynes' theory (1989, 2003), to which e.g. Caves et al., (2002, 2007) and Smerlak and Rovelli (2007) refer. But first I will start by opposing the full-fledged subjectivist interpretation to frequentism; this will suggest a hypothesis of why the 'subjective shift' is so tempting.

It is a popular and very tempting idea that, somehow, "probability depends on our knowledge, or on the subject" (cf. C2 above). When I throw a regular die, I consider the probability for any particular throw to show a six to be $1/6$. But what about my friend Alice who is equipped with a sophisticated camera allowing her to capture an image of the die just before it comes to a halt? For her the probability seems to be 0 or 1. Or imagine Bob, waiting for a bus, only knowing that there is one bus passing per hour. He might think that the probability he will catch a bus in the next five minutes is low (say $5/60$). Alice, who sits on a tower having a look-out over the whole city, might have a much better idea of the probability in case (she might quickly calculate, based on her observations, that it is close to 1). Are these not patent examples of the idea that a same event can be given different probabilities, depending on the knowledge of the subject? And is in that case probability not a measure of the strength of belief of the subject who attributes the probability values?

Examples as these are unlimited, but the above claims can both be explained and countered by referring to the boundary conditions that are explicitly part of detailed

²⁶ $R =$ 'I win a car by switching'; $P(R) = 2/3$.

frequentist interpretations, in this case the conditions of observation. Doing a normal die throw, and observing the result on a table as is usually done, corresponds to a well-defined ρ -system, with a specific initiating system, probing system etc. In the example, observer Alice does not measure the same probability: she does not measure the probability of finding a six on a table after regular throwing and regular observing, but of finding a six after measurement of whether a six will land or not on the table. The latter is a very different, and indeed fully deterministic, experiment; *at any rate, the observing/probing subsystem (including a high-speed camera) is very different*. A similar remark holds for the bus-case; the measurement system (even if just the human eye in a given location) is part of the ρ -system; one cannot equate observer (probing sub-system) Alice and observer Bob if their means of observation are widely different. Ergo, Alice and Bob do not identify probabilities of the same event or system: *that* is why they attribute different numbers.

For readers who are not convinced, let us go over the examples more carefully. One may consider the probability of event e_1 = “the outcome $R = 6$ in a regular die throw (probed as usual on a table)”. One could also consider the probability of e_2 = “ $R = 6$ after a camera has registered that $R = 6$ one microsecond before the die came to a halt”. The probability of $e_1 = P(e_1) = 1/6$, while $P(e_2) = 1$ (suppose so). Within my model, these probabilities are *not* different because of different subjective knowledge states. First note that also for e_2 one can define the probability without referring to ‘subjective knowledge’ or ‘information’; $P(e_2)$ could be measured by an automat.²⁷ So $P(e_1) \neq P(e_2)$, not because someone has a different strength of belief, but because $e_1 \neq e_2$: both probabilities concern different events, different experiments, and in particular different observing conditions and systems. Every probability can be seen as the result of a series of automated tests.

Now, since nothing is simple in the foundations of probability, it is at the same time true that $P(e_2)$ can also be expressed in a manner that invokes, implicitly or explicitly, the status of knowledge of some observer or ‘probability attributor’. $P(e_2)$ can be considered a conditional probability, namely the probability that $R = 6$ *if it is given* that $R = 6$ one μ sec before halting; in symbols $P(e_2) = P(e_1 \mid R = 6 \text{ one } \mu\text{sec before halting})$. And the phrasing “*if it is given that*” seems indeed equivalent, in this context, to “*if it is known that*”. This semantic equivalence may well explain why one is tempted to interpret conditional probability as dependent on the knowledge of the probability attributor.²⁸ On the view I defend here, such a subjective shift could be innocuous as long as one realises that it is a shortcut of thought—it is by no means the only interpretation, let alone the most encompassing one. Therefore, based on the explanatory power of **DEF1&2**, the natural inference is that subjective interpretations, attributing a probability value $P(p)$ to a proposition

²⁷ Such an automated experiment is this: let a robot launch a die, let it select by camera vision those trials that show $R = 6$ one μ sec before the die comes to a halt, and let it measure on that ensemble R again at full stop of the die (all this could be done by a machine). The relative frequency of these results will converge to 1, as our robot could determine.

²⁸ Van Fraassen (1980, p. 164ff) gives a detailed analysis of how subjective interpretations (probability linked to ignorance) originating from chance games slip into the objective statistics of physics. Also Gnedenko gives an enlightening analysis of this subjective shift (Gnedenko, 1968, p. 26ff).

‘p’, only have a coherent content if there is an unambiguous map from $P(p)$ to $P(e)$ as defined in **DEF1&2**, with ‘e’ an event occurring in a repeatable experiment (or if $P(e)$ is indirectly testable). Needless to say, this condition greatly restricts the use of ‘probability’ as a scientific notion. The idea that it could be generalised as a philosophical, speculative notion is briefly entertained in Sect. 6.

It is important to emphasise that conditional probabilities can be interpreted in an objective way, since they play an essential role in probability theory; on my view they are the basic notion. Recall that I stipulated that it is often useful to replace $P(R)$ by $P(R \mid C_i, C_f, C_e)$ where C_i, C_f, C_e are all relevant variables that describe the initial, final, and environmental experimental conditions, respectively. (As noted above, this replacement can only be done if the C-variables are constant for all trials in the experiment at stake: they must define this experiment.) Von Mises emphasised, and I agree, that *all* conditional probabilities can be seen as corresponding to series of automated experiments in which no intervention nor belief state of a human agent is needed, whether these probabilities are related to chance games as in the above case, or to natural phenomena (1964, pp. 22–24). But is this not a happy argument for the homogeneity of probabilistic phenomena? Under the usual interpretation all natural stochastic phenomena occur according to probabilistic laws also without a human pondering about them. Therefore, in accordance with an acclaimed rule of epistemic parsimony, this gives a head start to interpretations in which the same holds for human-induced events such as die throws.

To corroborate, let us have a closer look at Bayesianism as taught by Jaynes (1989, 2003). In Jaynes’ words (2003, p. 88): “We want probability theory to indicate which of a given set of hypotheses $\{H_1, H_2, \dots\}$ is most likely to be true in the light of the data and any other evidence at hand”. In Bayesianism the key role is played by Bayes’ formula, which Jaynes writes as follows (2003, p. 89):

$$P(H|DX) = P(H|X).P(D|HX) / P(D|X),$$

with X = prior information, H = some hypothesis to be tested, D = the data. The formula allows to derive the ‘posterior’ probability of H in the light of new data or information, $P(H|DX)$, as a function of its ‘prior’ probability, $P(H|X)$. “Eq. (4.3) [the above formula] is then the fundamental principle underlying a wide class of scientific inferences in which we try to draw conclusions from data” (2003, p. 89). Thus, (this type of) Bayesianism understands probabilities as attached to hypotheses of which the likelihood can be calculated in view of ‘background information, data or evidence’.

Now, if one investigates the concrete cases to which the above formalism is applied, there is little doubt that these can be recast in the vocabulary of the frequency interpretation. The translation would be immediate if in all cases ‘H’ and ‘D’ appearing in the Bayesian probabilities $P(H|DX)$ or $P(D|HX)$ would correspond to *outcomes of a random experiment*, and if X would describe the precise *initial, final and environmental conditions* (‘C’ in my notation). Then the meaning of the above formula would simply be the same as derived within my von Mises-type model. In all examples I examined, this mapping indeed appeared to be possible. Let me show

it in detail for one example, the one that Jaynes treats first and in most detail (2003, pp. 93–96). In this example we have:

$X = 11$ machines fabricate widgets, which pour out of each machine into a box (1 box per machine); 10 of these 11 machines produce 1 defective widget out of 6 (these are ‘good’ machines); one machine (the ‘bad’ one) produces 1 defective (bad) widget out of 3. We randomly choose one of the 11 boxes without identifying from which machine, and randomly pick a widget from it. Take then:

$D =$ The picked widget is bad.

$H1 =$ We chose a bad box.

$H2 =$ We chose a good box.

Jaynes then calculates probabilities as $P(H1|D)$, the probability that ‘hypothesis $H1$ is true if data D is known’. It is however clear that all the calculated probabilities (pp. 93–96) can be understood as probabilities of outcomes of an automated experiment, in which a robot randomly picks first a box, then a widget from that box, and finally identifies whether box and widget are good or bad. In the objective phrasing $P(H1|D)$ is the probability of *outcome* $H1$ (a bad box is picked) given *outcome* D (a bad widget is picked), i.e. the probability that a bad box is picked if it is given that the picked widget is bad, all in the experimental conditions described by X (the result is $P(H1|D) = 1/6$). Even if Jaynes prefers a subjective vocabulary, this is of course a most classic application of probability theory, which can well be done within a detailed frequency interpretation.

I did not do a systematic study of all Bayesian applications to scientific problems; but in the works I consulted (e.g. examples in Gillies, 2000; Jaynes, 1989, 2003; Tijms, 2004; von Plato, 1994), it appears that ‘hypotheses’ and ‘data’ or ‘evidence’ can be mapped to outcomes of experiments; their probabilities can univocally be given a numerical value as a relative frequency in a large set of experimental data. I believe this is not a surprise, nay quite obvious, because, again, scientific concepts need a means of verification via experiments. Therefore I conjecture that my above conclusion can be generalised: *if this link with experiments and DEF1&2 cannot be made, Bayesianism cannot lead to new results* (e.g. concerning the probability ‘that a certain theory is correct’; see Bunge (2008) and Tijms (2004), p. 252 for misuses of Bayesianism).

Occasionally, this is a conclusion to which Jaynes himself tends. When investigating whether Bayesianism can attribute a probability to ‘the hypothesis that Newton’s theory is right’, he concludes: “[...] there is little hope of applying Bayes’ theorem to give quantitative results about the relative status of theories” (2003, p. 139). Von Mises would have relinquished hope from the start: the hypothesis ‘Newton’s theory is right’ is not a possible outcome of a random experiment. As a consequence, Bayesianism à la Jaynes, to which recent authors in quantum mechanics refer (e.g. Caves et al., 2002, 2007; Smerlak & Rovelli, 2007), should not be termed subjective, *pace* these authors. In comparison to truly subjective interpretations of probability, it is in fact an objective model, even if phrased in subjective terms. This is again almost explicitly stated by Jaynes himself (2003, p. 45).

In conclusion, I contend that Bayesianism à la Jaynes can be translated into the frequency interpretation: the (background or prior) information/knowledge of the subjective phrasing corresponds to the precise experimental conditions of the frequency interpretation. By extension, I conjecture that all coherent variants of the subjective interpretation can be unified with von Mises-style frequency models. This is still a conjecture, since I did not investigate in detail all these subjective interpretations nor their application to problems. But besides by the reasons given above, this conclusion is corroborated by the fact that frequency models have a much greater explanatory power, as I argue in this chapter. Subjective interpretations face many unanswered questions, and are, by the same token, difficult to unify with natural science. (It may still be that they are a handy short-cut of thought and/or a superior conceptual tool for formalisation for certain *localised* problems; but that is another matter and needs to be investigated in more detail. For only one example of such a localised problem for which the subjective interpretation might be helpful, cf. Uffink, 2011.)

R4. A correct interpretation of probability is essential for understanding key problems of (the interpretation of) quantum physics.²⁹ The above conceptual analysis brings in focus a striking similarity between classical and quantum systems. The philosophy of quantum mechanics is a very vast field, so I can here only advance what I believe to be the essential arguments in support of the above claim. I must here necessarily remain succinct.

It belongs to the basic elements of the standard or Copenhagen interpretation of quantum mechanics that “the observer belongs to the quantum system”, or at least that the measurement detectors belong to it, and influence it. Let me cite Bohr in his famous debate with Einstein on quantum reality (Bohr, 1935). Einstein and two of his collaborators, Podolsky and Rosen (together EPR) had initiated this debate in 1935 (Einstein et al., 1935), with an article that is not only an exceptionally impactful paper in physics (Fine & Ryckman, 2020), but also one of the milestones of the philosophy of quantum mechanics. EPR contest in the article that quantum theory is a ‘complete’ theory: they argue that the probabilistic predictions of quantum mechanics must be explainable, must be completed, by additional variables. In short: quantum probabilities must emerge from a deeper stratum of reality—which is essentially what Laplace thought about probabilities. Niels Bohr, one of the fathers of quantum theory, was not amused, and replied shortly after.³⁰ The key point of Bohr’s counterargument

²⁹ This sub-section is based on older work, where a few more ideas can be found (Vervoort, 2011, 2012).

³⁰ For a detailed interpretation of the EPR article, cf. Fine and Ryckman (2020). In essence, EPR claim that certain (entangled) quantum states possess ‘elements of reality’ which cannot be described by quantum theory. Ergo, it must be incomplete (it lacks certain variables describing these elements of reality). In somewhat more detail, and simply said, EPR claim to have demonstrated that non-commuting properties as the position (x) and momentum (p_x) of a particle must exist simultaneously—in overt contradiction to Bohr’s complementarity principle. Bohr’s reply can be summarised thus: x and p_x cannot be measured simultaneously; ergo they do not exist simultaneously; hence quantum theory is complete after all. Who is right? That is still debated, but notice that Bohr’s answer is heavily anthropocentric and operationalist/instrumentalist, putting it a priori in serious

is summarised in following somewhat mystifying line: “The procedure of measurement has an essential influence on the conditions on which the very definition of the physical quantities in question rests” (Bohr, 1935, p. 1025). To explain the ‘essential influence’ that any measurement has, Bohr invokes the quantum of action linked to Plank’s constant h : any measurement of any property needs the physical interaction of a detector/observing system with the test object under study, an interaction that carries minimally one quantum of energy, and that potentially exerts a perturbing influence. Bohr’s phrase is one of the more famous quotes of the Copenhagen school, but even experts as John Stuart Bell, author of Bell’s theorem, have complained that it is incomprehensible (cf. Bell, 1981 p. 58, where Bell also identifies the above quote as the key ingredient of Bohr’s reply—I agree).

Incidentally, I could not make sense of Bohr’s sentence before I did the present study on probability. My interpretation of probability offers a natural explanation of what Bohr must mean: namely that *(the values of) quantum properties depend in a fundamental way on the observing conditions, or, if one prefers, on (the interaction with) the observing subsystem*. Quantum properties are probabilistic properties; and I argued throughout this chapter that any probability is associated with a p -system and depends inextricably on the observing subsystem/conditions. In classical systems one has to look a bit more carefully for examples to exhibit this in-principle fact, but in the quantum realm it becomes of *ubiquitous and manifest importance*. One is tempted to conclude: quantum systems are probabilistic systems *of a pure kind*—more on this in a moment. Just one example: the probability that an x -polarised photon passes a y -polariser obviously depends on x and y ; here the angles x and y are the variables that describe the initiating and probing conditions.

A few salient statements of A. Peres’ textbook on the foundations of quantum mechanics, nicely expounding the orthodox Copenhagen view, confirm this idea: “A [quantum] state is characterized by the probabilities of the various outcomes of every conceivable test” (Peres, 2002, p. 24), where these ‘tests’ should be understood as pertaining to an experimental set-up. A little further: “Note that the word ‘state’ does not refer to the photon by itself, but to an entire experimental setup involving macroscopic instruments. This point was emphasized by Bohr [...]” (Peres, 2002, p. 25).

We are by now ready to conclude that these statements can be generalised to *any* probabilistic system, quantum or classical. To a die throw one could obviously associate a ‘state’ consisting of the six discrete results and their probabilities. Also the probability of an outcome of a die throw depends crucially on the ‘entire experimental setup’ that is used to perform it. Compare to the following passage (Peres, 2002, p. 73): “The notion of density matrix—just as that of state vector—describes a *preparation procedure*; or, if you prefer, it describes an ensemble of quantum systems, whose statistical properties correspond to the given preparation procedure”. I termed the ‘preparation’ of this statement the ‘initiating/initialising’ of the probabilistic system. I believe that instrumentalist passages à la Copenhagen of textbooks

tension with scientific realism... and therefore possibly with science (recall the MCT-thesis). Let us not say more here.

as (Peres, 2002) all can be understood by using the same notions as probability theory already uses, explicitly or implicitly.

Thus we come to the perhaps surprising conclusion that, in this respect, quantum systems are not as exceptional as usually thought. This thesis can be elaborated, as I started doing elsewhere (Vervoort, 2011, 2012), and as I will further argue below. My position overlaps with a mathematical analysis made by L. Szabó, who comes to the conclusion that quantum probabilities are interpretable as conditional probabilities in a classical probability space (Szabó, 1995, 2001).

Note that I of course do not claim that every element of quantum mechanics has its counterpart in probability theory à la von Mises. But (the observable predictions of) quantum mechanics should comply with probability theory. Probability theory has no ‘uncertainty relations’, neither the corresponding ‘commutation relations’—two paradigmatic ingredients of quantum mechanics. Indeed, without again going in details, quantum mechanics posits that the ‘commutator’ of position x and momentum p_x does not vanish, i.e., in symbols: $[x, p_x] = x.p_x - p_x.x = i\hbar/2\pi \neq 0$ (here i is the imaginary number and $\hbar$ Planck’s constant, and x and p_x are operators). This is the formal basis for the well-known claim that position and momentum of a quantum particle do not exist at the same time, at least are not measurable at the same time (recall Bohr’s argument against EPR). In classical mechanics, on the contrary, the commutator is zero and both properties exist at the same time.

Now, I believe that even *these commutation relations are foreshadowed by an adequate interpretation of probability à la von Mises*. As just stated, the commutation relations stipulate, among other things, which properties A and B can be measured simultaneously. On a natural reading of von Mises, $P(A, B)$, the joint probability of A and B , is defined *only if A and B can be measured by an experimental set-up that allows to measure A , B , and A and B (combined) by using the same observing system*³¹ (Vervoort, 2011). This conclusion is coherent with a claim made above, namely that one can always specify the experimental conditions C for any probability and consider $P(\sim|C)$ as the more precise version of $P(\sim)$. According to the calculus, we have: $P(A, B|C) = P(A|C).P(B|A, C)$. To make this a legitimate formula, C must of course represent the *same* experimental conditions (variables) in the three probabilities, i.e., on my model, in the three corresponding experiments defining the three probabilities in the formula. Ergo, C must refer to equipment, in general conditions of measurement/observation, allowing to measure both A and B (‘simultaneously’ one might say).

Thus in physical probability theory the question of simultaneous measurement of two random quantities is already crucial—just as it is crucial in quantum mechanics. Not of all quantities A and B the joint distribution exists; one can find conditions C in which A and B cannot be measured together by the same equipment. We see

³¹ I extract this from von Mises’ exposition in his (1964) pp. 26–39, even if the text is not entirely explicit. The key procedure described there is the following: start from two collectives (probabilistic experimental series), one in which A is measured, one for B . If A and B are ‘combinable’, then one can construct a collective for the joint measurement of A and B , and determine their joint probability. On a straightforward interpretation of von Mises, the latter collective must correspond to an experimental series *using the same equipment as used for measuring A and B separately*.

here again that (via the commutation relations) quantum mechanics fills in, in a stringent manner, the conditions under which to apply *classical* probability theory in the subatomic realm.

To end, I do not resist the temptation to suggest a link with the notorious ‘quantum contextuality’ revealed by the Kochen-Specker theorem, of which the mathematical intricacies only begin to be understood (Budroni et al., 2022). I will just be suggestive; I will just pass the word to authors who are recognised experts in this matter. Here is what Budroni et al. in their review state (2022, p. 1): “A central result in the foundations of quantum mechanics is the Kochen-Specker theorem. In short, it states that quantum mechanics is in conflict with classical models [hidden-variable theories à la Einstein] in which the result of a measurement does not depend on which other compatible measurements are jointly performed. Here, compatible measurements are those that can be performed simultaneously or in any order without disturbance. This conflict is generically called quantum contextuality.” Another expert on the topic, N. D. Mermin, states³² (1993): “The [Kochen-Specker] theorems establish that in a hidden-variables theory the values assigned even to a set of mutually commuting observables must depend on the manner in which they are measured—a fact that Bohr could have told us long ago (although he would have disapproved of the whole undertaking).”

Could von Mises have told us even earlier? Has Bohr read von Mises? These are speculations, and we will probably never know, but once more, it seems that these descriptions of quantum contextuality find their explanation in a detailed physical interpretation of probability. I believe that such an interpretation makes many quantum riddles transparent.

Note, finally, that the results **R2–R4** are all derived by giving due attention to the experimental ‘conditions’ mentioned in **DEF1&2**. This is of course the main reason why they deserve to be made explicit.

4.6 Conclusion

In this chapter I have had a closer look at the decidedly subtle interpretation of probability, with the aim of proposing a unified model—a variant of von Mises’ frequency interpretation. I am well aware that some topics could have been explored in more detail, and that questions will remain. Notably, it would doubtlessly be interesting to study through the lens of my model other problems that were investigated in detail by others (for instance, the case studies in Beisbart & Hartmann, 2011). Unfortunately, doing so would make this chapter prohibitively long, but I believe this exercise can

³² For those interested in quantum nonlocality, the dense quote by Mermin continues thus: “And Bell’s Theorem establishes that the value assigned to an observable must depend on the complete experimental arrangement under which it is measured even when two arrangements differ only far from the region in which the value is ascertained” (Mermin, 1993). In this book I will not deal with the technically complex (and still utterly mysterious) notion of quantum nonlocality; but see some ideas in the Epilogue.

now well be done (and note that such case studies are not aimed at distilling a unifying framework). Next, it might be that the function ‘P’ of Kolmogorov’s theory can be interpreted in a way that is different from mine and yet consistent. But I believe that such interpretations should have a weaker explanatory power than the model here proposed. They might be applications of the frequency interpretation valid under certain restrictive conditions—a case in point is Jaynes’ Bayesianism.

To end, let us go once more over the main findings, first those that, hopefully, comply with the MCT-approach of Chapter 1, then those that are somewhat more conjectural.

4.6.1 *Synthetic Conclusions*

My main conclusions are that, in order to understand probability as a scientific concept, one cannot do without a frequency interpretation à la von Mises; and that probability theory, beyond the mathematics, is ultimately a physical theory, more precisely a theory about the objective world. However, it is a theory of a special kind: *it is so general that it can be abstracted into a formalised mathematical theory*. In this sense, *it seems to be the most general quantitative theory about the world we have*. It seems to me that the importance of this claim cannot be overstated; we will encounter it at several occasions in this book. In order to counter criticisms against von Mises’ interpretation (which constitutes, together with Kolmogorov’s formalism, the full-blown real-world theory just mentioned) I proposed a modified account based on **DEF1&2** and **C1-C6**. A key ingredient is that it only makes sense to define probability for *systems-in-context that exhibit frequency stabilisation*—in short: ρ -systems, experimentally repeatable in well-defined, partitioned conditions. Simply paying attention to these experimental conditions, by realising that they are a part of the ρ -system, was the key tool for solving classic problems and to come to a unified picture.

In particular, I argued that including the probing subsystem into the ρ -system allows to define probability in an objective manner, and to reframe Bayesianism à la Jaynes objectively. Similarly, the probing system of any quantum experiment, having probabilistic outcomes, is part of the ρ -system that determines the corresponding probabilities. Thus, a basic assumption of the orthodox interpretation of quantum theory finds its explanation in a more fundamental theory than quantum mechanics: (physical) probability theory. Many quantum puzzles are due to not realising that the probing system determines probability measures (on the ontic level). Realising that the initiating subsystem belongs to the ρ -system also leads to the solution of paradoxes, such as Bertrand’s paradox.

Throughout this chapter we encountered a tenaciously lingering epistemic or operationalist or subjective aspect in (the application of) probability theory. The baseline is this: since a probability value can be scientifically verified by repeatedly reproducing the corresponding ρ -system (or by inferring indirect but empirically verifiable

predictions), it must correspond to something objective, ontic: namely, to the objective chance of some event to occur, observed or not. Yet, we detected at several occasions an epistemic element in probability theory too, at least on the surface. Notably: *the ascription of a deterministic or indeterministic nature to a given system* does depend on the knowledge state of the attributor. But it appeared that this worry could easily be defused: the difference in ascription can be reframed as a consequence of an objective difference in experiments and p -systems (cf. **C2**). The conclusion seems clear: determinism or indeterminism refer not to test-systems per se, but to p -systems including an initialising subsystem. And a p -system is deterministic in an objective sense if it ‘generates’ only probabilities 0 and 1 (more precisely, a property X of a p -system is deterministic if the p -system generates only probabilities 0 and 1 for X). So there is also a simple criterion to decide whether a property/variable X of a system T is deterministic or not: it is only deterministic if a repeatable ‘ X -measuring’ experiment exists on T that would always lead to trivial probabilities for X . I further argued that the counterfactual flavour of von-Mises-like definitions is a manifestation of a harmless and unavoidable aspect of any physical/real-world theory. Finally, I argued that one can avoid the notion of limit-at-infinity by introducing the notion of precision in the definition.

4.6.2 *Lines of Further Research*

I made a few conjectures in this chapter that might be interesting to investigate further. Notably: (i) Not only Jaynes’ but all coherent subjective interpretations can be unified with von Mises-style frequency models (cf. **R3** above); (ii) The study of the probabilistic foundation of quantum contextuality can be pushed forward by a mathematical analysis (cf. **R4** above); (iii) The relation between precision-intervals and causal variables determining these intervals can be pushed forward by a mathematical analysis (**C4**). Conjecture (i) is already backed-up by the explanatory power of the account presented.

Surely of a more general philosophical interest is the following matter, which concerns the universal scope of probability theory and the origin of the subjectivist temptation. Going over the main arguments one last time seems to warrant intriguing even if somewhat speculative conclusions. First, note that the claim that probability is best defined relative to a precision δ corroborates the inclusiveness of probability theory: it can also be applied to systems that cannot be repeated *very* many times in exactly the same conditions. Now, this general applicability of probability theory seems further supported by following argument. All physical systems in the narrow sense (those in physics textbooks)—or if one prefers, series of experimental outcomes obtained on such systems—are covered by probability theory (possibly restricted to trivial probability values, 0 or 1). Since we have to assume that *all* real-world systems ultimately consist of physical building blocks, it seems compelling to think that probability is applicable, *in principle*, to *all* real-world events—including events involving free-willed humans! So, is it then legitimate, after all, to speak of say

the probability of ‘Alice says < hi, Bob > today between 4 and 5 pm’? Recall, I have argued that probability theory only applies to ρ -random systems. And most random-looking events created by humans are certainly not ρ -random, since we do not know conditions in which these events would frequency-stabilise. But what about all-knowing experimenters? Could they identify conditions of stabilisation for any property of any system and for any event? Are there *in principle* conditions for frequency stabilisation for any event, any property, and any system, including those studied in say human and social sciences? It seems, then, that I should specify my original claim and state that probability theory *in scientific practice* only applies to ρ -random, i.e. frequency-stabilising systems, *so when verifiable numerical values are demanded*. That is why I used the notion of scientific or empirical probability, the topic of this chapter. Perhaps it makes sense, *in philosophical arguments about principles*, to use the notion of probability in a wider sense. The notion of quasi-probability that I used a few times above could then also be justified by above rationale about the universality of the concept of probability.

Once we are launched on this track, we might even go a little further. Indeed, the uniquely wide scope of probability theory is also corroborated by the fact that it can be abstracted into a purely mathematical, axiomatised and self-standing theory—as Kolmogorov has shown. This is not the usual fate of physical theories, of course; in that sense it is not exactly a physical theory in the usual meaning. It seems to me that this idea warrants a conclusion of great importance: namely, that probability theory is almost on a par with logic regarding scope of application. I believe it is one of those theories that would survive even when other basic theories, say quantum mechanics, would be replaced by others.

Now, if, as hypothesised, probability theory (ultimately, even if not practically) applies to all physical systems in a wide sense, then it should also apply to us, experimenters. But then the operational touch, the use of the phrasing “it is possible to do an experiment” in **DEF1&2** might become unavoidable, or at least transparent. I suspect one can understand it this way: probability theory also describes us, as experimenters, via the ‘conditions’ that play such a key role in the ontology of probability. More specifically: *these conditions prescribe how we should do ‘good’ experiments, which lead, by definition, to objective measures*. Only when one repeats the ‘right’ experimental conditions—which can be extremely sophisticated, think of quantum experiments—probabilities can be measured, and can we conclude that the measurement was well done. Deterministic experiments leading to trivial probabilities—those that occupy most of physicists’ time in the laboratory—are just a special case. From this point of view, *probability theory would be the (most general) theory of scientific experimentation, of measurement*. (To the least, such a theory can immediately be derived from a von Mises-type model as ours. The key idea simply is that good measurements are operations that lead to frequency stabilisation as in **DEF1**.) This would be, at the meta-level, the reciprocal view to the subjectivist claim that probability theory is the theory of scientific inference. That probability theory is, as a mathematical theory, a ‘measure-theory’ is then a happy coincidence—or not.

Again, within scientific realism, this operationalist or epistemic flavour is certainly not in contradiction with the claim that probability theory tells us, at the same time,

something objective about the world, independently of human experimenters; probabilities tell us something objective about systems too. But then *there should be something in nature that makes systems behave according to the rules of probability theory; there is something that stabilises frequencies*. What could this something be? (I conjecture that a coherent answer is possible; some hints are proposed in the Epilogue, Chapter 9.) Other closely related questions are: Why is frequency stabilisation a ubiquitous property of nature? Does the operational flavour of the definition of probability point to our human cognitive limitations, as Laplace thought? Is nature ultimately deterministic rather than indeterministic?

It is my hope that the notions studied in this book will help answering at least some of these fascinating questions.

Appendix 4.1. Alleged Mathematical Problems of von Mises' Theory

Let us have a very brief look at some other criticisms of von Mises' mathematical theory, which is based on the concept of collective. One criticism, stating that von Mises' probability as a limit for $n \rightarrow \infty$ is not always well defined (see e.g. Richter, 1978) was rebutted in the main text. It becomes transparent when one realises that von Mises' theory is a physical theory, not a strictly mathematical one, as also argued in the main text (see von Mises' own arguments in his (1928/1981), e.g. p. 85). A tougher critique concerns the 'condition of randomness' that von Mises imposes on collectives. According to him, the randomness of a collective can be characterised as 'invariance under place selection': roughly, the limiting frequencies of a collective should remain invariant in subsequences obtained under 'place selections', certain functions defined on the original collective. But which and how many place selections are required?—von Mises' critics ask. An important result was obtained by A. Wald, who showed that for any denumerable set of functions performing the subsequence selection, there exist infinitely many collectives à la von Mises: a result that seems amply satisfying for proponents of von Mises' theory (see the reviews in von Plato (1994), Gillies (2000) and Khrennikov (2008) p. 25). A well-known objection by J. Ville can be shown not to be a real problem (see Khrennikov (2008) p. 27 and also Ville's own favourable conclusion reproduced in von Plato (1994) p. 197).

Although I believe one does not need the notion of collective and its calculus, note that von Mises' attempts to mathematically describe randomness led to interesting developments in the mathematics of string complexity, as produced by his 'competitor' Kolmogorov, and mathematicians as Martin-Löf. A general and extremely interesting result of these developments can be stated as follows: *real physical randomness defies full mathematical characterisation* (e.g. Khrennikov (2008) p. 28). This seems not really surprising; in any case, if a series of experimental results can be generated by an algorithm, it is not random by definition (it may of course

look random to the naïve observer). A real series of outcomes of coin tosses cannot be generated (predicted) by an algorithm; but it does show frequency stabilisation.

This remark allows to counter a curious critique by Fine (1973), who derives a theorem (p. 93) that is interpreted by the author as showing that frequency stabilisation is nothing objective, but “the outcome of our approach to data”. However, closer inspection shows that Fine’s argument only applies to complex mathematical sequences (of 0’s and 1’s) that can be generated by computer programs. But again, if a sequence can be generated by a computer algorithm, it is by definition not random, but deterministic, even if it looks random. The essential point of real series of coin tosses is that they cannot be predicted by numerical algorithms... *and* that they show frequency stabilisation. From this perspective, mathematical randomness or complexity, as discussed in relation to the complexity of number strings, has little to do with physical randomness.³³

Note that the model I proposed is immune to criticisms against the calculus of collectives, since I do not use it. I argued that von Mises’ randomness condition is not necessary for the theory, since the axioms of Kolmogorov’s measure-theoretic approach can be proven without using that condition (an explicit proof in Gillies (2000) p. 112).

Appendix 4.2. Probability of the Velocity of Molecules and Other ‘Theoretical’ Probabilities

To further clarify **DEF1&2**, it is instructive to verify these definitions in the case of a natural probabilistic property as the velocity of gas molecules. This is a case to which von Mises devotes attention himself (von Mises, 1928/1981 p. 20), since it is slightly subtle. The probability that a gas molecule has a certain velocity exists according to **DEF1&2**. Indeed, the system (the gas molecule in its environment), or more precisely the event consisting in the molecule having a certain velocity in the given environment, exhibits frequency stabilisation; according to physics it is—at *least in principle*—possible to repeat velocity measurements on the same or on similar molecules. The probabilities are for many types of gases given by the Maxwell–Boltzmann distribution, stating that the probability $P(v)$ that the gas molecule has a velocity in an interval dv around v is:

$$P(v) = 4\pi \cdot \left(\frac{m}{2\pi kT} \right)^3 \cdot v^2 \cdot \exp\left(\frac{-mv^2}{2kT} \right) \cdot dv, \quad (4.2)$$

where m is the mass of the molecule, k Boltzmann’s constant and T the temperature of the environment. (This implies that the probability density $\rho(v) = P(v)/dv$.)

³³ Except if the number strings are generated by a real physical number generator, which is almost never the case.

This case is somewhat subtle: the probabilities (2) are known to be a good description of reality since more than a century, even if in practice they were not directly determined by velocity measurements, due to technical limitations in instrumentation. How then did physicists come to accept (2) as the correct formula? Initially, Maxwell and Boltzmann had derived (2) based on theoretical arguments. But the essential point is that (2) could subsequently be tested in numerous manners in an *indirect* way: if one assumes (2) as the correct probability for velocities, one can use physical theory to derive various predictions for other variables (say kinetic energy) that are functionally related to velocity (kinetic energy = $mv^2/2$)—and it is these predictions which were tested so often that the starting hypothesis (2) got ‘scientifically generally accepted’, and indeed one of the pillars of statistical physics. This is, in essence, also von Mises’ justification.³⁴ In sum, when we say in **DEF1** (ii) “the relative frequency of the outcome R_j ($j = 1, \dots, J$) converges” we might say “the relative frequency of the outcome R_j ($j = 1, \dots, J$), or of a functionally related event, converges”. For our concern the most interesting point illustrated by this example is that the probability values (2) strongly depend on the environment; the essential parameter describing the environment is the temperature T of the gas.

I believe that, *mutatis mutandis*, a similar analysis can be made for the ‘theoretical’ probabilities that are considered by C. Beisbart. Beisbart (2011) investigates the interpretation of probabilistic properties as they are predicted by physical models, e.g. of Brownian motion and of the spatial distribution of galaxies. Without entering in detail, it seems **DEF1&2** can well be understood to also apply to these cases. For instance, a Brownian particle/system can be subject to repeated observations in constant-enough conditions; and experimental verification of stochastic properties as dwell-time or displacement radius could be done, and has been done, via the determination of frequencies as in **DEF1&2**. Somewhat more subtle are probabilistic parameters physicists use for describing spatial patterns of e.g. galaxies. If such a model describes the spatial distribution of galaxies in the probabilistic sense, via e.g. the probability $P_R(N)$ that a given radius R contains N galaxies (assuming for instance homogeneity), then it is again possible to apply the recipe of **DEF1&2**. One can pick ‘many’ (say n) different regions with radius R on the map, and count within each of these regions the number of galaxies. The experimental ratio “(number of regions containing N galaxies)/ n ” should approximate $P_R(N)$ within a workable precision if one increases n ; if not physicists would not accept the model as an adequate description.

³⁴ Here is von Mises’ verdict (1928/1981, p. 20): “It is true that nobody has yet tried to measure the actual velocities of all the single molecules in a gas, and to calculate in this way the relative frequencies with which the different values occur. Instead, the physicist makes certain theoretical assumptions concerning these frequencies (or, more generally, their limiting values), and tests experimentally certain consequences, derived on the basis of these assumptions. Although the possibility of a direct determination of the probability does not exist in this case, there is nevertheless no fundamental difference between it and the other [...] examples treated. The main point is that in this case too, all considerations are based on the existence of constant limiting values of relative frequencies [...]”.

Appendix 4.3. Illustration of the Relativity of (In)deterministic Ascription³⁵

In many cases the relativity of (in)deterministic ascription is obvious. Consider following thought experiment, a mechanised version of a die toss, consisting of say N runs (tosses). The test involves a $20 \times 20 \times 20 \text{ cm}^3$ cube made of soft plastic, positioned at the beginning of each toss with high precision in the middle of a drum-head. Underneath the drumhead a metallic pin moves quickly upward, imparting to the cube a vertical momentum so that it tosses around maybe once or twice; each such toss is initiated by pressing a button. The outcome or event 'e' is the number $\in \{1, 2, \dots, 6\}$ on the upper face, 'measured' after landing back on the drumhead. Suppose that Alice wishes to ascertain whether this experiment is genuinely (i.e. non-trivially) probabilistic and, if so, what the values of the probabilities $P(e)$ are; also suppose that the pin and its mechanism are screened off, so that Alice has no access to them and does not know their functioning. The best she can do is to toss the die very many times by pressing the button, and to make a table in which she notes, for each toss, the outcome e . In a trivial case she would find always the same result, say $e = 6$ (she is tenacious and goes until $N = 10,000$). In this case she would term the experiment or the system deterministic: she feels she can predict what would be the result of the $(N + 1)$ -th toss; so she assumes $P(6) = 1$ and $P(1) = P(2) = \dots = 0$. But consider now the case where each outcome looks perfectly random, unpredictable to her. Then she has to determine, based on her table, the frequencies $P_N(e) = \#_N(e)/N$, where $\#_N(e)$ is the number of times the outcome is e in N trials. Suppose she finds that for $N = 10,000$, $P_N(e)$ comes close to $1/6$ for all e , and moreover that, when comparing the $P_N(e)$ for $N = 100, 500, 1000$ etc. the $P_N(e)$ come closer and closer to $1/6$. Realising that this frequency stabilisation is, besides the unpredictability of individual outcomes, the hallmark of a (non-trivial) probabilistic system, she rather confidently concludes that the system is probabilistic and that the six probabilities $P(e)$ are $= 1/6$ (to good approximation). She has done her job as a physicist.

This experiment could be deterministic in disguise: an engineer, Bob, could have made the die and the system that commands the pin in such a manner that there is a perfect functional relation between the coordinates of the pin impact and the outcome 'e'. E.g., the pin hits $M = 20,000$ positions on the lower surface of the die, say distributed over a square grid. (Once the first M grid positions have been probed the next runs will repeat the same sequence, so M is the periodicity of the impact points.) If the system is well-calibrated and the die movement not too chaotic (the die is of soft material and rotates not more than once or twice), it is possible to establish such a functional relation between impact position and outcome, while at the same time ensuring that the outcomes 1, 2, ..., 6 arise in a random-looking sequence leading³⁶ to $P(e) = 1/6$ for all e .

³⁵ This example is extracted from a preprint (Vervoort 2020).

³⁶ This assumes of course that there are about $M/6$ grid positions that lead to each of the 6 outcomes e .

So this is a system that the engineer Bob and the ‘informed’ or ‘knowledgeable’ experimenter will identify as a deterministic system—each individual toss can be predicted in advance given the pin position—while the uninformed experimenter will deem it indeterministic. This is an example of relativity or subjectivity of (in)deterministic ascription regarding physical systems—relativity with respect to the knowledge state of the subject inquiring about the system. Notice again that this does not mean that probability values are subjective: both the informed Bob and the uninformed Alice will measure, when asked, the same probabilities $P(e)$ *if they do the same experiment on the same p -system*. In the situation sketched, Alice and Bob have done experiments with different p -systems: Alice with {knob + hidden pin; die; eye}, while Bob has used the die in the conditions {knob + controlled & observed pin; die; eye}. Bob is knowledgeable because he knows or has done repeated experiments to verify that each pin position leads to a constant outcome, for instance, pin position 565 always leads to $e = 1$, so that $P(e = 1 | \text{pin} = 565) \rightarrow 1$, etc.

Note also that there is, in principle, a way for Alice to discover the deterministic nature of the system, by noting outcomes and by making N runs with $N \gg M$, say $N = n.M$ ($n = 2$ or more), with M the periodicity of the impact points. If she does so, she will notice that also the outcomes repeat themselves with a periodicity M . She can then predict the individual outcomes with a likelihood that is proportional to n . In principle she can reach *quasi*-certainty about future outcomes, and that is all a physicist can ask for.

A less contrived example of probabilities emerging from, or reducible to, deterministic dynamics is the following. A typical numerical method in statistical mechanics is the Monte-Carlo technique, quite faithfully reproducing probabilistic features of a wide variety of real-world physical systems. Now, this numerical programming technique always uses, in practice, a pseudorandom number generator, simulating physical randomness in some variable. Such a generator is in reality deterministic: the generated numbers look randomly distributed but are actually generated by a complex function—deterministic by definition. Hence a dynamics that is fully deterministic can reproduce an enormous variety of systems of stochastic mechanics; these probabilistic systems can be understood to be deterministic under the surface.

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Chapter 5

A Unified Definition of Cause



Abstract The aim of this chapter is to show that an overarching definition of cause can be constructed, at least in a deterministic setting. Several well-known accounts, notably the counterfactual, manipulationist, and functional accounts can be unified. The functional interpretation is generally accepted in the physics community, but is unfortunately largely absent from contemporary analytic investigations—yet it is essential for understanding the ontology of the universe. The greatest tool for unification will appear to be the ‘representation-relativity of causal identification’ (this is a relativity not of causation as such, which is best understood as an objective feature of the world, embedded in the laws of nature). This perspectival relativity is akin to the one encountered when investigating Gettier’s problem. On the basis of this epistemological principle I will propose a modified counterfactual account, and show that it solves a remarkably wide array of problems. Finally, I will argue that there are good reasons to believe that this model, perhaps with minor modifications, can englobe probabilistic causation.

5.1 Introduction

A series of fascinating questions of philosophy and science involves the concept of cause. Does every event, every phenomenon, have causes? Or rather, are there some events that are not determined by causes but happen by pure chance, pure hazard? Is the principle of causation/causality true? Should every mature physical theory be causal? Is the difficulty in unifying relativity theory and quantum mechanics linked to the fact that the first is a causal theory, while the second an a-causal one? What is the link between the concept of cause and that of probability? Can we have a free will even if everything, including our thoughts and choices, is caused by preceding events?

If we wish to seriously address such questions, we need, first of all, a good definition of the concept of cause, beyond our intuitive notion. Interestingly, in daily conversations people know quite well how to use the word ‘cause’. I am definitely sure that the cause of the pain in my leg is the bite from my neighbours’ dog. The

cause of this broken window is my daughter having thrown a stone at it. A physicist may confidently say that the cause of the increase in electric current in a given circuit is the increase in voltage in a given potential source. So, we do seem to have an excellent command over the meaning of cause. From one point of view, this is obvious: without a grasp of the cause-effect relations that we encounter literally everywhere, we would surely not survive. Is it then not surprising that there is no consensus in the philosophy community about a good definition of cause?

One of the most famous attacks on the notions of cause and causation/causality (besides Hume's) is due to Bertrand Russell (1912), who claimed that physics does not only not need the concept of cause, but that such a concept cannot be defined in a physics context. (There are a few immediate objections to make to Russell's verdict in his seminal paper: 1) Russell only verified the solidity of definitions found in encyclopaedia; 2) It is not while the word 'cause' does not appear in the most abstract physical theories that it cannot be defined in a meta-theory of physics; 3) In their daily practice, physicists use the word cause all the time; 4) On closer inspection, Russell defends in this paper a specific definition of cause, namely the 'functional' one scrutinised below.)

Since then, it has become an utterly popular idea that cause is a multifarious concept, that different contexts require different definitions of cause. J. Norton puts it this way (2003):

How do we know which terms in the science to associate with the cause and effect? There is no general principle. (2003, p. 16)

The proliferation of different accounts of causation and the flourishing literature of counterexamples suggests no general agreement even on what it means to say that something is a cause. So perhaps we should also give up the search for a single principle and allow each causally well-behaved science to come with its own, distinct principle of causality. (2003, p. 6)

At a fundamental level, there are no causes and effects in science and no overarching principle of causality. (2003, p. 23)

N. Cartwright states in a similar vein (2004):

We currently have on offer a variety of different theories of causation. Many are strikingly good, providing detailed and plausible treatments of exemplary cases; and all suffer from clear counterexamples. I argue that, contra Hume and Kant, this is because causation is not a single, monolithic concept. There are different kinds of causal relations imbedded in different kinds of systems, readily described using thick causal concepts. Our causal theories pick out important and useful structures that fit some familiar cases—cases we discover and ones we devise to fit.

In the following I will greatly profit from detailed case studies done in a recent compendium on causation, written by L. A. Paul and E. J. Hall (2013). At the end of their reference book they conclude (p. 249):

After surveying the literature in some depth, we conclude that, as yet, there is no reasonably successful reduction of the causal relation. And correspondingly, there is no reasonably successful conceptual analysis of a philosophical causal concept. No extant approach seems able to incorporate all of our desiderata for the causal relation, nor to capture the wide range of our causal judgments and applications of our causal concept. Barring a fundamental change in approach, the prospects of a relatively simple, elegant and intuitively attractive, unified theory of causation, whether ontological reduction or conceptual analysis, are dim.

These are just a few excerpts illustrating the widespread agreement among experts that a unified, universal concept of cause does not exist. In this chapter I will argue that such pessimism is not warranted. True, to solidify my claim I will have to go through an elaborate and somewhat painstaking analytic-synthetic program; but without this effort any revision of the accepted wisdom would be illusory.¹ The proposed unification is restricted to deterministic causation,² which takes the lion's share in the literature, but I will suggest that a strong overlap with probabilistic causation seems to loom just around the corner.

Regarding the notion of 'reduction' in the last quote, Paul and Hall stipulate (2013, p. 7–8): "We will also assume a broadly reductionist outlook, according to which facts about which events cause which other events are fixed, somehow, by (i) the facts about what happens, together with (ii) the facts about the fundamental laws that govern what happens. [...] We will assume that a successful philosophical analysis of the causal relation ought to show exactly how facts about what happens, together with the laws, fix the facts about event causation." And on p. 256: "Thinking carefully about the different ontological stances one could take brings up another central issue for the metaphysics of causation: whether or not the project is reductionist, that is, does it reduce the causal to the non-causal. [...] Moderate and conservative realism lend themselves most naturally to reductionist interpretations: causal relations are reduced to the other, more fundamental entities that the realist admits in the world. On such views, causal theories are to be given in noncausal terms, and our ability to determine whether there is causation is not dependent on having independent cognitive access to any irreducibly causal facts."

It is useful to have the notion of reduction defined from the start. Indeed, the reductive/ontological demand, as just phrased by Paul and Hall, is one more challenge our effort of synthesis should take up; we will come back to it in Sect. 7, once the definition is constructed. Pleasingly, this chapter will show that building a precise definition goes hand in hand with uncovering the ontological/metaphysical grounds

¹ An older publication, which foreshadows the results derived in this chapter and which makes the link with the principle of causality, is (Vervoort 2013a).

² In the literature on causation, the causal relata (events, say) are usually considered to be either deterministic or probabilistic. Events happening in mundane contexts, say someone throwing a stone at a bottle or breaking a window or killing an opponent, are considered part of this 'deterministic' realm, to which definitions of cause should be applicable. Note that many of the events we encounter in daily contexts could also be called 'spontaneous' or 'self-caused' or 'free' (think of spontaneous decisions of humans); but still they usually fall in the literature under the class of events that should be covered by a definition of deterministic cause. I adopt this convention. A more thorough assessment of seemingly self-caused events is presented in the chapter on absolute free will, Chapter 6.

of causal claims. The ‘conceptual analysis’ goes hand in hand with the ‘ontological reduction’; the ‘descriptive project’ goes hand in hand with the ‘metaphysical project’, using terms from (Frisch, 2023). Thus, even if I will be rather summary when it comes to ontological questions about causation—as a classic scientific realist I *a priori* simply assume that causation is embedded in, emergent from, the laws of nature—, my definition will make the link with the ontology quite obvious, and indeed plainly corroborate the classic realist credo. (One reason why this can be done in a straightforward way is that our synthesis will encompass the notion of cause that is most effective in physics; then one only needs, for this argument, very familiar hypotheses of scientific realism, essentially that the laws of nature have ontological primacy.)

Before scrutinising the professional literature, I would ask the readers to halt a moment, and think for themselves: what would you consider a good definition of cause? Perhaps we should, at first, simplify the question and look for a definition that works in one field, say physics. Mind that this exercise should be conceived as follows, in accordance with the reductive demand above. When trying to construct a definition of cause, we could or should imagine communicating with a physicist who knows physics but not what ‘cause’ means; once we have given her our definition she should be able to identify the right causes of any given physical event or phenomenon—rather, of any description of such event that does not contain the notion of cause and its kin.

Now, what the ‘right causes’ are is supposed to be given by intuition, or by the intuition of the expert in the field. Clearly, such a methodology gives a huge weight to intuition. But this is justified by several facts:

- (i) Causal intuitions of experts generally coincide; a group of physicists or philosophers usually can agree about what the causes of an event/phenomenon are—at least if it is part of a reasonably well understood sub-domain of physics.
- (ii) Human brains are genuine ‘cause hunters’, much more so than equation hunters (and especially more so than philosophical insight hunters). This has an obvious evolutionary ground. Identification of causes is vital for human survival from our very first days on earth; we need to be able to identify cause-effect relations in order to guide our lives in all (new) situations encountered. Unsurprisingly, infants can identify causal relata from very early on (e.g. Leslie & Keeble, 1987).
- (iii) An encompassing definition can be built that coincides with intuition in simple causal situations, as I will show below.

In other words, in this heuristic process of testing whether a tentative definition of cause works, the philosopher’s practice cannot do without relying on her (or an expert scientist’s) intuition about causation in concrete situations. Now, even if we start in this construction process from an intuitive, pre-theoretic concept, success would come with an important pay-off: we would obtain a solid theoretical concept, one that can solve concrete problems transcending our intuition. Therefore, once we have identified a definition that works in the numerous and wide-ranging test situations investigated, *and* that moreover fits into a larger theory unifying the concept of cause

with other essential concepts³ (such as theory, law, probability, explanation, etc.), we can have sufficient confidence in the definition we found; we can then legitimately apply it to situations where mere intuition is insufficient or helpless. Even more importantly: we can then hope to be armed for addressing the questions mentioned at the beginning of this chapter—considered by some as among the most profound of philosophy.

(As a side remark, it seems that the above epistemic procedure reflects nothing more than what is done in any cognitive field deserving the attribute ‘scientific’. Intuitive concepts gain epistemic power through theory building, through logical integration in a synthetic theory. In physics, key concepts as force and energy also started their career as intuitive and imprecise notions. So, one might find here an example showing that the difference between natural science and philosophy is smaller than often thought, not only qua method but also qua verification. Both scientific models *and* causation models are tested via confrontation with empirical data: in the case of science produced by experiments, in the case of causation by intuition-based-on-experience; or one might say, by the brain, a machine fine-tuned over thousands of years to generate empirical cause-effect predictions.⁴ Sure, there is a difference in degree in precision and conclusiveness. But, as argued in Chapter 1 §3, this discrepancy can further be reduced using the synthetic method, here applied to published accounts of causation.)

Less intuitive situations occur for instance in probabilistic models such as the example studied in Appendix 5.1: there I will show that we can use our privileged model to do calculations and come to quantitative conclusions, going beyond intuition. Less intuitive situations also occur for instance in quantum mechanics—say in the famous Einstein–Podolsky–Rosen experiment in which two correlated electrons simultaneously acquire a definite spin value, even when only one is measured. What causes what in such an experiment? Is quantum mechanics compatible with the notion of cause (many physicists, the mythical Niels Bohr in the first place, believe it is not)? Of course, for addressing such difficult questions we need a solid definition.

As already mentioned in Chapter 1, another excellent reason why constructing a unified concept of cause is indispensable in our time, is that we urgently need it for assessing artificial intelligence (AI). AI already answers questions as “what is the cause of X in this neuron diagram” in a surprisingly correct manner, even in debated cases (Vervoort & Nikolaev, 2025). If philosophers want to keep ahead of their field, they need solid models to assess, train, and use AI for causal analysis—and surely soon for many more philosophical tasks.

Finally, we need the concept of cause for investigating the principle of causation, once venerated as a queen of natural philosophy, then vehemently decried by a host of

³ In this chapter I will elaborate in particular on the unification between the notions of cause, probability, and theory, as will become clear in the following. Since the link between (scientific) theory and law is quite clear, we have one more connection in the coherent network for free.

⁴ Moreover, many *indirect* empirical tests for our account of causation are conceivable, cf. e.g. Appendix 5.1; and consider the many inferences we make at different places in this book from our model of causation; several of these have empirical connections.

renowned philosophers, Hume, Mach, Russell, Popper, Bohr, van Fraassen, Norton, Cartwright to name a few. That is something we leave to the Epilogue, Chapter 9.

The chapter is organised as follows. A multitude of definitions of cause exists in philosophy (for recent reviews, cf. e.g. Paul & Hall, 2013; Psillos, 2014; Woodward, 2003). I will base my synthesis on prominent models, namely the counterfactual account (e.g. Lewis, 1973; Menzies & Beebe, 2020; Paul & Hall, 2013), the manipulationist account (e.g. Menzies & Price, 1993; Woodward, 2003, 2023), the physical or functional model (e.g. Mach, 1960; Russell, 1912; Schlick, 1949), and the probabilistic account (e.g. Cartwright, 1979; Hitchcock, 2021; Reichenbach, 1956; Suppes, 1970). I will start in Sect. 2 with the counterfactual definition and make a first synthesis for neuron diagrams (see also Vervoort & Nikolaev, 2025). Then I will focus on the definition of deterministic cause prevalent in physics (Sect. 3). In Sect. 4 I will revisit the manipulationist definition in order to disentangle it of its worries of anthropocentricity and circularity. In these two last sections I will introduce representation-relativity and show the key work it does, notably in unifying the functional and manipulationist definitions. Although the present chapter deals, strictly speaking, only with deterministic cause, in Sect. 5 I will propose an analysis of probabilistic causation, essentially to argue that under broad conditions it overlaps with deterministic causation. With the knowledge gathered in Sects. 3 to 5, we can then return to counterfactual causation, and make the second and final synthesis in Sect. 6.

A word on notation: I will denote with capital letters the definitions of counterfactual (**DEF-CF**), functional (**DEF-F**), manipulationist (**DEF-M**), and probabilistic (**DEF-P**) cause, and with lowercase letters the definitions restricted to neuron diagrams (**def-sn** and **def-mn**). These definitions constitute the core of the respective models. '**DEF-CF**' etc. denote the mature versions of these definitions, while '**DEF-CF0**' etc. are earlier and less mature versions. I will clearly indicate when additional clauses (e.g. a time clause) are implicitly understood in these definitions. (These implicit clauses are less essential and it is likely that they only play a role in folk applications of the notion of cause.)

5.2 A 'Counterfactual' Definition for the Neuron Diagrams of Paul and Hall (2013)

Let us first introduce the basic counterfactual definition of cause, which seems, in its essence, quite powerful. In most academic texts, this definition is assumed to concern event-causation; I will show in Sect. 6 how to generalise it to type-causation. Unless otherwise stated, I will assume that 'events' happen in definite space–time points; with caution this assumption can be generalised to include space–time regions,⁵ but

⁵ To be precise, whether events refer to space–time points or rather regions will be seen to depend on the context of inquiry, in other words the perspective the 'cause-inquirer' assumes when considering the event of which (s)he desires to know the cause (see following sections). Examples of events

we better start with the simple case. Note that this convention about what 'events' are corresponds to its use in physics (where events can be associated to properties assuming a value in a given space–time point or region), but also in probability theory as applied to physics.⁶ Suppose, then, that two events *X* and *Y* occur; we want to find out whether *X* is a cause of *Y*. According to the basic counterfactual account, in order to answer this question one has to evaluate *what would happen if X would not occur*. If in this counterfactual situation also *Y* does not occur, then *X* is a cause of *Y*.

This idea seems to correspond to deeply entrenched causal intuitions. I consider the cause of the fire in the forest to be the fact that I threw a match, because I know from experience that if I had *not* thrown the match, the forest would *not* have been set on fire. I believe that the cause of the premature death of my flowers is the fact that I did *not* water them, because experience tells me that if I *do* water them, they flourish. Thus, in such causal assessments, we imagine a counterfactual scenario about 'not-*X*' happening. *But not just any scenario*. A little thinking quickly makes clear that we change *X* to not-*X*, so to speak, and we evaluate the consequences, but at the same time we control our imagination and we do *not* change other events and conditions of the real scenario we are assessing *besides* those that are a consequence of not-*X*. In the example of the forest fire, the counterfactual scenario I consider is one without my match being thrown, *and in which 'all other things that can be held constant are held constant'*, so in which all other things, not related to not-*X*, are happening as they happened in the real scenario—*not* a scenario in which someone else throws a match, or a lightning sets fire to the forest etc. We should carefully control our imagination.

It may well be that an imaginative process like the one just sketched goes on in our mind (often automatically, unconsciously) when we assess (new) causal situations; more to the point, that something like this lies at the basis of a definition of the concept of cause. For now, let me just emphasise that it is far from trivial to make this clause 'all other things held constant', in short 'ceteris paribus', precise. Great efforts have been dedicated to it; most famous is D. Lewis's theory based on his 'possible world semantics' for counterfactuals (Lewis, 1973). But these efforts have been unsuccessful, according to now widespread agreement (Menzies & Beebe, 2020; Paul & Hall, 2013). I will argue in this chapter that if we modify the basic counterfactual definition judiciously, notably by making the 'ceteris paribus' clause precise, it becomes solid and can solve the existing problems. This task will ask quite some synthetic work: we need to scrutinise very many case studies on counterfactual causation, and make a detour over competing definitions.

Let us formalise the rudimentary model (**DEF-CF0**) as follows (in this section *X* and *Y* are events on token level; both are assumed to occur):

in space–time *regions* are: 'the drought in region *R* in summer 2023'; 'the magnetic field *H* in space–time region (*T*, *R*) assumes the values H_0 ', where *T* is a given time interval and *R* a given region in space and H_0 is a range of values each corresponding to the field in the space–time points in (*T*, *R*); 'the revolution in (*T*, *R*)', etc.

⁶ Recall from the chapter on probability that a probabilistic 'event' corresponds to some experimental outcome – a property of a system – assuming a certain value (in some space–time point or region).

X is a cause of Y
iff
 if X were not to occur, Y would not occur.

(DEF-CF0)

We have just seen hints that **DEF-CF0** might well need upgrading by making the *ceteris-paribus* clause explicit:

X is a cause of Y
iff
 if X were not to occur, Y would not occur (*ceteris paribus*).

(DEF-CF0')

For the time being, I leave the *ceteris-paribus* clause in **DEF-CF0'** undefined, or loosely defined as done above. And I specify that *X should happen before Y* (or at least, should not start after Y): there is an (implicit) time-clause in **DEF-CF0** and **DEF-CF0'**. In *all* my subsequent definitions I will assume this temporal requirement to be included. The reason is straightforward: our intuition commands it, and without this clause, a vast number of problems can simply not be solved. Thus, one conclusion of this chapter will be the very unspectacular thesis stating that effects never come before their causes. I believe (and will argue below) that there is no successful theory of bona fide causation/causality, be it in philosophy or physics, that concludes otherwise.⁷ Let us then bite the bullet and start solving problems.

There is a vast literature arguing that the simple **DEF-CF0** does not work in many cases: it leads to results that go against strong causal intuitions; the same is true for **DEF-CF0'**. Paul and Hall (2013) have compiled and discussed a majority of such problems, building on the work of many. They treat in great detail topics as 'early preemption', 'late preemption', 'tampering', 'overdetermination', 'causation by omission', 'double prevention', transitivity, etc. In a majority of cases, the authors study these problems with the help of 'neuron diagrams', which represent causal situations in a compact visual form.⁸ The book by Paul and Hall is of great utility, for I can use it to attempt a first synthesis: to propose a definition that works for all the cases studied in this book (or at least a great majority).

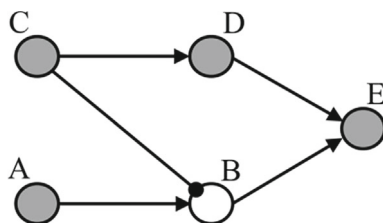
A representative neuron diagram is given in Fig. 5.1.

In the diagram of Fig. 5.1 five subsystems, represented by five neurons, are in causal interaction in the following way. Neuron C emits a stimulating signal (represented by an arrow) towards neuron D, which subsequently fires; D likewise sends a stimulating signal to E (all firing/ 'on' neurons are shaded). A stimulates B but B is inhibited to fire due to the inhibitory signal from C, represented by a line with a black

⁷ I exclude retrocausality, since there is too little evidence that it can solve problems in coherence with physics; on the contrary, it leads to well-known logical paradoxes that constitute a weighty no-go argument (more in Chapter 9).

⁸ Even if these diagrams can represent a sizeable portion of the problems discussed in the analytic literature, Paul and Hall emphasise that they surely do not capture all relevant features of causation. I agree; hence the importance of a further synthesis.

Fig. 5.1 Typical neuron diagram (reproduced from Fig. 1, Paul & Hall, 2013, p. ix), explained in the text. ‘Firing’ or ‘on’ neurons are shaded



dot or blob.⁹ The temporal order goes from left to right: C and A fire at t_1 , B and D react at t_2 , E reacts at t_3 , where $t_1 < t_2 < t_3$. Note that the (causal) connections, so the functioning or meaning of the diagram, can be described without using the word ‘cause’. Therefore, if we manage to construct a definition of cause that allows us to extract the cause of neuron events (like the firing of neuron E), we have a reductive definition.

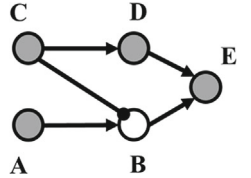
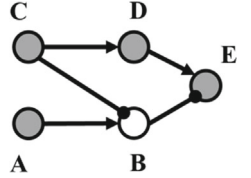
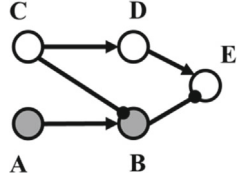
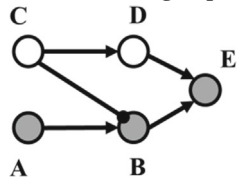
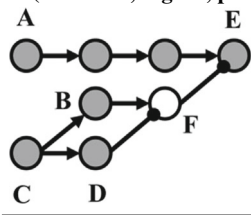
Looking at Fig. 5.1, most people would agree that the cause(s) of E’s firing are the firing of C and/or D. Or if they wish to be more precise, they could say that (the firing of) C is the ‘root’ cause at t_1 and D the ‘proximate’ cause at t_2 . Intuitively, the firing of A would *not* be identified as a cause of E’s firing, because its action is blocked by C. In the literature, the firing of A is called a ‘preempted back-up cause’. From now on, I will use following convention: capitals as X (Y, A, B, ...) will in principle refer to events, i.e. the X-neuron firing or non-firing, which I will sometimes abbreviate for clarity as $X+$ and $X-$, respectively. When no confusion can exist I will occasionally also indicate the X-neuron (the neuron at which the event X occurs) simply with ‘X’.

Actually, Fig. 5.1 represents one of the most discussed cases in the recent analytical literature, called early preemption. It is singularly interesting, because it constitutes a challenge to the basic counterfactual model; and, especially, we will see in the following that a great number of problems studied in Paul and Hall (2013) can be solved if one solves this case. Why does **DEF-CF0** run against intuition in this case? It works well when it comes to identifying $D+$ as a cause of $E+$: if D would not fire, E would not. Similarly, **DEF-CF0** leads correctly to the conclusion that $A+$ is not a cause of $E+$. But looking at the diagram and recalling what it represents, it seems clear that if C would not happen, E would still happen, due to the back-up neuron A. This scenario corresponds to Fig. 2 in Paul & Hall, 2013, p. ix, reproduced as diagram 4 in Table 5.1 below. Even if we call **DEF-CF0'** and its ceteris-paribus clause to the rescue, and examine what would happen if C would not happen ceteris paribus,¹⁰ E would still happen due to A. So the basic definitions **DEF-CF0** and **DEF-CF0'** lead to the wrong conclusion that C is not a cause of E.

⁹ In the words of Paul and Hall: “Arrows represent token-level stimulatory connections between neurons; lines ending with black dots represent token-level inhibitory connections” (2013, p. 10).

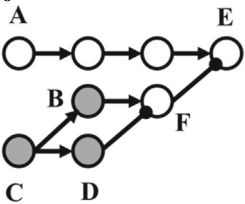
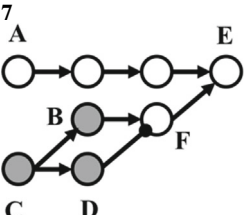
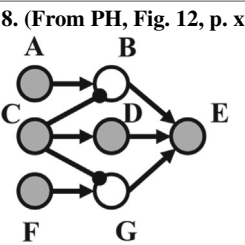
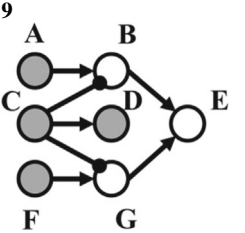
¹⁰ One notices here that the ‘ceteris paribus’ clause is C-relative (or, if one prefers, relative to the time at which C occurs, namely t_1 in Fig. 5.1). Indeed, what this clause exactly means seems clear: ‘all events occurring at the time of C and before, except C, remaining fixed’. That is how the counterfactual reasoning is usually done, for instance very often, even if not systematically, in (Paul and Hall 2013). I will *define* the ceteris-paribus clause along this line in the following.

Table 5.1 Neuron diagrams discussed in (Paul and Hall2013) (PH in the table), and variants thereof. The second column indicates the cause(s) at t1 of the last event in each diagram, as given by def-sn and def-mn, as well as the reason (either these causes correspond to ambivalent neurons or are obtained by the ‘simple counterfactual’ or SCF rule, cf. main text). The third column shows the causes, if any, as identified by Paul and Hall (2013). If the last neuron has a double border, as in diagram 10, it only fires upon reception of at least two stimulating signals

Diagram	Causes by <i>def-sn, def-mn</i>	Causes <i>from PH</i>	Comments
1. (From PH, Fig. 1, p. ix) 	C+ (C is E-ambivalent)	C+	Archetypical diagram representing ‘Early preemption’, cf. PH p. 9. C+ is the intuitive cause
2 	C+ (SCF)		Note that C is <i>not</i> an ambivalent neuron due to the two (not one) inhibiting connections in the lower path
3 	C- (SCF)		Variant not discussed in PH
4. (From PH, Fig. 2, p. ix) 	A+ (SCF)	A+	Variant of diagram 1, cf. PH p. 9
5. (From PH, Fig. 10, p. xi) 	A+ (SCF)	A+	“Double prevention”, considered a problem case (notably for transitivity), cf. PH p. 198, p. 216, p.224, p. 247

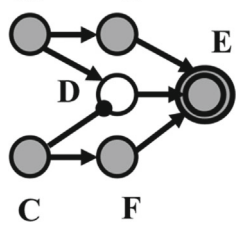
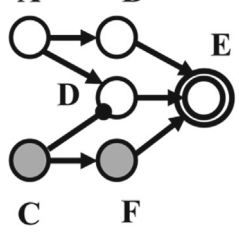
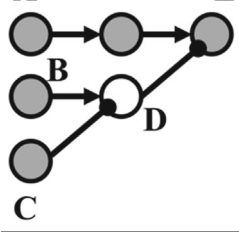
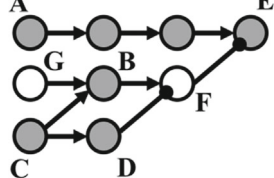
(continued)

Table 5.1 (continued)

Diagram	Causes by <i>def-sn, def-mn</i>	Causes from PH	Comments
6 	A- (SCF)		Variant not discussed in PH
7 	A- (SCF)		Variant not discussed in PH
8. (From PH, Fig. 12, p. xii) 	C+ (C is E-ambivalent)	C+	Case discussed in PH p. 88
9 	C+ (SCF)		Variant not discussed in PH

(continued)

Table 5.1 (continued)

Diagram	Causes by <i>def-sn, def-mn</i>	Causes from PH	Comments
10. (From PH, Fig. 14, p. xiii) 	C+, A+ (C is E-ambivalent, for A+: SCF)	C+, A+	Considered a problematic case, cf. PH p.90; PH converge to C+, A+
11 	A- (SCF)		Variant not discussed in PH
12. (From PH, Fig. 29, p. xvii) 	A + , C+ (SCF)	A+, C+	“Double prevention”, considered a particularly problematic case (PH, p. 202). PH converge to A+, C+
13. (From PH, Fig. 41, p. xx) 	A+ ; {G-, C+} (SCF)	?	This case is left unsolved by PH (p. 198, p. 225). The possibility that {G-, C+} is a joint cause is not discussed

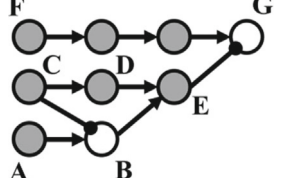
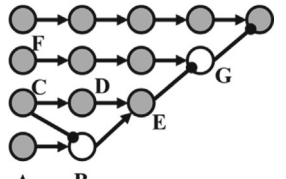
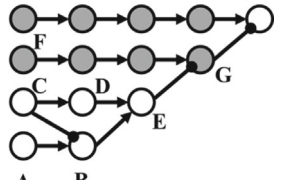
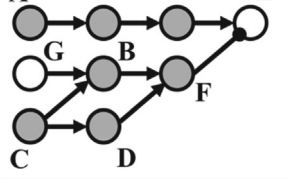
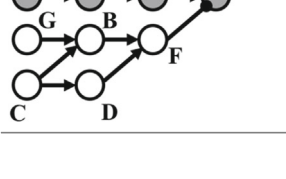
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Table 5.1 (continued)

Diagram	Causes by <i>def-sn, def-mn</i>	Causes from <i>PH</i>	Comments
14	G+, C- (SCF)		Variant not discussed in PH
15	A+, C+ (SCF)		Variant not discussed in PH
16	{A-,C-}; {A-,G+} (SCF)		Variant not discussed in PH
17. (From PH, Fig. 42, p. xx)	A+, C+ (A+: SCF; C is E-ambivalent)	A+, C+	cf. PH p. 199: “C+ and A+ are joint causes”. (Note: the term “joint causes” is better left for sets of causes, see def-mn. C+ and A+ are actually separate causes.)
18	A+, C+ (SCF)		Variant not discussed in PH

(continued)

Table 5.1 (continued)

Diagram	Causes by <i>def-sn, def-mn</i>	Causes from <i>PH</i>	Comments
19. (From PH, Fig. 43, p. xxi) 	C+ (C is G-ambivalent)	C+	“Redundant prevention”, PH p. 213
20. (From PH, Fig. 44, p. xxi) 	H+, C+ (H+: SCF; C is I-ambivalent)	C+	See PH p. 220
21 	F+, C-, A- (SCF)		Variant not discussed in PH
22 	C+ (SCF)		Variant not discussed in PH
23 	A+, G-, C- (SCF)		Variant not discussed in PH

(continued)

Table 5.1 (continued)

Diagram	Causes by <i>def-sn, def-mn</i>	Causes from <i>PH</i>	Comments
24. (From PH, Fig. 38, p. xix) 	C+ ; {F-, A+} (C is E-ambivalent; for {F-, A+}: SCF)	?	Case left undecided by PH, cf. p187
25. (From PH, Fig. 18, p. xiv) 	C+ (SCF)	C+	“Late preemption”, considered particularly problematic, cf. PH, p99ff
26. (From PH, Fig. 11, p. xii) 	{C+, A+} (SCF)	C+, A+	“Overdetermination”, considered problematic, cf. PH Ch. 3 §5

Here is a concrete example of early preemption (Hitchcock, 2007): “Assassin poisons Victim’s coffee. Victim drinks it and dies. If Assassin hadn’t poisoned the coffee, Backup would have, and Victim would have died anyway. Victim would not have died if there had been no poison in the coffee.” In this example Assassin’s action preempts Backup from poisoning Victim.¹¹ Intuitively, Victim’s death (Y in **DEF-CF0**) is caused by Assassin’s action (X), *not* by Backup’s, since that is preempted. But **DEF-CF0** gives the wrong result.

To solve this and other cases, we need to introduce some definitions. First, notice that there is something remarkable about Fig. 5.1, a feature that is absent from most diagrams from (Paul & Hall, 2013). There is an ‘ambivalent loop’ starting at the active neuron C: along the path C-D-E only stimulating signals are emitted towards the final event E+; along the path C-B-E, also starting at C and ending in E, there is an inhibitory signal—one that is backed-up by A. So, from C to E there is, so to speak, an ambivalent converging fork, consisting of one stimulating path, one inhibiting; with the extra feature that the inhibited neuron is backed-up (many philosophers

¹¹ To make the correspondence with Fig. 5.1, one could take, for example: C + = Assassin decides to put poison in Victim’s coffee (and notifies Backup); D + = Assassin puts poison in Victim’s coffee; E + = Victim drinks the coffee and dies; A + = Backup decides to put poison in Victim’s coffee; B + = Backup puts poison in Victim’s coffee (unless he gets a notification from Assassin).

noticed this feature, of course). Now, I will show in the following that *only* when such an ambivalent loop is present in a neuron diagram, one needs to go beyond simple counterfactual ideas. Moreover, I will argue that these exceptional cases can further be unified with all other diagrams, *and* with other major definitions through a higher-level theory.

Let us make this precise with a few definitions, *all relative to neuron diagrams as just defined* (all this could be done slightly more formally, but at this point there seems nothing to gain with it):

def-an (ambivalent neuron): A ‘Y-ambivalent’ neuron is: (i) either an active neuron from which two paths start towards neuron Y (i.e. converge on Y or on an active neuron that connects to Y in the forward direction), of which one path contains only stimulating arrows, and one path one inhibiting connection towards a (necessarily inhibited) neuron, which receives a (back-up) stimulating signal from an active off-path neuron; (ii) or an active neuron that stimulates (directly or via other stimulating neurons) an active neuron that complies with (i).

In Fig. 5.1 the C-neuron is an E-ambivalent neuron. Other Y-ambivalent neurons are the C-neurons in diagrams 8, 10, 17, 19, 20 and 24 in Table 5.1; ‘Y’ corresponds to the last neuron in these diagrams (usually E). In case (i) a ‘neuron that connects to Y in the forward direction’ is a neuron from which there is path to Y via connections that only go forward in time. Case (ii) in **def-an** is illustrated in diagram 24 (this is the diagram of Fig. 38 of Paul & Hall, 2013).

Using **def-an** we can propose a definition for cause in neuron diagrams. Actually, we can and should distinguish single-neuron causes and multiple-neuron causes; in (Paul & Hall, 2013) almost exclusively single-neuron causes are considered. Multiple-neuron causes refer to sets of events that are all simultaneous, so happen at the same time (concretely, in Table 5.1 I list causes at t_1 only). As will be seen in the following, when hunting for causes, by far the most useful clause in **def-sn** below is the counterfactual clause ‘ $\neg X$ implies $\neg Y$, ceteris paribus’: most of the time it is this ‘simple counterfactual’ (SCF) clause one needs, not the one about ambivalent neurons. It is stipulated that these definitions refer (exclusively) to neuron diagrams, which are assumed to represent actual situations, and that the causal identification has to be done by a (mental/theoretical/counterfactual) operation involving inverting (‘ $\neg$ ’) X, to be specified in the definition of the ceteris-paribus clause infra. Here is the definition:

1) def-sn (single-neuron cause): A single-neuron event X is a single-neuron cause of Y (also a single-neuron event) *iff* either the X-neuron is a Y-ambivalent neuron, or $\neg X$ implies $\neg Y$ (ceteris paribus).

2) def-mn (multiple-neuron cause): A multiple-neuron event $\{X_1, \dots, X_n\}$, consisting of $n \geq 2$ simultaneous single events, is a multiple-neuron cause (a set of ‘joint’ causes) of the single-neuron event Y *iff* none of $X_1, \dots, X_n$ is a single-neuron cause (in the sense of **def-sn**) and the conjoined $\{\neg X_1, \dots, \neg X_n\}$ form a minimum set of events that together imply $\neg Y$ (ceteris paribus).

Here ‘ $\neg X$ ’ can be read ‘not- X ’ and represents event X not happening, i.e. ‘ $\neg$ ’ operates the switching of $X+$ and $X-$. In **def-mn** the multiple-neuron event X consists of n simultaneous single-neuron events $X = \{X_1, \dots, X_n\}$; $\neg X = \{\neg X_1, \dots, \neg X_n\}$ corresponds to the set of events in which each X_i is switched. The ceteris-paribus clause stipulates how exactly the inverting has to be effectuated, in other words, how the associated counterfactual model has to be built. Here goes its definition¹² (X can be single or multiple events): given a (factual) neuron diagram (representing a scenario of factual interactions), the inverting of neuron event X corresponds to a counterfactual scenario, starting at the time of X , in which $X+$ and $X-$ are switched and all ‘forward’ consequences of this switching implemented ceteris paribus, so while keeping all other events, not related to X , fixed. ‘Forward consequences’ are those occurring after, not before, the time of occurrence of X ; in any case, the counterfactual scenario does not take into account what happens before X . (How to implement consequences of switching and which events are related to each other is simply stipulated by the rules that come with the meaning of the diagrams.¹³)

Thus, this clause or rule implies that in the counterfactual scenario (used to decide whether X is a cause of some posterior event Y) one can neglect all events that happen before X ; this is a mental, imagined scenario, in that it is a variant of the (model of) reality one wishes to causally dissect. As one example, the ceteris-paribus clause implies that in diagram 5 of Table 5.1 $F-$ is a cause of $E+$: $F+$ implies $E-$ if one keeps fixed all F -unrelated events in the path $A-E$.

One will note that these definitions certainly share some commonalities with other published accounts of causation. First of all, if one disregards ambivalent neurons, **def-sn** is quite similar to **DEF-CF0** and **DEF-CF0'**—at least under a broad interpretation, to be elaborated. Next, the idea of a ‘minimal set’ in **def-mn** resonates with ‘minimal sufficiency’ accounts¹⁴ (e.g. Paul & Hall, 2013, p. 16). Finally, the way we build the counterfactual scenario resembles to some extent the way it is done by other proponents of the counterfactual interpretation (e.g. passages in Paul & Hall, 2013, p. 20, 21, etc.) So **def-sn** and **def-mn** are inspired by ingredients of other models. But, to the best of my knowledge, these ingredients have not been combined in the way here proposed.

Using these quite straightforward definitions—they become in any case straightforward with some practice –, I have identified in Table 5.1 the causes of 26 representative diagrams, many of which are discussed in Paul and Hall (2013); the table also contains new diagrams not discussed there. Specifically, I have indicated the root cause(s) (so the causes at time t_1) of the *last* event of each diagram, as definitions **def-sn** and **def-mn** deliver them.

¹² I will make it very explicit; interestingly, it sounds a bit clumsy, presumably because our brain is trained to do this operation very quickly and very routinely. We do counterfactual thinking all the time, when planning, conceiving, deliberating, researching, and fantasising.

¹³ To take the simplest example, if two active neurons X and Y are only connected via a stimulating connection, $\neg X$ will imply $\neg Y$, a scenario in which both X and Y are off.

¹⁴ Actually, the requirement that the set be minimal is only needed for diagram 16 in Table 1.

These results should be the correct ones: they comply with intuition, and correspond often, but not always, to the ones favoured by Paul and Hall. Most importantly, while in (Paul & Hall, 2013) they are left as un-unified problem cases, here they all follow from one unified model. This is the main result of this section. As a consequence of this first synthesis, the proposed model can solve a majority of problems discussed in Paul and Hall (see **R1–R4** below), but not all; for some, I will need a second synthesis: a model unifying **def-sn** and **def-mn** with other powerful definitions.

As an example, the table indicates that in diagram 1 the root cause of E+ is C+, that in diagram 3 the root cause of E- is C-, etc. A third column displays agreement or disagreement with the verdict given in Paul and Hall (2013). Note that in the literature the causes at the initial time t1 are usually discussed, rather than at t2 etc., since causes closer to the last event are much easier to retrieve, since more intuitive—accordingly, the SCF rule applies to them. So I focus on the most difficult and most discussed case, even if our definitions work for all times, as can easily be verified.

Several points deserve to be highlighted before I comment on some interesting corollaries. First, the vast majority of causes of single-neuron events in these diagrams are themselves single-neuron causes (especially if one considers also causes at times $t > t_1$, not indicated in Table 5.1). So **def-sn** is more needed or useful than **def-mn**, which can explain why multiple-neuron causes are much less discussed in the literature. Yet, it will be seen that **def-mn** is definitely needed. And indeed, in real-life situations it is often as important to know joint causes as single causes, as any medical doctor, scientist, engineer or detective will confirm (see examples further on). Next, the table shows that there are only seven ambivalent neurons (the archetype being neuron C in diagram 1). Therefore, the vast majority of causes are identified with the ‘simple counterfactual’ (SCF in Table 5.1) clause stipulated in **def-sn** (namely: $\neg X$ implies $\neg Y$, ceteris paribus). Simple counterfactual inverting, much as the basic **DEF-CF0** has it, leads most of the time to correct causes! I will explain and corroborate this point in Sect. 6.

Let us now highlight a few further results (**R1–R4**) that follow from the above definitions. A full discussion of these and other problems discussed in the literature will have to await Sect. 6, after the final synthesis.

R1. Double prevention and transitivity. Diagrams 5 and 12 are examples of ‘double prevention’ (e.g. Schaffer, 2000), considered a particularly problematic case in the literature. Our model corroborates the intuitions of many philosophers (e.g. Hall, 2000; Hitchcock, 2001; Paul & Hall, 2013, Ch. 5) that transitivity is violated *in neuron diagrams* (as defined above and in Paul & Hall, 2013). Our model confirms that in diagram 5 C+ is a cause of D+, which is a cause of F-, which is a cause of E+ (all this follows straightforwardly from the definitions). Yet C+ is not a cause of E+, as again follows from the model. Other well-known accounts struggle with this case (Paul & Hall, 2013, p 82ff). I will come back to this key problem in Sect. 6: actually, the violation of transitivity appears to be an artefact of the neuron diagrams, rather than a fundamental property.

Note that my model also solves more complicated cases of multiple prevention as in diagram 20. A different and more complex route (‘de facto dependence’) to double prevention has been taken for instance in Hitchcock (2001), discussed in Paul and Hall (2013, p. 222ff). My model is robust against similar variants, some gathered in Table 5.1.

R2. Multiple-neuron or joint causes. Definition **def-mn** identifies in three diagrams (13, 16 and 24) multi-neuron or ‘joint’ causes. First of all, and interestingly enough, diagram 16 (a variant of 13, but not discussed in the literature) offers a weighty argument that such joint causes must exist. Indeed, it is intuitive that in this diagram there are *no* single-neuron causes of E- at t1. The simple counterfactual rule implies that neither A-, nor G+, nor C- are causes, in agreement with intuition. However, that same rule, the essence of **def-mn**, does identify {A-,C-} and {A-,G+} as joint causes, so sets of causes that together, in cooperation, cause E-. There *must* be a cause of E- at t1: these diagrams are conceived in such a way that if something happens at t2, t3, or t4, it must be because of something happening (or not happening) at t1. So if there are no single-neuron causes, the only possibility left is that joint causes exist; in diagram 16 these are {A-,C-} and {A-,G+}. Thus **def-mn** gives a unified solution to problem cases as diagrams 13, 16 and 24 (Note that, in diagram 16, the SCF rule on its own would imply that {A-, G+, C-} are joint causes; in all diagrams of Table 5.1, if X and Y are causes according to our definitions, {¬X, ¬Y} also invert the last event, and could, if one so desires, also be called a multiple-event cause. This is in agreement with the intuition that doubling causes can only guarantee the effect even more. But the minimality requirement of **def-mn** comes to the clearly better conclusion that one should look for minimal sets, in this case {A-,C-} and {A-,G+}).

Similarly, in diagram 13 (Fig. 41 in Paul & Hall, 2013) one finds that {G-, C+} is a multi-neuron cause of E+, a possibility not envisaged in (Paul & Hall, 2013). A+ is a separate cause, in a clear sense. Here is a further strong argument that such joint causes, as I defined them, should correspond to something real: the diagrams in Table 5.1 can be implemented electronically, using light-emitting diodes as neurons. In such electronic diagrams stimulating and inhibiting signals are carried by electric current, and the on and off states of a neuron correspond to the LED being on or off, so emitting or not. One can imagine circuits that have, next to each neuron-LED, a switch that allows an experimenter to deliberately switch the LEDs on and off—corresponding to the mental switching (‘¬’) of the definitions. Intuitively, hunting for causes (at t1) would also here be hunting for those events *or event sets* that, (deliberately) inverted, invert the state of the last LED. And it is quite intuitive, and could be tested in the lab, that in such a circuit materialising diagram 13, A+ and {G-, C+} are separate causes. A mundane but somewhat contrived example of this diagram can also be found, and comes to the same conclusion.¹⁵

¹⁵ A + represents the evil A setting fire to a forest, leading to the burning of a house (E +), the intended effect of A and his accomplice D. The other neurons represent fireguards that can, in principle, ‘activate’ (+) colleagues to extinguish fires, for instance by calling them by phone. Suppose only G and C sit in watchtowers and can, in principle, see the fire; F is the fireguard responsible of the zone of the fire and the house. D + stands for D, the accomplice, deciding to put

Finally, diagram 24 is also discussed at length by Paul and Hall (2013, p. 187ff). With **def-sn** and **def-mn** one finds that the causes are $C+$ and $\{F-, A+\}$, which seems a different verdict than the one proposed by Paul and Hall (2013, p. 188), who seem to leave the case undecided. (Diagram 24 is the only case in Table 5.1 for which one needs clause (ii) of **def-an** of ambivalent neuron.)

R3. Late preemption. This type of causal relation, schematised in diagram 25, is highly debated in the literature since D. Lewis's work, and considered one of the most problematic cases (e.g. Paul & Hall, 2013, p. 99, Menzies & Beebe, 2020). The typical example is that of C throwing a rock ($C+$) at a bottle and shattering it ($E+$); A also throws a rock but a little later, so that A 's rock sails through empty space but would have shattered the bottle if C 's hadn't. The intuitive cause of $E+$ is $C+$. Our model solves it in coherence with the previous cases: suffices to recall that events are located in space–time points, an assumption I made from the start, for reasons explained at the beginning of this section. Note that this assumption is well in accordance with the common-sense interpretation of the 'events' in the bottle-shattering scenario just sketched. Thus $E+$ stands for the shattering of the bottle at a given time t_2 (and at a given place); the shattering would not have happened *at* t_2 if A 's throw would have shattered the bottle, since it comes late. Thus **def-sn** works. (My last word on late preemption will be given by a model based on a further synthesis.) The definition similarly solves the example of a bell ringing twice (Paul & Hall, 2013, p. 101ff).

Note how simple this solution to late preemption is in comparison with other proposals, such as Lewis's and several other experts' (reviews in Paul & Hall, 2013, p. 99, Menzies & Beebe, 2020). Of course, I am only entitled to so much optimism because these solutions are consistent with results obtained within the next-level synthesis, and thus presumably with the MCT, the maximally coherent theory, as shown in Sect. 6.

R4. Overdetermination. A self-explaining example of the reputedly ambiguous 'overdetermination' (cf. detailed discussion in Paul & Hall, 2023, Ch. 3 §5) is given in diagram 26. Using the SCF clause of **def-mn**, one finds that $\{C+, A+\}$ are *joint* causes of $E+$. So, for this diagram, as defined in the literature and above, neither $C+$ nor $A+$ are separate causes. This is also the solution obtained with my final model. Tampering, black-box cases and other problems discussed in the literature within the neuron diagram paradigm will also be seen to be handled correctly by the final model.

F asleep ($F-$), but *only* if he receives a call from C confirming the fire, so that his evil action is not in vain. C does call ($C+$), thus deciding D to get active ($D+$). C also calls B , her boss, who calls F , in vain. G is an absent-minded guard, who normally should have called B . In this scenario $A+$ is clearly a cause of $E+$; the only other scenario (besides $A-$) in which $E+$ would not happen is when G would be active ($G+$) and *at the same time* C would not call D ($C-$) (so that D remains inactive). Hence, intuitively, $A+$ and the joint set $\{G-, C+\}$ are causes of $E+$, as the model stipulates.

For all diagrams of Table 5.1, the problem-solving capacity and relative simplicity of **def-sn** and **def-mn**, and, especially, their compatibility with an even more encompassing definition (Sect. 6), confer confidence that the solutions they deliver are the correct ones—to the point that one can rely on them in less intuitive cases, as represented in say diagrams 13, 20 or 24. There are several ‘experimental’ ways to further corroborate these definitions (independently of the absorption in a stronger model). As said, the definitions could be verified by electronic realisations of the diagrams. Next, neuron diagrams and these definitions can be programmed by a computer code. Assuming that the ‘ambivalent neuron’ clause can definitely be proven to be valid also for diagrams that are more complicated than those of Table 5.1, the definitions could then allow solving diagrams that are much more complicated than those in (Paul & Hall, 2013), say containing thousands of neurons instead of the 14 or less in the diagrams in Table 5.1. There must be diagrams that can only be solved in an impractically long time by humans, but in a fraction of a second by a computer. Finally, in (Vervoort & Nikolaev, 2025) and (Vervoort et al., 2023) it is shown that AI based on artificial neural nets (so-called Large Language Models, LLMs) already goes quite far in correctly identifying causes in neuron diagrams (as verified using **def-sn** and **def-mn**); unfortunately it is not fully transparent how LLMs do this.¹⁶ At any rate, assessing the evolution of AI’s causal reasoning skills is both a strategic and fascinating effort, where philosophy has a key role to play: only synthetic efforts can provide the ‘gold standard’ to evaluate AI’s causal verdicts against.

One obvious point to observe regarding definitions **def-sn** and **def-mn**, is that they have, of course, a touch of ad-hoc-ness. It seems clear that they cannot serve as general definitions of cause for general contexts—even if this is the ambition of the original **DEF-CF0** and **DEF-CF0’** from which they originate. It is the ‘ambivalent neuron’ clause that is responsible for this ad-hoc-ness. Therefore, one hopes that this clause can be disposed of in a second synthesis. To see how this could be done, let us look at other important accounts of cause: they might bring us the necessary inspiration.

5.3 ‘Functional’ Definition of Cause in Physics. Representation-Relativity

Within synthetic philosophy it is mandatory to be coherent with physics—with current physical theory—since physics deals with the ‘purest’, simplest systems and is therefore more easily systematised. Hence it is most likely to be true, most likely to represent something real, at least to some degree — according to the realist. To put it in romantic words, physics is philosophy’s best friend.

¹⁶ LLMs function in essence by statistical text completion, not by logical reasoning. But such models could be enhanced with logical reasoning models.

In physics it appears often easy to identify causes, whether on token (event) or type (property, variable) level.¹⁷ The cause of the increase in volume V of a gas in such-and-such conditions may be the heat H transferred to the gas; in short H causes V . (On token level this sounds less congenial: event e_1 causes event e_2 , where e_1 = the event consisting of an amount of heat h being transferred to a gas in a certain time lapse, and e_2 = the event consisting of a gas assuming a volume v at a certain time; h and v are numbers.) The lasing of a semiconductor diode may well be said to be caused by the electrical current injected in the diode, causing first a ‘population inversion’ in the electronic energy levels, and subsequently the emission of laser light. When evaluating a particle reaction, a physicist could say something in this style: “the production of particle X is caused by the collision of particles Y and Z ” (for instance, X may be a gamma ray, Y an electron and Z a positron).

Now, as several philosophers of physics have noted (e.g. Schlick (1949), Mach (1960) p. 325, Suppes (1970) p. 46, p. 67 and references therein, Bunge (2009) Chapter 4.1 and references therein), there are good reasons to believe that in physics deterministic causal relations are *functional* relations. Indeed, the volume V of a gas is a *function* of its temperature and therefore of the heat H transferred to it (in symbols: $V = f(H)$). This idea well parallels the intuitive idea that H causes V . The laser light intensity is a function of the injected current. The energy of particle X is a function of the energy of particles Y and Z .

And indeed, *associating causes with arguments in functions is precisely what physicists do in the only chapter of physics from which a definition of cause can immediately be extracted*. There is one chapter that explicitly deals with the notion of cause, namely the foundations of quantum mechanics, more precisely Bell’s theorem and its ancestor, the Einstein–Podolsky–Rosen paradox (Jammer (1974), Bell (1964)). There are a few other sub-disciplines that use the concept of causality, notably relativity theory, as we will see in Chapter 9. But no definition of cause can be distilled from them, at least not as immediately as from Bell’s seminal paper. (John Bell’s paper of 1964 is one of the most cited articles in physics—which is highly remarkable, since it has at least in part a philosophical content, related to the question of causation. Bell’s quasi-philosophical work laid the basis for new branches of science, quantum cryptography and quantum information theory.) Note again that the scarcity of the concept of cause in theoretical physics textbooks contrasts with the *practice* of physicists—in their daily work they use it all the time.

Let us have a quick look at the article in which Bell derives his famous inequality. [A brief intermezzo for those interested: Bell’s inequality allows for demarcating quantum from classical physics. Classical physics is causal physics, in a sense defined below. Bell has shown that (i) all classical theories must satisfy the Bell inequality in a certain experiment—at least if they are ‘local’, i.e. not relying on faster-than-light interaction/communication, as any known theory; (ii) quantum mechanics violates the inequality by a certain factor. In 1964 it was not clear what the experimental outcome would be: would the inequality be violated or not? The experiment, quite

¹⁷ The somewhat subtle interplay between definitions of cause on type and token level is well explained in many texts, e.g. in Woodward (2003), e.g. p. 39–41 and p. 74ff.

sophisticated, was done many years after Bell's analysis, and it demonstrated that the quantum prediction is correct—the inequality was violated by the amount calculated by Bell. The precise upshot is still debated, but the usual conclusion is that 'nature is not classically local' (cf. e.g. Myrvold et al., 2024). This interesting chapter of physics goes of course beyond the scope of this book, but we will briefly return to it in the Epilogue—discussing the principle of causality. My own preferred interpretation of Bell's results can be found in Nikolaev and Vervoort (2023) and Vervoort (2013b).]

Bell's article begins as follows: "The paradox of Einstein, Podolsky and Rosen was advanced as an argument that quantum mechanics could not be a complete theory but should be supplemented by additional variables. These additional variables were to restore to the theory causality and locality." For us, the question is: how do physicists believe that 'causality can be restored to the theory', or rather, how exactly do they conceive of the notion of cause? Bell does not give a fully explicit definition, but it is easy to make it explicit from his use of a few simple mathematical expressions: a physical variable or property Y is described in a deterministic manner, or is caused, if it can be expressed as a function of some other variable(s), say the set or n -tuple of variables X ; in symbols,¹⁸ if $Y = f(X)$.

So, for the physics community the question of what (physical) causes are seems largely unproblematic—at least in a deterministic setting. The secondary literature on Bell's paper is enormous, but as far as I know no physicist nor philosopher of physics has ever contested Bell's definition of (deterministic) cause. Of course, things are arguably simpler in physics than in other cognitive domains and in everyday life—where functional dependences usually cannot be established (in any case not with certainty, and even less by a precise mathematical expression as in physics). Now, since within synthetic philosophy there is such a huge incentive to be coherent with physics whenever possible, the aim of the remainder of this chapter could be summarised thus: *to look for a general definition, one that can also be applied in real life when functional dependences are not known or not known to exist, and that reduces to the definition of cause in physics.*

In this context, it seems interesting to note that the concept of 'function' is one of the most essential and most frequently used concepts of mathematics, where it is given a precise definition (e.g. Rudin, 1976; Royden, 1988, Ch. 1). Functions from a set X to a set Y associate ('map') to each element of X precisely one element of Y . In this sense, if the element of X and the function are known, the element of Y is also known—hence *determined*, one might say. This notion of determination is, of course, strongly resonating with that of causation (which explains Bell's interpretation sketched above). Functions are literally everywhere in mathematics, but also in physics; to the point that a majority of physics problems can be said to be chases for functions. This is quite obvious: solving the (deterministic) equations of physics allows one to discover how some quantity Y varies *as a function of* other variables X —how Y is determined by X . If the time-domain of these other variables does not

¹⁸ Bell actually writes $A = f(\lambda)$, where λ are the so-called 'hidden variables': yet unknown variables that would, if real, 'complete' quantum mechanics and make a causal theory of it. The expression $A = f(\lambda)$ is used in Bell's derivation of his inequality.

come after that of Y, physicists often call the variables X the causes of Y. In a slogan, *hunting for functions is hunting for causes*. The ubiquity of functional dependencies in physics (and mathematics) is then an intuitive argument for the ubiquity of causation, for the reality of the causal principle. But I will stop this digression here, even if I believe it is revealing and interesting (my last word on it in Chapter 9).

Let us now make the above ideas more precise, and construct an explicit definition. The simplest variant of functional definition of cause would be, on type level:

$$\begin{aligned} &X \text{ is a cause of } Y \\ &\quad \text{iff} \\ &\exists \text{ function } f: Y = f(X). \end{aligned}$$

(DEF-F0)

(As said, in this section on physics, X and Y denote physical properties, more precisely variables.) For reasons to be explained, I will not include so-called ‘physical constants’ among the variables, just as is usually done in physics (hence their different name). For instance, the gravitational constant g is usually not considered a variable but a constant, for self-explaining reasons. Constants occur in the equations used to derive functional expressions, so appear in the functions resulting from solving the equations, so could a priori be included among the causes; but this is not what is usually done. More on this point in a moment, when I discuss Maxwell’s equations as an example.

On token level **DEF-F0** becomes: “The event represented by variable X assuming a value x is a cause of the event represented by the variable Y assuming a value y iff $Y = f(X)$ (and hence in particular $y = f(x)$)”. (Note that such an expression only makes sense if x is one of the possible values of the property X and y of Y as they describe the system under scrutiny; so this is assumed.) One may write the definition on token level thus:

$$\begin{aligned} &X_x \text{ is a cause of } Y_y \\ &\quad \text{iff} \\ &\exists \text{ function } f: Y = f(X) \text{ (and hence } y = f(x)). \end{aligned}$$

(DEF-F0’)

To complete **DEF-F0** and **DEF-F0’**, which really are one and the same definition,¹⁹ a few add-ons are due, two rather trivial, and one essential. First, as already stipulated in the previous section, this definition has an implicit time-clause, stipulating that the time-domain of X (or more precisely of X_x , i.e. of X assuming the value x) should be anterior, or at least should not start after, the time-domain of Y (or Y_y). Without this clause, **DEF-F0** would imply that causation is symmetric. Mathematically speaking, functions (as they occur in physics) can usually be inverted, so that from the bare **DEF-F0** would follow that X is also a function of Y, and therefore caused by Y. But this goes against the usual intuitions: for instance, in the above example we accept that heat causes volume increase, but not the reciprocal.

¹⁹ Because, in physics, it makes no sense to consider properties X that would have no range of values x assigned to them; the two definitions imply each other.

It deserves some further study whether it is judicious to define the time-clause in its less stringent variant, stating that the time-domain of X (or X_x) 'should not start after' the time-domain of Y (or Y_y), rather than 'should be anterior'. This allows for the possibility of simultaneous cause and effect. But in many, and perhaps all, physical contexts the so-called *initial conditions* that define the time-evolution of the system under scrutiny allow one to identify an unambiguous and asymmetric cause-effect relation. These initial conditions are needed to solve the equations. Consider e.g. following case: what is the cause of the fact that this ball remains motionless on this table? We are in a physics context; we are looking at a physical event consisting of the movement of a ball on a table; more precisely we ask what is the cause of the event consisting of this ball having a vanishing 3-dimensional velocity on this table at time T or in time interval ΔT . Physics teaches us that the movement of the ball at time T depends on the forces being exerted on it at time T (in this case gravity and the equal reaction force of the table, adding up to a total force equal to zero) *and* on the velocity at some arbitrary moment *before* T (since the force and velocity at time T are zero the preceding or 'initial' velocity of the ball also must have been zero, according to Newton's second law of motion). So one could say the causes of the considered event (the non-moving of the ball at T) are the absence of total force on the ball and the absence of initial velocity. The vanishing force could be considered a simultaneous cause, but the vanishing initial velocity is a preceding cause; so as a whole the cause is anterior to the effect.

In quantum mechanics, there exist cases where simultaneous cause and effect seem, *a priori*, to make sense. If two particles are in a so-called 'spin-singlet' state, the spin X of one particle is always opposite to that of the other (Y), so that $X = -Y$, a particularly simple functional expression trivially implying that $Y = -X$. Do both spins cause each other? The other option is to stick to causes that come before their effects. According to the latter version of the definition, and according to the current status of quantum theory (judged to be not fully mature by Einstein and fans!), there would be no causes of X and Y : we do not know any function f and property Z such that $X = -Y = f(Z)$, where Z is anterior to X and Y . Many physicists even accept that there is a fundamental theorem that prohibits the existence of such a f and Z : Bell's theorem. Now, events having no causes is also strange, and the orthodox no-cause verdict may be due to the provisional nature of quantum theory. However, a more widely acceptable solution to this conundrum exists. Spins are quantum so probabilistic properties, and therefore the analysis in terms of functions may not be adequate. As we will see in Sect. 5, there should be a different definition of cause (**DEF-P**) that applies to probabilistic properties. If one analyses the above-sketched anticorrelation of the singlet-state using **DEF-P**, one comes to the satisfying conclusion that there *are* antecedent causes to X and Y : see Chapter 9 §1.

Now, I believe that the temporal aspect of causation is most interestingly discussed within the context of the most fundamental physical theories, as highlighted by Ladyman and Ross (2007, §5.4). The first point to observe is that most fundamental equations of physics are time-symmetric, as recalled by Russell (1912) and Ladyman and Ross. *This indeed suggests that the most general time clause in our definitions of cause should be: "the time domain of Y should not start before that of X ".* General

relativity, for instance, offers a cogent case where causation is mutual, and where cause and effect exist simultaneously and are interchangeable, so to speak. Einstein's field equations express the deterministic link between the spacetime curvature X and the energy-matter distribution Y . Both X and Y exist simultaneously, and it is not obvious what comes first, curvature or energy-matter; this is reflected in Wheeler's famous slogan "Spacetime tells matter how to move; matter tells spacetime how to curve" (Thorne et al., 2000). So at the most fundamental description of reality, *the safest move is to provide for mutual causation*. Note, on the other hand, that future research may come up with theories that do explain 'what came first' (on some natural interpretation). Note also that even the most fundamental theories, when they are applied to solve concrete problems, are normally supplemented with initial values that dictate 'which variable(s) come(s) first', exactly as in above mechanical example. Similarly, in electromagnetism concrete problems normally contain the time sequence that determines the solution: e.g., first an electrical field is switched on, then the magnetic field is created—even if Maxwell's equations are time-symmetric and both fields seems to act simultaneously in these equations. I will come back to some finer details about the time-clause in Chapter 9—which is, at any rate, not the essential ingredient of causation as noted at the end of Sect. 1.

The second somewhat trivial implicit clause in **DEF-F0** and **DEF-F0'** is that X and Y should refer to, describe, different systems, or at least different properties if the same system is scrutinised. Without this clause one would find that, for instance, the speed V of a system at time t_1 is a cause of its speed at $t_2 > t_1$, which sounds awkward. Indeed, in deterministic systems $V(t_2) = f(V(t_1))$ for some function f . Here is another example. Suppose a stone falls in a pond, causing circular surface waves to emanate from the point of impact. What is the cause of these circular surface waves? A more precise question is: what is the cause of this water surface profile (the height H of the wave crest) at space–time point²⁰ (x, y, z, t) ? Even more precise is: what is the cause of the event consisting of the wave assuming a height H at space–time point (x, y, z, t) ? According to **DEF-F0'**, supplemented only with the time-clause, any surface profile $H'(x', y', z', t')$ taken at a time t' before t would be a cause of H , because in principle the theory of fluid mechanics allows for calculating H as a function of H' . But this is again against our intuition: we usually do not say that an earlier part of the wave causes a later part; we would of course say that the wave and its parts are caused by the (impact of the falling) stone. This and other examples seem to indicate that the causal relation is between different entities, or, more generally, between different properties, whether these describe different entities or one and the same (as, perhaps, in the above example from general relativity). That the causal relation usually exists between different entities complies with the common and ancestral intuition that *things are not self-caused*. (This compelling conviction can surely find a scientific justification in the law of conservation of energy: energy, things, and changes are not created out of nothing, they come 'from something else'.) Thus, a second clause that I will assume to be implicit in **DEF-F0/F0'** is that the properties X and Y pertain to different entities; minimally, that property $X \neq$ property Y . Note that in definitions

²⁰ Or one could consider as event a set of wave heights at a region in space at a certain time.

as **def-sn** for neuron diagrams (or related definitions in the literature) it is also trivial that $X \neq Y$; one doesn't even bother to mention it. In sum, I believe this second clause is less essential; it seems that intuition easily takes care of it.

Let us now come to the third specification that needs to be included in **DEF-F0/F0'**—and this one is essential; it will be the key to the final synthesis in Sect. 6. One can uncover this extra specification via two angles of analysis. Here is angle 1.

According to the functional definition **DEF-F0**, the stone is indeed the cause of the wave, more precisely the characteristics of the stone (X , including for instance its diameter) are the cause of the wave characteristics (Y) (fluid mechanics predicts $Y = f(X)$ for some function f that can be calculated), in accordance with our intuitions. As always, this functional relation can be calculated from the equations governing fluids—an essential remark we will come back to later. But which characteristics exactly are 'the' cause(s); to what detail should one go? As is well-known, causal analysis can be done at different levels of precision. For instance, physicists may include among the causes of the precise wave profile the precise way in which the stone falls (e.g. the height X' from which it falls on the water surface), as well as the depth of the water basin (X''). They then need to use more precise equations that include these extra variables. X' and X'' are co-determining the *detailed* shape of the wave; fluid mechanics can predict their influence (if one takes these variables into account the calculations become quite involved). So X alone, or all three of X , X' and X'' can come out as causes, according to the same **DEF-F0**, *depending on the precision one asks*. Higher precision demands more precise, 'truer' model equations.

Thus, as we will corroborate in a moment, it seems that what one counts as cause depends on the details of the effect-event 'one' considers. Let me first make a remark on the use of 'one' in the above sentence. In principle, although this is slightly tedious and not really necessary, one could distinguish the person who inquires about the cause of effect Y (the 'cause-inquirer') from the person who identifies or assigns a cause X in response (the 'cause-assigner'); often these are one and the same person. This distinction is sometimes helpful in *conversations* about cause-effect identification: there is a (trivial) logical sequence. Indeed, in a sense, causes are not, as objects, 'ostentatiously out there' in the world. For causes to be explicitly identified, obviously someone has to inquire about them before the same or another person can venture an answer. Now, the point is that this inquiry imposes standards of precision, more generally a context. It seems that this precision or this context needs to be specified in **DEF-F0**, else it seems ambiguous and could give rise to different cause-assignments to the same Y . [Instead of stating "this inquiry imposes standards of precision" one could, in principle, say that the cause-inquirers impose standards of precision; and that the cause-assigners have to take these into account, if they wish to avoid identifying 'wrong' causes. But this seems obvious; I will put remarks along this line between square brackets.]

Let us briefly elaborate on the above example, since it is quite representative of physical practice. If one looks for the causes of the experimental event "the wave height Y assumes numerical value y ", where y has been measured with a precision instrument (e.g. using laser light) and thus has a small experimental uncertainty or error-interval δy_{exp} (a small fraction of y), the causes will include all variables X

$= (X_1, X_2, \dots)$ that allow²¹ one to write: $y = f(x) \pm \delta y_{\text{exp}}$ for some f (with x the corresponding values of X). If one only takes X_1 into account, it may be that from the equations follows that the calculated $f(x_1)$ does not correspond to the measured y : say $f(x_1) = y + \delta y_1$, where $\delta y_1 > \delta y_{\text{exp}}$. The fact that the calculation does not correspond to the experimentally measured value suggests that one has to include other variables/properties $X_2, X_3, \dots$ until calculation shows that $f(x_1, x_2, \dots) \in [y - \delta y_{\text{exp}}; y + \delta y_{\text{exp}}]$. Then all causes are satisfactorily identified. In sum, to first approximation X_1 is the cause of Y , but to a better approximation $X_1, X_2, \dots$ are the causes. All depends on the precision or details with which one considers Y_y .

Importantly, it seems that physics is not different from other sciences, or even common contexts. A non-expert police officer might be satisfied with identifying ‘a mental disorder’ as the cause of M ’s act of murder. An expert in criminal psychology, perhaps one who knows M , might list more detailed causes. In a discussion among physicists, the cause of a certain substance igniting might be identified as the heat transfer between the substance and a flame; chemists might find that the causes are to be found in a list of detailed chemical reactions. “What caused this war?” might be answered differently by a historian, a political scientist, an economist, a sociologist, or a psychologist. Actually, that the truth of a causal claim depends on the context within which it is under discussion, has been argued by several philosophers—not only in the case of scientific causal claims (see e.g. Menzies & Beebe, 2020, §4). There is convincing evidence that people’s causal judgements are sensitive to contextual factors (e.g. Clarke et al., 2015; Hitchcock & Knobe, 2009; Icard et al., 2017; Kominsky et al., 2015).

In physics, ‘context-dependence’ often, or perhaps always, boils down to ‘precision-dependence’, as illustrated above. Now, we can come to this conclusion—that something has to be added to **DEF-F0** to make it precise—also from a different angle of inquiry, angle 2, which is even more compelling.

When **DEF-F0/F0** stipulates that, for X to be a cause of Y , a function f needs to exist that connects them, one may ask: how do we know that the function exists? Who or what tells us? More precisely, *one might object that it happens that different theories/models exist for describing a given property Y or effect-event Y_y —we have studied such cases in some detail in Chapter 3 on Gettier’s problem —, different theories from which different functional dependencies, and hence causes, could be derived. At any rate, the function necessarily comes from, is derived within, a theory/model, essentially a set of equations.* [Namely, the theory/model that is assumed—by the cause-assigner—to describe the context of inquiry about the effect Y , a context imposed, in principle, by the cause-inquirer.]

Indeed, knowing that a functional relation between X and Y exists is almost always theory-based, or at least theory-laden (this thought is also entertained, but ultimately dismissed, by Suppes, 1970, p. 45; we will corroborate it forcefully later on). It happens extremely rarely that people establish the existence of a functional relation by direct empirical research—except in cases when scientists explore a fully

²¹ One could write: “that *correctly* allow one to write” etc. But this is understood. The expression $y = f(x)$ is correct if it follows from the appropriate theory.

new problem space. I will illustrate this rare case in the next section with an example of LEDs (light-emitting diodes) materialising neurons. But even then they need to make some assumptions, which one may construe as part of a rudimentary theory/model. The point is that if (the context of) Y is not unambiguously specified, people [cause-assigners] may adopt different theories/models, for instance corresponding to different sub-disciplines within physics or to different scientific disciplines. Recall the examples just mentioned, e.g. the case of physicists and chemists using different theories/models to analyse the same phenomenon. Remaining in physics, there may be a debate about which theory/model describes Y and in particular an event Y_y best.²² As discussed in Chapter 3, theories/models may ‘coexist’ in physics to describe Y_y . As we saw above, in physics it is usually the *precision* with which an event (Y_y) is considered [by the cause-inquirer] that determines the context. But there are exceptions: it may happen that different co-existing models lead to numerical predictions with the same precision. My conclusion from these insights from science is straightforward: *the most precise way to specify the context of Y is to explicitly define the model/theory that is considered* [that is of interest to the inquirers, and that therefore should be used by the assigners, if they aim at consistency].

Of course, in daily conversations, even among physicists, such a specification is often *not* made. Cause is an imprecise, pre-theoretic concept, loosely used by people (we are trying to make it a theoretical concept). When one asks “what is the cause of Y ?”, I believe it is usually understood, it is usually implicit, that *the best available theory/model* should be the basis for the causal hunt—this seems quite obvious. Now of course, if the question is asked in an absolutist state of mind it is perhaps understood that *the best possible theory in the absolute* should be used—to identify The Causes (this idea is elaborated in Vervoort, 2013a). But this notion of best possible theory in the absolute is a tricky one, since, to all likelihood, we do not have access to it; alas, we do not know The Theory of Truly Everything. The specification of which theory/model is meant is therefore often not superfluous. Note also that the demand to specify the theory/model can be seen as equivalent to the demand that Y be specified unambiguously enough (the specification of the theory can be considered as included in the specification of Y). These simple remarks echo the platitude that causes can be identified as precisely as the effects are described.

Thus, we come to the conclusion that, at least in physics, we can reach full clarity if we simply specify, when identifying a cause, what the theory/model is from which the cause and its corresponding function f is derived. This, then, leads straightforwardly to the following definition of functional cause, where T stands for a given theory or model:

$$\begin{aligned} &X_T \text{ is a cause of } Y_T \\ &\quad \text{iff} \\ &\exists \text{ function } f, T: Y = f(X) \text{ according to } T. \end{aligned}$$

(DEF-F)

²² For instance, superconductivity has still no fully agreed-upon theory (see Chapter 3), several theories are being studied and have proponents. In such a case, cause-assigners may come to different causal claims.

Or more simply (equivalently if well understood):

$$\begin{array}{c} \text{X is a cause of Y (according to T)} \\ \text{iff} \\ \exists \text{ function } f: Y = f(X) \text{ (according to T).} \end{array} \quad (\text{DEF-F})$$

This definition, which may look a bit awkward to some, is actually very simple, and can be read thus: “X is a cause of Y according to theory/model T *iff* T implies that $Y = f(X)$ for some function f ”. On token level the definition is equally straightforward.²³

In sum, by this move we have made explicit the last, and indeed essential, specification that has to be added to the bare **DEF-F0**: in order to avoid confusion, or identifying wrong causes, one has to specify the theory/model/set of hypotheses T that is used to derive the functional relation between X and Y. **DEF-F** is my final definition of functional cause. Recall that it has two more implicit and rather trivial clauses: a time-clause and an entity-clause. Note that ‘Y_T’ in **DEF-F** indicates the effect as understood, described, in the framework T; and similarly for X_T.

At this point I would like to propose a conclusion that goes widely beyond physics alone, one that concerns causal claims in *any* context. Readers who have read Chapter 3 on Gettier’s problem will have noticed that there is close parallel between that chapter and the present section on functional/physical cause. It is often a good idea to start from physics, the most basic of the natural sciences, to try and see whether conclusions valid there can be generalised to other contexts. In Chapter 3 it was argued that this move is the right one, and that the representation-relativity or theory-relativity of concepts as truth and knowledge should be generalised to everyday contexts—at least if one wants to solve somewhat sophisticated philosophical problems. This demand of generalisability is just a requirement that follows from the key desideratum of synthetic philosophy, namely to strive for maximally coherent philosophical frameworks, frameworks that work coherently in as many contexts as possible, and that are continuous with physics. We saw in Chapter 3 that this generalisation works if one understands ‘theory/model’ (well-defined concepts in physics and its philosophy) in a looser, broader sense, applicable to any context: namely essentially as ‘coherent set of hypotheses/assumptions’ or ‘conceptual framework’ or ‘coherent set of beliefs’ or sometimes even just ‘common-sense beliefs’. These are all terms denoting what one could call an *explanatory perspective* on, or a *representation* of, a piece of reality, of a collection of real events or facts (in this chapter I will denote them all by ‘T’, not ‘T*’ as in other chapters—since we need subscripts in the following).

²³ It reads: “ X_x is a cause of Y_y according to theory/model/hypotheses T *iff* T implies that $Y = f(X)$ for some function f (and hence $y = f(x)$)”. Note that to determine, on event-level, whether $x = f(y)$, where x and y are numbers, one does in general not only need a theory (equations), but also some contingent values of variables that describe the context of the contingent event Y_y . These values are usually called ‘initial conditions’ in physics (see the above example of the ball on a table); without them one cannot, in general, solve equations and calculate numbers. We may simply include these values as assumptions of the model T. (Note the analogy with Hempel’s scheme of scientific explanation, which also involves laws *and* particular facts.)

This rationale leads then to the quite straightforward conjecture that the clause ‘according to theory/model T’ of **DEF-F** might well be needed for a general definition of cause, not only in physics. I will call this feature the *representation-relativity of causal claims*, a notion of which the full importance will become clear soon. We will also encounter this notion when studying in the next sections the manipulationist and probabilistic interpretations of cause.

A last remark about **DEF-F** is implicit in what we said above, and essential for the ontology of causation. *DEF-F* allows us to understand the inextricable link between causation and the fundamental ontology of nature—by an argument that is, I hope, obvious by now. Indeed, by definition of the physicalism we have to adopt, all events are ultimately governed by the laws of physics. But, as we have seen, the corresponding law statements embody the functional relations between the variables of the corresponding theory T; solving the law-equations amounts to discovering the precise functional relations between these variables.²⁴ As another example, solving Maxwell’s equations in a given concrete situation in which a current J is switched on enables one to derive a functional relation between J and the magnetic field B that is created or caused: $B = f(J)$. One can always imagine or assume temporal domains of these variables (e.g. first comes the current, then the field); these assumptions can be considered as included in T. *DEF-F* then immediately allows us to extract cause-effect relations directly from the laws, fundamental or less fundamental, that constitute the core of T.

We will later come back to this essential idea, which highlights the key role **DEF-F** plays in a synthetic understanding of causation. One must therefore regret that the functional account seems underrepresented in contemporary scholarship.

5.4 ‘Manipulationist’ Definition of Cause. A Preliminary Synthesis

In this Section I will discuss a definition of cause that has become very popular in the last decades, the manipulationist theory, as developed by authors as Menzies and Price (1993), Woodward (2003, 2023), and Hausman and Woodward (2004). The standard reference I will follow is the well-known work of Woodward (2003, 2023). A lot of research has been invested in this theory, to which I cannot do full justice here. I will only treat selected aspects of the manipulationist theory, since I believe its problems, several of them summarised in Woodward (2023), can be solved by a more synthetic model. Also, I criticised some aspects of it elsewhere (Vervoort, 2014). Yet, it is an interesting model and, if properly understood, it resonates with the one I will ultimately adopt.

²⁴ Similar remarks hold for probabilistic laws: these allow for extracting correlations, which, as we will see in the section on probabilistic causation, are the key input of the definition of probabilistic cause.

Woodward's theory was, at least initially, developed for systems from physics and other disciplines based on structural equations, involving properties/variables²⁵ $X, Y, \dots$. The intuitive idea is straightforward: X is a cause of a Y *iff* X allows to manipulate Y . The key definition of this account, which I will call **DEF-M0**, is the following (Woodward, 2003, p. 51):

X is a cause of Y
iff
There is a possible intervention on X that will change Y [or the probability
distribution of Y]

(DEF-M0)

I will neglect for the time being the probabilistic case in brackets, to which I will return in the next sections. Actually, Woodward calls the above causes 'total' causes, and introduces also the concept of 'direct' and 'contributing' cause. However, I have argued elsewhere that for typical physical systems the notions of direct and contributing cause are inconsistent (Vervoort, 2014). Therefore, I believe that at least in physics the legitimate definition of 'cause' (simpliciter) is **DEF-M0**. Also beyond physics it captures the essence of the manipulability account; so I will focus on it. Several comments are due to get a clearer understanding of this definition.

First, in the manipulationist literature there is a quite involved debate as to what 'intervention' actually means (Woodward, 2023, §§4, 8, 10, 13). Woodward states (2003, p. 47): "More generally and slightly more precisely, we may think of an intervention on X with respect to Y as an exogenous causal process that changes X in such a way and under conditions such that if any change occurs in Y , it occurs in virtue of Y 's relationship to X and not in any other way." The problem with this definition (as well as with its more elaborate versions, cf. Woodward, 2003 p. 98, Woodward, 2023 §5 (M1–M4)) is that for the typical systems studied in physics the system equations are coupled, meaning that *each equation contains at least one variable that also belongs to at least one other equation* of the set. This creates a causal link between *all* variables, which renders it impossible that "if any change occurs in Y , it occurs in virtue of Y 's relationship to X and not in any other way" (details in Vervoort, 2014). Indeed, Woodward is inspired by authors as Pearl (2009) who consider systems that have *decoupled* system equations. Here "each equation represents a distinct causal mechanism which is understood to be 'autonomous' in the sense in which that notion is used in econometrics; this means roughly that it is possible to interfere with or disrupt each mechanism (and the corresponding equation) without disrupting any of the others" (Woodward, 2023, §5). But this does not work for typical systems studied in physics, and 'intervention on X ' must be defined differently for coupled systems in general.²⁶ Before doing so, I will make a brief detour about coupling; this also gives me the opportunity to illustrate another important feature of equations, or rather their

²⁵ Therefore, it is useful to remember that X and Y range over value domains, minimally $\{0, 1\}$, in physics normally intervals of the real numbers.

²⁶ The original definition may remain valid for systems/disciplines based on sets of decoupled equations, say econometrics etc.

interpretation by physicists: the difference between dependent/independent variables and system constants.

A paradigmatic example of coupling is given by Maxwell's equations of electromagnetism, describing a potentially infinite class of systems belonging to such subfields as electrostatics, magnetostatics, and electrodynamics (Jackson, 1999, Ch. 1). These are a set of four 'coupled partial differential equations', expressing the relation between the electric field (E), the electric displacement field (D), the magnetic field (B), the magnetic field intensity (H), and their so-called sources, namely the electric charge (ρ) and current density (J) (underlined variables are vector fields):

$$\nabla \times \underline{H} = (4\pi/c) \underline{J} + (1/c) \partial \underline{D} / \partial t.$$

$$\nabla \times \underline{E} = - (1/c) \partial \underline{B} / \partial t.$$

$$\nabla \cdot \underline{B} = 0.$$

$$\nabla \cdot \underline{D} = 4\pi\rho.$$

(The equations are only shown here to illustrate a typical feature of system equations—there is no need to go in any detail about the meaning of the variables. The operator ' ∇ ' effectuates derivation in the space coordinates, $\partial/\partial t$ is the time-derivative.) In above equations, 'c' is the speed of light. One notices that the set contains variables (D and B) that belong to different equations; more precisely, all equations contain variables that also belong to at least one other equation. This coupling makes mathematical solution difficult: none of the equations can be mathematically manipulated 'on its own' to derive a solution for a variable; they have to be handled in parallel.

One also sees that the role of the 'variables' and 'constants' in the system equations is clearly distinct. For any given physical system the constants are fixed, i.e. they keep the same numerical value independently of the values of the variables. In above example, 'c' is such a system constant: its value is assumed to be given (π is a dimensionless numerical constant without physical interpretation—still another case). 'Variables' play a very different role: within one system with fixed constants, the variables may assume different values. If in electromagnetic systems the source (or independent) variables J and ρ change (for instance due to human intervention, or due to changes in the environment), the other (dependent) variables will also change (always for fixed constants)—as the equations show. Interestingly, the terms 'dependent' and 'independent' variables used in physics (and mathematics) have a distinctly causal flavour; so scientists do seem to implicitly use them with a causal meaning in mind. Physicists usually call 'independent variables' those variables the values of which are chosen by experimenters or have another causal connotation (notably because these variables assume their values, in the real system represented by the equations, *before* the dependent variables do); while 'dependent variables' are those that then depend as effects on the independent ones (through the mechanisms described by the equations). In many experimental situations J and ρ are the independent variables (charges and currents are switched on first; this then creates/causes the fields). But in other situations the role of dependent/independent variables is reversed: e.g. a changing magnetic field can induce a current in a conductor,

which is then considered the dependent variable.²⁷ Notice that this cause-effect relationship between variables seems to correspond to the same basic intuition as the one underlying the manipulationist model, but even more conspicuously the functionalist interpretation of cause: the equations allow deriving functional relations between independent and dependent variables.

Let us now return to **DEF-M0** and the interpretation of intervention. In the first manipulationist theories the notion of cause was explicitly linked to human agency, but authors as Menzies and Price (1993) and Woodward (2003, 2023) strived at making the theory non-anthropocentric—so that causal connections can also exist and be identified when human manipulation is impossible, e.g. in geological or cosmological contexts (cf. e.g. Woodward, 2023 §§8, 10, 11, 12). But then one wonders: why not bite the bullet and rephrase **DEF-M0** in explicitly objective, non-anthropocentric terms? Objective means here: referring directly to what is supposed to happen in reality, without the detour over human intervention. If successful, it seems that such a move would greatly clarify the model.

Intuitively, and from the many examples given (Woodward, 2003, 2023), I believe it is clear how to interpret ‘intervention’ in a consistent way—consistent with physics, notably—while keeping a trace of manipulationist intuitions. I propose this interpretation: *an intervention on X is an operation that changes the value of a variable X—full stop*. Note now that the charge of anthropocentricity can at once be taken care of by simply stipulating that this ‘change operation’ can occur under the influence of humans *or spontaneously*, ‘*in the course of events*’ in which no human is involved (in nature system properties change all the time without any human being involved). Let us also stipulate that the change operation can happen in the real world *or* in thoughts: physically in the lab (or in the wild), or mathematically if the equations are at hand, or, *most of the time, theoretically*, i.e. in our mental representation of the system. As an example: using theoretical mechanics and its equations, I can in thoughts consider what would happen if the gravitational force X between earth and moon would change; I can numerically change X and see what the equations predict; this a theoretical change operation on X. This counterfactual, mental operation would immediately lead me to such conclusions as that X causes the movement of the moon, or of tides—an example tentatively considered in (Woodward, 2023, §10). With this objective interpretation of intervention, **DEF-M0** becomes transparent and is equivalent to **DEF-M0'**:

$$\begin{array}{c}
 X \text{ is a cause of } Y \\
 \text{iff} \\
 A \text{ (possible) change in } X \text{ will change } Y \text{ [or the probability distribution of } Y] \\
 \text{(DEF-M0')}
 \end{array}$$

²⁷ Of course, this naming does not really help solving the equations, so these names are not strictly needed. But they are still helpful to picture what is happening. Humans seem to automatically picture the world in cause-effect relations.

In this definition the thorny notion of ‘intervention’ has been abstracted away, while one can still see a trace of it in the notion of change. Note that it would be quite amazing that humans’ efficient causal intuitions would be based on a mental process involving such a complex notion as intervention given by (M1-M4) in Woodward (2023, §5). But **DEF-M0** is simple and user-friendly for causal inferences: mentally, theoretically imagining what Y would (roughly) do if one changes X is often immediate. And indeed, it is with this simpler interpretation that the examples of cause-effect relations given in Woodward (2003, 2023) make sense; I will mention a few below.

DEF-M0 is now a definition that directly concerns the objective world, in which the realist believes. But also anti-realists may accept it: for them the ‘change’ would be in thoughts. More precisely, **DEF-M0** refers to an agent-independent reality, which can in principle be analysed via our theoretical representations; that is also how definitions in natural science work. Interestingly, note that the revised **DEF-M0** does not only solve the anthropocentricity problem, but also the well-known problem of circularity to which manipulability theories are vulnerable (e.g. Woodward, 2023, §6). Causation is now defined in terms of change, which can mathematically be defined (if X and Y are variables), and which (therefore) could be seen as logically or metaphysically prior to the notion of cause. As so often happens, unification leads to problem-solving on several fronts.

In **DEF-M0** I have left the ‘possible’ that figures in the ‘possible intervention’ of **DEF-M0**. As said, various authors debate about the question of which interventions are possible (Woodward, 2023, §10). Now, with what we have learned in the previous section on functional cause, it seems clear what is a possible intervention or a possible change: *that is of course determined by the theory/model/set of hypotheses one uses to think about Y and its potential causes* (see an example below). At first sight ‘possible’ might seem superfluous in **DEF-M0**—a change is a change and is always possible in thoughts—but not so fast, the clause is still useful to avoid false causal claims. For instance, without the clause, the speed of light ‘c’ could be identified as a cause of the field values in electromagnetic problems: changing c in Maxwell’s equations would change the fields, as the equations show. But the theory of electromagnetism (especially if combined with relativity theory) teaches us that ‘c’ better be treated as a constant, so that a change of c does not really qualify as a realistic possibility—c is one of the few ‘universal constants’, to boot, with a touch of sanctity! Hence c is no cause after all, according to **DEF-M0**, as it should be. (But everything is relative to the perspective, rather theory, one uses: those few theoretical physicists who study theories in which the speed of light *can* vary—such heretics do exist, in small numbers—would presumably have nothing against the statement that ‘c causes the fields’...).

So, we see here again how the representation-relativity that emerged in the functional model comes to the surface. It becomes even more manifest and unavoidable from another angle: the truth of ‘a change in X will change Y’ (or analogous counterfactuals) can usually only be ascertained if we have access to adequate hypotheses/theories describing X and Y. In practice, one very rarely evaluates whether X changes Y by direct manipulation of reality—I will give an example in a moment –; one

causation as a fact of the world. That is an old idea, and one further reason why the idea of theory-relativity is natural: theories can describe (quasi-)lawful behaviour.

Many problems of the manipulability account can be solved with the representation-clause in **DEF-M**, as will become fully clear in Sect. 6, when I will generalise it further. But here is already a flavour of a few of these solutions.

First of all, manipulationist causal claims necessitate the evaluation of counterfactuals. Woodward asks (2023, §5): “how should one understand (what is the appropriate interpretation of or semantics for) the counterfactuals in question? Without attempting to answer this question in detail, it seems plausible that if interventionist counterfactuals are to be useful in elucidating causal claims, their semantics must be different from the familiar Lewis/Stalnaker possible world semantics”. I decidedly agree, and as is hopefully clear from the above paragraphs, the only definite way to evaluate manipulationist counterfactuals is to use/apply the appropriate set of assumptions/theory—if one wants to identify appropriate causes. Specifically, when using **DEF-M** for identifying causes within a specified *T*, the manipulationist counterfactual must be evaluated, inferred, *using this T*. (Recall e.g. the example of the counterfactually changing gravitational force, the effect of which is inferred, evaluated, within Newtonian mechanics.) This resonates well with what Woodward states at places.²⁹ I will give a more formal analysis of this matter in Sect. 6. In a slogan: *theories are evaluation machines for counterfactuals*. Let me admit: I know of no other way to coherently evaluate them.

As another example, Woodward analyses a classic case of preemption: “Gunman one shoots (s1) victim, causing his death *d*, while gunman two does not shoot but would have shot (s2) also causing *d*, if s1 had not occurred” (Woodward, 2023, §7). The author deals with this case by referring to *two* interventions. But using **DEF-M** one readily comes to the (correct) conclusion that s1 is a cause of *d*, simply because even our common-sense hypotheses/knowledge tell us that if s1 would not have shot, victim would not have died the same way or at the same time³⁰ in the given situation. At least under a ‘*ceteris paribus*’ assumption (we will see in Sect. 6 that this clause should indeed be added to the definition for full disambiguation). Instead of referring to two interventions, one can simply apply **DEF-M**, where now *T* = common-sense hypotheses/knowledge (about how people are killed by shooting). This example may sound a bit contrived, but recall that the manipulability account is best restricted to situations described by variables from scientific theories.

²⁹ In (2023, §5): “We are making use of a *possibility constrained* notion of intervention. Use of this notion raises the difficult question of how the relevant notion of possibility should be understood. I will suggest below that the best way of making sense of this notion is in terms of some notion of conceptual or mathematical (or if you like, “metaphysical”) coherence—roughly speaking, the issue is whether there is an appropriately empirically grounded theoretical/mathematical apparatus that allows for a coherent description of the possible intervention in question and allows us to determine what would happen if the intervention were realized.”

³⁰ Recall that events are best conceived as being located in space–time (a suggestion coming from physics).

At this point we are in the position to make an important step of unification. Indeed, by now it will be clear that there is a strong link between the manipulationist definition **DEF-M0** (*a fortiori* my modified version **DEF-M**) and **DEF-F**, the functional definition dear to physicists. It is not a surprise then that the notion of functional dependence is implicit in a large part of Woodward's manipulability theory. For instance Woodward (2003, p. 39):

(...) on a manipulability account of causation, it is most perspicuous to think of causal relationships as relating variables or, to speak more precisely, as describing how changes in the value of one or more variables will change the value of other variables. This is also the way that many scientists think about causal relationships.

It is transparent that the 'relationship between variables' of this quote is covered by the definition of function: if one knows the functional relation between X and Y , one also knows how a change in the value of X relates to a change in the value of Y . Functional dependence occasionally appears explicitly (Woodward, 2003, p. 42):

"(...) I generally focus on the deterministic case, in which (...) if variables $X_1 \dots X_m$ are all of the direct causes of Y , then Y may be written as (2.2.1) $Y = F_Y(X_1 \dots X_m)$."

So, Woodward's definition of total cause **DEF-M0** and **DEF-F** strongly resonate. If an explicit argument is needed: a cause that satisfies **DEF-F** is (often or always) a cause on the manipulability account. If a functional relationship exists between X and Y , it can in principle be exploited to manipulate Y by manipulating X : that follows from the definition of function, and see the many illustrations in Woodward (2003). Conversely, if there is a possible intervention on X (i.e. an operation through which X is set/changed to a certain value) that changes Y , i.e. if X is a manipulationist cause of Y , and if we restrict X and Y to deterministic systems, the system variables X and Y should be functionally related, even if one does not know the formal expression of that function. How else would they be related?

If Woodward's **DEF-M0** is already strongly resonating with **DEF-F**, my modified **DEF-M**, if properly understood, is indeed equivalent³¹ with it in the case of deterministic systems, be it that **DEF-F** is then more precise.

It is important to note that **DEF-M** has the advantage that it naturally applies to a broader class of contexts and systems than **DEF-F**, say from sociology, economics, everyday life etc. In such contexts functional relations may not be known, neither scientific theories in a strict sense, but empirical co-variation between X and Y may be known—e.g. from experience or from tests involving manipulation of X . If there are good reasons (in principle condensed in a conceptual framework, a set of hypotheses T) to believe that this co-variation is not spurious, not an artefact, but can

³¹ A cause that satisfies **DEF-F** satisfies **DEF-M**. If a functional relationship exists between X and Y , a change in X will (almost always) be followed by a change in Y (for functions occurring in physics, a change $\Delta y = 0$ for $\Delta x \neq 0$ arrives with probability equal 0, except in the rare cases where Y lies in a horizontal plateau as a function of X). So the right-hand-side (RHS) of **DEF-F** implies the RHS of **DEF-M**. Conversely, if there is a relation à la **DEF-M** between X and Y in deterministic systems it must be functional; hence the RHS of **DEF-M** implies the RHS of **DEF-F**. (This argument assumes that relations between variables of deterministic systems are functional relations; but that is widely accepted.)

be repeated, i.e. has some constant character, then **DEF-M** allows one to conclude that X is a cause of Y (according to T). I will make this more precise in my final synthesis.³² As mentioned, in some cases T is rather trivial, for instance in the above experiment involving LEDs representing neurons.

Another hint to the generality of **DEF-M** is that it could, perhaps, be applied to probabilistic systems. Let us chiefly investigate probabilistic causation in the next Section—even if it is a priori not part of the subject of the present chapter, focusing on deterministic cause. (Since Sect. 5 remains conjectural to some degree, I might have relegated it to the Epilogue, but it fits best in the present chapter.)

5.5 Definition of Probabilistic Cause. Another Preliminary Synthesis

Let us then briefly treat the probabilistic definition of cause, as studied by authors as Reichenbach (1956), Suppes (1970), Cartwright (1979), Hitchcock (2021). The analysis will be lighter than those presented for other definitions, and I do not claim that a final model of probabilistic cause is given here; more synthetic work is needed to claim this with full confidence. Still, I will propose an account that synthesises the models of several authors; most relevant is that my account can be unified with the other definitions of cause in this chapter *under one widespread interpretation of probability*. This is the deterministic interpretation, which is widely even if not universally accepted for classical systems, but is under debate in the quantum realm.

The key idea from which we start is H. Reichenbach's well-known analysis of the connection between causation and probabilistic correlation or dependence (Reichenbach, 1956). Interestingly, with his analysis Reichenbach answered a question raised by the father of modern probability theory, A. Kolmogorov, in his reference work *Foundations of the Theory of Probability* (1933/1956, p. 9):

In consequence, one of the most important problems in the philosophy of the natural sciences is – in addition to the well-known one regarding the essence of the concept of probability itself – to make precise the premises which would make it possible to regard any given events as independent. This question, however, is beyond the scope of this book.

Kolmogorov asks whether there exist general conditions allowing one to know in advance (without doing a statistical test) whether two variables X and Y are *independent* (de-correlated) or not; in symbols, whether their joint probability satisfies following equation, for all values of X, Y:

$$P(X, Y) = P(X)P(Y|X) \stackrel{?}{=} P(X)P(Y). \quad (\text{Ind})$$

Or equivalently:

³² I assume, as always, that the appropriate temporal relation holds between X and Y.

$$P(Y|X) \stackrel{?}{=} P(Y) \quad (\mathbf{Ind})$$

$$P(X|Y) \stackrel{?}{=} P(X). \quad (\mathbf{Ind})$$

An equality sign corresponds, by definition, to probabilistic independence. (In the following X and Y stand for variables. In order to define independence on event-level it suffices to replace in **(Ind)** X by X_x and Y by Y_y as introduced in the previous section. Note that on type-level **(Ind)** should hold for all values of the variables, values that are defined by the system/situation under scrutiny.)

When dealing with a probabilistic question related, for instance, to the tossing of two dice, or in general to the joint occurrence of two stochastic events, one has to assume or guess whether the events/variables/systems are independent or not. Generally speaking, in real-world situations there is no rule; the only way to be sure is to experimentally verify **(Ind)**. (Note that we find here an argument in favour of the objective interpretation of probability: the factorability of $P(X, Y)$ is an experimental matter, not one of subjective guesses.) Now, even if no rule exists, one often correctly intuitively whether events/variables are dependent or not. For instance, two die throws can under normal circumstances safely be treated as independent. The interesting question is: on what basis do we intuit this? Surely by our intuitive feeling that these events are not *causally connected*. This is the basis of Reichenbach's acclaimed analysis.

According to Reichenbach, then, X is statistically dependent on Y if and only if either (i) X causes Y (or v.v.), or (ii) there is a common cause Z of X and Y , or (iii) both (i) and (ii) occur. The second part of Reichenbach's thesis states that such a common cause (case (ii)) has the property that it 'screens off' the correlations between X and Y . More precisely: if the correlation between X and Y is exclusively due to a common cause (there is no direct causation), and if Z is a complete common cause for X and Y (i.e., Z is the set of all variables acting as common causes), then X and Y must be conditionally independent given Z (Hitchcock & Rédei, 2020; Allen et al., 2017). Mathematically, *under the condition stated*:

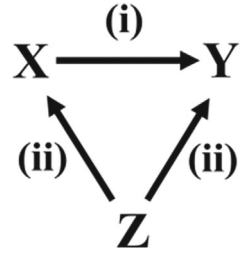
$$\text{If } P(X, Y) \neq P(X)P(Y) \text{ then } \exists Z : P(X, Y|Z) = P(X|Z)P(Y|Z), \quad (\mathbf{Reich1})$$

where Z is a complete common cause. (Recall that the variables X, Y, Z range over value domains, denoted $\{X\}$ etc.; the consequent in **(Reich1)** holds for all values of the variables.) One says that ' Z screens off X from Y '. The screening-off in the consequent in **(Reich1)** also occurs in the case of a non-common-cause causal scheme of the type: $X \rightarrow Z \rightarrow Y$ (as Reichenbach noted).

Let us write Reichenbach's thesis in a compact form (condition (iii) above is understood):

$$\begin{array}{c} X \text{ is statistically dependent on } Y \\ \text{iff} \end{array}$$

Fig. 5.2 Causal connections
of type (i) and (ii) in
(Reich2)



either (i) X causes Y (or v.v.),
or (ii) there is an earlier common cause Z of X and Y.

(Reich2)

Figure 5.2 symbolizes these conditions (i) and (ii).

Reichenbach's principle, sometimes called the 'common cause principle', seems to hold true, at least for classical systems: it has widespread empirical support³³ (cf. Pearl, 2009; Spirtes et al., 2001).

Note now that from (Reich2) one can immediately deduce a definition of probabilistic cause (if one assumes, as usual, that the appropriate temporal order exists between X and Y):

X is a cause of Y
iff
(i) Y is probabilistically dependent on X
and (ii) there is no earlier common cause Z of X and Y.

(DEF-P0)

In mathematical terms, using the definitions (Ind) and (Reich1):

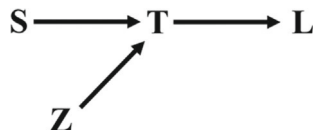
X is a cause of Y
iff
(i) $P(Y|X) \neq P(Y)$,
and (ii) $\nexists Z(\text{earlier}) : P(Y|X,Z) = P(Y|Z)$.

(DEF-P)

Trivially, for DEF-P to be applicable we need to assume that a probability function P is defined for variables X, Y, Z. Note that (i) should hold for at least some value of X and Y, and (ii) for all values of X, Y, Z (an application to a concrete case is given in Appendix 5.1). The key specification that one only considers 'earlier' variables Z prohibits the vacuity of the definition: without it, one could take $Z = X$ so that (ii) would never be satisfied. 'Earlier' means 'taking its values earlier than X and Y', as

³³ Some authors have contested the universal validity of Reichenbach's principle in classical systems (cf. Hitchcock & Rédei, 2020). In quantum systems it might not hold, because 'common causes' of quantum properties might be precluded by no-go theorems as Bell's theorem. But that is only so under certain heavy assumptions; e.g. nonlocal or superdeterministic common causes would still be allowed by the theorems. For readers interested in the philosophy of quantum mechanics, my view on this topic may be found in e.g. (Vervoort 2013b, Nikolaev and Vervoort 2023, Sect. 6); and see Chapter 9. My position surely is speculative, but there are at present no non-speculative views available on this matter.

Fig. 5.3 Model linking lung cancer to smoking and other factors



one demands of a common cause. This specification also allows one to consider the causal chain $X \rightarrow Z \rightarrow Y$ as a case of $X \rightarrow Y$ causation, as intuition tells us: in this chain Z *does* screen off X from Y but Z is not ‘earlier’ than X .

A definition closely related but not identical to **DEF-P** has been proposed by other authors, notably P. Suppes (1970) and Cartwright (1979). One difference is that these authors use ‘>’ instead of ‘ $\neq$ ’ in (i), according to the intuition that ‘causes raise the probability of their effects’. But this creates several unnecessary problems, as also seems to be the conclusion of Hitchcock (2021, §2.5). Cartwright (1979) proposes following definition: “ X is a cause of Y iff $P(Y|X, Z) > P(Y|\sim X, Z)$ for every background Z ” (where $\sim X$ is the complement of X). But as noted by Hitchcock (2021, §2.4), this would lead to the wrong conclusion that smoking is not a cause of lung cancer in a model, depicted in Fig. 5.3, linking smoking (S), lung cancer (L), the presence of tar in the lungs (T), causes of tar in the lungs other than smoking (Z). This model assumes that lung cancer is only caused by tar in the lungs, which can have different origins (causes), one of them being smoking.

Since T screens off L from S , $P(L|S, T) = P(L|\sim S, T)$ so that we get the wrong result (Hitchcock, 2021, §2.4). Using **DEF-P** we obtain for condition (i): $P(L|S) \neq P(L)$, and condition (ii) is also satisfied: $P(L|S, Z) \neq P(L|Z)$ and $P(L|S, T) = P(L|T)$ but T is not ‘earlier’ than S . Ergo S is a cause of L . Definitions à la Cartwright (1979) also get Simpson’s paradox wrong (Hitchcock, 2021, §2.4), while it is again solved in a straightforward way by **DEF-P**.

My goal in this Section is not to provide a complete analysis of a definition of probabilistic cause à la **DEF-P0** or **DEF-P**; I will restrain myself to showing that **DEF-P** can be unified with other definitions. This is a new result and it shows that **DEF-P** has, to the least, the potential to lead to a final solution for characterising probabilistic causation. This synthesis can be done under one widespread interpretation of probability, namely the assumption that probabilities emerge from a deterministic stratum, as for instance Laplace believed, one of the fathers of probability theory.

I would emphasise that this assumption is actually a ubiquitous guidance principle in the statistical theory of causal modelling, elaborated, notably, by mathematician and philosopher J. Pearl (2009). On this view, if a probability $P(Y|X) \neq P(Y)$ exists (satisfies Kolmogorov’s axioms and, I will add, the criteria exposed in the previous chapter), then there must exist a *function* from (X, Λ) to Y , where Λ are unobserved/uncontrolled (and possibly unknown) variables. (X, Λ) are termed the causes of Y : this is in agreement with our functional definition **DEF-F**. To justify this interpretation, Pearl states (2009, p. 26):

In this book, we shall express preference toward Laplace’s quasi-deterministic conception of causality and will use it, often contrasted with the stochastic conception, to define and analyze most of the causal entities that we study. This preference is based on three considerations.

First, the Laplacian conception is more general. Every stochastic model can be emulated by many functional relationships (with stochastic inputs), but not the other way around; functional relationships can only be approximated, as a limiting case, using stochastic models. Second, the Laplacian conception is more in tune with human intuition. [...] Finally, certain concepts that are ubiquitous in human discourse can be defined only in the Laplacian framework. We shall see, for example, that such simple concepts as ‘the probability that event B occurred because of event A’ and ‘the probability that event B would have been different if it were not for event A’ cannot be defined in terms of purely stochastic models (Pearl, 2009, p. 26).

As is well known, this deterministic interpretation of probability is appropriate for many classical probabilistic systems, systems that are known to be deterministic under the surface.³⁴ So, in this interpretation probability theory is an (immensely useful) tool needed because of our ignorance of certain uncontrollable variables that affect stochastic systems and processes; and probabilistic properties are (in principle) deterministic functions of these hidden variables.

Now, under this interpretation it appears that **DEF-P** is essentially equivalent to **DEF-F**. Here is a proof sketch. First, let us ignore for a moment the clause ‘according to T’ in **DEF-F**. Under the deterministic interpretation of probability, if $P(Y|X)$ is defined:

$$\exists f_1, \Lambda: Y = f_1(X, \Lambda) \text{ for all values of } X, Y.$$

Condition (i) in **DEF-P** states then that there is no function f_2 such that $Y = f_1(X, \Lambda) = f_2(\Lambda)$. One might say, Y is an ‘irreducible’ function of X. Condition (ii) states that, if there are ‘earlier’ variables Z such that $Y = f_3(X, Z, \Lambda)$, there is no function f_4 such that $Y = f_3(X, Z, \Lambda) = f_4(Z, \Lambda)$. Again, Y must be an irreducible function of X (and possibly of Λ and other variables). But that is what **DEF-F** tells us, be it slightly less explicitly. Conversely and analogously, we can reinterpret **DEF-F**, i.e. add some trivial clauses, so that it implies³⁵ **DEF-P** for probabilistic systems—always under the deterministic interpretation of probability.

Finally, what about the clause “according to T” in **DEF-F**? After what we learned in the previous sections, the first hypothesis that comes to mind is to simply add it to **DEF-P**—if it would make it more precise. There are indeed excellent reasons to do so. *Which* variables Z one considers in condition (ii) of **DEF-P** always depends

³⁴ Examples can be found e.g. in fluid mechanical systems (see the previous chapter on probability). Fluid mechanics is governed by deterministic equations, essentially the Navier–Stokes equation; yet there exists an enormous variety of probabilistic features in fluids, gases, eddies, diffusive systems etc. Another example are the probabilistic properties of statistical mechanics simulated by the Monte-Carlo technique mentioned in the previous chapter: all these properties, reproducing experimental probabilities, are simulated with pseudorandom number generators which are actually based on deterministic functions.

³⁵ In some detail, introducing the set of hypotheses T: suppose that according to some T X and Y are stochastic variables that are ultimately reducible to functions, and suppose **DEF-F** and T imply that X is a cause of Y. Then $\exists f_1, \Lambda: Y = f_1(X, \Lambda)$ for all values of X, Y considered in T, which is condition (i) of **DEF-P**. We can now add the trivial assumption that T stipulates that Y has a genuine functional dependence on X, not one that gets cancelled when one fixes some other variable Z. Then one must have (ii) of **DEF-P**. Ergo, a cause on **DEF-F** in T is also a cause on **DEF-P** in T; **DEF-P** is more precise.

on some hypotheses T (a model, a theory,...) about the piece of reality under causal scrutiny. In general, not all models contain the same variables, of course. Furthermore, whether the two probabilities in (ii) are equal or not is, ultimately, an experimental matter that can be tested—according to the frequentist interpretation elaborated in the previous chapter. But as I highlighted there, such a test is subject to a certain precision; and this precision should be specified, at least in principle, in T : one must include it in the assumptions leading to the verdict ‘ X is a cause of Y ’.

The fact, then, that **DEF-F** appears to be a special case of **DEF-P**, namely when **DEF-P** is applied to systems that are deterministic (above or under the surface), confirms that **DEF-P** is more general, as it should be. Deterministic properties are a special case of probabilistic properties, having probabilities equal to 1 or 0. Moreover, note that there is no principled reason why **DEF-P** could not work for *irreducibly* probabilistic systems (assuming that these exist; quantum systems are usually believed to be of this type). An illustration of the utility of this definition is given in Appendix 5.1, where a stochastic variant of an emblematic neuron diagram is studied (without any assumption of hidden determinism). Interestingly, **DEF-P** again leads to results that coincide with those found by other definitions, in the limits where the latter apply.

A further advantage of **DEF-P**, pending its final corroboration, seems to be that it could be used for defining precise measures of the causal influence of one variable on another, or between sets of variables. In complex systems with many variables it may be very useful to know which variable has the strongest influence on another one. Such a measure of the ‘influence’ of X on Y could be defined as follows:

$$\text{Inf}(X \rightarrow Y) = \sum_Y \sum_X |P(Y|X) - P(Y)|.$$

Here the sums run over all values of X and Y . $\text{Inf}(X \rightarrow Y)$ is only equal to zero if each term is equal to zero, in other words, if X is *not* a cause of Y according to **DEF-P**. In Appendix 5.1 this proposal is tested on a concrete example.

In conclusion, although more work is needed to compare **DEF-P** with other theories of stochastic causation, I believe that arguments exist to conclude that: (i) **DEF-P** has a genuine potential for problem-solving (I leave it as an open question whether it still has to be upgraded in a further synthesis); and (ii) that *probabilistic and deterministic causation can be unified*.

Pleasingly, **DEF-P** is objective, has an immediate connection to reality: according to it, causal connections are ‘measurable’ in a scientific way! At least, if one knows the relevant probabilities with a sufficient precision to verify conditions (i) and (ii) of the definition. That is not only dear to the heart of the realist, but is also needed for a synthetic philosophy resonating with science. At the same time, **DEF-P** (and **DEF-M** and **DEF-F**) only work for contexts that can be formalised (to some degree) and described by variables (**DEF-M** was at least originally developed for such contexts).

Table 5.2 Definitions of cause encountered in this chapter. The first two definitions lack the ‘according to T’ clause

X is a cause of Y <i>iff</i>	if X were not to occur, Y would not occur (ceteris paribus)	DEF-CF0’
A single-neuron event X is a single-neuron cause of Y <i>iff</i>	either the X-neuron is a Y-ambivalent neuron, or $\neg X$ implies $\neg Y$ (ceteris paribus)	def-sn
X is a cause of Y <i>iff</i>	$\exists$ function f : $Y = f(X)$ (according to T)	DEF-F
X is a cause of Y <i>iff</i>	a (possible) change in X will change Y [or the probability distribution of Y] (according to T)	DEF-M
X is a cause of Y <i>iff</i>	(i) $P(Y X) \neq P(Y)$, and (ii) $\nexists Z(\text{earlier}) : P(Y X, Z) = P(Y Z)$ (according to T)	DEF-P

As said, **DEF-P** only works concretely³⁶ for systems for which one can define probabilities with a reasonable precision. The conditions under which that is possible are given in the previous chapter.

But in many real-life situations such formalisation is not possible, and yet one would like to find causes there too. The model that can most straightforwardly be applied to daily-life contexts is the counterfactual one (**DEF-CF0’**); it has therefore the potential to be the most general variant. Let us therefore return to the counterfactual definition, and see whether we can make it compatible with the other definitions, in a final synthesis. The ideal would be that a modified counterfactual definition would turn out to be the encompassing model, from which the other definitions can be derived as special cases.

5.6 Final Synthesis. Representation-Relative Counterfactual Cause

In their most robust formulation, the definitions we encountered in this chapter are gathered in Table 5.2 (only **def-mn** is not given, since similar to **def-sn**).

I have already argued that **DEF-F**, **DEF-M**, and **DEF-P** strongly resonate, and coincide under rather broad conditions, notably for deterministic (and formalisable) systems. A priori **DEF-M** and **DEF-P** have the potential to have a wider applicability and encompass probabilistic systems. But **DEF-M** cannot be a good definition for probabilistic systems, as it stands. Indeed, “a (possible) change in X will change the probability distribution of Y” can be interpreted as equivalent to clause (i) in **DEF-P**. But it is well known that correlation (as in (i)) is not sufficient for causation; therefore

³⁶ One might still consider that a definition as **DEF-P** holds true for *all* systems and events – a hypothesis that assumes that probability theory describes ultimately all events in the universe. That might well be a valid assumption, but it is a metaphysical one; it cannot directly be verified; and, as argued in the chapter on probability, it doesn’t help to define objective, numerical probability values for most events.

DEF-M needs another clause, such as clause (ii) of **DEF-P** (supposing this definition can be validated as an overarching definition for probabilistic systems). In any case, **DEF-M** must be upgraded for probabilistic systems. More importantly, one senses that it should be possible to establish a close link between **DEF-M** and **DEF-CF0'** (and **def-sn/mn**) for deterministic systems. The key difference is that in **DEF-CF0'** there is an essential *ceteris-paribus* clause, while in **DEF-M** representation-relativity, i.e. the 'according to a given theory/model T' clause, plays a fundamental problem-solving role; this clause also appears in **DEF-F** and **DEF-P**. Allow me to insist, *without these clauses these definitions lose their problem-solving capacity*.

The problem we encountered with the counterfactual model, was that for neuron diagrams it leads to the contrived definition **def-sn**, which definitely loses some of its esthetical attraction due to the ambivalent-neuron clause. How can we get rid of this clause, by using what we learned from the other models? Could it be that we can eliminate the ambivalent-neuron clause by simply adding the representation-relativity clause to **DEF-CF0'**, and unify by the same token **DEF-CF0'** with the other definitions? In our effort of synthesis, this is the first idea that comes to mind; it is the simplest to try; and it works! (This is a patent example of how unification often solves problems on different fronts.) In the following I will show that using the representation-relativity clause alongside the *ceteris-paribus* clause in **DEF-CF0'** directly leads to the desired final synthesis, so that we do not need **def-sn** and **def-mn** anymore: they will be subsumed under the final counterfactual definition (**DEF-CF** below).

Recall that I have already hinted in previous sections, notably at the end of Sect. 3 on functional cause, that the representation-relativity or theory-relativity of causal claims might be fundamental also beyond science. Just as in Chapter 3, I argued in Sect. 3 that in everyday contexts 'theory/model' should be understood in a broader sense, namely as 'coherent set of hypotheses/assumptions' or 'conceptual framework' or 'coherent set of beliefs' or sometimes even just 'common-sense beliefs' (in other chapters I often designate all these notions by 'T*', but in this chapter I will use 'T' for simplicity). Also beyond science sets of hypotheses are needed to derive causes; examples abound. A detective trying to solve a murder case better have a 'theory' about it: perhaps not always a set of scientific theories, but in any case a more or less detailed and interconnected set of assumptions, describing more or less detailed and interconnected mechanisms, processes, motives etc. Without such a coherent set of hypotheses, the detective's identification of causes (the culprit) will likely be wrong. Similarly, a medical doctor, when looking for causes of complicated health problems, needs to know a set of theories and hypotheses representing mechanisms etc.; without these her diagnoses will be doubtful at best. The better the theories she knows, the better—more adequate and precise—is her identification of causes. And different doctors often state different causes: one more hint to the representation-relativity of causal claims. As already noted, representation/theory-relativity (in the broad sense) resonates with the context-relativity discussed in the literature (e.g. Menzies & Beebe, 2020 §4, cf. Section 3). But it makes it more precise, since it is decidedly better at problem-solving.

In sum, the investigations in this chapter suggest to upgrade **DEF-CF0** to **DEF-CF** in the following way:

X is a cause of Y (according to T)
iff
 if X were not to occur, Y would not occur (cet. par., according to T).
(DEF-CF)

DEF-CF is my final definition of representation-relative counterfactual cause. Let us now study its efficacy. T is, as always, some *given* set of hypotheses/model/theory about a factual situation involving X and Y. As in most counterfactual theories, X and Y are most easily understood as actual events; but I will argue that the generalisation to types is well possible. Here are some specifications (**S(i)–S(iii)**) completing the definition:

S(i). The definition stipulates, simply, that in order to identify the causes of Y (according to the representation or perspective T of Y), one needs to use T for inferring (mentally, counterfactually) which X, if not occurring or occurring differently than in the factual situation under scrutiny, would lead, according to T, to Y's not occurring or occurring differently. So, it should be understood that 'if X were not to occur' must be interpreted broadly enough, notably as 'if X were to happen differently than it actually happens' (note the link with **DEF-M**). What 'different' depends ultimately on the perspective T (see illustrations below, e.g. when we deal with 'tampering' in **RR6**).

S(ii) (compatibility-with-T). Definition **DEF-CF** can be interpreted as a neutral definition of cause-within-T, where T might be any theory, even implausible, fanciful, etc., but also as a methodological rule of how to identify '(sufficiently) true/correct/reasonable' causes – which is what we are normally after. In the latter case T is supposed to be an appropriate representation, a reasonably adequate model of the reality in which X and Y occur or are supposed to occur; T should be *explanatorily compatible with, applicable to*, X and Y. One might also say: T should 'cover' X and Y, much as in Hempel's covering-law model of scientific explanation; I will make the parallel with Hempel's model explicit below. T should also be compatible with the counterfactual situation in which X does not occur (not-X or $\neg X$); not-X should be possible or at least analysable within T; else we cannot derive correct causes of it. (It seems harmless to stipulate that a theory T being compatible with Z implies, among other things, that the theory allows to infer what would happen if Z would not happen or would happen differently. As an example, take the neuron diagrams: one can imagine each neuron being on *or* off, but not being blue – that is a hypothetical fact Z that is not covered by, not compatible with, the model of what neuron diagrams are. Take any 'theory-about-Z'.) In other words, in order to identify realistic causes, once presented with an actual Y, one should consider a T that contains realistic hypotheses/models of the situation in which Y occurs. One should 'apply' T to Y/not-Y and X/not-X. This is what happens in real life: when inquiring about a cause, one studies the situation within some (hopefully adequate) conceptual framework. In a scientific context, T may be a (reasonably efficient) scientific theory. In daily contexts, T is often simply the usual (and very large) set of assumptions that people normally have about reality, or a subset of it that applies to the causal situation under scrutiny: an *explanatory story* about the causal situation. For future use, let us call this set T_C , the 'common beliefs'. (The notion 'explanatory story' is particularly helpful to deal with the 'switching' problem under **RR1** below, for instance.) T_C might also be enriched with scientific theories. It might be enriched with particular knowledge/hypotheses about the

particular actual situation. *All that matters is that the perspective T be specified.* [It is again easy to cast this in the inquirer-assigner parlance. As argued in detail in Sect. 3 on functional cause, in the logical sequence of causal conversations it is the cause-inquirer who comes first, explicitly or usually implicitly, with a perspective T in mind; in logical principle the cause-inquirer imposes T, one might say. T is reflected in the way the effect Y is phrased. Sure, the representation T is often basic and implicit, semi-conscious, perhaps. For maximal coherence, the cause X, as proposed by the cause-assigner, should then be within the domain of application of T. But in conversations maximal coherence is not always respected: the cause-assigner might come with her particular, different T; that surely happens in real life. The ultimate requirement of a precise use of the word ‘cause’ is that T be specified.]

S(iii). The *ceteris-paribus* clause has the same meaning as given in Sect. 2 in **def-sn** for the neuron diagrams. As discussed there, this clause is of key importance when investigating whether $\neg X$ leads to $\neg Y$: in this (in practice almost always) mental process of investigation/deliberation one should keep fixed what can be kept fixed (according to T) when changing X. Many examples of the *ceteris-paribus* application are given below and in Sect. 2 on the neuron diagrams.

Although straightforward, some readers might ask for an explicit account of how to evaluate the counterfactual in **DEF-CF** and of the precise meaning of the clause ‘according to T’. Here the following equivalent variant of **DEF-CF** might be helpful: “*X is a cause of Y according to T iff T implies that $\neg X$ leads to $\neg Y$ (cet. par.)*”, where ‘ $\neg X$ ’ stands for ‘X not occurring or occurring differently’, much as it was defined in **def-sn** for neuron diagrams. As mentioned above, the implication in ‘according to T’ or in ‘T implies’ or in ‘leads to’ that is needed for using **DEF-CF** is akin to the implication in Hempel’s scheme of scientific explanation. While Hempel’s scheme is based on subsumption under laws (in short: laws and particular facts imply the explanandum), in **DEF-CF** the subsumption is under T. But as illustrated by the examples provided, there is no essential difference; for that matter, non-trivial Ts often contain laws, quasi-laws, generalisations based on regularities, rules, etc. This being said, the implication-from-T must be interpreted more widely than as ‘necessary logical deduction’: ‘most probable inference’ as in informal logic is fine (see e.g. the example in **RR3** below). The latter type of inference resembles abduction, which is not a type of logically necessary deduction.

If one wishes to go in greater detail, one needs to distinguish event and type causation. Event causation needs the representation T to be applied to particular events; type causation to types. Also Hempel’s model of explanation works for events and types. According to Hempel, in order to explain event Y (why does Y occur?), one should subsume Y under at least one law L and some particular event(s) X. For instance, Y = my flowers died; L = plants need water to survive; X = I did not water my plants (neither did someone else). Hempel’s scheme says, in short: *L and X imply Y*. One might say: one has to *apply* the law L to the event X and draw (according to the laws of reasoning) the conclusion, which is Y. Very similarly, in my account of ‘X causes Y according to T’, one has to apply T to the effect Y, and verify or infer (complying with the laws of reasoning) which events X of the supposedly real scenario would, if not occurring, imply the non-occurrence of Y. Subsuming under a representation T (a set of coherent hypotheses, a model,...) is just a generalisation of subsuming under the laws L of Hempel. Again, this procedure necessarily involves

perspectives T that apply to Y and X ; T must tell a coherent story about Y and X ; else one finds incoherent causes.

In conclusion, we have the following correspondence:

(Hempel) X explains Y (according to L) *then* X and L imply Y

(DEF-CF) X causes Y (according to T) *iff* $\neg X$ and T imply $\neg Y$.

(In Hempel's scheme I use the implication 'then' only, since the reverse implication is problematic.) Some might prefer the latter expression as the final version of **DEF-CF**, if a time and *ceteris-paribus* clause are included, and if implication is understood as specified above. The close correspondence between both accounts resonates with the idea that causation and explanation are closely related, and that one can characterise T as an *explanatory* story involving X and Y .

A priori, if we can corroborate it, **DEF-CF** would have vast advantages over classical counterfactual definitions, in which the truth conditions of counterfactuals remain vague (Paul & Hall, 2013)—recall e.g. the complex metaphorical account of D. Lewis. In **DEF-CF** the truth of ' $\neg X$ leads to $\neg Y$ according to T ' is usually unambiguously decided: it simply depends on whether within T it is true that $\neg X$ implies $\neg Y$. In principle this is objectively verifiable by anyone knowing T . Logical or at least rational inference and reasoning are verifiable, decidable. A next important advantage is that **DEF-CF** works not only on token level, as the original counterfactual models, but as easily on type level, just as Hempel's model of explanation works in both cases. I will illustrate this claim by examples, and come to a conclusion about this matter in **RR7** below.

I now show how **DEF-CF** solves problems of existing accounts (results **RR1-RR7**); in particular, how it subsumes **def-sn** and **def-mn** (**RR2**).

RR1. Context-relativity of causes. Representation-relativity is related to the context-relativity discussed in Sect. 3 and in the literature (e.g. Hitchcock & Knobe, 2009, Paul & Hall, 2013 p. 183ff, Menzies & Beebe, 2020 §4). A problem case discussed under different guises deals with the cause of my flowers having died (Y) (e.g. Paul & Hall, 2013 p. 183, Menzies & Beebe, 2020 §4). My gardener did not water them (X_1), but also the Queen did not (X_2). Under **DEF-CF0'** and the basic counterfactual models, X_1 is a cause, but so is X_2 , counterintuitively. How does **DEF-CF** solve this case? One has to start with representing the situation/scenario in which X and Y occur. Suppose, then, that the roses died (Y) and Mr. Vervoort's gardener did not water them enough (X_1). Normally, one ponders about this tragic situation through the lens of our common beliefs, T_C (let us stipulate this). At any rate, notice that X_1 is a realistic possibility according to T_C , the common beliefs about the habits of people and gardeners. We all know that people, even the most conscientious, occasionally omit to do their usual tasks. Next, it straightforwardly follows from T_C , containing also some folk biology, that if Mr. Vervoort's gardener would have watered the roses enough (*ceteris paribus*), they (probably) would not have died. Our usual beliefs license us to conclude that watering roses enough is vital for their flourishing, and therefore that not watering enough is often lethal. So

DEF-CF comes to the conclusion that X_1 is a likely cause of Y *according to* T_C . T_C is applicable to, covers, can be used to assess, the situation involving Y and X_1 . However, T_C does not cover X_2 . T_C cannot coherently conclude that if X_2 would not have happened, so if the Queen would have watered Mr. Vervoort's garden, the roses would not have perished, because according to that same T_C the Queen will, alas, never water Mr. Vervoort's garden. It is specification S(ii) of **DEF-CF** ('compatibility') that stipulates that (correct) causes can only be identified by X -compatible conceptual frameworks T , and that in such T s also $\neg X$ must be realistic (else X cannot be a realistic cause). As stipulated in S(ii): obviously, X cannot be a cause according to T if $\neg X$ cannot realistically happen according to T .

In conclusion, according to **DEF-CF** the Queen not watering Mr. Vervoort's roses is *not* the cause of their perishing. For maximal disambiguation of causal claims, it is necessary to add here 'according to T_C ' or something along this line. It could be, after all, that according to the goofy story of the popular writer W , the cause of Mr. Vervoort's roses' untimely death *is* the fact that the Queen did not water them. All that matters is that T be specified.

We can now relate to another discussion in (Paul & Hall, 2013): as they and several other authors have observed, causal investigations are often 'infected by the normative', especially when causal omissions are involved (see examples on p. 183ff, including the above queen-and-gardener case). Paul and Hall conclude: "It remains an open question how best to give a precise characterization of the phenomenon these examples illustrate" (p. 184). Within my account the answer seems straightforward: they illustrate representation-relativity. More explicitly: causal investigations, especially when omissions are involved, are often assessed through the lens of representations that include normative models. In the queen-and-gardener story we look at things through the lens of a T_C that surely includes some normative considerations (the 'cause' is interpreted as the 'culprit', after all): common-sense beliefs about the tasks and habits of gardeners and queens. It seems quite possible that people engage in causal investigations relatively more frequently when norms are at stake and culprits need to be found; at any rate, *these norms can well be part of the perspective through which one assesses causal scenarios*. This then automatically infects the causal claims, in accordance with **DEF-CF**.

Here is another example (Paul & Hall, 2013, p. 183): "Billy and Suzy are playing soccer with each other, but on rival teams. One of Suzy's teammates successfully kicks a goal. Both Billy and Suzy were nearby, and either one of them could easily have blocked the goal. But we credit the goal in part, at least, to Billy's failure to block it, and not to Suzy's failure to block it". To answer the question "what caused the goal?" we look at the game through the perspective of our usual model T_C of how soccer is played, which contains the rule or norm that only adversary players try to block goal attempts of a given team. Hence according to T_C , only Billy's ineffective blocking is part of the causes³⁷; according to that same T_C , the option 'Suzy blocks' makes no sense (in the given situation).

³⁷ Because if Billy had blocked efficiently – *an option compatible with (the norms of) T_C* – the goal would not have been made.

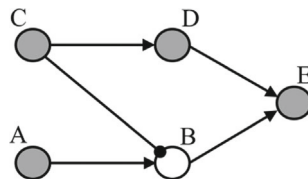
A last subtle example is a case of ‘switching’ (Paul & Hall, 2013, p. 232): “A train rushes along a track towards the point where the track forks. Suzy flips a switch, directing the train along the left-hand side of the fork. If Suzy hadn’t flipped the switch, the train would have taken the right side of the fork. A few miles later, the left and right tracks converge, and a mile after this point Billy robs the train. Assume that taking the left side of the fork rather than the right makes no difference to the train’s manner or time of arrival. Does Suzy’s flip cause Billy’s robbery?” Paul and Hall ask whether, due to the so plausible principle of transitivity, Suzy’s flip might be a cause, since it causes the train to be at the place of robbery, which in turn is necessary for the robbery to be possible. Within **DEF-CF**, Billy (or his act) is the cause according to T_C , more specifically T_C focused on/restricted to common, indeed trivial, knowledge about robberies. We all agree that if Billy hadn’t stolen stuff, there would have been no robbery (*ceteris paribus*, in the situation depicted). On the other hand, Suzy’s flip is the cause of the train being at the place of crime, *according to a different T_C* ’ focused on/restricted to the simple facts of railway transport and switches. Note that $T_C \neq T_C'$, more precisely, our usual beliefs about robberies (T_C) do not cover facts of railway transport, are not applicable to the event: “the train was at place P”. Put differently, in the situation depicted, the explanatory story about the train trajectory differs from the explanatory story about the robbery. Note that T_C surely has a specific normative dimension about *human* responsibility, not about train locations.

This example suggests an important conclusion: *if one wishes to appeal to transitivity between consecutive causes, one should consider causes that are relative to the same perspective, that are causes within the same perspective*. I will corroborate this idea under **RR6** below (it is somewhat reminiscent of the consistent-referencing principle introduced in the chapter on Gettier’s problem). The example also suggests that the neuron diagram in Fig. 45 in (Paul & Hall, 2013, p. 232) is *not* a good representation of the real-life scenario, since in the neuron diagram the switching does seem to come out as a cause. That neuron diagrams (as usually defined) are often indeed not a good model of reality will convincingly be shown in **RR2** and **RR6** below.

It is remarkable how many problems in the literature somehow hinge on context-dependence, subtle or less subtle. With some exercise, one can straightforwardly solve these through representation-relativity.

RR2. Early preemption, bis. Final verdict for all cases in Table 5.1. Let us look again and in detail at the surprisingly complex problem of early preemption, as treated already in Sect. 2. We will see in the following that this problem is subtle indeed, and tells us a lot about the pertinence of neuron diagrams and their interpretation. Let us start with the example from (Hitchcock, 2007): “Assassin poisons Victim’s coffee. Victim drinks it and dies. If Assassin hadn’t poisoned the coffee, Backup would have, and Victim would have died anyway. Victim would not have died if there had been no poison in the coffee.” As we saw in Sect. 2, **DEF-CF0** gives the wrong result (namely that Assassin is not the cause of Victim’s death). What does **DEF-CF** say? I will first swiftly go over the solution, and then elaborate in detail.

Fig. 5.1bis The neuron diagram believed to represent real cases of early pre-emption (reproduced from Fig. 1, Paul & Hall, 2013, p. ix).



Let us assume that we have no extra knowledge, and that we assess Victim's death through the common beliefs, T_C . Actually, on close inspection, T_C with **DEF-CF** give us the correct result. T_C tells that if Assassin had not poisoned the coffee *ceteris paribus*, Victim would *not* have died. 'Ceteris paribus' means: keeping fixed literally everything that can, according to T_C , be kept fixed, including Backup's action, in this case: his not acting. Indeed, common beliefs tell us that the scenario in which Backup does not do his evil deed *even if he had an agreement with Assassin* is well conceivable; Backup has free will, after all; he can change his mind. So thanks to the ceteris-paribus clause we come to the right conclusion; more precisely, we come to the conclusion that believers in T_C prefer. Or so says my model.

Why is this solution not more transparent, why has it been discussed so often in the literature, as witnessed by the large number of proposed solutions in Paul and Hall (2013)? I believe that part of the explanation is that the neuron diagram that is used to analyse the case (Fig. 5.1 repeated as 5.1bis), as well as the succinct wording à la Hitchcock, are too crude, not really appropriate to describe the factual situation under investigation. Let us have a closer look.

Hitchcock's version (version-1) is a condensed wording using a conditional ('if Assassin hadn't poisoned the coffee, Backup would have, and Victim would have died anyway'). This is not a sequential description of (supposedly) actual events. In the facts, *either* Assassin acts *or* Backup (or both); the rest is speculation, so to speak. Things can be rephrased as a description of what actually happens in the following way, for example (version-2, where A = Assassin etc.): "A and B conspire to kill V by poisoning his coffee. They agree that normally A will do the job, but if A's husband would be around at the fatal time t_m they have chosen for the murder, she will send a message to B so that he can replace her. Around t_m no husband is to be seen, and A administers the poison, killing V." Now, the fact that B would have done the job if A hadn't is, for ordinary people with limited knowledge, a speculative assumption. But in Hitchcock's story this assumption is declared from the perspective (T_H) of an all-knowing cause-inquirer—a story-teller—who has unfailing access to B's thoughts and knows that B would *necessarily* do the job. That is far from what humans normally can know. (Recall Chapter 3 on Gettier's problem, where we found out about the representation-relativity of 'to know'.) Arguably, ordinary people assessing the situation and looking for the culprit do not rely on T_H and the description in version-1 (even if verbally presented with version-1), but on T_C and a description in the style of version-2. T_C and a story à la version-2 give at least minimal information about *events and mechanisms in the real world*. At any rate, according to my account of causal identification, when people identify 'sufficiently true' causes,

they rely on sufficiently realistic descriptions of Y and sufficiently realistic conceptual frameworks, say T_C ; and they use the *ceteris-paribus* clause skilfully (or at least their causal inferences are in logical agreement with such a skilful use). Now, according to T_C it is perfectly conceivable that A's inaction leads to a living V and to a non-acting B. In symbols with obvious interpretation, we have the supposedly real scenario $A \& \neg B \& \neg V$; inverting A *ceteris paribus* leads, according to T_C , to $\neg A \& \neg B \& V$. Here $\neg B$ is kept fixed while it can be held fixed; this is the *minimally different version* (from the supposedly actual scenario) that complies with realistic 'laws' of T_C . Again, according to our usual beliefs there is no link of necessity between two people's actions in the situation sketched; for instance because they have free will (or have so according to usual beliefs!). In sum, it is quite realistic that B remains inactive even when A doesn't do the job.

On the contrary, in T_H , Hitchcock's representation, there is an implicit link of necessity. For cognitively limited beings as we are, it is difficult to assess who is the cause/culprit assuming a link of necessity and T_H (suffices to re-read T_H a few times). Perhaps A and B are *joint* causes according to T_H : that is indeed the conclusion I will come to in a moment, by using **DEF-CF** applied to T_H . One can imagine a far-fetched scenario in which there *is* a link of necessity between A and B, say via wireless-connected implants introduced in their brains (with their accord) by an evil surgeon. Suppose this mechanism ensures that if A does not kill V, B *necessarily* does. Then it is much more intuitive to call both A and B causes: they have sealed their fate in unison and have become one automat, so to speak. Interestingly, in court, in the normal interpretation of version-2 it seems likely that the main responsibility would be attributed to A, who did the fatal act; in court free will surely is considered real. But in the case with brain implants a court might well conclude that B's role is equivalent to A's, or nearly so.

Thus, it seems it is only this kind of scenario-cum-necessity that is well represented by the diagram in Fig. 5.1 and Fig. 5.1bis, under the usual rules (T_N) given in Sect. 2 (caution: in Fig. 5.1 and Fig. 5.1bis A and B do not correspond to the A(ssassin) and B(ackup) of above story). By definition, T_N (or 'T-neuron') contains the mechanical, abstract rules of the working of the diagram as given in Sect. 2. Indeed, the mechanisms at work in the neuron diagrams are such that there is also a link of necessity: in Fig. 5.1 C- necessarily implies B+ and E+, *ceteris paribus*. That is how the abstract diagram and its connections are defined; that is what T_N stipulates; in the abstract diagram there is no further 'real-life' interpretation given to its elements. So we concluded in Sect. 2 that **DEF-C0'** containing the *ceteris-paribus* clause did not deliver the intuitive answer (that only C+ is the cause). **DEF-C0'** (perhaps enriched with what we learned³⁸ with **def-mn**) leads to the conclusion that {C+, A+} are joint causes. But I would argue, along the above lines, that the intuitive cause is the cause according to T_C , namely C+. This is an illustration of how manifest representation-relativity is.

³⁸ Namely that 'X' in the definition can correspond to *sets* of single-neuron events.

In conclusion, **DEF-CF** can well help to interpret this case—and at the same time explain why it is complex. Indeed, for analysing early preemption and the diagram of Fig. 5.1, **DEF-CF**, which is at first sight very similar to **DEF-C0'**, leads to following much deeper analysis:

- (i) If one [first the cause-inquirer, in principle] interprets the diagram in abstracto, i.e. *according to* T_N , the strict mechanistic prescription of how the diagram works, the causes of $E+$ are $\{C+, A+\}$ (according to T_N).
- (ii) If one interprets this diagram in concreto, e.g. as representing version-2 above, i.e. *according to* T_C , then the cause of $E+$ is $C+$ (according to T_C).

It is key to note that, while I needed in Sect. 2 the clumsy ‘ambivalent loop’ clause in **def-sn** to come to the correct conclusion, **DEF-CF** leads to it in a simple, and above all, unified way. One can further corroborate this analysis in a quantitative way using **DEF-P**, the definition for probabilistic systems, as shown in Appendix 5.1.

Another conclusion I draw is that the neuron diagrams are, against appearances, a too idealised model of most realistic cases of early preemption! (If true, philosophers would have fallen prey to a fallacy frequent also among scientists, namely to misrepresent a complex reality by overly simplistic models. Isn’t that the history of science, and of philosophy, in a nutshell?) The diagram interpreted with T_N is a good model of T_H but not of the more relevant T_C .

To model concrete real-life cases of early preemption as version-2 above, one could propose a new type of connection and diagram as in Fig. 5.4, replacing Fig. 5.1: a double connection that ensures that one interprets the diagram as it should according to intuition (T_C).

In Fig. 5.4 the double connection with dot from $C+$ to $B-$ indicates that it is ‘maximally blocking’, i.e. remains blocking even if $C+$ is inverted to $C-$. It is this diagram that is a good model of a real-life case of preemption as version-2 above: it leads to the conclusion that only $C+$ is the cause of $E+$.

Can one solve all diagrams with ambivalent neurons in Table 5.1 (diagram 1, 8, 10, 17, 19, 20, 24) in the same way, i.e. by putting a maximally blocked neuron in the blocking branch (see another example in Fig. 5.5) ? Yes, to the extent that these diagrams are intended to represent real-life cases, as it always seems to be the case in (Paul & Hall, 2013). Just as in realistic cases of early preemption, we can assume that for real-life cases involving ambivalent loops intuition *and* T_C tell us that the blocked neuron in the blocking branch of the ambivalent fork can remain blocked

Fig. 5.4 An upgraded version of diagram 1 in Fig. 5.1. A double connection with dot keeps its blocking even when the emitting neuron is off

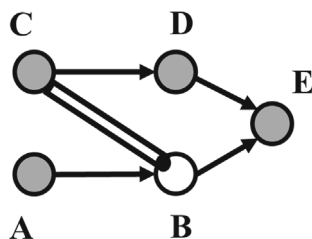
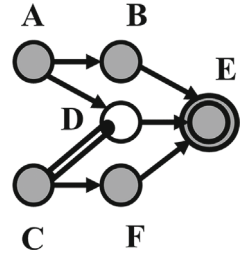


Fig. 5.5 A variant of diagram 10 in Table 5.1. With the maximally blocking C-D connection one finds that the causes of E+ are A+ and C+. The double connection ensures that one interprets this diagram as it should, i.e. according to T_C



in the counterfactual scenario: it does not switch on according to T_C (or suppose so). Specifically, if we look at real-life interpretations of diagrams with ambivalent loops through the lens of a T_C in which the blocked event does not *necessarily* gets unblocked in the counterfactual scenario, we find the intuitive results of Table 5.1. In (Vervoort & Nikolaev, 2025) it is shown explicitly that this ‘maximal blocking interpretation’ leads to the intuitive causes in all these diagrams; the definition given by Vervoort and Nikolaev is a slight reformulation of **def-sn** in terms of this notion of maximal blocking; it leads to the same results as in Table 5.1.

Another example with an ambivalent loop is given in Fig. 5.5, which is the upgraded version of diagram 10 of Table 5.1 (Fig. 14 in Paul & Hall, 2013). With the interpretation we gave to the double connection, one does not need the ambivalent-neuron clause and **def-sn** to find the intuitive answer (that A+ and C+ both are separate causes of E+); one simply needs the ‘simple counterfactual’ rule of **DEF-CF**... and stipulate that these are the causes under T_C for instance or under $T_{N'}$, where $T_{N'}$ is T_N combined with the mentioned interpretation of the double connection. (Under T_N {C+, A+} are the *joint* causes, which is in this case only a marginally different answer.)

Of course, drawing double (maximally blocking) connections in all ambivalent loops is just a trick; we do not need this visual tool to come to the desired result; we just need **DEF-CF**, just as for early preemption. So the important news is that the intuitive results of Table 5.1 can be derived from **DEF-CF** applied to the diagrams, with $T = T_C$ for instance, more precisely, any perspective that does not presuppose a link of necessity as in T_N , the mechanistic interpretation. As in early pre-emption, *it is the ceteris-paribus clause combined with the representation-clause that does the work to understand what happens in ambivalent loops.*³⁹ Diagrams without ambivalent neurons were already solved by the SCF-clause of **def-sn** and **def-mn**; but that is part of **DEF-CF**; which therefore solves all cases of Table 5.1.

RR3. Double prevention, bis. We had already shown in Sect. 2 that **def-sn** gives the right conclusion for diagrams 5 and 12, cases of double prevention (corresponding to Fig. 10 and 29 in Paul & Hall, 2013). Then **DEF-CF** should also come to the right

³⁹ When applying **DEF-CF** to the diagrams, one should also recall what we learned from **def-sn** and **def-mn**: that X in the definition can – of course – refer to several simultaneous events, and that minimal sets must be selected – this is only important for diagram 16, see Sect. 2.

conclusion: this is immediate for diagrams that are solved by the simple counterfactual rule (SCF in Table 5.1) of **def-sn**, since that is also the basis of **DEF-CF**. As we have learned, it is wise to apply **DEF-CF** to real-life cases, and be cautious with the diagrams. Paul and Hall give following example (2013, p. 175), corresponding to diagram 12: “David makes the coffee (A), and fills his cup (E). Meanwhile, Steffi scoops up James the cat (C) as he lashes his tail wildly (B); her quick action prevents a disastrous spilling (D). Most of us want to say that Steffi saved the coffee—that is, C was among the causes of E.” **DEF-CF** applied with T_C comes to the same conclusion: we all can well imagine that if Steffi had not scooped up James on the loose (*ceteris paribus*), the coffee would have been spilled. That is a logical consequence of T_C , containing some folk physics and biology, applied to the situation depicted—not a logically necessary conclusion perhaps, but at least a very likely one. (This type of inference resembles abduction, which is different from logically necessary deduction. But note that we can rephrase the example and trim down and specify T_C in such a way that that conclusion would be a necessary conclusion, in Hempel-style deduction. So there is no essential difference.)

We still have to solve the transitivity problem related to diagram 5 (see Sect. 2). We can best do that once we have studied ‘tampering’ below.

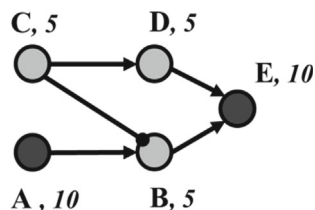
RR4. Late preemption, bis. Definition **def-sn** already solved this problem, since it applies to events located in space–time points. Here is the example of Sect. 2 again (cf. diagram 25 in Table 5.1): C throws a rock ($C+$) at a bottle and shatters it ($E+$); A also throws a rock but a little later, so that A’s rock sails through empty space but would have shattered the bottle if C’s hadn’t. The intuitive cause of $E+$ is $C+$. This is also the verdict one obtains with **DEF-CF** (applied with T_C and recalling specification $S(i)$): if C would not have thrown a rock (*ceteris paribus*), the bottle would have shattered *differently* (at a different time, with a different momentum of impact, with a different pattern of shattering, etc.). That is a consequence of T_C , which includes some rudimentary physics. If, on the other hand, A would not have thrown a rock (*ceteris paribus*), the bottle would still have shattered in the same way. Hence C’s throw is the only cause (according to T_C).

Again, we come to a verdict in a straightforward way, to be compared to the much more complex accounts described in the contemporary literature.

RR5. Overdetermination, bis. We had already solved overdetermination using **def-mn**, using its SCF-clause. Hence **DEF-CF** must also solve it in a straightforward way, with as additional advantage that a subtle explanation can be given, differentiating between different causal perspectives people may have. In this case, both a real-life description in terms of bullets and bottles (within T_C) and a purely mechanistic description (within T_N) based on diagram 26 would lead to the conclusion that $\{C+, A+\}$ are joint causes of $E+$, since only if both do not occur E does not occur (according to T_C and T_N). Neither $C+$ nor $A+$ are separate causes.

RR6. Tampering and transitivity. ‘Tampering’ is a very interesting case discussed in detail by Paul and Hall; it is diagrammatically represented by Fig. 5.6 (reproducing Fig. 15 of Paul & Hall, 2013, p. 92ff). This is a variant of early preemption, in which

Fig. 5.6 ‘Tampering’, a variant of early preemption (idem as Fig. 15, p. 92, Paul & Hall, 2013). Neurons can fire with different intensities, indicated in the diagram



now neurons fire with different intensities, indicated in the diagram, according to a sum/subtraction-rule⁴⁰ (subtraction when the signal is inhibiting). For instance, B fires with intensity 5 because it receives a +10 intensity from A and a -5 intensity from C. Let us add that the neurons can only fire with intensity 5 and 10 (intensity = 0 corresponds to being off). With this, Fig. 5.6 is self-explaining. What is the cause of E firing?

Paul and Hall state that “C and A are joint causes”, which they consider the obvious intuitive verdict (there is no real justification given). Sure, on the one hand, one intuitively feels that somehow C also ‘does something’ in the firing of E. Moreover, if transitivity holds—and this is considered by many an obvious feature of causation—, then C must be a cause (since E’s intensity is caused by D and B, and D can only be caused by C, intuitively).

DEF-CF gives again a finer analysis. Just as in the case of early preemption, one can look at the question from at least two perspectives, T_N and T_C . First, let us consider the matter under T_N , the mechanistic rules of the neuron diagram as just defined in detail. Then actually only A+ is a cause of E+, against above verdict, and against transitivity! Indeed, note that C, as a kind of ambivalent neuron, contributes to E in its upper branch always exactly as much as it subtracts from the signal from A in the lower branch; this happens for any of its possible intensities (0, 5, 10); in this sense it always exactly neutralises itself. (An analogous remark holds for the classic early-preemption diagram.) Indeed, we can write (i_X is the intensity of neuron X):

$$i_E = i_D + i_B; i_D = i_C; i_B = i_A - i_C; \text{ hence}$$

$$i_E = i_C + (i_A - i_C) = i_A.$$

Since this holds for any value of the intensities, as specified in T_N , the intensity of E is always equal to that of A, whatever the intensity of C is. **DEF-CF** gives, accordingly: A is a cause, but not C, since if C would fire differently (*ceteris paribus*), E would fire in the same way (according to T_N). Thus, transitivity does again not hold in the mechanistic neuron model T_N , just as it is violated in diagram 5 (cf. **R1** in Sect. 2).

Now, we already saw when treating early pre-emption that intuition does not always fit to T_N , and that the neuron diagrams must be treated with caution when it comes to concrete real-life situations. I believe we can now, finally, expose the crux of the problem. I believe the heart of the matter lies in the idea that ‘*perfect ambivalence*’, ‘*perfect causal self-neutralisation*’ as in Fig. 5.6 is rarely a good

⁴⁰ This rule is reminiscent of how artificial neural nets in AI work.

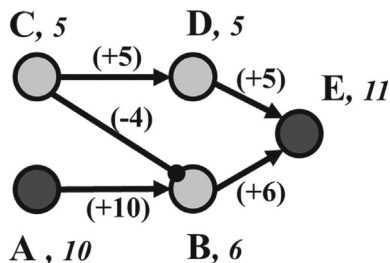


Fig. 5.7 A variant of tampering (described by model T_N'), itself a variant of early preemption. Neurons can fire with different intensities; and stimulate or inhibit with different strengths, the numbers in brackets. In this example, the strength of the emitted stimulating signal is equal to the intensity of the emitting neuron, while inhibiting signals can be weaker (see neuron C)

*representation of what happens in the real world.*⁴¹ The same remark applies to the original diagram Fig. 5.1, the emblematic interpretation of early preemption. Perfect self-neutralisation means: annihilation with numerical precision. To better see the severe limitation perfect ambivalence implies, we can generalise T_N and upgrade it to T_N' to take into account numerical differences in the causal action of C in the upper and lower path. In T_N' we simply stipulate that $i_D = i_C$, and $i_B = i_A - i'_C$ ($i_C \neq i'_C$): C emits a different intensity to D than to B; it stimulates with a different strength than it inhibits. Does this not look much more natural? To the least, it is more general: mathematically, T_N is a special case of T_N' (when taking i_C *exactly* equal to i'_C). One obtains as the defining equations of T_N' (neurons can fire with any numerical intensity, see details in the caption of Fig. 5.7):

$$i_E = i_D + i_B; i_D = i_C; i_B = i_A - i'_C; \text{ hence} \\ i_E = i_C - i'_C + i_A \neq i_A.$$

And now **DEF-CF** comes to the conclusion that, in general, both A and C are causes (according to T_N'), as intuition has it, and in accordance with transitivity. In sum, T_N' surely is a better model than T_N for many realistic situations, and therefore more likely to be compliant with T_C (of course, all depends on the realistic case in question).

A numerical example that operates according to the rules of T_N' is given in Fig. 5.7. Here the inhibiting strength of C = $i'_C = 4$. In the factual diagram $i_E = 11$, but if $i_C = 0$, $i_E = 10$, and if $i_A = 0$, then $i_E = 1$. In general, if A or C fire differently (cet. par.), E will fire differently, as stipulated by the equations: **DEF-CF** concludes that A+ and C+ are causes (according to T_N').

This conclusion is corroborated by **DEF-M** and **DEF-F**,⁴² which also arrive at this intuitive verdict. That is excellent, even if not a surprise anymore since we already

⁴¹ It is not impossible though, e.g. it seems realisable with electronics and LEDs: but these are idealised systems. Note that any, even small, deviation from an ideal electronic realisation would be represented by T_N' rather than T_N (see text).

⁴² Clearly, $i_E = f(i_C, i'_C, i_A)$ for the function $f(i_C, i'_C, i_A) = i_C - i'_C + i_A$. In other words, both A and C cause E.

know that they are equivalent in formalisable deterministic cases. Note also how this case of tampering illustrates how ‘occurring differently’ (mentioned in **DEF-CF**) depends on the model (recall specification $S(i)$ of **DEF-CF**). A neuron firing differently means ‘firing with a different intensity’; to the least, this may be what the cause-inquirer has in mind. Which intensities are allowed must be specified in the representation T .

At this point a few important conclusions can be drawn. A key finding of our investigation is that D. Lewis’s neuron diagrams, with their original rules (T_N), are often not a good representation of reality. The problem appears when the diagrams involve ambivalent loops, but also in other cases, cf. ‘switching’ in **RR1** above. I came to this conclusion when discussing early preemption and tampering, and I proposed more functional variants of these diagrams (using blocking connections or variable-strength connections) that do conform to intuition (T_C). Concomitantly, we have seen that the double-prevention diagram 5 and the tampering diagram in Fig. 5.6, with the usual interpretation, violate transitivity—at least when applying **DEF-CF** (cf. Sect. 2, **R1** on double prevention). This should not be a surprise, if these diagrams are not realistic models. Something unnatural must have slipped into them. It seems that the problem stems from the ambivalent loops, from the perfectly self-annihilating processes involved. These might indeed well represent only a theoretical possibility, but not one that maps to (most) real processes. (For what it is worth, in physics self-annihilation is anathema, at least in all systems I know of.) Perfect self-annihilation implies, or is another interpretation of, the unnatural link of perfect necessity we suspected to be the key problem when discussing early preemption (**RR2**).

This analysis coheres with the subtle analysis **DEF-CF** provides of the notion of transitivity.⁴³ From this definition it follows that *transitivity should hold, at least when the causal events are covered by the same T* . This is a logical consequence of the definition, namely of following reasoning, consisting of applying **DEF-CF** to the causal chain $X1 \rightarrow X2 \rightarrow X3$ considered from perspective T . Loosely, this leads to: if $X1$ does not occur, $X2$ does not occur (according to T); if $X2$ does not occur, $X3$ does not occur (according to T); therefore, if $X1$ does not occur, $X3$ does not occur (according to T). Thus transitivity must hold, *at least in a context in which all events in the transitive chain are described by the same model T* . Ergo, the neuron diagrams in which transitivity is violated (under T_N) are not realistic, are not good models of real processes. Or put differently, the violation of transitivity is an artefact of certain neuron diagrams, as originally defined. This conclusion is in agreement with a conjecture made under result **RR1** about switching, where we already suspected that transitivity only holds within the same T . Here the diagram under T_N implied transitivity, while the real-world scenario did not, due to the fact that two different perspectives are involved.

⁴³ I will do the analysis of transitivity here solely on the basis of **DEF-CF** (in view of its problem-solving capacity this may suffice). To increase the robustness of my conclusion, it would be interesting to compare these findings to other philosophers’ work on the topic (e.g. Hitchcock 2001); I leave this to further work.

RR7. Token versus type causation. There is a long-standing discussion in the literature as to whether causation relates singular events (tokens) or general types (properties, variables, ...). According to **DEF-CF** the answer to this classic question of metaphysics seems clear: both are possible, since perspectives/representations can be applied to both events—we have seen many examples—and general types. We have already argued so when discussing the correspondence between our account and Hempel’s scheme of scientific explanation, which also applies to events and types. As an example of type causation, we saw in Sect. 4 that ‘gravitation causes tides’ according to the manipulability account, **DEF-M**. **DEF-CF** arrives at the same conclusion, since if the (strength of) gravitation would be different (*ceteris paribus*), the tides would be different, for all we know from the theory of mechanics. ‘Smoking is a cause of lung cancer’ is true (according to usual beliefs and medicine), since if one does not smoke, the chance of lung cancer is reduced (*ceteris paribus*). Sure, if statistics are available the latter example might be better dealt with using a definition like **DEF-P**; but we have seen that this definition correlates with **DEF-CF**—to the point that unification seems a very realistic possibility. Other examples are easily constructed, and corroborate that **DEF-CF** covers type causation. [As before, the cause-inquirer decides, in principle, whether event or type causation is meant by the way ‘Y’ is phrased.]

RR8. Black-box cases. A last example I will mention are the black-box cases discussed in (Paul & Hall, 2013, p. 161ff)—cases for which there is no clear outcome. These become transparent if one realises that, to come to an unambiguous causal verdict, one needs to specify a model, a representation; if that is absent or overtly ambiguous as in black-box cases, one cannot make univocal causal claims. Clearly, this is one of the main lessons to draw from **DEF-CF**.

In conclusion of this section, we see that **DEF-CF** solves a majority of cases discussed in (Paul & Hall, 2013) and in other reference works. Importantly, it does so in resonance with other definitions of cause discussed in this chapter. If compared to the other accounts existing in the literature, **DEF-CF** derives these solutions in a simple, unified way. This gives reasonable confidence that other problems, not discussed here, can also be solved by our modified counterfactual account.

5.7 Conclusion

The goal of this chapter was to propose an overarching definition of cause by synthesising prominent accounts of causation. I focussed on the deterministic case—in principle all causal situations that are not described by probability theory. Table 5.3 summarises (sanitised versions of) the definitions encountered and indicates in the last column their application space, i.e. the nature of the causal relata. I have included my tentative definition for probabilistic causation.

The first fact to observe is that these definitions—based on intense research efforts by very many philosophers—have a pleasing homogeneity: broadly speaking, they

Table 5.3 Definitions of cause encountered in this chapter. The last column indicates the scope of the causal relation. **DEF-CF** is most general

DEF-CF	X is a cause of Y (according to T) <i>iff</i>	if X were not to occur, Y would not occur (cet. par., according to T)	X and Y are events or types described/covered by a general representation T, scientific or mundane.
DEF-F	”	$\exists$ function $f: Y = f(X)$ (according to T)	X and Y refer to events or types (variables) of a deterministic (formalised) model T in which functions are defined (or hypothesised).
DEF-M	”	a (possible) change in X will change Y [or the probability distribution of Y] (according to T)	(At least in the original manipulationist account) X and Y are types (variables) of a (scientific) model T.
DEF-P	”	(i) $P(Y X) \neq P(Y)$, and (ii) $\nexists Z(earlier) : P(Y X, Z) =$ $P(Y Z)$ (according to T)	X and Y refer to events or types (variables) of a probabilistic (formalised) model T in which probabilities are defined (or hypothesised).

express the same idea, be it in different guises. This is obviously an important sign that unification might well be possible. In scientific contexts, **DEF-F** and **DEF-P**, supposing the latter definition can be validated as a general definition for probabilistic systems, are the most precise, when applied respectively to deterministic and probabilistic systems. But surely **DEF-CF** is the most encompassing definition, which is clear from its scope: it applies most widely, as indicated in the last column in Table 5.3. By the same token, it has the largest problem-solving capacity, as exposed in detail in the previous section. It is therefore my privileged candidate for a solution of the problem of defining cause.

To be applied with precision, the other definitions need a formalised model T of a certain part of reality, at least in principle. Such models are often not known, say in social and human sciences and in everyday contexts (even if they could be hypothesised to exist). For instance, in a complex interpersonal situation, one may suspect that a certain fact or factor pertaining to person A (an act, a trait of character, a habit, ...) has causal influence on the behaviour of person B; but usually we do not have a formal (deterministic or probabilistic) model at hand describing such a mechanism. Then **DEF-F** and **DEF-P** cannot be used with confidence, or become speculative. On the other hand, **DEF-CF** is broad enough to apply.

In some more detail, I have argued that **DEF-F**, **DEF-M**, and **DEF-P** strongly resonate, and coincide under broad conditions, notably for deterministic formalisable

systems described by variables. **DEF-M** is a definition of manipulationist inspiration that I introduced to disencumber the original version of its anthropocentricity and circularity worries; this makes its proximity to **DEF-CF** obvious. **DEF-M** and **DEF-P** have the potential to have an applicability beyond deterministic variables, if they can be shown to work for probabilistic systems⁴⁴; but they still are in principle, as originally understood, restricted to formalisable/scientific contexts T. **DEF-CF**, our champion, can go beyond and be applied to daily-life contexts.

Accordingly, all three other definitions, at least when restricted to deterministic systems, can be seen as special cases of **DEF-CF**, i.e. *applications of DEF-CF to restricted domains of reality*. For instance, it is easy to show that for deterministic variables X and Y **DEF-F** and **DEF-CF** are equivalent: the proof is similar to the one given for the equivalence of **DEF-F** and **DEF-M** at the end of Sect. 4. That then also makes **DEF-CF** and **DEF-M** equivalent under the condition stated.

Even if these investigations focus mainly on deterministic causation, it is fascinating that they show that a close connection with probabilistic events/properties/systems seems at least a cogent possibility. I have already argued that **DEF-P** coincides with **DEF-F**, and hence with **DEF-CF**, when it is restricted to probabilistic systems with a deterministic (hidden) basis. But note that there seems no obstacle to interpret, *or define*, **DEF-CF** as reducing to **DEF-P** for probabilistic systems. Indeed, one can for instance generalise **DEF-CF** thus: “if X were not to occur or to occur differently (*ceteris paribus*; deterministically or probabilistically), Y would not occur or occur differently (according to T)”. The specification “deterministically or probabilistically” simply stipulates that there are two classes of events/types each corresponding to a slightly different interpretation of the definition. We already know how to interpret it for deterministic systems. Now, for (formalisable) probabilistic systems this definition is most naturally interpreted as **DEF-P**, it seems. Indeed, if probabilities exist for X and Y (according to some assumptions or model T), then the fact that “if X were to occur differently, Y would occur differently” can naturally be interpreted (on type level) as meaning that $P(Y|X) \neq P(Y)$, which corresponds to clause (i) of **DEF-P**.⁴⁵ Next, it is the *ceteris-paribus* clause in **DEF-CF** that points to clause (ii) of **DEF-P** (!) Indeed, “all other things being equal” means, in formalisable models: “all other earlier⁴⁶ variables (Z) being fixed”. In other words, “if X were to occur differently (*ceteris paribus*), Y would occur differently” means: “if X were to occur differently (keeping all other earlier variables Z fixed), Y would occur differently”. In a probabilistic context, fixing variables means conditionalising on them, so the latter expression means in probabilistic language “ $P(Y|X, Z) \neq P(Y|Z)$ for all earlier Z”; which is equivalent to (ii) in **DEF-P**.

⁴⁴ **DEF-M**, as it stands, cannot be a good definition of probabilistic cause, as shown in the previous section.

⁴⁵ On token level, “if X = x were to occur differently (probabilistically speaking), Y = y would occur differently (probabilistically speaking)” can, if X and Y are stochastic variables, naturally be interpreted as: $P(Y = y|X = x) \neq P(Y = y)$, since an equality sign would mean that Y = y would occur independently of the value of X.

⁴⁶ Recall from the discussion of **DEF-CF0'** and **def-sn** that the *ceteris-paribus* clause is defined as stipulating that *earlier* events are to be kept fixed.

In sum, it seems that **DEF-P** is a natural interpretation of **DEF-CF** for probabilistic systems; one could then *define* the probabilistic reading of **DEF-CF** to mean **DEF-P**. I will leave these ideas here as avenues of future research. In order to further analyse probabilistic causation and test **DEF-P** I introduce in Appendix 5.1 stochastic neuron diagrams—perhaps they are an interesting tool for further research. The most pressing open question is of course whether **DEF-P** already represents the MCT as it stands.

It is interesting to note that the unification achieved in this chapter is derived by a methodology akin to the paradigmatic unification of theoretical physics. There genuine progress is ascertained if a theory with wider applicability can be shown to encompass other efficient models as special cases. That is essentially the outcome of the present study; as in science, the wider model then has a greater problem-solving capacity than the local models. This logical-epistemological state of affairs parallels the chronological steps in the research method used in this chapter. We started with investigating the widest-scope model, the counterfactual one, and found a partial solution applicable to neuron diagrams. However, this solution (**def-sn** and **def-mn**) looked contrived and seemed to beg not so much for further analysis as for further synthesis. This was done by looking at other contexts and accounts: we found that the physical model **DEF-F** and the modified manipulationist model **DEF-M** were crucially representation-relative, a feature hidden in neuron diagrams, since, at first sight, ‘the representation is the diagram’. We could then return to the counterfactual model with this knowledge. The first and simplest move, just adding the representation clause to naiver counterfactual models, appeared to work.

Surely the newest idea in this chapter is the introduction of the notion of representation-relativity or theory-relativity (or theory*-relativity if we re-introduce the wider notion of theory* used in the other chapters of this book). As a postulate or rule, representation-relativity of causal identification and of causal claims states that causes are necessarily relative to conceptual frameworks, perspectives, theories*; or one might say: the identification/ascription of causes is representation-dependent. (Even if causes are relative, causation is best seen as absolute, as ontological: see arguments above and below.) This postulate was used in the solution of almost all problems, related to context-relativity (**RR1**), early (**RR2**) and late (**RR4**) preemption, double prevention (**RR3**), overdetermination (**RR5**), tampering (**RR6**), transitivity (**RR6**), token-versus-type causation (**RR7**), black-box cases (**RR8**) and others. This is remarkable, if one considers the plethora of divergent solutions for each of these problems proposed in the literature. I believe that this situation illustrates the need for a synthetic reflex in future philosophical practice: if philosophers only propose analytic solutions to problems without a holistic approach, taking the whole problem space into account, there seems no limit to the proliferation of ever new problems.

As already stated, many philosophers believe that cause is an absolute, ‘invariant’ concept (Menzies & Beebe, 2020): that there is only one true cause (or one set of true causes) for a given effect. I admit that this is a quite compelling intuition; people normally use the word ‘cause’ in this sense, without specifying anything more, say a theory. As argued in great detail in this chapter, I believe this intuition is not really adequate from a precise philosophical perspective. But notice that such an

invariantist understanding can well be subsumed by my model: it corresponds to the belief that causes are always, in all contexts, identified within the *best available or best possible theory/model/set of hypotheses*. If this is (implicitly) understood, there is no need to consider representation-relativity. Fair enough, but to solve problems, one better make the implicit explicit.

Menzies and Beebe (2020 §4) state that “an advantage of invariantism is that it allows us to conceive of causation as fully objective, mind-independent”. Not only invariantism: as mentioned in the Introduction of this chapter, I conceive of *causation* as objectively embedded in the (laws of) the universe and therefore mind-independent; I have to make this assumption to be in accordance with scientific realism. The discussion under **DEF-F** might help sceptical readers to convince themselves that there are good arguments for this embeddedness-in-laws. As we saw in Sect. 3, causes can directly be extracted from the laws; they are immanent in them. So not causation is mind-dependent; but causal identification is representation-relative; and one could say that causal identification, and the causes-according-to-T, approach objective ‘truth’ to the extent that T is an adequate representation of reality. The better T, the better the causal identification can be. In sum, if one looks at the small print, representation-relativity characterises causal identification (and psychological and logical aspects of causation), rather than causation as an ontological feature of the universe.

These ideas connect immediately to the ‘reductive challenge’ discussed by Paul and Hall, as already mentioned in the Introduction. Recall their statement (2013, pp. 7–8): “We will also assume a broadly reductionist outlook, according to which facts about which events cause which other events are fixed, somehow, by (i) the facts about what happens, together with (ii) the facts about the fundamental laws that govern what happens. [...] We will assume that a successful philosophical analysis of the causal relation ought to show exactly how facts about what happens, together with the laws, fix the facts about event causation.”

This challenge is met by having shown that **DEF-CF** reduces to **DEF-F** in deterministic physical systems (Paul and Hall consider deterministic causation, cf. 2013, p. 8). Indeed, **DEF-F** implies that causes can directly be extracted from the laws, the fundamental equations (recall the discussion in Sect. 3, especially at the end). Usually, in any case when dealing with event causation, one also needs particular facts about what happens, the ‘initial’⁴⁷ and/or ‘boundary’ conditions in technical terms, in order to solve the equations and determine causal, i.e. functional, relations with numerical precision. In sum, the equations, *together* with contingent temporal relations, *together* with **DEF-F** enable one to determine cause-effect relations. Just as Paul and Hall demand, “facts about what happens together with the laws fix the facts about event causation”.

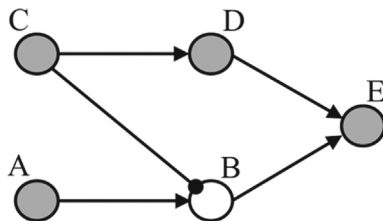
⁴⁷ In some detail: a differential equation of order n for a property $Y(X, t)$ considered as a function of X and time t needs n initial values (at time t_0) to be solvable, namely: $Y(X, t_0), \dots, Y^{(n-1)}(X, t_0)$, where $Y^{(n-1)}$ is the $(n-1)$ th time-derivative of Y . Thus, these initial conditions contain the information about the time domains of the variables $X, Y, \dots$: this information is needed for causal identification.

Therefore, as announced in Sect. 1, it indeed seems that building a precise definition goes hand in hand with uncovering the ontological/metaphysical grounds of causal claims; that the ‘conceptual analysis’ goes hand in hand with the ‘ontological reduction’; that the ‘descriptive project’ goes hand in hand with the ‘metaphysical project’ (Frisch, 2023). If one prefers: the latter is just in reach of the former. Now, to many, it is of course not a surprise that a solid definition *immediately* connects with what goes on in reality, with the deeper ontology! (Think also of the chapters on probability and free will.) I hope that those who are in doubt will find in this chapter a counterargument to their doubts. At the same time, a full discussion of this parallelism is not in the scope of this chapter, but we will come back to it in Chapter 9, dedicated to lines of future research.

I believe that the above ideas and results inevitably imply that we should have a new, more open-minded look at the old principle of causation/causality—let us interpret it here as stating that *everything happens according to laws of cause and effect*. Clearly, by showing that a unified notion of cause is a very real possibility, we have resolved one crucial objection to the principle (namely, that it cannot be consistently defined). Next, if everything happens according to the laws of nature (the laws of physics), and if, as **DEF-F** ensures us, the laws of nature inherently embody cause-effect relations, then it immediately follows that everything happens according to laws of cause and effect. Ergo, the principle of causation would be true, after all! (More in Chapter 9 §1.) Clearly, one should include in this argument stochastic laws; but here **DEF-P** ensures us that they too, via the correlations they embed, are causal laws.

Indeed, if the ontology of causation and the causal principle are now in reach, we better start with distinguishing a probabilistic and a deterministic principle of causation. But these seem to work largely analogously, just as deterministic and probabilistic cause seem so prone to unification. To lift one tip of the ontological veil, and looking first at deterministic systems, **DEF-CF** immediately points to the *ontological necessity* of (deterministic) causation, the necessity that is usually considered an ingredient of the principle of (deterministic) causation. Indeed, according to **DEF-CF**, a cause is *necessary* for its effects in the precise sense that without the cause, the effect would not happen, *ceteris paribus*, i.e. in the situation considered. This is so at least according to some representation or theory T; and if T is the best theory we have, we might as well conclude that the cause (or the corresponding event/property) is ontologically necessary. One could also say: causes *produce* their effects, in a sense that seems slightly more metaphorical. In probabilistic systems, and recalling the chapter on probability, it is collections of objects, probabilistic ensembles, ensembles of p -systems that necessarily comply with the necessary laws of probability theory. (The natural next thought is that this necessity is induced by underlying deterministic laws governing the individual systems. But let us not digress too far from the topic of this chapter.) Concomitantly, my conjectured generalised **DEF-CF** implies, for probabilistic systems, that a cause is necessary for its effects in the sense that the probability distribution of the effects would be different if that of the cause would be different, *ceteris paribus*. In this precise sense, there is ontological necessity in probabilistic causation too.

Fig. 5.8 The neuron diagram believed to represent real cases of early preemption



But I will leave it to future work to delve deeper into the perspective that **DEF-CF** and scientific realism open to the ontology of causation. First ideas can be found in Chapter 9 §1. Let us end by recalling that the identification of causality and lawfulness has, of course, a very long tradition. Specifically, the causal principle stating “all events occur by laws of cause and effect (because the laws of nature can be construed as laws of cause and effect)”, which follows from our synthetic analysis, has been defended in different guises since antiquity, by the amazing atomists Democritus and Leucippus among others, and by several authors since, Helmholtz, Mach (perhaps) to name only them⁴⁸ (for a condensed history, cf. Bunge (2009), pp. 252–253).

Appendix 5.1. Stochastic Neuron Diagrams and Test of DEF-P

In this Appendix I propose a probabilistic interpretation of neuron diagrams. I will illustrate the procedure using the diagram of Fig. 5.8.

If one interprets this diagram (diagram 1 in Table 5.1) in a probabilistic manner, one can test **DEF-P** for probabilistic systems, and at the same time corroborate its coherence with **DEF-CF**. In stochastic neuron diagrams symbols have a different meaning: A, B, C, ... are now variables that can assume different values each with a given probability. The simplest case corresponds to all variables being dichotomic; it is this case that can directly be compared to the original neuron diagrams as defined in Sect. 2. For instance, $C = 1$ corresponds to the C-neuron being on, and $C = 0$ to being off, idem for the other neuron-variables. In the probabilistic interpretation one attributes a probability to each value, e.g. $P(C = 1) = c_1$, $P(A = 1) = a_1$, etc. (Then $P(C = 0) = 1 - c_1$, etc.) The deterministic case can be modelled as a special case of

⁴⁸ Helmholtz wrote (around 1881): “The principle of causality is nothing but the assumption of the lawfulness of all natural phenomena” (translated by Bunge (2009), p. 252). Mach put it this way (around 1883): “The business of physical science is the reconstruction of facts in thought, or the abstract quantitative expression of facts. The rules which we form for these reconstructions are the laws of nature. In the conviction that such rules are possible lies the law of causality. The law of causality simply asserts that the phenomena of nature are dependent on one another” (quoted in Bunge (2009), pp. 252–253). As Bunge remarks, from this passage one could infer that Mach does not consider laws as being immanent in nature but as rules of procedure, which is at odds with the position we defend.

a general probabilistic scenario, by taking all probability values equal to 0 or 1. For instance, in Fig. 5.8 $P(E = 1) = 1$ if the diagram is interpreted deterministically and if $P(C = 1) = P(A = 1) = 1$. (One can imagine this as the outcome of measurements on a series of identical diagrams, where C, A, D, and E are always on.) As we have seen in Chapter 4, probabilistic modelling is only appropriate if the real situation represented by diagram 1 can be repeated or occurs sufficiently many times in sufficiently similar conditions. Therefore, *strictly speaking*, the probabilistic diagram 1 is not really a good description of, for instance, the Assassin-Backup case, which is not really a probabilistic scenario, but a deterministic one-time murder case. We do not have statistical data about such a gloomy scenario; but let us imagine we have.⁴⁹ (At any rate, I believe the best interpretation of the Assassin-Backup case is provided by **DEF-CF** as applied to the deterministic scenario discussed in the above sections. Here I only wish to illustrate **DEF-P**, which I conjecture to be the privileged account for genuinely probabilistic situations.)

This being said, it seems we can build a probabilistic model T_P that avoids some of the problems we encountered with T_N , the usual interpretation of diagram 1. We saw that T_N is not appropriate when applied to many realistic scenarios—say the Assassin-Backup case. According to the usual interpretation (T_N), diagram 1 stipulates that neuron B always fires if C is off and A on. But as we argued in Chapter 5, Backup’s decision B, which he makes as a free being after a message from Assassin, might change (for a variety of reasons) even if he had initially decided to use the poison ($A = 1$). So let us associate a probability to Backup’s decision B after the message: $P(B = 1|C = 0, A = 1) = \beta$, which may differ from 1, its value in the original interpretation T_N . This is the key parameter of the model: it represents Backup’s ‘reliability’, i.e. his readiness-to-poison after Assassin’s message that she cannot do the job, and after his initial agreement. To keep the calculations simple, let’s only attribute probabilities to A, C and B; D and E react deterministically as in the original T_N . The probabilities $P(B|C, A)$ are chosen so as to remain close to the original deterministic interpretation. For instance: $P(B = 1|C = 1, A = 1) = 0$ expresses that Backup never acts if C acts, just as in T_N .

In detail, the model T_P is defined by following assumptions:

- (A1) $P(C = 1) = c_1$, hence $P(C = 0) = 1 - c_1$;
- (A2) $P(A = 1) = a_1$, hence $P(A = 0) = 1 - a_1$;
- (A3) $P(B = 1|C = 1, A = 1) = 0$;
- (A4) $P(B = 1|C = 0, A = 1) = \beta$;
- (A5) $P(B = 1|C = 1, A = 0) = 0$;
- (A6) $P(B = 1|C = 0, A = 0) = \epsilon$, assumed to be a very small number, normally

0

From (A3)–(A6) follows that we also know the complements, e.g. from (A4):

$$P(B = 0|C = 0, A = 1) = 1 - \beta.$$

⁴⁹ It seems one can justify the probabilistic interpretation by assuming that there is some theoretical even if unknown probability that C is active, that B acts as promised, etc.

We now wish to answer the question: what is/are, according to **DEF-P**, the cause(s) (at t_1) of $E = 1$ in model T_P ? Recall **DEF-P**: “ X is a cause of Y iff (i) $P(Y|X) \neq P(Y)$, and (ii) $\nexists Z$ (earlier) : $P(Y|X, Z) = P(Y|Z)$ ”. Since we are interested in the causes at t_1 , clause (ii) is not needed, or, rather, vacuously satisfied. Simply applying the definition, one finds that (the variable) C is a cause of (the variable) E iff $P(E|C) \neq P(E)$ for some values of E and $C \in \{0, 1\}$. From the rules of probability theory follows that the test-condition $P(E|C) = P(E)$ is equivalent to:

$$(A7) \quad \sum_{\{A\}} P(E|C, A) P(A) = \sum_{\{C, A\}} P(E|C, A) P(C, A) = \sum_{\{C, A\}} P(E|C, A) P(C) P(A)$$

Here the sum $\sum_{\{A\}}$ runs over $A = 0$ and $A = 1$, and $\sum_{\{C, A\}}$ stands for the sum over the four possibilities $\{C = i; A = j, \text{ with } i, j \in \{0, 1\}\}$. To verify (A7), we need $P(E|C, A)$ for all values of the variables, for instance $P(E = 1|C, A)$, where $P(E = 1|C, A)$ stands for any of the four probabilities $P(E = 1|C = i, A = j)$ with $i, j \in \{0, 1\}$. This can be calculated as follows, using the fact that $E = 1$ can only occur if $C = 1$ (and hence $D = 1$) or if $B = 1$:

$$\begin{aligned} (A8) \quad P(E = 1|C, A) &= P(C = 1 \text{ or } B = 1|C, A) \\ &= P(C = 1|C, A) + P(B = 1|C, A) - P(C = 1, B = 1|C, A) \\ &= P(C = 1|C, A) + P(B = 1|C, A) - P(B = 1|C, A) P(C = 1|B = 1, C, A) \\ &= P(C = 1|C, A) + P(B = 1|C, A) - P(B = 1|C, A) P(C = 1|C) \end{aligned}$$

With (A1–A8) and the other inputs of the model one finds, taking $\varepsilon = 0$ for simplicity:

$$(A9) \quad P(E = 1|C = 1) = 1$$

$$(A10) \quad P(E = 1|C = 0) = a_1\beta$$

$$(A11) \quad P(E = 1) = c_1 + (1 - c_1) a_1\beta$$

These results can be interpreted (and can also be calculated without using the full-blown formula (A7)). (A9) is obvious: if C is on E is by assumption always on. (A10) gives the probability that E fires given that C is off, which is equal to the probability that A fires (a_1) times the probability that, in these conditions, B fires (β). (A11) gives the probability that $E = 1$ as a sum of upper (via D) and lower (via B) path contributions, which are mutually exclusive.

Thus, in the probabilistic model T_P we come, using **DEF-P**, to the realistic conclusion that in general $P(E|C) \neq P(E)$, i.e., C is a separate cause of E , *unless* $c_1 = a_1 = \beta = 1$. Under the latter ‘exceptional’ deterministic condition (A and B deterministically always on and B deterministically always poisoning), which corresponds to the original diagram under T_N , C is *not* a separate cause; then $\{A, C\}$ are *joint* causes, as one can also calculate using **DEF-P**.⁵⁰ This verdict is in agreement with **DEF-CF** applied to the original diagram under the deterministic interpretation T_N , which led to $\{A+, C+\}$ being joint causes (cf. Sect. 6, **RR2**). The original diagram T_N corresponds to T_P with $c_1 = a_1 = \beta = 1$. So in this case **DEF-P** leads to the correct result obtained with **DEF-CF**: C and A are not separate but joint causes.

⁵⁰ One calculates that $P(E = 1|C = 0, A = 0) = 0 \neq P(E = 1)$.

We interpreted this in Chapter 5 in legal terms: Assassin and Backup mechanically sealed their fate, so that both are guilty.

In a similar manner one calculates that in the probabilistic scenario A (Backup) is a cause of E *unless* $\beta = 0$ or $c_1 = 1$ or $a_1 = 0$, which is again coherent. For instance, if C always fires, E always fires independently of A; and the reliability $\beta = 0$ means that A (Backup) never puts the poison. In the case that $c_1 = 1$ and a_1 and β some value between 0 and 1, say 0.5, **DEF-P** leads to the natural conclusion that only C (Assassin) is a separate cause. This is also what **DEF-CF** gave under the ‘commonsense’ interpretation T_C of the deterministic diagram (recall Sect. 6). Thus T_P seems a more natural model than T_N ; T_P appears to better capture typical human reflexes of conspiring accomplices.

This can also be seen by using the measure for the causal influence of X on Y proposed in Sect. 5:

$$\text{Inf}(X \rightarrow Y) = \sum_Y \sum_X |P(Y|X) - P(Y)|.$$

One finds for instance that:

$$\text{Inf}(C \rightarrow E) = 2 c_1 (1 - \beta a_1).$$

This seems again coherent. For instance, the causal influence of C on E is the higher, the higher c_1 is and the lower the chance is that B fires.

However, the above claims need to be consolidated by further research, notably by testing **DEF-P** on more realistic stochastic models, and by comparing it to what other theories (Bayesian accounts, causal modelling,...) predict. The goal of this appendix is solely to illustrate **DEF-P**, and to suggest that further work on stochastic neuron diagrams might be interesting for testing **DEF-P** as the final solution to probabilistic causation.

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Chapter 6

On Absolute Free Will



Abstract In this chapter I investigate the question of whether absolute free will exists: the human capacity to make choices independently of external influences. To answer this question, one needs to take physics and philosophy of physics into account. Within this framework, it follows from rather straightforward arguments that absolute free will cannot exist. I then prepare the next chapter where I defend the idea that from a socio-humanitarian perspective ‘relative’ free will can exist, a capacity necessary for moral responsibility. This notion is compatible with neuroscience but makes abstraction of physics. These two chapters constitute my bi-perspectival account of free will.

6.1 Introduction

The question whether humans have free will is an all-time classic of philosophy; a staggering number of famous minds from virtually all philosophical and scientific disciplines have expressed their opinion about it. No doubt that most of us *feel* free often, or most of the time; and few people doubt that we have some freedom, some capacity to choose between different actions—say between lifting now my left hand, rather than my right one. And yet a great number of philosophers, physicists, neurobiologists, psychologists etc. have dared to question this belief in free will—even if this belief seems a key precondition not only of our legal system and our habits of blame and praise, but perhaps also of the very dignity of human life. To my taste, there is perhaps no philosophical topic with such profound philosophical ramifications and potential implications—on the theoretical level, but ultimately also for (a practical philosophy of) society—, and yet so little agreement among the experts. Therefore, discussing free will in only two chapters of a book seems, a priori, a risky enterprise. I better, then, specify what the precise goal of these chapters is. They aim, in the first place, as most of the preceding chapters, at conceptual clarification using the same method of synthetic problem solving as used throughout this book. (This applies especially to the next chapter, in which I will define the more complex notion of relative free will.) Thus, I will try to synthesise definitions using

expertise from (analytic) philosophy starting from Aristotle, and to construct a model that can hopefully solve more problems and answer more questions than competing models with a comparable scope. I do not claim to have studied all competing models/theories, and I obviously do not pretend to provide a complete theory of free will answering all questions from all disciplines related to it. My model can be completed along many axes, notably a neuroscientific one. But at any rate, these two chapters will be an illustration of the general method of philosophising recommended in this book.

The problem of free will does not only have a definitional component, but, just as those of probability and cause, also an ontological one. The most pressing and interesting question is, of course: *Does free will exist? Do we have free will?*—at least sometimes. For such a tremendously contentious problem as free will, it should be no surprise that my answer is: *that depends on how one defines free will*. Of course, we want the strongest possible definition, at least for starters. Now, we should have learned from the chapter on Gettier's problem that a proposition as 'free will exists' depends on the conceptual framework, the representation, the theory T* one considers. *Definitions take their meaning within conceptual frameworks*—the Quinean idea we derived in Chapter 3. Similarly, the expression: "I believe to do an act of free will" depends on how one defines the subject 'I', within which perspective.

So here is a key starting idea of this chapter, one that strongly resonates with the other chapters of this book: there are different perspectives, theories, regarding the notions of free will and the 'I' that feels free. (This idea is akin to one defended by List (2019), even if we infer different conclusions of it, cf. next chapter.) Now, not only are there different perspectives, *I claim that to have a full view on free will one needs to use at least two conceptual frameworks*: the perspective based on physics and its philosophy, and the perspective from social theories (which are closer to common sense, and ideally include or are at least compatible with neuroscience). I will argue later why also the second perspective is needed. Thus, I will develop a bi-perspectival account of free will, one that rests on the assumption that a full philosophical theory contains two perspectives that are *complementary* (and that seem, a priori, even to lead to contradictory conclusions).

My 'complementarity of perspectives' is a notion inspired by the famous complementarity introduced by Niels Bohr in the philosophy of quantum physics (Bohr, 1948). Since Bohr's philosophical ideas are debated (and often considered vague), I will attempt my own interpretation of his complementarity. Bohr advanced the powerful idea that (values of) properties of quantum systems are inextricably related to the measurement context; different experiments reveal different properties; and these experiments are complementary in that they cannot be done simultaneously¹ on a given system but yet jointly describe the system fully. (Recall how I interpreted these claims within my frequency model of probability, cf. Chapter 4, R4 in Sect. 5). It seems that Bohr thought of complementarity in a broader sense: not only in physics but in many other disciplines complementary concepts and perspectives are needed

¹ The fact that certain measurements cannot be done simultaneously is formally described by the commutation relations of quantum mechanics (cf. R4 in Chapter 4).

for a full description of a phenomenon. This is Bohr's conception of complementarity as I understand it; let me specify mine in a bit more detail.

To answer existence questions, ideally, one considers the best possible, or rather the strongest available theory; and an (accepted) concept defined within, or compatible² with, such a theory. Regarding the free will problem, one can ask the ontological question "does free will exist?" within T1 = natural science (the collection of 'best' neurobiological, physical, etc. theories) and its philosophy. But one can also ask it within a global 'best' theory that describes society, the human psyche, legal and moral practices, etc.—call this collection of theories (T2) a complete social theory of the human, or a socio-humanitarian theory, for lack of a better name. It is understood here that such a theory makes abstraction of physics (or at least of what physics tells us about free will); common sense is also such a perspective that is usually unencumbered by advanced physics and its philosophy. I believe that both perspectives T1 and T2 are complementary and needed in the free will debate.

The reader should not be distracted by the above lines: *in fine* they just say that 'free will' can have different meanings; let us look at them, and then decide in which sense free will can exist. I will start in this chapter with the notion of free will as illuminated by natural science including physics. In the next chapter I will mainly look at free will from a socio-humanitarian point of view.

6.2 Does Absolute Free Will Exist?

Both within natural science, socio-humanitarian theories, *and* common sense, one might try to define free will along the following line—as is often done in all these quarters:

Free will is the capacity to choose between different possible actions undetermined by past external events (or at least partially undetermined by such events).

I will call this the 'absolute' definition of free will. Such a conception posits the existence of spontaneous, self-generated actions, or better of self-generated decisions leading to actions, actions/decisions that are not (fully) determined by past events external to the mind of the free-willed subject. Let us now see whether this definition makes sense from a scientific point of view, in other words, whether such an absolute free will can exist. I will give two arguments, first a slightly elaborate one, then a much simpler one. (The broad line of these arguments is well known; I can only hope to provide some extra clarity.) Science is the perspective we should privilege according to synthetic philosophy, if it can be used at all; its verdict, if any, should be closest to truth.

² As an example: the notion of cause is not really defined within physics proper; one does not really need it in the theory; yet one can construct, within the philosophy of physics, a definition that is compatible with physics: see our chapter on cause.

According to modern neuroscience, actions and decisions originate in the brain, and these decisions are ‘materialised’ by neuron circuits, which are biological and hence ultimately physical systems. Physical systems are described by properties (variables, see e.g. the chapter on causation). Then, let us assume that the above definition makes sense, and consider a subject *S* who can (sometimes) choose between decisions *d*₁, *d*₂, *d*₃,... leading, respectively, to actions *a*₁, *a*₂, *a*₃.... Since the common-sense definition speaks of undetermined decisions, let us, to verify the soundness of the definition, inquire what could determine, a priori, these possible decisions or choices. Such determinants can generally be classified in two groups: on the one hand there might be *external influences* (present or past), denoted {*e*₁, *e*₂, *e*₃, ...}, such as external stimuli, encounters with other people, past education, etc. All these can influence decisions. On the other hand, the definition is a priori compatible with the existence of *internal, self-generated influences* (past or present)—here one might think of spontaneous thoughts or perhaps internal stimuli, let us denote them as {*s*₁, *s*₂, *s*₃, ...}. Such internal, self-generated influences are supposed to be undetermined by past or present events that took place outside the subject *S*.

Since the ultimate theory is supposed to be a physical theory (whether accessible to humans or not), I may consider the physical properties/variables that are the physical pendants of the {*d*_{*i*}}, {*e*_{*i*}} and {*s*_{*i*}}. For instance, expressed as physical variables, each of the *d*_{*i*} may represent the electrical activity of the neuronal circuit(s) that is (are) associated with decision *i* (whether in the following the variables are singletons or *n*-tuples plays no role: the formalism below is exactly the same in both cases³). Similarly, in a physical description {*s*₁, *s*₂, *s*₃, ...} are variables that describe self-generated internal brain processes/events (materialised again in neuronal circuits), and {*e*₁, *e*₂, *e*₃, ...} represent past or present external influences.

Within physics, to find out what determines the possible decisions {*d*₁, *d*₂, *d*₃, ...}, and therefore the possible actions {*a*₁, *a*₂, *a*₃,...}, I may consider the following probability:

$$P(d_1, d_2, \dots | s_1, s_2, \dots, e_1, e_2, \dots),$$

or, if one prefers, each of the $P(d_i | s_1, s_2, \dots, e_1, e_2, \dots) \forall i$, which is the probability that a certain decision/action *d*_{*i*}/*a*_{*i*} is chosen by *S* given the determining factors {*s*₁, *s*₂, ..., *e*₁, *e*₂, ...}. (Note that one does not need to make a distinction between the *d*_{*i*}-variables and their corresponding *a*_{*i*}-variables: for a judicious choice of variables one can assume that $P(a_1, a_2, \dots | s_1, s_2, \dots, e_1, e_2, \dots) = P(d_1, d_2, \dots | s_1, s_2, \dots, e_1, e_2, \dots)$.) Note also that a probabilistic description is general: it encompasses both probabilistic and deterministic variables; the deterministic case is a special variant of the probabilistic case (cf. the chapter on probability). A deterministic description would be appropriate if the considered decision-making of *S* is deterministic—either because the physical universe would ultimately be deterministic, or because *S* is very determined to choose only one of the *d*_{*i*}/*a*_{*i*}.

³ This is so because in probability theory the expression $P(X|Y)$ can be defined when *X* and *Y* are single variables or sets/*n*-tuples.

Now, the absolute definition assumes that a free-willed subject S can, at least sometimes, choose between possible decisions/actions $\{d1, d2, d3, \dots\}$, and that these are *not only* determined by past external influences $\{e1, e2, e3, \dots\}$, but also by self-generated $\{s1, s2, s3, \dots\}$ —at least one of such. In probabilistic language, this exactly means that the probability of choosing $\{d1, d2, d3, \dots\}$ *depends* on the $\{s1, s2, s3, \dots\}$, i.e.

$$P(d1, d2, \dots | s1, s2, \dots, e1, e2, \dots) \neq P(d1, d2, \dots | e1, e2, \dots). \quad (6.1)$$

Let us simplify notation: our absolute definition, assuming it corresponds to something real, implies that there are decisions/actions d such that they are determined by one or more self-generated influences s , so that e does not screen off s :

$$P(d|s, e) \neq P(d|e). \quad (6.2)$$

Inequality (6.1) and the equivalent (6.2) immediately follow from the definition of probabilistic dependence (see preceding chapters). But the conclusion is now immediate: according to physics, there are no self-generated events/properties, there are no events that would be fully determined within S . No physical events pop out of nothing: one of the most widely accepted, and massively corroborated, principles of physics, mentioned already by the epicurean atomist Lucretius in his *De Rerum Natura* (1st C. BC). In sum, there are no influences/events/variables ' s ' and decisions ' d ' that make (6.1) or (6.2) true; hence our initial assumption that absolute free will exists cannot be true.

[One might contest: surely there are internal spontaneous thoughts that have only internal determinants, say prior percepts, feelings, thoughts, But, granted this were a legitimate description, also such internal prior events cannot be undetermined; they in turn cannot pop out of nothing. (This is particularly clear if one recalls that mental events are, physically speaking, electrochemical brain processes, changes in currents: these must come from somewhere.) That means that there must be (assuming for a moment that internal spontaneous influences s exist) an (internal) variable s^1 such that $P(s|s^1) \neq P(s)$. And this implies, mathematically, that also $P(d|s^1, e) \neq P(d|e)$, which is an instance of (6.2), and which follows from the assumption that absolute free will exists and that d is an absolutely free decision. According to a well-known recursive argument, this reasoning can be applied to earlier stages of determination involving variables $s^2, s^3, \dots, s^N$ each occurring at earlier times and determining their predecessors, so that one ultimately arrives at $P(d|s^N, e) \neq P(d|e)$, where s^N are events that occurred before S 's birth. Ergo, our starting assumption leads to the contradicting conclusion that any decision must be determined by only external events, occurring certainly outside of S 's mind. These (zillions of) external events are, physically and mathematically speaking, the ultimate determinants of any choice—even if we can of course not retrace them.]

Some might object: but there could be quantum events at the origin of a decision—these are random, uncaused, according to the standard interpretation of quantum mechanics. Even if this were the case, such quantum events and the systems in

which they occur—say a collection of electrons—also come from somewhere. They depend on, i.e. they are correlated with, many variables: their environment (the wave function that describes them is determined by the environment, recall e.g. Bohr’s position above). For instance, they depend on macroscopic variables describing the neurons and other matter with which they interact, and hence with the past state of these systems. Again one arrives, by a similar reasoning as above, at the conclusion that even quantum events in the brain are ultimately depending on external events. (Moreover, quantum events are random, and real free choices are, in the usual sense, voluntary and conscious, and therefore certainly not random. This is a further argument showing that quantum randomness cannot be at the origin of free will, against what is often thought.)

In conclusion, from a physical point of view, there are no self-generated choices, choices that have no origin outside of a subject’s mind. Decisions and choices are always necessarily determined only by events that ultimately lie outside S’s mind. That is what neurobiology and especially physics tell us. *The conclusion is clear: the absolute view of free will—which is also the common-sense view—cannot be sound.*

This is an unfathomable idea—a dangerous one according to many! And yet I believe it is among the most important conclusions the mind can come to, even if reason seems to stand here in blatant opposition to common sense, to what we feel, and to the general regimentation of modern societies. It is a paradigm case of an idea that needs prolonged, careful professional efforts to be understood; and of which the consequences could have, if understood and applied by society, enormous impact. There are philosophers who have taken up the grand challenge to reflect on the consequences, and have shown the direction in which to go. The great Spinoza is the first who comes to my mind. In his *Ethics* he developed a systematic theory, based on the absolute view, which coherently answers countless questions from so many philosophical chapters—a framework that has therefore a greater chance of correctness than accounts of smaller scope. But several other philosophers, all free will sceptics, deserve to be mentioned (a partial list of contemporary philosophers is Strawson, 1986; Pereboom, 1997; Ch. 18, 2014; Smilansky, 2000; Waller, 2011; Caruso, 2012; Vilhauer, 2012). I will briefly comment on the implications of the absolute perspective in the next chapter.

Of course, the above argument could have been much shorter (we could simply have noted from the start that self-generation is a physical heresy), but I found it useful to spell it out in some detail. Note that the only non-trivial assumptions that I made are (1) scientific realism, in particular physicalism/physical reductionism, the idea that ultimate reality is physical and can be described as usual by variables, and (2) the ‘*nihil ex nihilo*’ principle. (Remark: of course, no-one can seriously believe that humans can construct a physical theory ‘of literally everything’ that would allow us to predict, on the basis of the properties of the ultimate constituents of the universe, all the thoughts and decisions that go on in a given human’s brain. And yet, all these thoughts, however ethereal, are embedded in physical systems and these material *pendants* of our thoughts are therefore ultimately, *in principle* (not in practice!), describable by physical theory, by the tools of physics. The latter very widespread hypothesis, which I have to assume to remain within the best available

scientific knowledge and therefore the MCT, is all I need, besides the ‘*nihil ex nihilo*’ principle.)

Actually, there is a very short argument, which should *in fine* be equivalent to the above one. It goes as follows. According to the scientific realism I must adopt, any action starts from, is controlled by, neurobiological processes in the brain, hence ultimately by physical processes (even if we cannot reduce all of neurobiology to physics in practice). Humans are ultimately physical systems, and physical systems are governed by the laws of physics, originating at the Big Bang. However counterintuitive and unpoetic it may sound to many, *humans are therefore bound by the necessary laws of physics*. It plays no role whether these laws are deterministic or probabilistic: via these laws, whether deterministic or probabilistic, we are ‘correlated’⁴ with the very far past, we entirely depend on it, in every respect. Still in other words, our every feeling, thought and choice is *generated* by physical events that lie in the far past. As regards choice-making, on the purely physical level, there is strictly no difference between humans, falling stones, and quantum objects as electrons: we follow a path generated by the past; just as stones and electrons do not self-generate their histories, we do not—because we are as physical as they are. Ergo, from the physics point of view, the concept of self-generated and undetermined, absolutely free choices and actions, the concept of free will that seems such a cornerstone of humanity, must be an illusion. With some drama, one might say: it is the universe that chooses for us.

We have obtained an argument against absolute free will that reminds us, of course, of arguments proposed by many other philosophers. In the contemporary literature, one often encounters a version that is called the ‘Consequence Argument’, due to van Inwagen (1983). However, van Inwagen’s argument rejects free will on the basis of *determinism*—as do almost all free will sceptics. My above argument substantially generalises, so strengthens, the Consequence Argument, since it only assumes physicalism/physical reductionism, which encompasses deterministic *and* indeterministic systems. My stronger consequence argument is thus the following, simply replacing determinism by physicalism in van Inwagen’s rationale: “If *physicalism* [the hypothesis that the building blocks of everything in the universe are physical] is true, then our acts are the consequences of the laws of nature and events in the remote past. But it is not up to us what went on before we were born, and neither is it up to us what the laws of nature are. Therefore, the consequences of these things (including our present acts) are not up to us.” My argument locates the origin of the tension with free will in the fact that our acts *originate in, are generated by*, the physical world outside and anterior to us; it denies that we are the ultimate source of our actions.

Interestingly, this dramatic conclusion based on physics is very similar to the conclusion that neuroscientists have derived from various detailed neuroscientific findings (cf. e.g. Walter, 2001; Soon et al., 2008 and see next chapter). This should not

⁴ Note that statistical correlations may well exist in a given system, even if they are entirely undetectable. Measuring/detecting correlations is subject to measurement uncertainties, limitations of experimentation. For an illustration in billiards, see (Nikolaev and Vervoort 2023).

surprise us if there is coherency between natural sciences, all of which philosophers and scientists believe to be ultimately reducible to physics. (For the same reason, the physics-based argument is stronger.) But it is pleasing to contemplate that the web of knowledge is coherent.

6.3 Does a Relative Form of Free Will Exist? Where Does the Illusion of Free Will Come From?

This is all I will say here about the physics-based perspective, within which absolute free will does not exist. But it seems—and now we move in less certain territory—that there exists a type of free will from the perspective of socio-humanitarian theories, if one makes abstraction of physics—resonating with what common sense would tell. I will in this Section go over some grounds, notably from neuroscience, that seem to explain why such a relative free will could exist (a notion to be defined precisely in the next chapter), and where (the illusion of) free will comes from. I will do this in a very sketchy way; these reasons are surely known by philosophers and scientists, especially those who reject the reality of absolute free will; and they can be discussed in much greater detail in the light of neuroscience (see e.g. List, 2019; Walter, 2001 and references therein). I mention them here to justify the work of the next chapter, which is to define relative or ‘socio-humanitarian’ free will.

From the common-sense and felt perspective, the conclusion that absolute free will is impossible is of course entirely counterintuitive, nay unbelievable—to the point that, when one defends this idea in the presence of staunch believers in free will, one may find oneself sound quixotic. It seems that in my mind/head/brain, I can freely envisage so many options, I could take so many different decisions, I can guide my life and have influence on the environment and other people. I can write my life story, as the saying goes. Sure, but all that is not in contradiction with being subject to the laws of physics; which must still be at the basis of all such capacities. All such capacities and the way we perceive them must be emergent from the uniquely complex system that a brain is—so complex that we only have a vague idea of what is going on in it. Analogously, the famous nematode *Caenorhabditis elegans*, having about 300 neurons, and especially somewhat more developed worms, seem to guide their movements and seem to have choices in going left or right; *and from a certain point of view this is the case*; and yet it is a scientific triviality to state that they are ‘only’ physical systems subject to the laws of physics. *The impression I have that ‘I’ can choose, depends on what one considers (and feels) the ‘I’ to be.* More precisely, the proposition “The subject S can sometimes choose freely” depends on the theories with which one looks at these concepts, including that of subject. In the physical sense, the ‘I’ is a collection of atoms of which the temporal and spatial boundaries are defined by human convention, and for which the concept of free will makes little sense, as argued above. But from the perspective of what I know, grasp, feel about my ‘I’, with my very limited knowledge about what really goes on, ‘I’ can

spontaneously do a whole lot. At first sight, it is beyond doubt that 'I' can choose *and* that it is 'I' who chooses—not the universe! Because I necessarily have a very limited view of what the 'I' really is; and yet, this limited view is the only one that counts—almost always, except when philosophising.⁵ So yes, the 'I' is (part of) a well-identified physical body—having a national ID—, with a very personal brain, with a sensorimotor system that can make this body act, and with neural networks in the prefrontal cortex that embed the personal beliefs, the personal goals, the personal memories etc. that are activated when I must act—actions that I *therefore* perceive as my very personal actions. (But again, this is not in contradiction, rather to the contrary, with the fact that these brain mechanisms must obey the laws of the universe. If they did not, they would be erratic.)

Similarly, some actions demand deliberation, when different possibilities *seem* to offer themselves to me. In my head, I can represent a given real-world situation and different choices I can make in this situation, by using models, ensembles of hypotheses, theories that apply to these choices and the situation: I can play around with hypothetical consequences of these choices (hypothetical consequences that are, on the abstract logical-epistemological level, inferences from the models I use to represent the situation and the choices), and evaluate these consequences, leading to a final choice and action. (It is this key idea that will be at the basis of my definition of relative free will, cf. next chapter.) Surely, to all of us this deliberative thinking has a strikingly ethereal and free character, I seem to be able to steer it in all kinds of directions: surely because it seems to only involve my 'mind', not my body, nothing physical—I can do it without moving, just sitting in the armchair. But this disembodied-ness can only be an impression, an illusion; which might well be at the origin of the ineradicable belief in free will. As said, the intricate links between, on the one hand, 'mental' processes we call 'deliberating', 'choosing' etc. and, on the other, the underlying neurobiological mechanisms only begin to be explored in the fascinating and relatively new science of the brain—neuroscience.

What is also useful to remember is that the emergence of life from inert matter is still a mystery; no surprise then that the emergence of consciousness and free will from inert matter are even a greater mystery. Over millions of years of evolution, neuron cells have learned, through natural selection, to remember and to store what we call 'knowledge', 'beliefs', 'reasons', 'preferences' etc. in sophisticated neural circuits; and brains have learned to activate these circuits when their owners are confronted with what they perceive as possible future pathways. The faculty of thinking can be seen as an emergent property of a biological supersystem, acquired via natural selection, and ultimately based on the laws of physics. All this is not new.

All this being said, there is of course a bona-fide sense in which subject S makes choices based on *her* neural circuits embedding *her* beliefs, preferences etc. Concomitantly, there is a sense in which the subject acts upon and hence influences the external world—even if part of the larger whole. (Even in a strictly physical sense it is of course legitimate to consider that it is subject S that interacts in this and

⁵ And when using philosophical insights to build ethical hypotheses, and hypotheses for a better, more just society (cf. the last section of the next chapter).

that way with the environment and produces changes in it.) It seems therefore that concepts as agency make sense and correspond to something real. Therefore, I believe that it makes sense to define a capacity that one could still call free will (so that we all know what we are talking about) and that corresponds to this all-pervasive impression, this unshakeable belief we have to be free-willed. This impression emerges from something real, after all, since people do seem to denote roughly the same thing with the words ‘free will’. As I announced at the beginning of the chapter, such a definition, then, is the definition within a general socio-humanitarian perspective and informed common sense, which make abstraction of physics and its philosophy. I believe it is useful to define such a relative free will, because it is such a vital ingredient of legal practice, of daily habits of blame, praise, punishment, and perhaps even of (the philosophy of) happiness. Indeed, this concept does a lot of work, it solves many problems: it will appear to be intimately linked to other key concepts, notably consciousness, control, agency, responsibility.

However, I will immediately add that a full theory of free will should include both perspectives: the absolute and the relative view are complementary and need, ideally, both be used for handling the deepest problems. These deepest problems are ontological and have an ethical dimension, so go far beyond definitions, and cannot be treated in any serious detail in these chapters. Some examples are: How to take the absence of absolute free will into account in our ethical, legal etc. philosophies and practices? (I believe that society needs a great change in mindset here.) To what extent are people praiseworthy and blameworthy? Do people need to be punished (since in an absolute sense they had to do what they did)? What is the sense of personal responsibility? etc. I will briefly comment on these problems at the end of the next chapter and in the Epilogue.

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Chapter 7

On Relative Free Will. The CMT Model



Abstract It makes sense to study what ‘free will’ means in daily conversations, law, and practical philosophy—even if it is a shaky concept. Thus, to complete my bi-perspectival model of free will, I propose a compatibilist account of relative free will that attempts: (1) to provide a synthesis of a variety of well-known models, capable of addressing problems of the latter; (2) to account for the fact that free will comes in degrees; (3) to interface with neurobiology. I argue that (relative) free will comes in degrees, as suggested by neurobiology. A concept that can precisely ‘measure’ the variability of free will is—once more—that of ‘theory*’, or rather, the capacity to make assumptions/hypotheses and apply them. This model highlights that free-willed actions are consciously monitored by the agent, through beliefs, assumptions, generally theories*—hence the CMT model (for Conscious-through-Monitoring-through-Theories*). I end by adjoining the absolute view to the relative one, and sketch some implications of this bi-perspectival account.

7.1 Introduction

In the specialised philosophical literature on the free will debate, two widespread positions are ‘hard determinism’ and ‘libertarianism’ (cf. Fischer et al., 2007; Kane, 2005, 2012; Mele, 2009; O’Connor & Franklin, 2022; Pereboom, 1997, 2014; Walter, 2001; Watson, 2003; List, 2019). Hard determinists reject free will on the hypothesis that all events, including our choices, are determined (fixed) by the laws of nature and the past; libertarians accept free will as a given and reject determinism. Clearly, hard determinism largely corresponds to the physical reductionism I adopted in the previous chapter. However, as argued there, strictly speaking I believe the dichotomy is not between determinism and libertarianism, but between physics / physicalism and libertarianism: my argument against absolute free will is in essence only based on invoking the laws of physics, whether deterministic *or* probabilistic. Fundamental randomness in nature (as posited by the usual interpretation of quantum mechanics) is of no more help for absolute free will than genuine determinism. But for the discussion in the following sections we can equate determinism and physicalism;

we do not really need to make the difference, since we study free will from now on mainly from a socio-humanitarian angle.

Many professional philosophers are neither hard determinists nor libertarians but ‘compatibilists’, maybe even a majority as claimed in (Pereboom, 1997, p. 242): according to this position, determinism and free will are not in contradiction. My own model of free will could be termed compatibilist, in the sense that I believe that even in a physical universe in which there is no absolute free will, there is something that can be called relative or socio-humanitarian free will. But it is better to call this view bi-perspectival: according to it, the absolute and relative perspectives are complementary and both are needed for a fuller theory of free will. Absolute free will as the capacity to choose undetermined by past events does not exist; but there exists a capacity that is a placeholder, and that intervenes in legal, social, and humanitarian theories, and in common-sense conversations.

It deserves to be mentioned from the start that I consider *human* free will, not animal free will if such a thing would exist. More precisely, I focus on the free will that is, in philosophy, very often considered to be *a necessary condition for moral responsibility*. (It is interesting in itself that philosophers have usually looked at the question of free will from the perspective of moral responsibility. Our model developed below suggests that the essential and more fundamental correlate is consciousness, rather than moral responsibility. Note that this common bias among philosophers is well in agreement with the claim that free will is a socio-humanitarian notion.)

Our goal is to construct a ‘minimal’ model in the sense that we aim to identify the minimal set of necessary and sufficient conditions to term an act ‘free-willed’ or ‘free’.¹ In this endeavour we use classic tools of analytic philosophy. We might not be very typical in our attempt to synthesize theories of free will. With synthesis, we mean, as in the rest of this book, the attempt (or its result) of showing how different established theories can be derived from one and the same unified model; more realistically, how the essential ingredients of different accounts resort under one umbrella model. In the free will debate, millennia old, analytic philosophy has invested such enormous efforts and has made, especially the last fifty to sixty years, such progress in conceptual clarity, that efforts at synthesis seem now well justified. This is so on the condition, of course, that the resulting model can solve problems where isolated accounts remain silent or unconvincing. We insist that our model should be judged on its results, namely the fact that it can tackle problems that have been identified in competing modern theories, as shown in Sect. 6. Each of the ingredients or components of our model will appear necessary to address these problems.

Our most conspicuous influences come from A. J. Ayer (1954/1997), and, especially, M. Mahner and M. Bunge (1997), as explained in Sects. 3 and 4. Mahner and Bunge develop an advanced biophilosophical account of consciousness that resonates strongly with neuroscience, and is therefore well in the scope of synthetic philosophy.

¹ From now on I reproduce parts of Vervoort and Blusiewicz (2020a); I will therefore mostly use the plural ‘we’. I have made many add-ons and modifications, especially in Sects. 5–8, for instance to make the link with other chapters of this book.

All three authors construct theories of which essential ideas can be traced back to such fore-runners of the philosophy of free will as Aristotle and Aquinas. In Sect. 5 we will show in detail how these models can be combined into our favourite model, which we term the CMT-model of free will (for ‘Conscious-through-Monitoring-through-Theories*’). We intentionally construct the account in a detailed, step-by-step way, to further illustrate the method of synthetic philosophy.

Our naturalized approach was also inspired by, and can be compared to, the work of philosopher and neuroscientist Henrik Walter (2001), who proposes a component theory of maximal libertarian free will: a theory with essentially three requirements that a libertarian would demand of a ‘free’ act (in short, alternativism, intelligibility and agency). Walter does not only support his claims by advanced neuroscientific findings, but has also done an extensive literature review to show that his component theory covers a wide spectrum of the most relevant modern theories and issues on free will; hence the relevance of benchmarking our model to Walter’s.²

We will argue that one advantage of the CMT-model is that it captures a feature of free will we believe is essential, namely that it comes in degrees. The variability of free will has been hinted to by neurobiologists (e.g. Brembs, 2011), and analysed by philosophers (O’Connor, 2009); we will present our own arguments in Sect. 2. Our model suggests a qualitative ‘measure’ for this variability of free will (Sect. 5). The most important results are gathered in Sect. 6, where we show that this model can synthetically interpret key properties of recent accounts (and accordingly solve their problems), notably those of Frankfurt (1969, 1988), Wolf (1990), Fischer and Ravizza (1998), Walter (2001), List (2019) and other compatibilist models reviewed in O’Connor and Franklin (2022). In Sect. 7 we briefly sketch how our model could be instrumental for some scientific disciplines, in particular neurobiology and computer science (AI). Finally, in Sect. 8, the conclusion of Chapters 6 and 7, I comment on how the relative perspective must be completed with the absolute one.

7.2 Background Assumptions. A Preliminary Observation: Free Will Comes in Degrees

Our background assumptions are those adopted in the rest of this book, but let us mention them explicitly to make this chapter self-contained. First of all, we assume that our mental activity, our thoughts, choices, feelings etc., have a neurological, and ultimately chemical-physical basis in the brain: mental states are embedded in

² Walter himself is not a libertarian; in his book he tests the three ingredients of libertarian free will against natural science, in particular neuroscience. He argues that neuroscience does not really support alternativism (the hypothesis that ‘at the atomic scale’, i.e. taking *all* physical facts into account, a human brain has, at any given time, real alternatives to choose between—cf. Walter, 2001). So Walter is not in favor of strong libertarianism but shows that versions of intelligibility and agency do find support in neuroscience; these concepts will be seen to have their pendants in our minimal model. In this chapter we assume from the start that alternativism is eliminated, for reasons explained in the previous chapter.

neural (super)networks, mental events/acts are brain processes (many relevant neuroscientific insights can be found e.g. in Walter, 2001; many essential implications for philosophy are studied e.g. in Mahner & Bunge, 1997 and List, 2019). Based on the famous Libet experiment and several other neuroscientific experiments (showing for instance that ‘free’ choices can be predicted up to ten seconds before the subject’s conscious action, cf. Soon et al., 2008 and Sect. 7), many, but certainly not all, neuroscientists favour a deterministic interpretation in the free will debate. We arrived at a similar physicalist conclusion in the previous chapter, from a different but concurring angle, forcing us to look for a compatibilist model.

Technically speaking, our goal is to construct a definition of a ‘free-willed’ act by a human, in the form of necessary and sufficient conditions. Hence we aim at identifying the conditions $C_1, C_2, C_3, \dots$ such that ‘Person P does act A of his own free will’ if and only if $C_1, C_2, C_3, \dots$ are satisfied. The C_i are all necessary conditions and they are jointly sufficient to imply free will. As announced, and remembering Occam’s razor, we are looking for the *minimal*, most economic set $\{C_1, C_2, C_3, \dots\}$.

But we would also like our model to say something about a feature of free will that seems important, namely its variability. Consider your favourite dog, Snoopy, playing in the garden, and suddenly seeing the neighbour’s cat appear. You shout: Snoopy, stay put! For a fraction of a second Snoopy remains motionless, throws a guilty glance at you, and then hurries after the cat. Has Snoopy free will? Could Snoopy have done otherwise, has he alternatives? Perhaps in a certain sense he has (cf. the previous chapter), but most philosophers agree that free will, and certainly the free will needed for moral responsibility, is essentially a human thing (if it exists at all). But granted, at the same time it seems quite intuitive to consider that Snoopy possesses an embryonic sort of free will. Has a new-born child free will? A foetus? No, most would say. When, then, does a child start having free will?

If adults can have free will, and if it is linked to, or emerges from, brain activity—a brain activity that grows continuously more complex when a foetus and new-born child evolve into an adult—one naturally arrives at the conclusion that free will comes in continuous degrees. This seems an innocuous conclusion within a neuroscientific perspective, which has been noted by philosophers³ (O’Connor, 2009). Another way to understand this variability of free will is to realize that animals have continuously evolved from bacteria over invertebrates, fish, amphibians, reptiles, small mammals to primitive human species to modern *Homo sapiens*, and that somewhere in this evolution, paralleling a steady increase in brain complexity, free will must have appeared, not suddenly, but evolving in the same continuous way as the brain did. Since this hypothesis will be instrumental in the following, we term it the ‘degree-thesis’. Simply put: free will comes in degrees; in other words, the mental capacity of free will is something that admits of (continuous) variations or gradations, something that a human (and generally speaking an animal) can have to greater or lesser extent—moreover an extent that can vary with time within an agent (O’Connor, 2009).

³ O’Connor (2009) arrives at the conclusion of variability of free will by an independent route, notably by observing that agents can be, to greater or lesser extent, consciously aware of the factors guiding their choices. This observation can well be fitted to our final model developed in Sect. 5.

7.3 A Basic Definition of Free Will (Ayer Model)

Let us now start with our main task and look for the simplest non-trivial compatibilist model of free will; this simplest model can however not solve certain problems and will have to be completed in a following stage. A considerable part of the academic debate on free will has turned around the ‘principle of alternative possibilities’ (PAP) (Frankfurt, 1969; Kane, 2005). According to the PAP, a person P does act A of his/her own free will only if P could have acted otherwise than doing A (the latter condition is a necessary condition for free will). Most libertarians embrace the PAP as evidence that free will excludes determinism. But it is well known that there are considerable problems with the characterisation of free will provided by the PAP. Notably, Frankfurt has provided well-known counterexamples: cases in which P could not have acted otherwise, while P still seems to act of free will⁴; hence the PAP would be a wrong characterisation of relative free will (Frankfurt, 1969, 1988). These arguments have been criticised in turn (e.g. Widerker & McKenna, 2003); an overview of the most recent developments in this lively debate can be found in O’Connor and Franklin (2022). We believe it is fair to say that this matter is left undecided in the literature.

A classic compatibilist attempt to defuse the worries related to the PAP was given in a well-known article by A. J. Ayer (1954/1997), following a long tradition tracing back to Aristotle (Nicomachean Ethics, Book III), Hobbes, and Locke. Ayer’s article famously argues that the capacity to choose as invoked in the PAP may well be guaranteed by the simple requirement that subject P acts *without constraint*. Thus, in its simplest expression, an act of free will is, for Ayer, an act that is not under constraint. In some more detail, we can extract following analytic definition of free will from Ayer’s article:

DEF-A: ‘P acts of his own free will’ *iff*

- i. P is not compelled by other persons;
- ii. there is no habitual ascendancy over P by other persons; and
- iii. P acts voluntarily.

Ayer leaves the concept of ‘voluntary act’, again a notion that goes back to Aristotle, imprecise; therefore we will have to come back to it in a moment. But from the text it follows that at least one ingredient of ‘voluntary’ is: not being under constraint of an abnormal mental condition such as a mania, a neurosis, etc. (Ayer, 1954/1997). Can an act that is not constrained and thus free according to (i)–(iii) also be free in the sense of the PAP, in other words, could the unconstrained P also ‘have done otherwise’? Ayer answers in the affirmative; he construes ‘P could have acted otherwise’ as follows: A) P was not constrained in the sense above (no-one compelled P and the act was done in a ‘normal’ psychological condition) and B) P *would* have acted otherwise *if P had so chosen*.

⁴ In any case, with moral responsibility.

Sure, this construal is far from universally accepted (cf. the review in O'Connor & Franklin, , §2.2); but what is essential for us is that it appears that DEF-A can be supplemented in order to address worries of much more sophisticated recent theories, as argued in Sect. 6. Therefore, we take DEF-A as a minimal analytic starting point. (Even if this is not essential at this point, one could then also adopt Ayer's construal of 'could...' as 'would... if...' as plausible, also because it is compatible with determinism (Ayer, 1954/1997) and physicalism. Note that the same holds for clause A): P can be unconstrained *in the quite common sense* (i)–(iii) even if everything is determined/physical. In Ayer's words, free will should be contrasted with constraint, not determinism.) For the time being we retain: there are respectable arguments that an act of (relative) free will can be characterized in terms of absence of constraint in the sense (i)–(iii), with the proviso that we would like to know more about what 'voluntary' means in (iii).

Still, it is clear for other reasons that DEF-A cannot be the whole story. We can repeat an argument we already invoked above: free will should be something more than just 'unconstrained will that has the ability of choosing between options', because it seems that even animals could have this type of will—recall Snoopy the playful dog. But as most philosophers we ascribe relative free will only to humans—always within a relative, socio-humanitarian perspective, in which free will and morality are related.

What, then, is the feature that distinguishes humans most clearly from animals? There is a long philosophical tradition related to this question, but it is presumably uncontroversial to identify consciousness, the capacity to think, rationality, as the key element(s) of distinction. Then, something of this should enter the definition of free will. And indeed, *consciousness and/or rationality seem implicit in Ayer's concept of voluntariness*, as an ingredient of free will, as also follows from early analyses of the concept of voluntariness by Aristotle and Aquinas (cf. Pereboom, 1997, Ch. 1 and Ch. 5). We therefore should find a model for this essential but admittedly somewhat vague concept.

7.4 Voluntariness and Consciousness: Mahner–Bunge Model

To analyse voluntariness, the most developed naturalised model we are aware of is proposed in Mahner and Bunge (1997)—'naturalised' meaning: in agreement with science, in particular (neuro)biology. In their *Foundations of Biophilosophy* (1997) the authors present a penetrating and wide-ranging approach aiming at providing an axiomatic theory of key concepts of biology including psychobiology. Their approach and methods are eminently science-compatible, and their background assumptions coincide with ours (cf. Section 2 and below). In Chapter 6, on psychobiology, Mahner and Bunge (MB henceforth) propose an analysis of concepts as mind, mind–body interaction, consciousness, self, voluntariness, free will etc. In principle we only need

their definitions of voluntariness and consciousness in order to complete Ayer's DEF-A, but we will give here a succinct overview of other notions too as conceptualised by MB: this can complete our own sketchy versions in the previous chapter. Sure, MB's axiomatic and partly formalized approach is only one possible theoretical framework for the above concepts; but their hypotheses and theory are systemic and heuristically powerful. Again, the ultimate justification of the final model we will construct lies in its problem-solving capacity—and in its relevance for science (Sects. 6 and 7). We will only retain the essence of MB's theory, and occasionally make small changes as indicated.

As said, MB closely link the mind with the neuronal states and networks in the human brain; mind and soul are not conceived of as immaterial, spiritual, perhaps immortal entities. In this theory the mind of an animal is construed as the union, the *set* of all mental processes of its brain.⁵ Mahner and Bunge propose following definition (MB, 1997, p. 205):

DEF-MB-1. Let P denote a plastic neuronal supersystem of animal b of species K . Then the *mind* of b during the period τ is the union of all the mental processes (specific functions π_s) that components of P , i.e., plastic neuronal systems n , engage in during τ . More precisely,

$$M(b, \tau) = U_n \pi_s(n, \tau).$$

The other key concepts, plastic neuronal (super)system, mental process, mental function, are also defined in Mahner and Bunge (MB 1997, Ch. 6). However simple, this theory allows one to address some of the old, paradigmatic problems of the philosophy of mind. One such question is: where is the mind located? Strictly speaking, the mind is nowhere, since it is a set, hence a conceptual object; only brains, whether minding or not, are somewhere (MB p. 207). Further, according to this model there can be no real mind–body dualism, thus avoiding an old stumbling block of philosophy, neither mind–body interaction—as opposed to brain–body interaction. MB explain in following passage (MB 1997 p. 206):

There can be no mind-matter interaction because – unlike individual mental processes and brains – mind and matter are sets, hence conceptual objects. However, it does make sense to speak of 'mental-bodily interactions' provided this expression is taken to abbreviate 'interactions among plastic neuronal systems, on the one hand, and either committed neuronal systems or bodily systems that are not part of the Central Neuronal System on the other'. Thus, there are interactions between sensory and motor areas, between ideational neuronal systems and external receptors, between the cortical and subcortical regions of the brain, between the brain and the endocrine and immune system, and so on. Because mental events are neural events, and because the causal relation is defined for pairs of events in concrete systems (recall Sect. 1.9), we have:

Corollary 6.5.: Mental events can cause nonmental events in the same body, and v.v.

⁵ Of course, in this context the notion of set, a concept of formal logic, is not the most user-friendly one, at any rate other less formal conceptualisations are also possible. We do not really need the notion of mind, but give it as illustration.

Consequently, disturbances of nonmental biofunctions may influence mental states and, conversely, mental events such as acts of will may influence nonmental bodily states. This is what neurochemistry, neurology, psychiatry, psychosomatic medicine, psychoneuropharmacology, education, and propaganda are all about.

In a similar vein and building on the above assumptions one can then define consciousness, termed the highest of all brain functions by Bunge and Mahner (MB 1997, p. 209). First they define (we only slightly modify their phrasing):

DEF-MB-2. A *conscious mental process* / choice / act (an act conceived as a mental process, or linked to a mental process) is a mental process / choice / act that is *monitored* (recorded, analysed, controlled, or kept track of) by some other mental activity in the same brain.

Simply put: *for a mental process or act to be conscious, it must be thought about by a higher part of the brain*—a thesis that seems corroborated by neuroscience.⁶ And further (MB 1997, p. 209):

DEF-MB-3. The *consciousness* of an animal *b* is the set of all the states (or, rather, processes) of the brain of *b* in which *b* is conscious of some perception or thought in *b* itself.

(This is not a circular definition thanks to DEF-MB-2.) MB explain:

According to this convention, an animal can only be conscious of some of its own higher mental processes: not just feeling, sensing, and doing, but also thinking of what it perceives or thinks. An animal conscious of mental process *x* (in itself) possibly undergoes (either in parallel or in quick succession) *two* different mental processes: *x* – the object mental process or content of its consciousness, and thinking about *x* – i.e., being conscious about *x*. (MB 1997 p. 208-209)

It seems natural to think of this pair of mental processes as embedded in two different neural circuits: one embedding process *x*, and one embedding thinking about *x*. Using DEF-MB-2, one can now construct a definition of the voluntary act as invoked for instance by Aristotle, Aquinas and Ayer (MB 1997, p. 210).

DEF-MB-4. An animal act is *voluntary* (or intentional) *iff* it is a conscious purposeful act.

‘Purposeful’ is left undefined, but is self-explaining; and, for that matter, it seems less essential to us since it seems that a choice or conscious act can always be associated with a purpose. And finally (MB 1997, p. 211):

⁶ Neuroscientific findings show that the cognitive center of the brain, the prefrontal cortex, is connected to large parts of the brain and seems to function as the ‘integrator’ of information embedded in very many other mental processes. This is the neurobiological intuition that lies at the basis of MB’s intuition that consciousness of *x* is related to ‘thinking about *x* by a higher level part of the brain’—say the prefrontal cortex.

DEF-MB-5: An animal acts of its own *free will* iff

- i. its action is voluntary; and
- ii. it has free choice of its goal(s)—i.e., is under no programmed or external compulsion to attain the chosen goal.

In essence, we find in clause (ii) the absence of constraint that is the main condition for Ayer. And ‘voluntary’ in clause (i) and in DEF-A (iii) is now explained by DEF-MB-4: it means conscious and purposeful.

Thus, one finds that Mahner, Bunge and Ayer (MBA) ultimately derive a very similar definition of free will. Rather than just showing the final result, we have followed MB in a step-wise manner, to highlight that their definition is based on a solid psychobiological theory. Most importantly, we will show below that an upgraded MBA-definition can well be related to a majority of compatibilist theories on free will. Let us for the moment omit the notion of purpose in voluntary, since it seems less essential and can be seen to be contained in the notion of choice or choice-act (so we stipulate it is implicit in the definition below). Then one can unify Ayer’s and MB’s models as follows.

DEF-MBA. Act A by animal b is ‘free-willed’ or ‘free’ (is made of b’s own free will)

iff

- i. the act A is *unconstrained* (no programmed or external compulsion), and
- ii. the act A is *conscious* in that the act (linked to a mental process) is *monitored* (recorded, analysed, controlled, or kept track of) by some other mental activity in the brain of b.

In this way we have specified, with MBA, that no-constraint is essentially absence of constraint by external agents and by (externally or internally) programmed influences, where internal constraints can be manias, neuroses, etc. We can now define not only ‘free acts’ but also ‘free will’, namely as the *capacity* to perform free-willed actions in the sense of DEF-MBA.

It is worth mentioning that DEF-MBA luckily is in a long historical lineage. Notably, it comes close to Aristotle’s view on free will, but it is also meaningfully linked to Kant’s (see further). Aristotle conceived of free will as a capacity to make choices that are unconstrained and *not made of ignorance* (cf. Pereboom, 1997, Ch. 1). The latter idea can well be linked to clause (ii) of DEF-MBA, as we will detail in a moment. Thus, we see appearing in this model of free will the component of consciousness that we intuitively suspected from the start to be an ingredient of free will—simply by recognizing or assuming that ‘real’ free will cannot be attributed to lower animals and is typically a human thing.

A second essential, science-based hypothesis we started from is that free will comes in degrees. We posited this degree-thesis essentially by recognizing that there must be a (continuous) evolution in free will from primitive humans to modern humans, and from a new-born child to a knowledgeable adult. The next question to ask is then: which concepts in DEF-MBA admit of degree? Constraint, but more conspicuously the notions of *analysis* and *control* in clause (ii).⁷

In the next section we look in more detail into this variability. We will introduce a concept that captures *how* one monitors a free act (on the logical-epistemological level), providing by the same token an interpretation of the variability of free will.

7.5 Elaboration of the MBA-Model: The CMT-Model

If free will comes in degrees, a natural question is: can one then define a measure of the variability of free will? A related question that brings this matter sharply to the point is: can we define a maximal form of free will—maximal in the sense of optimal, most adequate? The second question might seem complex and touchy; it is rather absent from the modern debate, but it was very present in the views on free will of Spinoza, Kant and many other historical philosophers. We will use the second question as a heuristic tool to try to answer the first one.

To link this problem to recent debates, let us look at a case mentioned in (Griffith, 2013 p. 33), a case that pervades the contemporary debate under different guises. Suppose a young girl, Trina, lives in a closed community that taught her since her early childhood that stealing from people outside the community is praiseworthy, the thing to do. Suppose that on her first day at school Trina happily puts her worldview to practice and steals things from her young schoolmates. Does Trina do these acts of her own free will? As argued by Griffith, in this case many people would accept that Trina was brainwashed (to some degree), and that she is not really blameworthy because she did not know any better. Thus, many believe that Trina has no real free will. This is also what the CMT-model says: while stealing, Trina can surely be conscious of it (she may perform some more or less conscious analysis of some of her deeds, there may be a little voice in her head saying: “stealing from these girls is cool”), but at the same time there is a programmed constraint acting in the sense that she was brainwashed by parents and community. So it is clause (i) that implies that her act of stealing is not free-willed. But again, it seems clear that the notion of *degree* of free will is helpful here: one could say that Trina’s behaviour is free to some degree, but not an optimal one. She is constrained by some (harsh or mild) form of brainwashing; and, especially, she monitors, analyses, controls her deeds by a presumably sub-optimal hypothesis or worldview, namely that stealing is praiseworthy. To make a link with what follows: she does not seem to rely on the

⁷ In general, the ‘being conscious’ of clause (ii) can vary; this comes close, or can at least interestingly be compared, to the account of O’Connor (2009).

optimal assumptions to monitor, analyse and control her deeds—assuming there is any conscious reflection involved at all.

Now to our question whether a maximal/optimal form of free will could be defined: DEF-MBA does suggest a possibility to qualify optimal/adequate free will. Answering this question amounts to construing the optimal forms of analysis and control that consciousness, our brain, can perform when monitoring an act. Using insights gained in previous chapters, we believe the most synthetic concept to identify such optimal analysis and control is that of ‘representation’ or more precisely ‘theory’—*the very concept that has been immensely useful in the preceding chapters!* We again use ‘theory’ here in a very broad sense, including scientific and academic theories (ethical, philosophical, sociological, political, physical, ...) but more generally also belief systems, including every-day beliefs and assumptions/hypotheses; (coherent) bodies of information; worldviews; etc. In Chapter 1 the notation ‘theory*’ was introduced to refer to theories in this broad sense.

The link with analysis and control is then not far to seek: it may be said that one always analyses and controls a conscious act, choice or decision *with reference to, or within*, a theory*. This seems obvious when making decisions that need the input of expert intellectual or theoretical knowledge. But even when making a banal choice on an idle Thursday evening, say between whether to go to the cinema or rather to visit a friend, one ‘analyses’ or ‘contemplates’ both alternatives within certain beliefs, using certain assumptions—for instance assumptions about the satisfaction each activity will provide. Sure, the analysis may be barely conscious and ultra-rapid in this example; but we already agreed that being conscious comes in degrees (more examples are given below.) In any case, in order to identify optimal analysis and optimal free will we better look not at banal cases but at ethically or intellectually demanding ones. And then it seems quite clear what optimal, most adequate analysis means: namely analysis within the ‘optimal theory*’—the best theory* that we have or have not at hand; the best-informed or most adequate assumptions on which to base our free choice or decision. Sure, in many cases it is not clear what the best theory* is, but then, in many cases it is. (Recall from the chapters on Gettier’s problem and causation that the notion of ‘best possible’ or ‘best available’ theory is very instrumental and natural even if it usually is an idealisation.⁸) In sum, according to the above analysis maximal free will is free will using the optimal beliefs, assumptions, theories*; free will that is based on using the optimal theories and assumptions in our decision making. In an ethical context: the best moral theories and principles. (Note

⁸ In cases when the ‘optimal theory’ is unknown, one may consider ‘optimal theory’ as an idealised, hypothetical concept, something as a hypothetical extrapolation of existing provisional theories. In natural science for instance, it is current practice to talk about ‘future better theories’ (such as the elusive Theory-of-Everything); in physics mathematical theorems are formulated regarding the features of these ‘more optimal’ future theories, such as Bell’s theorem. In sum, there seems little doubt that this notion of optimal theory is operationally useful: one often can well know which theory or assumptions are better than others (for a given end); the optimal theory (in the absolute sense) corresponds to the extrapolated, hypothetical end product of this progress. Philosophy and natural science are full of this abstract concept, even if we surely will never know the ultimate theories.

that there is a whole multidisciplinary field dedicated to optimal decision making: decision theory (e.g. Berger, 2013).)

We note that this conclusion closely fits to what Kant thought about free will in his *Foundations of the Metaphysics of Morals* (1786/1983). Here Kant states, notably: “a free will and a will under moral laws become one and the same” (Kant, 1786/1983 BA98). Thus for Kant, real free will is, in essence, *will under moral law*. In our parlance: will in accordance with (monitored, controlled by) adequate moral theories*.

Let us give a few more examples of how assumptions/beliefs/theories intervene in free and therefore conscious choices (from Vervoort & Blusiewicz, 2020b). When freely deciding which suit to choose to go to work, I may ‘assume’ or I may ‘use the belief’ or ‘be in the belief’ that suit A is more appropriate for today’s work than suit B. Or, when deciding of my own free will to help my neighbour, if I have Kantian inclinations, I may base my decision on the view that altruism is a source of fulfilment; or I may base it on the belief that the act will be rewarded by something more mundane—say a babble plus beer. The more intellectually demanding the context is, the more elaborate the assumptions I adopt: in some cases these better be part of a well-grounded theory. For instance, when a spaceship commander decides in an unforeseen situation to freely press the full power button at a precise time, he hopefully does so on the basis of assumptions that take serious mechanical calculations into account. If he does not, and hits the button in a panicky reflex, one hesitates to call his act genuinely free—in agreement with almost all philosophical interpretations.

There is another perspective from which to understand the close connection between the notions of free will/consciousness and that of theory*. If being conscious of *x* involves thinking about *x*, then there seems to be an unavoidable link with truth. Thinking that has no connection at all with (approximate) truth is erratic, and therefore at odds with genuine free will, supposed to be compatible with moral responsibility and goal-oriented agency. Next, we have seen in the chapter on epistemology and Gettier’s problem that the truth of propositions is inextricably perspective- or theory*-dependent. There is no more or less adequate thinking that does not logically, epistemologically, and presumably ontologically (see Sect. 7) ‘need’ (the manipulation of) coherent sets of hypotheses. In a slogan: there is no thinking that is not based on theories*. Many might agree that this is, in hindsight, a truism—but one with surprisingly far-reaching consequences, as developed in this book.

We were trying, initially, to identify a measure for the variability of free will. The answer will be clear by now: what varies in our capacity to perform actions that are free according to DEF-MBA is the *adequacy of the theories** we use to monitor these actions. In other words, the assumptions and belief systems involved in monitoring a free act are more or less adequate for guiding these acts. It is in this sense that one can say that Trina in the above example does not seem to use the optimal assumptions to monitor and control her deeds.

In conclusion, we will adopt the MBA-model with the extra assumption that, in clause (ii), *the monitoring (analysing, controlling) of the conscious brain involves theories*, and that these theories* have an adequacy that admits of a degree*. For the reasons explained, we only attribute free will to humans, not to animals in general,

so we introduce in the definition the notion of ‘agent S’. We call our enhanced MBA-model the CMT-model of free will. It defines free will as the capacity to perform actions that are Conscious-through-Monitoring-through-Theories*, and is condensed in following definition:

DEF-CMT. Act A by agent S is ‘free-willed’ or ‘free’ (is made of S’s own free will)

iff

- i. the act A is *unconstrained* (no programmed or external compulsion), and
- ii. the act A is *conscious* in that the act (linked to a mental process) is monitored (recorded, analysed, controlled, or kept track of) by some other mental activity in the brain of S. The latter monitoring occurs through representations, i.e. theories*, which have a varying degree of adequacy (cf. text).

Note that our model of free will is condensed and minimal; it is also a minimalistic model of consciousness via clause (ii). Compared to the MBA-model, DEF-CMT conceptualises, from a logical-epistemological point of view, *how, by which means*, internal mental monitoring is done. By the same token, it allows us to interpret the variability and other debated properties of free will, as shown in the next section. Experts of the highly debated, and often considered elusive, concept of consciousness—studied in philosophy, neurobiology, cognitive science, computer science, etc.—will hopefully appreciate the simplicity of the model, condensed in clause (ii). They might also recognise that DEF-CMT can be classified as a variant of the well-known higher-order-thought (HOT) theory of consciousness (Rosenthal, 2005). Thus, even if we came from a very different direction (and were initially not aware of HOT-theories), we ended up with a definition that lies in a respectable tradition. This is one more justification of the use of the notion of ‘theory*’.

Minimalism is particularly instrumental when one wishes to study a concept as free will or consciousness in different disciplines, say neuroscience or artificial intelligence (AI), as is chiefly done in Sect. 7 and (Vervoort & Blusiewicz, 2020b; Vervoort et al., 2021). For example, Vervoort et al. (2021) investigate the conditions under which artificial intelligence could, one day, become conscious. Starting from DEF-CMT, framed with notions that make sense in computer science, these authors propose more or less concrete routes for further interdisciplinary research on enhancing the cognitive skills of future AI.

In the following section we argue that the CMT-account can subsume other recent theories of free will, and solve problems of the latter. We focus on compatibilist accounts as ours. Clearly, for some problems the CMT-model needs to be used in combination with the perplexing conclusion of the previous chapter—the non-existence of absolute free will. The fact that both perspectives are needed for certain key problems is condensed in the idea of complementarity.

7.6 CMT-Model Compared to Other Compatibilist Theories

R1. Frankfurt's theory. A first well-known theory to consider is Frankfurt's 'hierarchical mesh theory' of free will (Frankfurt, 1969, 1988). In a nutshell, according to Frankfurt an action or choice A is free-willed if it is object of, if it meshes with, a 'second-order volition' or desire—a higher desire about the first-order desire to do A. Then A is really (rationally) desired, in agreement with one's real self, with one's second-order (rational) desires; in other words, A flows from the 'will one wants'—a reflective capacity animals likely do not have.

A priori it seems clear that this model resonates well with our model: it suffices to note that second-order volitions are part of the general beliefs, worldviews etc. agents use to guide and control their life and actions. Action A meshes with a second-order volition in that it is consciously monitored (assessed, analysed etc.) with help of a worldview, a belief system, assumptions of life, in other words theories*. So there surely seems place for a partial overlap between both models. However, it is well known that cases as Trina's, the brainwashed girl, are a threat to Frankfurt's theory (for a recent overview and references, see e.g. Griffith, 2013, Ch. 4; O'Connor & Franklin, 2022). Trina may well act in accordance with higher volitions, really believe in what she does, and therefore be entirely free according to Frankfurt's model—a conclusion most people would disagree with. The CMT model solves this problem: Trina is brainwashed and therefore not unconstrained (condition (i)); *and* she monitors (assesses) her deeds through questionable, likely inadequate beliefs (condition (ii)). One could say she has a limited form of free will. Next, there is also a well-known infinite-regress problem threatening Frankfurt's theory (why stop at second-order and not include higher-order volitions ?) This problem is absent from our model for obvious reasons.

R2. Wolf's theory. A next interesting account is Wolf's 'Reason View' of free will (1990). Wolf's is also a mesh theory, but whereas in Frankfurt's model free will is a subjective matter (an act is free as long as there is a mesh between the agent's choices and his personal, subjective, higher-level desires), Wolf adds that these personal desires should also have a 'connection with the world outside'—they should have an objective dimension, they should 'connect with the True and Good' (Wolf, 1990). So, in order to have free will, one should have the right, objective reasons to do things, reasons that connect to the True and Good.

The key ingredient of this account can well be subsumed by ours. Our model says that an action should be consciously monitored, guided by a (sufficiently) adequate theory*, ethical or other. This can be understood, in Wolf's parlance, as expressing that the agent should have adequate *reasons* for her or his act, reasons that 'connect with the True and Good', in the sense that they are involving, embedded in, backed-up by, adequate theories*. Note that 'adequate theory' is a more instrumental and precise concept than 'adequate reasons', especially if one wishes to interface with (computer) science, as explained in Sect. 7 and (Vervoort & Blusiewicz, 2020b; Vervoort et al., 2021); and we avoid the somewhat metaphorical 'connection with

the True and Good’.⁹ Furthermore, it has been objected to Wolf’s theory that agents can be manipulated or programmed into accessing the True and Good (e.g. Griffith, 2013, Ch. 4). We avoid this problem via clause (i) in DEF-CMT.

In sum, in spirit there is an obvious link between Wolf’s theory and ours. It may be that relying on ‘theories’ and ‘consciousness’ as we do, rather than on ‘reasons’ (Wolf) or ‘intelligibility / rationality’ as for instance Walter does (2001), is in last analysis a matter of taste (and logical construction of the theory). Still, we submit that the variability of free will, our essential starting assumption, is conceptualised most precisely via the concept of theory*.

R3. Fischer and Ravizza’s theory. A much debated and sophisticated theory is Fischer and Ravizza’s ‘Reasons-Responsiveness View’, putting an emphasis on the ‘guidance-control’ involved in a free act (Fischer & Ravizza, 1998; Fischer et al., 2007). We will here summarise the essence of this theory in a condensed manner. In a nutshell, an agent has free will if her actions and choices are sensitive to, respond to, reasons, where it is emphasized that this reasons-responsiveness should not rely on luck, once in a while, and that it is *not* responsiveness under compulsion or neurotic disorder. Rather, the agent should be responsive to reasons through a systematic (cognitive) mechanism that ensures guidance-control over her actions.¹⁰

To make the link with the CMT-model, first note that the concept of conscious control is explicitly part of our basic starting model, the MBA-definition (clause (ii)). The conscious monitoring of an act via adequate beliefs thus seems to imply reasons-responsiveness and guidance-control (or it could be construed to imply this). So, there is a partial overlap of the theories. However, Fischer’s and Ravizza’s Reasons-Responsiveness view is subject to the same criticism as Frankfurt’s theory: it implies that the brainwashed girl Trina steals of her free will, since she may well act from a practical and un-excessive reasoning mechanism. This is a worry we are clearly immune against, cf. the argument given above.

Modern theories of free will pay much attention to the notion of control. As said, our model tells us *what kind of control it is, or how this control is realised from a logical-epistemological point of view*—the point of view that is needed for philosophical problem solving.

R4. Other compatibilist accounts discussed in O’Connor and Franklin (2022). **Sourcehood.** It is instructive to study the recent review article (O’Connor & Franklin, 2022), providing a thorough overview of key ideas and positions in the contemporary free will debate, including of their historical lineage and interconnections. We submit they can all nicely be interpreted within the bi-perspectival account presented in this and the previous chapter. Let us go over the most relevant of them.

⁹ Instead of ‘connection (with the True and Good)’ we use the notion of ‘monitoring (via theories*)’, which can be given a firm neurobiological interpretation: the monitoring is materialized by two neural circuits being physically connected (cf. above and Sect. 7).

¹⁰ In slightly more detail, free-willed acts should be governed by a *moderately* reasons-responsive brain mechanism (say, enacting deliberation), i.e. one that is reasons-receptive and *weakly* reasons-reactive. Fischer and Ravizza (1998) introduce these criteria with the aim to exclude excessively dogmatic and phobic acts from being free-willed.

The authors recall that nowadays two of the most commonly cited features of free will are the ‘freedom-to-do-otherwise’ and ‘sourcehood’ (O’Connor & Franklin, 2022, § 2.3–2.4). Sourcehood-accounts have been motivated or inspired, notably, by the Frankfurt-style thought experiments mentioned in Sect. 3, and by the work by Fischer and Ravizza. First, according to the account proposed in the previous chapter, in the absolute sense I have no genuine freedom to do otherwise. But in the relative sense, seen from the perspective that is surely most vivid to us, I often do have (the impression to have) the possibility to choose between, say, greeting acquaintances I meet on the street, or menacing them with a knife (!). In my abstract, imagined world-in-thoughts: of course. *I have this (theoretical) possibility to choose when I can imagine, i.e. represent, different options, different (sequences of) events starting from my actions, and evaluate these options; and both the representation and the evaluation involve theories**—even if these can be extremely banal and barely conscious (but fully unconscious reflex-like actions are not really free). When I then choose one option, it is ‘I’ who chooses, seen from the perspective that I can grasp. In this relative but extremely vivid sense, I am the source of my actions, of my greeting instead of menacing. And the control I exercise is mediated by, or derives from, a form of cognitive *understanding* I have of what happens to me. In this sense, one could say that *free will derives from adequate representing, or understanding* (of what happens to me). But again, in a complementary, truer sense, the absolute one, my choices are determined, correlated with countless preceding events in a chain that starts entirely outside me. For example: I do not menace people, because I am educated, conditioned, by education, society, law, etc.; and all the innumerable factors of education, society, law etc. that have influenced me in my ‘choice’ not to menace are materialised by events in the real world; and these are all necessarily correlated with events occurring long before my birth. My capacity to do otherwise is, as a matter of principle, only a theoretical, metaphorical one, based on my capacity to represent the world.

Thus, one sees that the concepts of freedom-to-do-otherwise, sourcehood, and control are intimately related on our account: they can be interpreted in a synthetic move, sketchily presented above and at the end of the previous chapter. Competing compatibilist accounts are reviewed in O’Connor and Franklin (2022, § 2.2–2.4). The authors pay much attention to, notably, the case of phobic agents debated in the recent literature. They conclude: “it seems compatibilists have yet to furnish an analysis of the freedom to do otherwise that implies that phobics clearly lack the ability to choose or do otherwise that is relevant to moral responsibility and yet some merely determined agents have this ability” (O’Connor & Franklin, 2022, § 2.2). The solution provided by DEF-CMT to this problem seems clear. Much as in the case of the brainwashed Trina, both clauses (i) and (ii) prescribe the solution; they do so in a coherent manner. Most relevant is clause (i): phobic agents are under inner compulsion and therefore not really free—even if this comes with a degree. (And if they are not really free, they are not really morally responsible, since free will is necessary for moral responsibility.) Clause (ii) is coherent with this conclusion, and adds something about the quality of the limited free will phobic agents have: phobia imposes representations of reality that are usually considered of a less adequate type.

Non-phobic agents are equally unfree in the absolute sense, but are monitoring their acts with theories* that are considered, comparatively, more adequate (or that could be so); and they are not under inner compulsion. One might compare this simple solution to the more complicated ones proposed by authors as Vihvelin (2004) and others reviewed in O'Connor and Franklin (2022, § 2.2).

Similarly, published accounts dealing with the 'Manipulation Argument' are reviewed and found wanting. An example of this argument is Mele's 'Zygote Argument' (2006, Ch. 7), involving a goddess who manipulates the zygote of person S1 to make sure that S1's child, S2, will kill some specific person S3 when adult. Suppose S2, as a result of a moderately reasons-responsive mechanism à la Fischer and Ravizza and on the basis of motivations with which S2 is identified (as demanded by 'identification accounts', a special class of sourcehood-accounts¹¹) does commit the programmed murder. Many judge that S2 is not free nor morally responsible for murdering S3 even though this agent satisfies both the reasons-responsiveness and identification criteria: a problem for a variety of compatibilist models that hinge on these criteria. O'Connor and Franklin state (2022, § 2.4): "The general problem raised by this line of argument is that whatever proposed compatibilist conditions for an agent S's being free with respect to, and morally responsible for, some action ϕ , it will seem that agents can be manipulated into satisfying these conditions with respect to ϕ and, yet, precisely because they are manipulated into satisfying these conditions, their freedom and responsibility seem undermined." As a consequence, authors as Fischer and Ravizza have later tried to upgrade their model by reinforcing their free will, or sourcehood, criteria with an extra clause, stipulating that a free agent should take or have taken responsibility for the act under scrutiny. But "[t]he problem with this reply is that we can easily imagine [the goddess] creating [S2] so that his murdering [S3] is a result not only of a moderately reasons-responsive mechanism, but also a mechanism for which he has taken responsibility" (O'Connor & Franklin, 2022, § 2.4)

As above, DEF-CMT solves cases as these via clause (i): S2 is manifestly manipulated and therefore not free. It is as simple as that. The number of problems that can be solved by a bi-conditional (and bi-perspectival) account is large, so we hope that philosophers will agree with us that this seems a promising, synthetic way out of the mentioned conundrums.

R5. Walter's work. The neuroscientist and philosopher H. Walter (2001) has proposed a 3-component theory of *libertarian* free will, so it cannot directly be compared to ours. But since this theory is based on neuroscientific investigations by Walter's and other teams, it is still revealing to have a look at the similarities and differences. A summary is given in Table 1.1., p. 43 (Walter, 2001). Each main ingredient (briefly, alternativism; intelligibility / rationality; and agency / origination) can be understood according to Walter's classification in a minimal, moderate or maximal

¹¹ There is a school of philosophers that tries to stabilise classical compatibilism via the requirement of self-determination: a free act should be self-determined (rather than determined or undetermined), i.e. brought about by motives with which the agent is identified (for a list of representatives, cf. O'Connor & Franklin, 2022, § 2.4).

version. Comparison shows in a straightforward manner that the CMT-model, if we append to it our thesis of the non-existence of absolute free will, corresponds to Walter's alternativism in its minimal interpretation, plus intelligibility in its moderate interpretation. Our model does not explicitly include agency, but we define the free act of an *agent*, and therefore agency can be understood as implicit in this notion of agent *and*, especially, as made explicit by our 'consciousness' clause (ii). In other words, we can define genuine agency as the capacity of a subject to act freely in the sense of DEF-CMT. Along these lines, we have discussed in the previous chapter and under **R4** above how agent-based origination / sourcehood can be interpreted in a universe that is, ultimately, 'only physical'. (The reader will have understood that we use shortcuts as 'there is nothing but the physical' with a tremendous grain of salt: from the physical so many spheres or disciplines or problem-spaces emerge, say the psychological sphere, or the social one—which each need their domain-specific representations, theories, to be more or less understandable.)

Readers interested in neuroscience will enjoy discovering how Walter—as one representative of a number of neuroscientists—connects philosophical concepts to contemporary neuroscience. In the footsteps of scientist-philosophers as Mahner and Bunge, his work belongs to those rare but very welcome efforts of genuinely interdisciplinary research.

R6. List's theory. Finally, it is instructive to compare our bi-perspectival view to the sophisticated and scientifically well-informed compatibilist account by C. List (2019) (which will be represented here in an utterly sketchy manner). List starts from the assumption that following three conditions are sufficient for free will, in short: intentional agency, alternative possibilities, and causal control. His essential move for arguing that physical reductionism is not undermining these three conditions is to advance that the latter belong to other epistemic 'levels' than physics and hence cannot be reduced. The best theories we have to describe and explain for instance intentional agency and causal control are such scientific theories as psychology and neuroscience, and according to these the mentioned conditions for free will do very well exist. Since philosophy should adopt the best scientific theories at hand, it logically follows that free will exists.

I partially agree: relative free will exists. Indeed, one will note quite some similarities with my bi-perspectival account—notably List's starting point to look at things from different theoretical perspectives or levels –, with the difference that I include physicalism in the maximally coherent beliefs to adopt. True, it is debatable whether physicalism is a scientific thesis in the strict sense; but it is a thesis that answers immensely important (philosophical) questions for which non-physicalism remains powerless or unconvincing—such as: on what other elementary constituents neurobiological systems would be based, if not on physical ones? Therefore, even if I agree with List that *on a practical level* neuroscience and psychology and any of the socio-humanitarian sciences can clearly *not* be reduced to, described by, physics, physical reductionism still holds as a philosophical principle; we must assume that *in the absolute* all facts of the psyche and of society are *in principle* consequences of the laws of physics. Adopting reductionism leads then quasi-immediately to my

absolute view on free will, which is absent from List's theory. But this elaborate theory can surely enrich my minimalistic relative account, the CMT-model.

7.7 Proposal of a Few Interdisciplinary Lines of Research, Mainly About Consciousness, Natural and Artificial

Before concluding, let us briefly indicate some avenues of research suggested by the CMT-model in neuroscience and computer science. I conjecture that this model could be instrumental in tackling neuroscientific questions related to consciousness—considered an essential but at the same time highly elusive concept in neuroscience (Stern, 2017).

I think here in the first place of the process of monitoring by a neuronal superstructure, presumably in the prefrontal cortex, that should represent a theory*. This abstract, logical-epistemological notion plays such a central role of problem-solving in almost all chapters of this book, that I suspect it must have a rather immediate real-world, i.e. neurobiological, correlate. From some point of view, this is an obvious assumption; but whereas theories-in-the-mind have a very ethereal touch, in the brain they might be more solid, structured, lawful than meets the eye—and it is this lawfulness that might be interesting to study empirically. Perhaps it is interesting and feasible to search for the neuronal correlates for rudimentary theories*—say, law-like assumptions/hypotheses—in primates. One guesses that they are related to memory-circuits, or to the 'mirror neurons' that have become fashionable lately. Indeed, neuroscientists have associated these mirror neurons with the 'theory of mind' of primates, the cognitive base for recognising the 'self' and 'the other' (Gallese & Goldman, 1998). In the vocabulary of this book, if primates have rudimentary psychological beliefs, i.e. a rudimentary psychological theory*, altering the corresponding neurons could dramatically change their behaviour; *and* some of these neurons might be involved in any basic cognitive behaviour. In sum, it is tempting to conjecture that any behaviour in any animal that points to embryonic cognitive capacities should be associated with (presumably dedicated) neuronal networks or mechanisms that materialise sets of hypotheses / theories*. Perhaps this umbrella concept can serve as a guide and synthetic problem-solver in neuroscience too. At this point this is of course not more than a conjecture.

Next, one can analyse the well-known Libet experiment through the lens of our account, as is done in (Vervoort & Blusiewicz, 2020b). Libet's experiment, in which an electroencephalography (EEG) scan of a test person is made, records the so-called 'readiness potential': an electrical signal generated by the firing neurons of the scanned brain. Every voluntary action, such as moving a hand, induces such a slowly rising electrical potential. During the experiment, the test persons are asked to spontaneously flex their hand, whenever they feel like it, and note the time when they make this 'free' decision. The actual time of movement can be measured electronically with high precision; both times coincide reasonably well. Libet's quite

surprising and famous result was that the brain activity as recorded by the EEG started almost half a second before the time at which the test person consciously decided to move. Libet and most neuroscientists interpret this result as showing that it is the brain that decides, rather than a free mind. More precisely: unconscious brain processes are at work before the conscious impression of choosing manifests itself; it is not the mind that commands the body, but rather the other way around. Of course, this interpretation is not unanimous, but even more spectacular results have been reported since. For instance, using fMRI (functional magnetic resonance imaging), Soon et al. (2008) reported that ‘free’ choices of test persons (namely the choice to press a left or a right button) could be predicted up to 10 s (!) before the test person consciously made the decision to pick one or the other button. The authors state: “We found that the outcome of a decision can be encoded in brain activity of prefrontal and parietal cortex up to 10 s before it enters awareness. This delay presumably reflects the operation of a network of high-level control areas that begin to prepare an upcoming decision long before it enters awareness” (Soon et al., 2008).

Such neuroscientific findings corroborate our theoretical views in a few respects, and can be interpreted with our model. First of all, the verdict that absolute free will does not exist seems to resonate with the experimental detection of unconscious processes that would generate conscious decisions. If everything is ultimately physical, then high-level neurobiological processes, e.g. associated with conscious decisions, must emerge from physical constituents and laws; and this emergence seems not only a logical-epistemological truth, but an almost tangible neurobiological process unfolding over time-scales of the order of a second in the brain. (One could also see here, it seems, an illustration of how causation on the physical level, going from basic components, say atoms, to more complex ones, say crystals,¹² is paralleled by causation on the neurobiological level, which also goes from more basic elements, say individual neurons, to more complex neuronal superstructures, say those involved in consciousness.) At any rate, if everything is physical, consciousness must be correlated with, caused by, preceding unconscious mechanisms; and this may well be what one observes in the experiments. Now, even if there is no absolute free will, the CMT account stipulates that in the experiments by Libet and Soon et al. the decision events¹³ are nevertheless free *in the relative sense*, since these decisions are unconstrained and conscious, by construction of the experiments. At the same time, these choices / decisions are clearly of a rather banal type, involving utterly minimalistic hypotheses (such as: ‘at some point I am supposed to press a

¹² The laws describing elementary systems are usually believed to explain, or to ‘generate’, the laws describing more complex systems. In short, the laws of the component systems ‘cause’ the laws of the composed systems. It seems that what is said here in a metaphorical way can well be made precise using the notion of cause discussed in Chapter 5. The first step is to acknowledge that a more precise statement is: the properties X of the elementary constituents of matter are described by laws etc. To these properties or variables X one can then apply the formalism developed in the chapter on causation.

¹³ Recall that the CMT-model allows one to test whether individual acts, or events, are made of free will.

button’ and ‘this object here is a button’). It might be interesting to perform experiments in which decisions involve more elaborate cognitive ingredients. In sum, on the CMT model, the test person in Libet’s and similar experiments is free at the moment of conscious choice, even if this conscious and free act is determined by previous causes—as the neurobiological data might suggest.

In artificial intelligence (AI) research, a much debated question is: can future computers and robots simulate or emulate consciousness (and/or free will)? If possible at all, the CMT model suggests that one of the key properties a computer / AI should have to emulate consciousness is the capacity to ‘use’ higher-order theories—and this includes the capacity to adequately apply theories to (all) real-world situations and to act accordingly. Some will conclude that we are very far from this possibility (but see Vervoort et al., 2023 for ways to monitor progress, via a test for theory-based, or hypothetical-deductive, reasoning in AI). This suggests the following line of research: can machines learn to acquire and use theories*, and which types and how? Vervoort et al. (2021) present first ideas, and the outline of an interdisciplinary research program. Interestingly, recently cognitive scientists have also come to the conclusion that mastering theories is a key goal for AI. In the words of Lake et al., (2017, abstract):

We review progress in cognitive science suggesting that truly human-like learning and thinking machines will have to reach beyond current engineering trends in both what they learn and how they learn it. Specifically, we argue that these machines should (1) build causal models of the world that support explanation and understanding, rather than merely solving pattern recognition problems; (2) ground learning in intuitive theories of physics and psychology to support and enrich the knowledge that is learned; and (3) harness compositionality and learning-to-learn to rapidly acquire and generalize knowledge to new tasks and situations.

And Lake et al., (2017, p. 2):

The alternative approach [alternative to pattern recognition] treats models of the world as primary, where learning is the process of model-building. Cognition is about using these models to understand the world, to explain what we see, to imagine what could have happened that didn’t, or what could be true that isn’t, and then planning actions to make it so. The difference between pattern recognition and model-building, between prediction and explanation, is central to our view of human intelligence. Just as scientists seek to explain nature, not simply predict it, we see human thought as fundamentally a model-building activity.

These ideas obviously align with ideas developed in this book. Actually, a few teams of physicists have started exploring whether artificial neural nets can abstract basic physical theories from data, and use these theories for prediction (two examples are discussed in (Vervoort et al., 2021)). However, the regularities that these deep learning models can detect and extract from data are not really ‘theories’ in the strict sense, but rather specific laws, say Newton’s law of gravity, as applied to certain numerical domains. This is still very far from abstracting and mastering Newton’s full theory of mechanics—a huge step in generality and predictive power. But if in the future AIs can be built that master theories of physics, psychology, sociology, ethics, axiology,¹⁴

¹⁴ Robots and AI in general might well be taught a ‘theory of their values and goals’: know what their objectives are, know the value-hierarchy; they might also have the practical/common-sense

etc., then I do believe that such AIs would possess rational consciousness (Vervoort et al., 2021), in accordance with DEF-CMT. Of course, sentience, or feeling/emotion-based consciousness, is something else: AI does not feel, even if some types of AI (chatbots) can already very well simulate, in words, having feelings.

7.8 Conclusion of Chapters 6 and 7

In these last two chapters I developed a bi-perspectival account of free will, one that starts from recognising that free will can have different meanings, or, more precisely: that free will can be understood from two perspectives, an absolute one (that includes all our best theories, including physics and its interpretation) and a relative, socio-humanitarian one (that we can grasp incomparably much better). Within each of these perspectives, free will has a different meaning; I proposed the definition of free will that is best compatible with each perspective. In Chapter 6, I considered the absolute perspective, and arrived at an astonishing conclusion, one that is known since antiquity but that continues to stupefy and disconcert many people, including philosophers and scientists. This conclusion is that genuine free will, as the capacity to *really* have options of choice, is an illusion. I argued that this conclusion follows from the most solid theories we have, so should be considered as part of the maximally coherent network of ideas, the MCT. In other words, for this extraordinarily counterintuitive conclusion to be wrong, there must be something wrong in some of our most basic, most fundamental principles of science or logic—and this is much more unlikely than that our folk views on free will are incorrect. In Chapter 7 I developed a definition of the type of free will (or, more generally, the human capacity or condition) that is necessary for attributing agency and moral responsibility to a human—the type of free will that people usually consider relevant in philosophy, in our social and psychological attitudes, in legal systems, in daily life.

My relative account, heavily drawing on well-known analytic models and on the work by Mahner and Bunge (1997), stipulates that a free act is unconstrained, and consciously monitored through sets of hypotheses—called theories*. It thus highlights the rational aspect of free will. The model is compatible with the fact that free will comes in degrees, a hypothesis that is natural within a biphilosophical approach. Note that my model is independent of whether the universe is deterministic or indeterministic. To fix ideas, let us look a last time at a concrete example. According to the bi-perspectival account the following three facts can be true at the same time:

knowledge/theories to implement these. The actions and in particular the goals of such advanced AI hopefully will be under benevolent human supervision. But there is a risk that this will not be the case (Bostrom, 2014). Note also that having a theory of what one knows (and is) amounts, on our view, to (rational) self-consciousness. I do not see principled reasons why AI could not be taught to master such a set of theories.

- (i) a murder may be committed by an act of free will (in that the act was unconstrained and accompanied by a more or less conscious reflection based on some beliefs, assumptions etc.);
- (ii) but the free will of the murderer is likely to be of a corrupted type (her moral theory* is likely not adequate);
- (iii) in any case, the murderer *had to commit the murder*, in the sense that the murderer's decision resulted from external events that are, scientifically speaking, *ultimately* beyond any agent's control.

The theory stipulates that claims (i) and (ii) should not be adopted without *also* considering (iii), which follows from the absolute perspective; (iii) should *seriously be taken into account*—more on this in a moment.

This view can be considered compatibilist, although it is more precise to consider it bi-perspectival: there are two meaningful conceptions that are complementary. Which of these complementary perspectives to adopt depends on the context; for the hardest (notably philosophical) problems both need to be used. I do not really assume that free will and determinism both exist; I claim that *a form of* free will, relative free will, exists at many instances in many people even in a physical universe. Recall that on the absolute account, free will is not only eliminated by determinism, but by the much more general and widely accepted thesis of physical reductionism, or physicalism. Determinism is a more debateable hypothesis than physicalism; the latter must be accepted within the MCT.

Some philosophers may contend that it is better to reject free will altogether (e.g. Pereboom, 2014; Waller, 2011), and to not introduce a superfluous notion as relative free will. I well sympathise with this view, and I am not entirely sure what the best direction is. But it seems that ultimately both approaches can converge, and that the choice is only a pragmatic matter.

My ideas lie, of course, in a long philosophical tradition—free will has fascinated thinkers since antiquity. I will here not attempt a historical overview [a synthetic account of key ideas of historical thinkers can be found in O'Connor and Franklin (2022, § 1)]. Among the antique thinkers to whom my view connects best, I have mentioned several times Aristotle (with connections to Plato, of course, when it comes to the notion of the quality or adequacy of free will)—I should mention many more. But if I would have to choose two thinkers to whom I feel particularly close, I would mention Chrysippus and Spinoza.

I find it remarkable¹⁵ that already the Stoics and Epicureans had intuited that all stuff of the world, including human souls, must obey natural laws. The Stoic Chrysippus (c. 279–c. 206 BC) is one marvellous exponent of the ancient line of thinkers who came to recognise that everything that happens lies in a network of cause-effect relations, put less metaphorically: that everything happens necessarily

¹⁵ I find it revealing that ancient philosophers could intuit, based on logical thinking and elementary (pre-)scientific observations, what physics and neuroscience have confirmed two thousand years later. On a speculative level, this seems to me evidence that the universe is *logical* (much remains to be said here!). It also corroborates the idea that philosophy can work in harmony with science, and function as proto-science.

in a way that is correlated with the past. And yet, Chrysippus thought that our actions can be ‘up to us’ in a certain sense—here my account is again very close to his in several respects. Chrysippus thought that choices are up to us when they are co-determined by convictions that make up our personality, and by deliberation based on the options we imagine. So, our choices are determined, but what matters for moral responsibility is *in what manner* they are determined (O’Connor & Franklin, 2022, § 1). I wholeheartedly agree. I will end this book with commenting on Spinoza’s view on free will in the Epilogue, Chapter 9.

An essential ingredient of my view on free will is that both the absolute and relative perspective must be used when one wishes to interpret this world (more or less) adequately, when one wants to deal with society or propose ideas to improve it. I believe that philosophers should participate actively to this grand effort. I believe that the common-sense view on free will is not enough; and that, however unfathomable, the absolute perspective has urgent relevance. I would go as far as contending that an appropriate view on free will can help us in tackling some of the challenges of our time, enormous as they are.

Now, this book is dedicated to problems for which it is possible to come to well-founded and thoroughly argued solutions—say, the problem of *defining* the concept of free will. To my regret, I cannot propose here a well-founded theory of all the psychological, societal, etc. implications of my bi-perspectival view; of how to deal with the extraordinary idea that we do not really have free will. As several philosophers have excellently argued before me, I believe the consequences for our views on moral responsibility, ethics, law, retributive punishment, praise, blame, desert, reward, happiness, etc. are considerable (to name only a few, Strawson, 1986; Pereboom, 1997, Ch. 18; 2001; Waller, 2011; Caruso, 2012; Vilhauer, 2012). Even if I have not worked out a coherent theory—which would take another book or book series –, I feel it is important to mention at least a few intuitions (often in line with those of the mentioned philosophers). I will do this in a pointillist manner in the Epilogue (Chapter 9§2), which is dedicated to conjectures that resonate with Chapters 1–7 without being fully developed. I hope this exercise will show that scientific philosophy can provide a basis for addressing practical problems with potentially immediate importance for society.

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Chapter 8

General Conclusion



Abstract I review a few results of the book to highlight that synthetic philosophy is theory-based philosophy, in which theory relativism plays a key problem solving role. The potential for a fruitful use of AI in synthetic problem solving is recalled, in philosophy, and possibly in other disciplines of the humanities and social sciences.

In this book I have proposed a philosophy that starts from one basic hypothesis or postulate—obvious to many—, namely that science and a science-compatible philosophy are possible. Of course, scientists and philosophers often accept this hypothesis as an evident, immediate given; it is only for defusing radical scepticism that one even needs to state it (and promote it as an explicit postulate). Yet it appears that this assumption, innocuous though it may seem, has surprisingly weighty implications and ramifications.

The history of science has shown that science works—on the logical level—by electing some principles as basic, by constructing theories around these basic principles, and by deriving predictions, answers and solutions from or within these theories, using, essentially, logic or mathematics.¹ History has also shown that there is sometimes a hierarchy between theories (within one problem space): it happens that effective approximate theories (often called models) are derived from more fundamental theories; or, the other way around, that several approximate theories are synthesised into one more fundamental theory. In a given epoch, the most fundamental theories are not derived from theories at an even higher level of abstraction; it is not guaranteed that they ever will be. One of the most concrete and quasi-quantifiable characteristics of such fundamental theories is that they allow one to answer in their problem space a maximal number of questions in a coherent manner. Having made from the start the assumption that a science-compatible philosophy is possible, this state of affairs in science suggests that also in philosophy the quest for the MCT, or the maximally coherent web if one prefers, is the best guarantee for finding solid answers; and that this MCT can be hunted for by searching for the sub-theories (resorting under it) that solve the most problems in their sub-spaces; the MCT is then the final synthesis.

¹ Sure, on the practical level, on the level of discovery, the basic postulates can be said to be inspired, or induced, by empirical observation and experimentation – ultimately.

One other consequence for philosophy, at least for synthetic philosophy, is that some principles should be accepted as basic, unprovable. That is how science and scientific philosophy work.

The problem of induction in science, studied in Chapter 2, can be phrased thus, in two more or less equivalent ways:

- (i) There is no proof, no justification for (the universal, stable-in-time character of) laws (allowing us to use them for prediction), or
- (ii) There is no proof, no justification for the fact that any given theory of natural science can work, i.e. can have predictive (and explanatory) power.

The principle of induction can be phrased thus:

- (i) There are laws of nature (with a universal, stable-in-time character), or
- (ii) Science can work (by inducing laws from observation).

Hence, the problem of induction is obviously immediately solved or dissolved if one accepts the principle of induction as a basic principle of philosophy of science: one of these most fundamental principles that cannot be proven, i.e. derived from a more general theory. Accordingly, following the other basic tenet of synthetic philosophy just recalled, namely that scientific and philosophical validity of principles and theories is measured by their problem-solving power, adopting the principle of induction solves manifestly more problems and questions than rejecting it.² I therefore believe it is part of the MCT, and elected it as a first principle of synthetic philosophy.

There indeed seem to be good reasons to adopt the principle as basic, in the sense of maximally general, not derivable from other principles. The principle of induction is not a law of science proper, so should be one of meta-science. And, to the best of my knowledge, in the works of philosophers there is no hint of an idea that could point to a more general philosophical principle characterising (the existence of) all laws of nature than the statement “there are scientific laws”. “There are scientific laws of nature”, or if one prefers, “laws of nature (or scientific law statements describing laws of nature) can be induced, generalised, starting from empirical research” has then a legitimate status of first principle, to use Aristotle’s term. From this perspective the claim that there is no justification of the existence of scientific laws and their universal, stable character, is irrational—even when acknowledging that these law statements are defeasible.

Thus, the two intimately related concepts of law and theory, as they are well understood in science and philosophy of science, stand at the core of synthetic philosophy. Is it then a surprise that the notion of theory, or of coherent set of hypotheses (‘T*’), or, slightly more generally and metaphorically, of perspective or representation, plays such a crucial role in this book? The idea that *all concepts and problems* (of science and philosophy) *receive their meaning within a perspective, more precisely a theory*, appears to have a remarkable problem-solving capacity in philosophy—a claim that

² Examples of problems/questions that can coherently be solved/answered: the problem of induction; the question why different experimenters can observe the same numerical results when doing the same experiments; all particular instances of the general question: “Why does this scientific prediction come true?”.

is extra remarkable in view of the blatant obviousness this idea has for many of us. All chapters of this book can be seen as an elaboration or illustration of this idea.

In Chapter 3, on Gettier's problem, we laid an independent basis for this thesis, initially coming from a linguistic angle, by arguing that there is a fundamental representation-relativity in the concepts of truth, knowledge and justification. Such a principle of epistemological representation-relativity appeared to be the most natural inference from a solution to Gettier's problem and other epistemological problems. Initially, I somewhat restricted this principle to the use and meaning of the expressions 'to know', 'proposition *p* is true' and 'justified' (to do justice to language habits and to solve Gettier's problem). But the great problem-solving capacity of this principle, confirmed in the subsequent chapters, left me no other choice than to adopt it as the key to disclose the fundamental nature of truth and knowledge and justification. *It says us what truth and knowledge and justification really are.* Concomitantly, I adopted a relativistic account of truth (a mixed coherence—correspondence view), in which the concept of absolute/objective truth can be recovered as a natural limit case. This account appeared to resonate deeply with Quine's holism. It also led to a view on the essence of knowledge that can pleasingly be condensed in simple phrases, such as: *knowing implies assuming*, and *the atoms of knowledge are theories*.

This is not all. I believe that this principle has amassed so much credentials by now that it allows us to infer, without much scruples, other conclusions about cognition-related matters, for instance about the nature of understanding and explanation (but I will leave this as a topic for further work). To give one more example: I think we can now assert rather confidently that truly powerful AI, AGI (artificial general intelligence), must be AI that masters theory-based reasoning (in all domains of interest, cf. Vervoort et al., 2023).

Chapter 4, on the interpretation of probability, did not immediately use the notion of representation-relativity; nevertheless, keeping theory-relativism in mind was essential here too. If all theses and propositions are theory-relative, then also the question of what probability is. If the question is asked from the perspective of mathematics, the answer is simple: probability is what satisfies Kolmogorov's axioms. But in this book we are interested in the interpretation of probability beyond mathematics. I argued that one cannot do justice to this utterly subtle problem if one does not acknowledge that probability theory is, above all, a physical theory. It describes ensembles of real-world systems, satisfying certain conditions (cf. Chapter 4). (And it does so arguably in the most general, the most abstract way possible, which makes the link with the concept of cause—recall **DEF-P** in Chapter 5—and, especially, with the principle of causality—that other utterly general principle, cf. Chapter 9—perhaps less surprising. This points to a fascinating topic for further research: unify probability theory and the principle of causation.) Simply realising that probability theory can be seen as a physical theory was the source of the solution to most of its problems—to the least, it inspired these solutions. In sum, theory-relativism reminded us also in this chapter that it is essential, when starting to study a problem, to identify the right perspective from which to address it.

Chapter 5, on the notion of cause, illustrated the utility of representation-relativity in a particularly conspicuous manner. Positing the theory-relativity of causal claims

appeared to be the tool for resolving a wide variety of questions and conundrums documented in the literature. Problems related to context-relativity (Ch. 5, **RR1**), early (Ch. 5, **RR2**) and late (Ch. 5, **RR4**) preemption, double prevention (Ch. 5, **RR3**), overdetermination (Ch. 5, **RR5**), tampering (Ch. 5, **RR6**), transitivity (Ch. 5, **RR6**), token-versus-type causation (Ch. 5, **RR7**), black-box cases (Ch. 5, **RR8**) could all be solved using one and the same assumption of theory-relativity. It seems to me that if one principle can solve a series of problems that have hitherto been dealt with in a disconnected manner, its soundness becomes rather inevitable.

One might wonder: is representation-relativity of causal claims really the same representation-relativity as the one encountered in Chapter 3 on Gettier's epistemological problem? It should be, as a simple argument shows. The representation-relativity of 'true' (and propositional truth) means that 'p is true' can only definitely be evaluated via a certain T*; without such a T* the sentence has no definite meaning. Now, a causal claim as 'X is the cause of Y' is logically equivalent to 'It is true that X is the cause of Y'. Since the latter proposition is T*-relative, the causal claim itself must be. In sum, in principle we could have anticipated such a perspectival aspect of causal attribution from the start. Actually, I had not made this anticipation, but I re-discovered representation-relativity in the exercise of problem solving. In any case, it does not come as a surprise; and the results of Chapters 3 and 5 corroborate each other. Recall that I emphasised in Chapter 5 that representation-relativity characterises *causal identification* (and psychological and epistemological-logical aspects of causation), rather than *causation as an ontological property of the universe*. Within scientific realism, one better assumes that the causes, identified with an as-adequate-as-possible T*, correspond to something real, independent of the human mind; that reality is not generated by a theorising mind.

In Chapters 6 and 7 on free will representation-relativity played an essential role too, in many respects. By now, this should be obvious: these chapters deal in the first place with the definition of a concept; and concepts (of science and philosophy) find their meaning in conceptual frameworks. I submitted that free will is perhaps best studied via a complementary, bi-perspectival approach. Indeed, the rejection of free will that seems an unavoidable conclusion within a physical perspective, is so unfathomable and harshly counterintuitive that it seems justified to save the concept within a socio-humanitarian perspective. Without such a concept it may be difficult to constructively dialogue with researchers in practical philosophy and with people and society at large, for instance when the notion of personal responsibility is at stake. This complementary view implies, when thinking practically, a subtle act of balancing between perspectives, a balancing of which I did not clearly define the principle (although I suspect it turns around global wellbeing, admittedly a somewhat vague idea for the time being).

Again, the notion of theory* plays not only a heuristic role, as in the chapters on the problem of induction and probability, but intervenes also directly in the definition of free will, as in that of cause. (Relative) free will is, on my account, a capacity that is essentially cognitive; and therefore mediated through sets of hypotheses, theories* (in the wide sense)—as any cognition is. In hindsight, the main steps of the argument are straightforward, and can be summarised thus: the key condition for free will is

consciousness; we can only be conscious or aware of X if we have an (even utterly vague) idea of what X is; i.e. if we know at least something about X (minimally, *that it is there*); and knowledge is mediated through theories*. This is the essence of the CMT account of relative free will.

In closing, I would like to repeat a conclusion already drawn in Chapter 1, namely that the method promoted here, problem-solving-through-unification, or unification-through-problem-solving, has a genuine potential for being enhanced by AI, present and future. The synthetic method often starts with establishing a cartography of problems existing in a given problem-space. Most of the relevant problems have extensively been dealt with in the philosophical literature; and since a few years Large Language Models (LLMs) have been developed that ‘have read’ large parts, and perhaps most, of the literature. At any rate, such LLMs can in principle be trained with the complete philosophical literature, or a great part of it (which opens the interesting possibility of training dedicated models). These computational tools work, at their basis, by statistical text completion, implying that they can in principle identify the problems (in a given space) that are statistically considered most relevant in the philosophical literature, moreover in a phrasing that is adequate with high probability (relative to the screened corpus). Hence, they can in any cognitive space discussed in the literature provide a cartography of problems that has a very high probability of relevance; an example is given in Appendix 1.1. This AI-generated cartography can then be a very solid tool for further—provisionally human—synthesis. Such LLMs can also directly propose solutions to problems. For the time being, these solutions should not be expected to be synthetic, new, creative ones—solutions that significantly extend the web of knowledge (to start with, LLMs are not yet really ‘reasoning’, cf. Arkoudas, 2023; Vervoort et al., 2023). But that may well change in the near future. Therefore two conclusions impose themselves. First, the potential for constructive interaction between philosophy and AI is huge. And note that it is tempting to conjecture that the method we proposed for philosophical problem solving can also be used, *mutatis mutandis*, in other disciplines of the humanities and social sciences. This would especially be so if these disciplines are relatively mature, i.e. theory-based and oriented towards problem solving. Second, very soon we will need synthetic philosophy to assess the outputs of AI. If we do not have synthetic theories at our disposal, the intellectual authority might change sides at some point.

True, for the time being the reader may estimate that the use philosophers can make of AI is still rather modest. (For sure, I did not showcase spectacular results obtained thanks to AI.) When commercial LLMs became efficient, this book was almost finished, so I have not presented here full-scale *systematic* studies of using them as an aid for problem solving, let alone studies with statistical relevance. But I believe that such more systematic work is now possible and that it should be highly revealing. Such large-scale studies could for instance compare the cartographies given by different LLMs, or compare AI-generated cartographies with human-generated ones, based on polls among experts. My own not very systematic tests (e.g. in Chapter 2, Appendix 2.1) have convinced me that such comparisons should, to

high likelihood, show a great convergence between models, which is further evidence that AI-generated cartographies should indeed provide a solid basis for synthesis.

Another conclusion of this book is that synthetic philosophy should well be able to constructively interact with science, *in a bi-directional way*. This seems rather obvious if knowledge is conceived of as a coherent web spanning all cognitive fields. Surely there is a typical dynamics. In the first stage, philosophy looks at what mainstream scientific results tell us about a certain problem space, in order to start grounding philosophical theses. Here are a few obvious examples of such widely accepted scientific findings relevant for philosophy:

- (i) Quantum systems are essentially probabilistic and a-causal;
- (ii) Cognition happens in interconnected neural networks in the brain;
- (iii) There are few social law statements.

Now, once philosophy has inferred a synthetic theory about the related philosophical problem-space (say about (i) causation or (ii) consciousness or (iii) the nature of scientific knowledge and laws), it may then well propose new interdisciplinary ideas, coherent with the synthetic theory found, ideas that can guide further scientific research. The coherence requirement of synthetic philosophical theories implies that their interdisciplinary inferences or conjectures, even if speculative, should have at least a reasonable chance of scientific relevance. Examples of such conjectures, backed-up by synthetic philosophy, could be:

- (i) Quantum systems might be causal after all; theories that causally complete quantum mechanics are a worthy topic of investigation.
- (ii) Consciousness emerges from a special kind of neuronal interconnection, involving a hypotheses-generating network;
- (iii) There must be many more social laws, after all.

Note that especially claims (i) and (iii) seem, at least at first sight, in tension with mainstream science. On the other hand, the case can be made that theses (i–iii) have massive support from a synthetic point of view. So, I submit that the virtuous path may be: science → philosophy → science. I tried to propose a few examples of how philosophy can inspire science in this book; a further (perhaps more ambitious) example, essentially idea (i) above, is considered in the Epilogue, based on what we learned about the archetypical problem of causation in Chapter 5.

In sum, I believe that a dose of optimism is warranted as to the guiding role philosophy can play for foundational scientific problems.

A last conclusion from Chapter 1 that I would like to repeat is that in principle *all* philosophical branches can be part of the coherent web, and be the object of synthetic investigation. For such an integration to succeed, these branches should be made coherent among themselves and with science. To paraphrase and extend Ladyman and Ross's verdict (2007, p. 28), synthetic philosophy is the enterprise of elucidating coherence networks across the sciences *and all branches of philosophy*. Indeed, the MCT-postulate, felicitously having a semi-quantitative nature, operates for all chapters of philosophy—even if it is surely applicable with greater confidence in disciplines closer to established science. For instance, I see no principled reason

why a unified ethical theory could not be constructed, starting from the cartography of problems attached to ethical proto-theories (virtue theories, deontological theories, utilitarianism, ...). I believe that such a construction could proceed, in essence, by counting solutions provided by candidate master theories, and possibly by using sciences as neurobiology and neuropsychology to partly justify these solutions, in general to inform problem-solving. Sure, such an effort seems considerably more onerous than the investigations done in this book. A few hints of how synthetic philosophy could be relevant for very practical problems are given in the Epilogue. Again, the same methodology might well apply to other disciplines of the human and social sciences.

As a final conclusion, in what seems a new era of artificial intelligence, with its huge possibilities but also potential risks—such as an uncontrolled multiplication of fake information—I believe it is crucial that philosophers maintain an unshaken faith in the accessibility of truth. Philosophers have a large responsibility in shaping the global perception about truth-finding (Boghossian, 2006); and fear of knowledge does not help. Therefore, in the challenging times in which we live, I believe philosophers have a key role to play in showing how difficult problems can be solved: the planet needs their contribution. True, things are somewhat subtle: objective truth seems only asymptotically reachable, relatively, via perspectives; but these can be assessed for their objective robustness, possibly using AI as an aid. I therefore hope that others will apply the method of synthetic philosophy to other, or the same, problems of philosophy, and beyond. A possible start might be to hunt for weaknesses of my provisional solutions, and to propose a new synthesis.

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Chapter 9

Epilogue: Lines of Further Research



Abstract In this chapter I extract more speculative claims from the preceding chapters. The aim of this exercise is to show that synthetic philosophy, even if dealing in the first place with theoretical issues, inevitably interfaces with problems of practical importance (Sect. 2), as well as with scientific ones (Sect. 1). Section 1 aims at rehabilitating the principle of causation and at showing that it offers a most synthetic solution to Bell's theorem. Section 2 dips a toe in the intricate and intriguing universe of questions related to human agency in a causal world. The theses in this chapter are not derived with the same rigour as most of the preceding ones; they are therefore not necessarily part of the MCT. Yet, since they are based on synthetic results, I believe they are, to the least, interesting topics for further research. Most arguments are presented in a succinct manner.

9.1 On the Principle of Causation

In Chapter 5 on cause, we started by noting that many key philosophical questions hinge on the concept of cause. The question that immediately comes to mind is: are all events caused, do all events have causes—or are some events spontaneous, uncaused, a-causal, undetermined? In other words, is the principle of causation true? In the wake of the shatteringly negative verdicts of giants of philosophy and physics as Hume, Mach, Russell, Bohr, Heisenberg, Pauli and many more, few contemporary philosophers seem keen on defending a principle that once was a queen of natural philosophy.

Yet it seems, if we are right that a unified concept of cause exists, that the queen is happily back in the race of fundamental principles. (I ask the reader to measure for a moment the dramatic touch in this revival race. In several communities, say in quantum physics and in many important schools of philosophy of science, the queen is deemed dead and utterly uncongenial to boot.) Now, the existence of a synthetic definition amounts to a genuine, be it not conclusive, argument in favour of (1) the 'reality' of causes—more precisely, of the idea that causes correspond to something real—and, especially, (2) the truth of the principle of causation—the

idea that the universe operates according to laws of cause and effect. Hypothesis (1) is trivial within our realist background assumptions. Realists assume that coherent theories, for instance about the notion of cause, have the best chances to correspond to something real. But (2) has become very unpopular it seems.

Let me define the principle of causation/causality as follows, as is often done: “All events have causes in accordance with necessary laws of nature”; or “Every effect is produced through lawful necessity by a cause” (Norton, 2003; other versions exist, cf. e.g. Norton, 2003; Bunge, 2009). With the help of the modified counterfactual definition of cause of Chapter 5 (**DEF-CF**), it is clear what this principle means, also the ‘necessity’ invoked: namely, that anything occurring in our universe is caused by other events, following the laws of nature; and that these other events, the causes, are necessary *in the most basic logical sense* that if they would not occur, the effects would not occur (in the same way). (We added that the optimal causal identification is done by the optimal theory giving us the laws, or rather, the provisionally optimal law statements. More generally, causal identification is representation-relative, even if causation is not.)

My argument in Chapter 5 for the reality of the causal principle was straightforward enough. If we make the most likely assumptions, namely (i) that everything happens according to the laws of nature (the laws of physics, which are either deterministic or probabilistic), and (ii) that a synthetic account of cause, based on **DEF-CF** (**DEF-F** and **DEF-P**), shows that the laws of nature inherently embody cause-effect relations, it immediately follows that everything happens according to laws of cause and effect. As argued at the end of Chapter 5, these laws, both deterministic and probabilistic, can well be understood as having a character of necessity. For this and other reasons explained in the following, I believe that it is high time to rehabilitate the causal principle; that it says something essential about things in themselves.

Clearly, my view is in opposition with old but vivacious and very respected schools of empiricist, positivist and other philosophers, Hume, Kant, Mach, Russell, Nagel, van Fraassen, Cartwright, Norton, Ladyman and Ross, to name a few. In a highly influential and often-quoted passage, Russell asserted (1912, p. 1): “All philosophers, of every school, imagine that causation is one of the fundamental axioms or postulates of science, yet, oddly enough, in advanced sciences such as gravitational astronomy, the word ‘cause’ never occurs. [...] The law of causality, I believe, like much that passes muster among philosophers, is a relic of a bygone age, surviving like the monarchy, only because it is erroneously supposed to do no harm.” In a modern version, Norton states: “What I do deny is that the task of science is to find the particular expressions of some fundamental causal principle in the domain of each of the sciences” (Norton, 2003, p. 1).

But I believe that Russell and Norton make the same category mistake. It suffices to note that the *principle of causality is a principle of metascience, of philosophy, not immediately of science*. I believe this remark defuses much of their arguments. Sure, the laws of nature, or their statements, are the building blocks of science, not the causal principle; *but it does belong to philosophy to characterise these laws, and to analyse which properties they all share*. And in this respect, the principle of causality is utterly instrumental and seems universal—as countless philosophers

have intuited since antiquity. Put differently, the task of the sciences is to find the laws and regularities of their respective disciplines. These laws describe reality on the most fundamental level.¹ But, 1) these laws allow us to identify causes in a unified manner (recall from Chapter 5 how causes can be deduced from, notably, the laws of physics); and, especially, 2) *it is the task of philosophy to recognise what the lawful ontology expressed by these laws amounts to*. Notably: the necessitarianism, expressed in the causal principle, that these laws entail and the consequences of such a counterintuitive but eminently relevant worldview (recall Chapters 6 and 7 on free will). In sum, if one rejects the causal principle as a *metaprinciple*, one is at high risk of rejecting essential *philosophical* theses and theories.

This being said, it seems that authors as Russell, Norton and others underestimate the importance of the notion of cause *even in physics proper*. I have already mentioned the use of cause in Bell's theorem in Chapter 5§3, and I will return to it below. Next, in relativity theory it is accepted that effects can only be influenced by causes in their past light cones, and that causes cannot have effects outside their forward light cones. The notions of cause and effect are used in a pre-theoretic way, but they comply with the definition proposed in Chapter 5 (which is nothing else than a synthetic reformulation of intuition). In some more detail, in relativity theory it is of fundamental importance that causes happen before their effects; if not, logical paradoxes as the grandfather paradox would follow; and such logical paradoxes are considered impossible (this result is part of any book on relativity theory). Interestingly, the logical verdict is (I believe righteously²) accepted as the ultimate ground for judging about physical matters. That causes do not happen after their effects (for all frames of reference) is a consequence of the assumption that causal influences do not travel faster than light. In this sense, causation/causality³ plays an essential role in relativity theory in discarding faster-than-light influences; the non-existence of such superluminal causal influences has solid empirical support (e.g. Mattingly, 2005). In sum, physics does give a prominent place to the notion of cause, after all. I believe philosophy should do so *a fortiori*.

¹ It is not really necessary here to make a distinction between the laws of fundamental physics (relativity theory, quantum mechanics and the yet elusive theory unifying them), the laws of 'less fundamental' physics and the laws of other sciences. Indeed, as soon as these laws are backed-up by theories, my definition of cause allows identifying causes derived from these laws. This is a pleasant consequence of unification.

² It seems that logic can be seen not only as a theory of correct reasoning, but also as the most fundamental theory of physics, or, rather, of 'what happens in the universe'. Reasoning (as applied to matters in the real world) can be called 'correct' if there is an appropriate link with reality, after all. Furthermore, logical consistency is shared by all our bona fide theories, be it physical or mathematical. These ideas resonate with (Russell, (1912/2001, Ch. 8).

³ More precisely, 'relativistic causality' as it is usually called is the hypothesis that observed preferred orders of physical processes (say, first the throwing, then the shattering) are observer-independent (for inertial observers)—so are not altered by transformation between inertial frames. This hypothesis is withheld in general relativity, and leads to the rejection of superluminal signalling. I recommend as further reading the analysis of L. Motle (of 25.01.2011) in the following Stack Exchange discussion: <https://physics.stackexchange.com/questions/3871/is-causality-a-for-malised-concept-in-physics> (accessed 10.08.2024).

The ambitious and interesting analysis of causation by Ladyman and Ross (2007, Chapter 5) leads them to a view that comes close to Norton's, as they acknowledge themselves (2007, p. 272). Interestingly, they make a distinction between fundamental physics and special sciences, which include non-fundamental physics. (Ladyman and Ross take 'serious candidates for quantum gravity' to constitute fundamental physics. Since these theories aim at unifying general relativity and quantum (field) physics, but are not yet constructed in any final form, I include quantum and relativity theory in the fundamental theories—for definiteness, and as is usually done.) Referring, notably, to work by Earman and Roberts (1999), Ladyman and Ross state:

However, we think that Earman and Roberts are correct in their claim that fundamental physics discovers something of a kind that special sciences don't; and we call this kind of something a universal real pattern. We argued in the preceding sections of this chapter for Russell's and Redhead's thesis that these real patterns are not causal. They are instead structural, in the sense articulated in Chapters 2 and 3. In light of our agreement with Kincaid about what special sciences do, this means there are, according to us, two respects in which the achievements of fundamental physics differ from the achievements of special sciences: the former are exceptionless, and they do not consist in the identification of causal factors (2007, p. 283).

And a bit further:

Saying that special sciences furnish causal explanations, while fundamental physics aims at discovering universal laws [as Earman and Roberts (1999) do], comes very close to the view we articulated in 5.4. Special sciences, according to us, use causal ideas as heuristics for locating real patterns. Fundamental physics is in the business of describing the structural properties of the whole universe. These properties are not causal relations. Special sciences and fundamental physics are thus mainly different from each other in a way we find it very natural to express by saying: fundamental physics aims at laws, whereas special sciences identify causal factors (2007, p. 286).

(Note, first, that one can of course make an immediate link between our scientific realism equipped with theory relativism and Ladyman and Ross's 'ontic structural realism': our best theories reveal the laws of nature, and therefore 'structure' in this obvious sense.) Pace Ladyman and Ross, the account of causation provided in Chapter 5 shows that one does not need to distinguish between fundamental and non-fundamental science, and even everyday discourse, when it comes to analysing causation. These all lie in an epistemic continuum, as do the laws and regularities on which they are based. A unified approach is possible thanks to the notion of theory-relativity: these various cognitive contexts come with their theories*, and it is these theories* that dictate what the causes are according to **DEF-CF**. Of course, in the sciences these theories are based on laws, fundamental and less fundamental (a qualification that strongly depends on the historical epoch, for that matter); so it is these laws that will rule the causal attribution. The key point is this: the recipe immanent in **DEF-CF** works well also for laws presently considered as fundamental—take for instance Einstein's field equations of general relativity. These equations relate the local spacetime curvature (described by the Einstein tensor) with the local distribution of matter and energy (described by the stress–energy tensor); they can be written as a

set of nonlinear partial differential equations, i.e. deterministic equations. Although exact solutions to these equations can only be found under simplifying assumptions, they show that any relativistic variable X occurring in the equations is a *deterministic function* of other properties Y . Hence **DEF-F** or **DEF-CF** show that the mutual co-variation that is the essential criterion of causation is well present.

True, Ladyman and Ross make an interesting point, when they remind us that in fundamental laws the time-asymmetry typical for (mundane) causation is absent (2007, §5.4). They refer to theories for quantum gravity as String Theory or Loop Quantum Gravity. These theories are, however, unfinished, which makes the discussion somewhat undefined; but we can equally well refer to relativity theory, quantum mechanics or electromagnetism. To go back to the above example of general relativity, the dependence between relativistic variables X and Y is *perhaps* most generally interpreted as a mutual influence, so mutual causation, as captured by Wheeler's slogan "Spacetime tells matter how to move; matter tells spacetime how to curve" (recall the discussion in Chapter 5§3). To be clear, the potential worry against my account raised by Ladyman and Ross's interpretation is this: **DEF-F** and **DEF-CF** might not be applicable to general relativity and other fundamental theories in that a classic time-clause ("causes come before their effects") may not be trivial to apply here. What comes first, spacetime curvature or matter-energy?

Now, as already concluded in Chapter 5§3, if the time-clause only stipulates that effects should not come before their causes, the definitions allow for simultaneous mutual causation—which solves the problem. And more importantly, to solve the equations in practice, one needs defined initial conditions, and these do impose a temporal asymmetry (which is not deducible from the laws/equations themselves). For instance, given an initial distribution of matter and energy, as well as the initial/boundary conditions, the field equations determine the future evolution of the gravitational field and the motion of objects within that field. So it is natural, at least possible, to conclude that the given energy and matter distribution *causes* the evolution of the field and the motion of the objects under study. It is also natural to say that "the collision of black holes causes gravitational waves", etc.

Again, the time-clause surely is not the essential ingredient of causation (so many things occur before so many other things without being causes, after all). Finally, let me repeat that the notion of causality is important in general relativity too: the principle of 'relativistic causality', mentioned above in the context of special relativity, remains valid in general relativity and ensures that superluminal influences are excluded—as observed.

I believe we can conclude from this excursion that the time-clause does more work in pragmatic contexts, or in folk applications of the notion of cause, when the cause-inquirer is interested in solving practical problems or asks agency-related questions—such as the eternal: "Who or what did this?" In fundamental physics one wishes to identify (discover, derive) the pertinent equations and to apply and solve them; one is usually not obsessed by the practical question "what causes this?", neither by the philosophical question "is there a cause?". But, once again, that should not withhold us from doing philosophy and note the connection of necessity between properties, things, phenomena described by these laws, whether fundamental or less

fundamental. That is, I believe, the essential message contained in the principle of causality.

Let me emphasise that the causal principle is not the same as the principle of determinism, claiming that everything is caused in a deterministic manner (for reviews, see Earman, 1986; Hoefer, 2024). Philosophers of physics widely agree that physical systems and events are either deterministic or probabilistic; but it is undecided and a matter of debate whether physical events, systems, and variables ultimately *all* belong to *one* of these two categories. Among physicists there is perhaps still a lingering ‘orthodox’ position stating that the ultimate building blocks of reality are probabilistic (because quantum); but, as said, things are much subtler upon deeper scrutiny (e.g. Earman, 1986; Nikolaev & Vervoort, 2023; Vervoort, 2013, 2019; Wuethrich, 2011). Now, importantly, the above-mentioned principle of causation very well applies also to probabilistic systems. Laws of nature can be deterministic or probabilistic; and I have argued that the notion of probabilistic cause can surely be made solid. Thus, the principle can be made more general in this way: “All events have causes in accordance with necessary laws of nature, whether these laws are deterministic or probabilistic”. As argued in Chapter 5, there is necessity in probabilistic laws of nature too: in probabilistic systems it is *collections* of sufficiently similar systems that necessarily have to obey the laws of probability theory (cf. Chapter 4). Concomitantly, series of stochastic events (generated by a p -system) cannot be fundamentally erratic or wildly indeterministic: they too have to show frequency stabilisation, which is an iron law just as scientifically testable as any deterministic law of physics—even if the probabilities extracted from the frequency measurements come with an unavoidable uncertainty, as almost any experimental property. From probabilistic laws, say the laws of quantum mechanics, one can extract causal relations using **DEF-P** or its ilk, in essence: X is a cause of Y iff $\{(i) P(Y|X) \neq P(Y), \text{ and } (ii) \text{ there is no earlier ‘off-screener’}\}$ (see an illustration below). Here too one sees the necessity: the way Y and X occur must follow regularities. X is necessary for Y in this sense: if X would occur differently, i.e. with a different probability distribution, Y would, in general, occur differently (that is the meaning of (i)). And accordingly, many and perhaps all stochastic systems studied in physics are characterised by ensemble-equations that are perfectly deterministic: think of the equations of Schrödinger, Fokker-Planck, Liouville, etc.

Of course, in a spirit of synthesis, one immediately wonders whether these two kinds of necessity, respectively characterising deterministic and probabilistic systems, are not one and the same thing. Sure, we now enter into speculative territory, but this necessity in nature seems, to me, to be an utterly fundamental property: it seems to make probability theory *and* logic possible (as disciplines that describe reality). It seems to be a prerequisite not only of the universal nature of the laws of physics but also of the applicability ‘to reality’ of the theories of probability and logic. I have argued elsewhere that probability theory is, in a sense, the most fundamental physical theory we have (Nikolaev & Vervoort, 2023; Vervoort, 2019). If this feature of necessity is so fundamental, it is likely a unifying principle. It is then tempting to assume that probabilistic systems are ultimately deterministic ‘under the surface’—in deterministic systems the necessity has its simplest expression. The

necessary, rule-based nature of the behaviour of collections of ρ -systems would then derive from the deterministic necessity in the individual components; this would seem to be the philosophically most pleasing conclusion, complying with Occam's razor—there is only one category of events and systems, namely deterministic ones. But this matter may well be undecidable; physical science seems not to give enough guidance. At any rate, *it seems that the more science-compatible question is whether nature complies with the general principle of causality, comprising deterministic and probabilistic causation, rather than the question whether nature is ultimately deterministic or irreducibly probabilistic.*

Kant famously saw in the principle of causality a mere tool for organising our experience; a product of the synthetic faculty of imagination, rendering experience possible.⁴ In his steps many philosophers considered and still consider the principle at best as a heuristic maxim, a (more or less unavoidable) rule to guide scientific research (“search for the causes, in order to find the laws”). E. Nagel for instance claims that the principle is a “maxim for inquiry rather than a statement with definite empirical content” (1961, p. 320). But again, within scientific realism one quickly realises that such an interpretation does not seem to go to the core of things. Yes, the principle is also a heuristic rule of great validity, but for a good reason: *because it corresponds to something real*. If not, why would it be of such heuristic value, why would it organise our perceptions so well? I believe it is time to recognise the failure of philosophical theories that do not adequately answer this question, and the many questions related to it.

Surely the most cogent attack against the idea that everything has causes, comes from the interpretation of quantum mechanics, especially the interpretation that is often called the ‘orthodox’ or ‘Copenhagen’ interpretation, originally due to Niels Bohr and Werner Heisenberg. Famously, both these fathers of quantum mechanics were eager to deny that causation plays a role in quantum theory (e.g. Jammer, 1974). I will give here a very condensed review of my position on this matter, and I ask the reader to remember that I stroll here far outside the scope of this book. In the next paragraphs my goal is to show (to readers interested in the philosophy of physics) that the philosophical results of the previous chapters have an intimate link with topical research in physics, and suggest new interdisciplinary research projects.

For starters, two short remarks. First, few experts in the foundations and philosophy of quantum mechanics still believe that the Copenhagen interpretation has prevalence over other interpretations; the weaknesses of this interpretation have now been studied in detail (cf. e.g. Becker, 2018). And second, there is an interpretation, or rather an equivalent re-formulation of quantum theory, namely the de Broglie-Bohm theory, which explicitly assigns causes to quantum events (Bacciagaluppi & Valentini, 2009; Holland, 1995). This theory is sometimes called the causal interpretation of quantum mechanics.

⁴ It is noteworthy that Kant dedicates a remarkable number of pages of the Critique of Pure Reason—to his claim stating that although causality is not in the real (noumenal) world, we necessarily reason as if it were. This idea hurts the realists’ feelings.

Now a little more substance. The debate of whether (hidden) causes exist for quantum events is equivalent to the ‘hidden-variables’ question—i.e. the question whether (yet unknown) physical properties could exist that explain quantum theory. The hidden-variables question is most clearly analysed in Bell’s theorem (cf. Chapter 5§3). The meaning of this theorem can be summarised thus, according to the standard view: *only nonlocal hidden variables could exist (and reproduce quantum mechanics)*. But nonlocal causes violate Lorentz-invariance and are thus in tension with relativity theory, as explicitly emphasised by John Bell⁵; they are therefore considered unlikely by many. (The de Broglie–Bohm theory is considered almost always as nonlocal, in agreement with the standard interpretation of Bell’s theorem: the causes it assigns to quantum events involve nonlocal interaction. And indeed, although this is a topic of ongoing research, there are difficulties to make the de Broglie–Bohm theory relativistic. Yet, I am not sure whether this strongly nonlocal nature is the correct conclusion, or rather an overhasty interpretation of the rather unintuitive mathematics, as briefly explained below.)

Now, the interpretation of Bell’s theorem I just gave is the standard one. But it is highly debated (e.g. Myrvold et al., 2024; Vervoort, 2013). I admit that I suspect that this standard interpretation is erroneous (unless one understands ‘locality’ in a different way than is usually done)—ultimately always for synthetic reasons. In order to come to the standard view that “only nonlocal hidden variables could exist”, one has to make an extra assumption that is usually left implicit because considered trivial, an assumption often called ‘free will’ (!). In technical terms this is the assumption that there is no correlation between the hidden causes (called λ) of the quantum properties and the detector settings (say α) in experiments. I will call this the ‘no-correlation’ argument or hypothesis. Bell and many people have claimed that assuming that there would be correlations (due to common causes⁶) between λ and α , would violate the assumption of free will. How could there be common causes between some particle property λ and detector setting α , if the value of α can be freely chosen by an experimenter? That seems to amount to a cosmic conspiracy in the eyes of many.

But is this argument based on free will sound? I believe that, to counter it, *we can invoke exactly the same reasoning we used to invalidate absolute free will in Chapter 6*. If everything is ultimately material, physical, then also the choice α in the head of some experimenter is part of the physical. The neuronal processes that embody the choice cannot pop out of nothing, and must *in fine* be correlated with what happened at the Big Bang, according to an identical argument as we made in Chapter 6. The same must be the case for the properties λ . Both α and λ must

⁵ Indeed, this is how Bell concludes his seminal paper of 1964: “In a theory in which parameters are added to quantum mechanics to determine the results of individual measurements, without changing the statistical predictions, there must be a mechanism whereby the setting of one measuring device can influence the reading of another instrument, however remote. Moreover, the signal involved must propagate instantaneously, so that such a theory could not be Lorentz invariant” (Bell, 1964).

⁶ I do not consider the possibility of retro-causation (and other more exotic possibilities), as in the rest of this book. Indeed, if our conclusions in Chapter 5 are correct, the philosophical problems connected with causation can be solved while assuming good old forward causation. Retro-causation is defended for instance in Price (1996).

therefore be caused by, correlated with, somehow connected with, the variables that describe the Big Bang (in full agreement with our concept of cause, **DEF-CF** or **DEF-P**). Ergo, there is a correlation, a connection, between λ and α , after all. This ab-initio correlation can, of course, not be calculated (nor measured). We do not have a Theory-of-Absolutely-Everything at hand, but it is still natural to assume that such a theory exists in principle. Then it is also only natural to assume that the events happening at the Big Bang (or, if one prefers, the properties of the elusive ultimate theory describing it) are the common causes of λ and α (this idea is elaborated in somewhat more detail in Nikolaev & Vervoort, 2023; Papatryfonos & Vervoort, 2025, Sect. 4). Note that it is good practice in physics to assume the existence of properties even if we do not have access to them. That is how statistical mechanics and statistical thermodynamics can be constructed: these theories assume the existence of particles having positions and velocities, even if we do not know them.

(At this point, let me risk the following side-remark: the solution I propose here seems straightforward to me. To the point that I sometimes suspect that people have overlooked it for psychological reasons. There is a tenacious mistrust in the physics community of anything that is non-calculable, of anything that looks more philosophical than physical. Indeed, the assumption that everything is correlated with what happened at the Big Bang is not strictly part of physics; one cannot calculate these correlations; it is philosophical, if one wishes. And yet, it is an utterly natural hypothesis, explicitly or implicitly made by scientists and philosophers all the time. So, to me, there is a puzzling element in this debate. At the same time, I am well aware this matter is unusually subtle; so a dose of agnosticism seems advisable.)

Now, if one rejects the free-will or no-correlation assumption one rejects the standard interpretation of Bell's theorem: then there *can* be causes λ of quantum properties after all, not only nonlocal ones. The causal principle can then well be upheld, Q. E. D. One can also say: *the causal principle gets further corroboration by offering a solution also to this problem of the interpretation of Bell's theorem.* The interpretation that I have briefly sketched here is often called 'superdeterminism' (because it posits that freely-chosen variables have causes retracing to the Big Bang); but it is actually simply a (consequence of a) modern variant of the causal principle. Note also that I do not need the assumption that the λ are deterministic variables; they may be probabilistic. Therefore I prefer the term 'supercorrelation' as coined in (Vervoort, 2013) or 'contextuality' (cf. Papatryfonos & Vervoort, 2025, Sect. 4).

In sum, within my interpretation, *Bell's theorem is a consequence of physical reductionism, or, equivalently, of the principle of causality, and expresses the universal connectedness of things, rather than the nonlocality of quantum mechanics and nature.* What is mind-boggling about the theorem is then that it allows us to detect this ab-initio connectedness of (almost) everything. (The idea that the principle of causality implies universal connectedness, i.e. universal dependence, is not new and can for instance be found in Mach's writings,⁷ see a condensed history in Bunge

⁷ Mach wrote (around 1883): "The business of physical science is the reconstruction of facts in thought, or the abstract quantitative expression of facts. The rules which we form for these reconstructions are the laws of nature. In the conviction that such rules are possible lies the law of

(2009), pp. 252–253.) It is important to note that this interpretation does not rely on nonlocality. Relying on the causal principle has the great advantage of doing away with this cumbersome notion, which is in immediate tension with relativity theory.⁸ Many people have come to believe that Bell’s theorem and the experimental results show that quantum theory *itself* (and therefore nature) is nonlocal. This seems a stark inference of Bell’s theorem, even if we consider its orthodox interpretation, which only states that local hidden variables are impossible, not that nature is therefore nonlocal (for a position embracing nonlocality, cf. Maudlin, 2011). What gives extra credit to my interpretation, is that the alleged nonlocality of quantum mechanics and of de Broglie-Bohm theory is utterly mysterious: *it cannot be used to superluminally influence, to change the quantum statistics of some system at a space-like separated interval*, as is well-known (e.g. Myrvold et al., 2024, §8.2). What kind of nonlocality is this then? It is instructive to realise that even Bell changed his view on nonlocality (Myrvold et al., 2024, §8.3). Despite more than half a century of research, it is still not clear whether the ‘nonlocality’ of quantum physics induces an insuperable incompatibility between the two pillars of physics. On my view, it doesn’t. On my view, nonlocality can be explained and replaced by a notion with a clear and bona-fide meaning, namely universal connectedness or non-separability (or universal correlation), which, in turn, is a direct consequence of the causal principle. On this interpretation, the alleged nonlocality of quantum mechanics and de Broglie-Bohm theory would amount to an optical illusion of sorts. But let us remain cautious: only a unification of both theories should provide the final verdict.

A causal interpretation of Bell’s theorem has important logical consequences, obviously for philosophy, but also for physics. In philosophy, embracing the causal principle led, to my taste, to one of the most powerful philosophical systems of all time, namely Spinoza’s philosophy (cf. some comments in the next section). In physics, it has a clear heuristic role: *it incites to look for new local hidden-variable theories, compatible with relativity theory*. If the unification of quantum mechanics and general relativity theory remains the Graal of physics, then the result that local hidden-variable theories are possible after all, is obviously good news—since it seems difficult to unify a local theory as general relativity with a nonlocal one. What I have in mind here are *effective* hidden-variable theories that violate the free will or no-correlation condition (and hence bypass Bell’s no-go theorem) while this violation can be explained by the real but hidden and intractable common causes describing the Big Bang or other early times (more in Nikolaev & Vervoort, 2023; Papatryfonos & Vervoort, 2025). There are several candidate theories that, even if unfinished, seem to comply with this scheme. The de Broglie–Bohm theory (Holland, 1995), the ‘double-solution’ theory of de Broglie (Jamet & Drezet, 2024) and theories à la ‘stochastic electrodynamics’ (De la Peña et al., 2015) might indicate an

causality. The law of causality simply asserts that the phenomena of nature *are dependent* on one another” (cited by Bunge (2009), pp. 252–253).

⁸ The reasoning behind the derivation of Bell’s inequality makes it clear that it can only be violated either by nonlocal, i.e. superluminal, hidden variables or hidden variables violating no-correlation (perhaps there are more exotic options, but they seem unlikely). Again, nonlocality means: involving superluminal influences.

interesting direction to take. These theories hypothesise the existence of a new property, a (hidden) pilot-wave or background field, that is supposed to explain quantum mechanics. This extra field does depend on the detector settings (see e.g. Holland, 1995, p. 78–79), so that Bell’s no-go theorem is indeed circumvented. And, as said, I doubt that this violation of the no-correlation assumption should be called nonlocal or superdeterministic—at most ‘contextual’ or ‘soft-superdeterministic’ (Nikolaev & Vervoort, 2023; Papatryfonos & Vervoort, 2025), a harmless variant. Such effective theories can be perfectly physical (de Broglie-Bohm theory surely is) and explain quantum mechanics (perhaps the de Broglie-Bohm theory can be elaborated to bring this explanation to its logical end), while they are, on my hypothesis, justified by a hidden ‘superdeterministic’ substratum that we only need as a heuristic or explanatory assumption—or, of course, for our philosophical understanding. Again, in a similar way classical mechanical particles and classical mechanics are presupposed in order to explain and construct statistical thermodynamics, even if the individual mechanical properties are intractable. Other hints that de Broglie–Bohm theory is in fact local come from the so-called hydrodynamic interpretation of quantum mechanics (Vervoort & Gingras, 2015).

It seems to me that this same scheme of reduction also applies to other theories aiming at deriving quantum mechanics via violation of no-correlation (Donadi & Hossenfelder, 2022; ‘t Hooft, 2016, 2021). Especially relevant is the ‘Cellular Automaton’ theory of Nobel laureate G. ‘t Hooft, (2016, 2021), who has invested sustained efforts in constructing a theory that could, ultimately, unify quantum mechanics and general relativity. The Cellular Automaton theory is always deemed superdeterministic; but of course no attempt is made in this theory to derive the choices of experimenters starting from the degrees of freedom of the Big Bang! Again, it seems to me that the superdeterminism of this theory is entirely physical and harmless, against charges made in the literature (these charges are answered in e.g. Andreoletti & Vervoort, 2022; Hossenfelder & Palmer, 2020). One is left wondering whether the negative connotations that many physicists and philosophers still entertain about superdeterminism have unduly diminished the interest for this essential work by ‘t Hooft and others.

But let us return to the principle of causation. The main upshot of the above excursion in quantum mechanics is that the causal principle can well be maintained, also taking quantum theory into account. It is not that the principle needs to be saved with artificial assumptions, in a nostalgic move to save an old principle. There are positive reasons: the principle not only incites to do further research in quantum physics as just argued⁹; I believe it has also a greater explanatory power than its competitors—even if restricting the discussion to physics and its philosophy. If correct, this would be the main reason why it deserves to be accepted as a plausible candidate, and indeed the best provisional solution. I believe the causal principle can not only explain the mysterious ‘nonlocality’ of quantum mechanics along the lines sketched above; it can

⁹ The competing assumption, so the ‘no-hidden-variables’ assumption, seems to be an obstacle to the unification of quantum mechanics and general relativity.

also answer foundational questions of probability theory. This would bring a particularly weighty argument into the balance. Indeed, above I have suggested at a few occasions that probability theory is, in a sense, the most fundamental physical theory we have. It has a broader scope than quantum mechanics and classical statistical theories, since all these theories comply with probability theory. Moreover, general relativity theory is a special case of a probabilistic theory with probabilities 0 or 1. Now, according to the general view on theory-building and problem-solving exposed in Chapter 1, any meta-principle that can coherently answer questions raised by this encompassing theory, should, a priori, be of a unique unifying strength. And indeed, there are fundamental questions raised by probability theory that, to the best of my knowledge, can only be answered by the causal principle. I will just mention these questions here, but not repeat my answers (cf. Nikolaev & Vervoort, 2023; Vervoort, 2019). The first question is: How to interpret probability and statistical correlation in a coherent manner, valid for quantum and classical systems? The second question is: How can one explain the ubiquitous occurrence of the Gaussian probability distribution in classical and quantum systems? As far as I know, if one rejects the causal principle one cannot answer these fundamental questions—just as it becomes difficult to interpret Bell’s theorem.

(The above considerations also suggest following idea that deserves, I believe, further investigation: could it be that nonlocality, superdeterminism, contextuality, and retrocausality are just superficially different sides of the same multifaceted coin: different conceptual interpretations of counterintuitive mathematical theories and theorems? This idea is entertained in Papatryfonos and Vervoort (2025), where it is argued that causality offers the better synthetic prospects.)

Before concluding, I would add one less important point. We just had a look at the question of whether quantum variables could be caused by, could be emerging from, deeper-lying properties—a question we scrutinised from the point of view of some alleged best theory, some absolute angle, as is always done (this led to the abstract notion of hidden-variable theory and Bell’s analysis). But, interestingly, one can even answer the question by looking from the angle of quantum physics itself (which is often believed to be more provisional). Indeed, in Chapter 5 it was shown that causal identification is in principle representation-relative. So, the question “Does every event have a cause?” could in principle be understood relative to some specific theory, say quantum theory—even if it is clear that normally the best possible theory is targeted. (If we would append the Copenhagen interpretation to quantum theory, the doctrinal answer would be ‘no’.) Now, *even within quantum mechanics itself it is well possible to attribute causes to quantum events—causes that are in agreement with DEF-P of Chapter 5*. Quantum events are probabilistic, and we have seen in Chapter 4 that probabilities are expressed as functions of the measurement context, possibly representative of some natural environment. More precisely, it was argued there that any probability can be understood as a conditional probability, conditional on the variables describing the (measurement or natural) context, cf. Chapter 4, §3 and §5 R3. This immediately implies that the variables describing this context qualify as causes, according to DEF-P. Let us look at an example. The Bell-correlation $P(s1, s2|a, b)$ —the paradigmatic example that is studied to show that the spins or

polarisations (s_1 , s_2) have no causes!—can well be understood as expressing that the detector settings (a , b) are causes of (s_1 , s_2). Indeed, (a , b) are set before the measured values (s_1 , s_2); and within quantum mechanics there are no off-screening common causes between (s_1 , s_2) and (a , b). Hence, by **DEF-P**, (a , b) are causes of (s_1 , s_2) *according to quantum mechanics*! This verdict does not hurt our intuition: the measurement specified by (a , b) involves an interaction with the particles and is thus one of the causes of the measurement result. A similar reasoning can be held for any quantum event, characterised by a quantum probability. In sum, if one accepts **DEF-P**, even within quantum mechanics the (probabilistic variant of the) principle of causality is valid, against what is often believed.

This concludes my short and surely incomplete defence of the principle of causation/causality. I have to admit that the arguments amassed above in favour of the principle leave me with a sense of bewilderment. There is a strong current against the principle, prevailing notably in philosophy of science and (philosophy of) quantum physics,¹⁰ where causality is almost universally rejected. Yet, I can only repeat that this principle is an immediate consequence of two assumptions that seem utterly cogent by now: the existence of a synthetic definition of cause, and physicalism, the assumption that the ultimate building blocks of the universe are physical—a hypothesis that can hardly be surpassed in generality and acceptance. Unless I overlooked something essential, this gives the principle of causality an air of inevitability as a philosophical principle. As said, for its explanatory power in philosophy, one of my favourite references is Spinoza's philosophy, briefly commented on in the next section. But in physics too, the principle has explanatory and in any case heuristic potential. One can imagine several lines of interdisciplinary research, some mentioned above, which would make little sense without assuming the principle. The 'no causality' or no-hidden-variables hypothesis appears remarkably arid in comparison. It seems to dictate that physics ends with quantum mechanics, that quantum mechanics cannot be reduced to a more fundamental theory. But until now science has refused to die.

9.2 Agency in a Causal World. Consequences of the Non-existence of Absolute Free Will

In this section I consider some lines of further research starting from the findings obtained in Chapters 6 and 7 on free will. There I argued that any human decision and choice is dependent on—i.e., caused by, necessitated by—previous external causes. I argued that this conclusion is a direct consequence of physicalism.

¹⁰ As only one example, in some recent quantum experiments it seems that the cause-effect order can be reversed or is indefinite (Goswami et al., 2018). But these results are disputed (Vilasini & Renner, 2024). Recall that the time-clause in my definitions allows for simultaneous mutual causation; and it is not the most essential component; there is a more fundamental component.

As is well known, the non-existence of absolute free will seems to have profound consequences for a variety of chapters of practical philosophy, ethics and philosophy of law, related to moral responsibility, the practice of blame and punishment, etc. For my quick survey in this Epilogue, I will start by noting that most of the latter disciplines and concepts concern or have an intimate relation with the concept of happiness, so with human feelings—or so they can be construed. So does a principle I wish to promote here, even if cursorily; a thesis I believe to be of great importance. This well-known principle, a version of utilitarianism, states that *the ultimate criterion to judge ethical acts is the happiness or wellbeing they procure*. (I will not go in any depth into the meaning and possible implementation of the maxim.) Exercising blame or praise, imposing punishment, etc. have impact on happiness, wellbeing of individuals and society. Therefore, when one wishes to connect a theoretical analysis of the concept of free will to the best theories for such humanitarian and social concepts (responsibility, praise, punishment,...) and their respective disciplines, one needs to take into account psychological, feeling-related facts. In other words, what one derives of a *theoretical* model of free will (I tried to provide the philosophical core), must be linked to, and have relevance for, psychological, feeling-related realities and their conceptualisation, which are studied in *practical* philosophy and human and social sciences. This means that my bi-perspectival model must be integrated in, made relevant for, psychological and social facts that are not fully theoretically understood neither fully rational, but governed in large part by feelings and emotions. I conjecture that the principle that allows, ultimately, to make this integration—this next step of synthesis—is that of eudaimonia or happiness maximisation.¹¹ I submit that integrating the absolute view on free will in practical philosophy should and can be done with as ultimate guiding rule, or aim, the maximisation of global and individual eudaimonia or wellbeing.

At any rate, it seems to me that this constraint of practical philosophy, this feeling-ladenness of psycho-social reality, explains why the relative view on free will remains most important for most practical issues; and why the absolute view can only help in *moderating*, rather than revolutionising, more classic views based on the relative perspective. Let me repeat: the *felt* experience of having choices is very real. As we all know, feelings, emotions, perceptions, leave often a much stronger impression of ‘realness’ than theoretically known things. Eating this pie feels decidedly good, even if I know it is not beneficial for my health. Thus, it seems we should not hope for excising or radically changing a (well-informed) view of relative free will in practical philosophy, neither in our usual attitudes and habits of life. Perhaps one can see the (gradual) incorporation of the absolute view as an ideal that advanced societies should strive to. And I dare to add, as a tribute to philosophers as the Stoics

¹¹ Recall that in natural science, general principles are a frequent, maybe the ultimate, motor for theory-synthesis. For instance, Einstein developed his relativity theory by applying symmetry principles to the lower-level Newtonian mechanics. Thus he crafted a theory with wider applicability starting from a more restricted one, by making the latter obey some general principles. That seems an inspiring approach also in philosophy.

and Spinoza: not only societies, but also individuals, if eudaimonia, wellbeing, or at least peace of mind,¹² is what they are after.

Indeed, as several philosophers have argued, the absolute view incites us, for instance, to be moderate in praise and blame; to refrain from excessive punishment; to ban capital punishment; to reorient retributive punishment as much as possible to re-education (as some countries already do); to replace hatred by understanding; etc. (As said, in this Epilogue I will express my opinion in a sketchy way. All these ideas have been explored by many philosophers; all these ideas are obvious to very many citizens all over the world; and all these ideas are unbelievably wrong to others. My point is that they are linked to fundamental synthetic theses, so that they could probably be studied using the tools of synthetic philosophy.) Thus, the absolute view teaches tolerance, towards others and ourselves—not to find excuses for the inexcusable, which would in the long run decrease wellbeing, but to find some solace when one faces the *fait accompli*, or when all reasonable efforts have had adverse effects. Notice that, in this way, our view professes nothing more than what ancestral wisdom surely has taught since prehistoric times to humans in countless places. Mild forms of blame and punishment might be justified for the well-known reasons of deterrence. In this context, let me refer for instance to D. Pereboom's lucid analysis of how hard determinism can well be compatible with normal interpersonal relationships and morality (Pereboom, 1997, Ch. 18). This is how Pereboom's classic article ends—I hope the close resonance with the view expressed above and in Chapters 6 and 7 is clear:

Given that free will of some sort is required for moral responsibility, then libertarianism, soft determinism [compatibilism], and hard determinism, as typically conceived, are jointly exhaustive positions [...]. Yet each has a consequence that is difficult to accept. If libertarianism were true, then we would expect events to occur that are incompatible with what our physical theories predict to be overwhelmingly likely. If soft determinism were true, then agents would deserve blame for their wrongdoing even though their actions were produced by processes beyond their control. If hard determinism were true, agents would not be morally responsible—agents would never deserve blame for even the most cold-blooded and calmly executed evil actions. I have argued that hard determinism could be the easiest view to accept. Hard determinism need not be of the hardest sort. It need not subvert the commitment to doing what is right, and although it does undermine some of our reactive attitudes [such as deep resentment for wrong-doing], secure analogues of these attitudes are all one requires for good interpersonal relationships. Consequently, of the three positions, hard determinism might well be the most attractive, and it is surely worthy of more serious consideration than it has been accorded.

In the economic and political sphere too the implications are, potentially, of enormous weight, if they can be enacted effectively—and it will be obvious that this a fantastic challenge. But is this a reason not to work on it? The absolute view teaches that all things and events are correlated with the events at the Big Bang. There are therefore, in a sense, no events and choices really up to us, really originating in us. And at the same time all events are, for the same reason, interconnected, correlated among

¹² In times of hardship, happiness is not the ideal that can realistically be strived to, but peace of mind might be.

each other¹³: an idea that was already part of ancient, notably oriental and Greek, philosophies, for instance certain schools of Buddhism. These ideas suggest that there is a sound perspective from which earthly success and wealth are the fruits of the working of the universe—chance, one might say—and only in a relative sense of the actions of successful and wealthy people. The universal causal determination of the absolute view is manifesting itself in everything, but in this case most visibly in the good genes, the good upbringing, the good friends, the good brain, etc.—at least one of these—that the successful and wealthy enjoyed having at some point in their lives. As all things related to free will, success and wealth should also be approached in a bi-perspectival way. On the political and economic level, it seems to me that the conclusion can only be that advanced societies should strive to more equity (perfect equality seems not realistic¹⁴), in order to compensate for misfortune. And this should be done by the most effective means, for instance by a more just taxation than is now in vigour (Piketty, 2014).

I do not need to remind the reader that our world urgently needs solutions: we face enormous challenges—think of the problems due to the global climate change, which I believe are intimately related to the inequity problem. Everything here is a matter of balance, where eudaimonia maximisation seems to me to be the ultimate guiding principle. I conjecture that future AI (deep learning / machine learning) can become extremely useful in assisting us in making strategic decisions, including in politics and economics; I submit that it could do so by applying the principle of eudaimonia maximisation. Here is one more ambitious line of further research, perhaps already started by some. Of course, whether those in power will accept to be guided by AI in political and other decisions is a wholly different matter.

Along the same lines, to those who find the idea of physical reductionism depressing, I would say the following. The partial control we have over our lives, relative free will, is very real: it is real to our feelings, even if it is not to our theoretical understanding. The felt sense is, in the practice of life, surely the most important. So, by all means, let us continue living pro-actively, with the motivating feeling of being the authors of our existences. After all: *our decisions and actions do make differences in the world*—that is difficult to deny. And they make differences in wellbeing, the key guiding principle; therefore they are, in principle, assessable, ‘measurable’.¹⁵ Thus, there are quite objective indicators that can tell us if we are on the right track

¹³ Correlations can, in practice, most of the time not be detected, but under certain conditions they can; and they can even be detected in a numerically precise way (see examples in Nikolaev & Vervoort, 2023).

¹⁴ Here is one example of how more equity can be promoted in a very practical way. The author has worked for a well-known German company, Carl Zeiss, where, until about twenty years ago, the highest salary (of the best paid employee, the director) could be at most ten times higher than the lowest salary. Carl Zeiss, a paragon of German technological innovation, was constantly elected as one of the best companies to work for in Germany. Alas, contemporary economic discourse has convinced people that the admirable ‘factor of ten’ rule is not realistic anymore in modern times.

¹⁵ For practical philosophers who wish to test their ideas empirically, it is pleasant to know that happiness, peace of mind, etc. can be assessed, ‘measured’ by various experimental methods of psychology, neurobiology, molecular biology, etc. So, for instance, the hypothesis of Spinoza and the Buddhists that altruistic actions are, on average, increasing even the authors’ wellbeing—an

with these actions: we can obtain feedback in different guises. As said, I believe the ultimate touchstone is our happiness, or its correlates, say peace of mind, more realistic in times of hardship. My view is therefore all but fatalistic; fatalism likely does not bring us much good. Much to the contrary, we saw that our free will can grow, can be made more effective, notably by learning, by rendering the perspectives with which we look at the world and ourselves sounder. Contrary to what some philosophers have claimed, in a solid bi-perspectival view hope is warranted: the hope that I will grow as a human, for instance by educating myself, by learning. And the hope that the universe, or fate, will be reasonably favourable to me.

Of course, there is a kind of subtle balancing advocated here between two perspectives. But that is also done in science—recall Bohr’s complementarity –; and every parent and every teacher knows that some constructive forms of illusion are helpful, nay needed. Just as in science, in a more elaborated theory, in a next stage of synthesis, this balancing might be given an entirely solid basis.

As a tribute to Spinoza, it seems to me that I can hardly do better than to cite O’Connor and Franklin from their condensed summary of Spinoza’s view on free will—the philosopher who has given, in his *Ethics* (1677 [1992]), perhaps the most systematic philosophy based on determinism—or what I would call physicalism in modern terms. Here is the apt summary (O’Connor & Franklin, 2022, § 1.2):

[Spinoza] endorses a strong form of necessitarianism in which everything is categorically necessary as opposed to the conditional necessity embraced by most compatibilists, and he contends that there is no room in such a world for divine or creaturely free will. Thus, Spinoza is a free will skeptic. Interestingly, Spinoza is also keen to deny that the nonexistence of free will has the dire implications often assumed. As noted above, many in the modern period saw belief in free will and an afterlife in which God rewards the just and punishes the wicked as necessary to motivate us to act morally. According to Spinoza, far from this being necessary to motivate us to be moral, it actually distorts our pursuit of morality. True moral living, Spinoza thinks, sees virtue as its own reward (Part V, Prop. 42). Moreover, while free will is a chimera, humans are still capable of freedom or self-determination. Such self-determination, which admits of degrees on Spinoza’s view, arises when our emotions are determined by true ideas about the nature of reality. The emotional lives of the free persons are ones in which ‘we desire nothing but that which must be, nor, in an absolute sense, can we find contentment in anything but truth. And so, in so far as we rightly understand these matters, the endeavor of the better part of us is in harmony with the order of the whole of Nature’ (Part IV, Appendix).

I wholeheartedly agree. Spinoza does not dwell on ‘possible worlds’ and the idea that ‘things could have been otherwise’, or that ‘(the laws of) nature could have been otherwise’. We live in *this* world, ruled by causal laws that embed or imply necessity, as I argued in the previous section; and it is not conducive to insight—nor to eudaimonia (!), according to Spinoza—to assume contingency in events or laws. I doubt that the popular assumption that ‘laws are contingent’ solves problems in a maximally coherent way. (Sure, laws can be said to be logically contingent; they could be different from a purely logical point of view; but such a hypothesis has no bearing at all on the universe as it is.) To the contrary, as the reader will have

idea counterintuitive to some –, should be empirically testable. Of course, this demands tightly controlled, complex experiments, which seem not frequently done.

understood, it seems to me that Spinoza's necessitarianism offers coherent answers to a multitude of problems in a most parsimonious manner. It is simple and radical, and does not introduce redundant categories and hypotheses.

To end on a bi-perspectival note: things happen as they must; but we can, and often should, be part of the action.

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